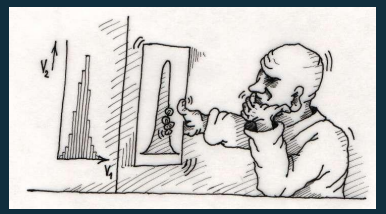
01

生物统计学：

生物信息中的概率统计模型

课程导论



2026年秋

有关信息

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Phone: 18627968927

课程网页

http://www.microbioinformatics.org/Biostatistics.html

QQ群：1130392068

考评

10%

课堂讨论

20%

课后作业&随堂测验

70%

终结性考试

课程成绩=课堂讨论（10%）+课后作业&随堂测验（20%）+终结性考试（70%）

LECTURE 01

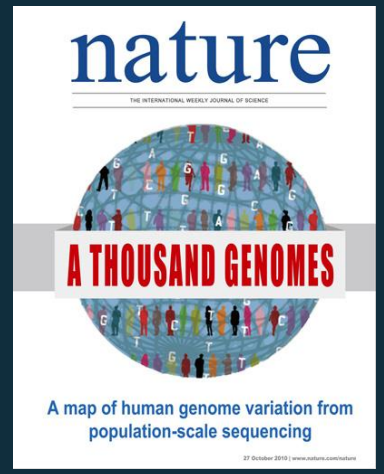
生物统计学：生物学视角

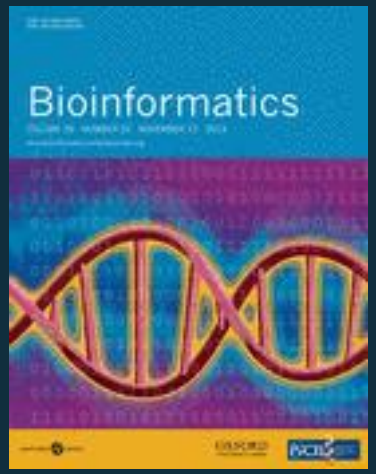
生物信息学 @HUST

Ideogram
Contig
HsUni6
Genes_seq
Symbol
ASB11
PHKA2
MAGEB1
FTHL17
PHF8
LOC169981
NLGN3
SLC16A2
ELF4
LOC340581
LOC266694
SPANXC
MTMR1
CETN2
ATP2B3
生命是序列的，生命是数字的！

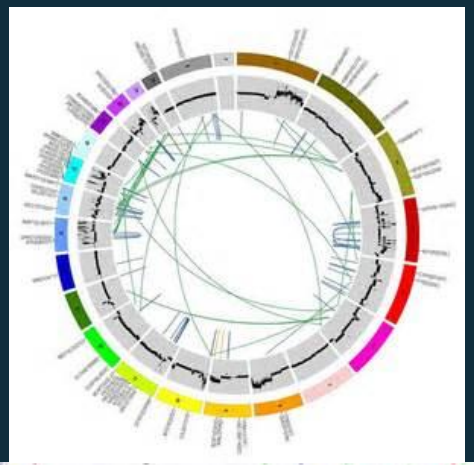
nature
Individual genomes from Africa and China
Acute myeloid leukaemia genome
Designer nucleases for gene therapy
Tracing gene flow across Europe
BREAK GLASS WHEN READY
YOUR LIFE IN YOUR HANDS
Instructions for the personal genome age
NATUREJOBS
Mentor awards

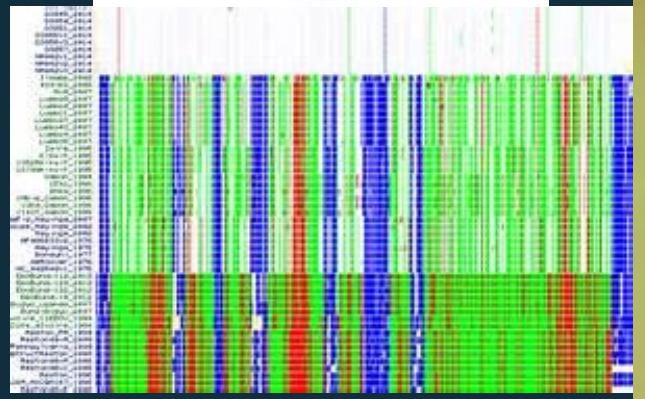
Science

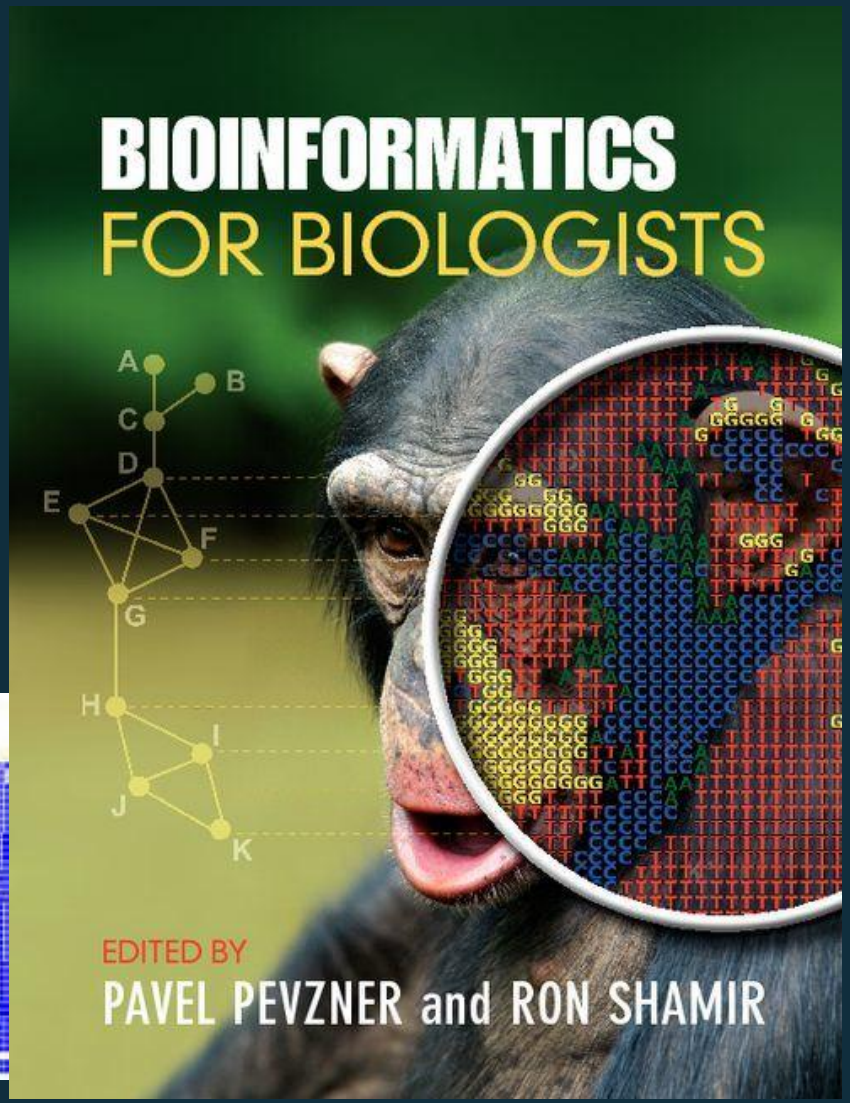

nature
A THOUSAND GENOMES
A map of human genome variation from population-scale sequencing


Bioinformatics

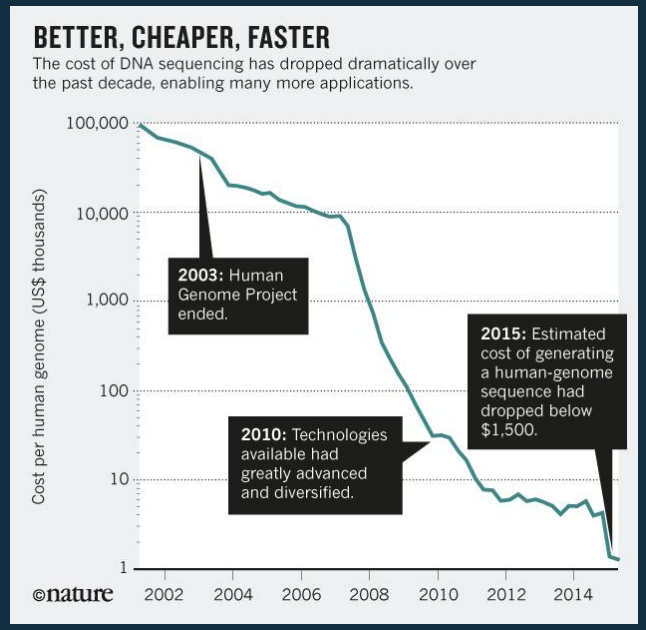
生命是序列的，生命是数字的！

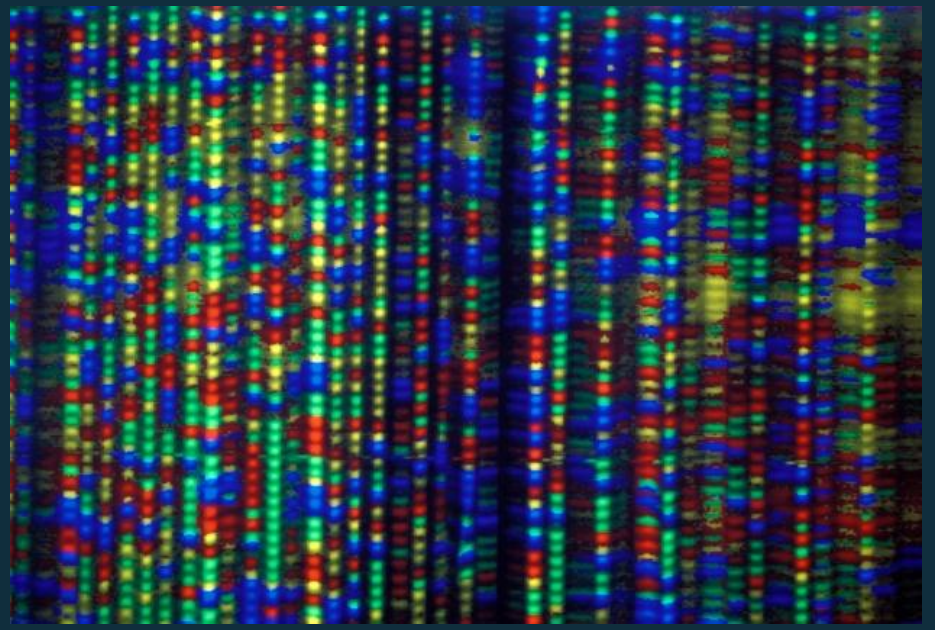




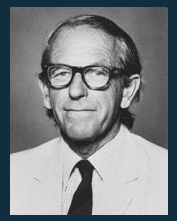


DNA测序与生物信息学

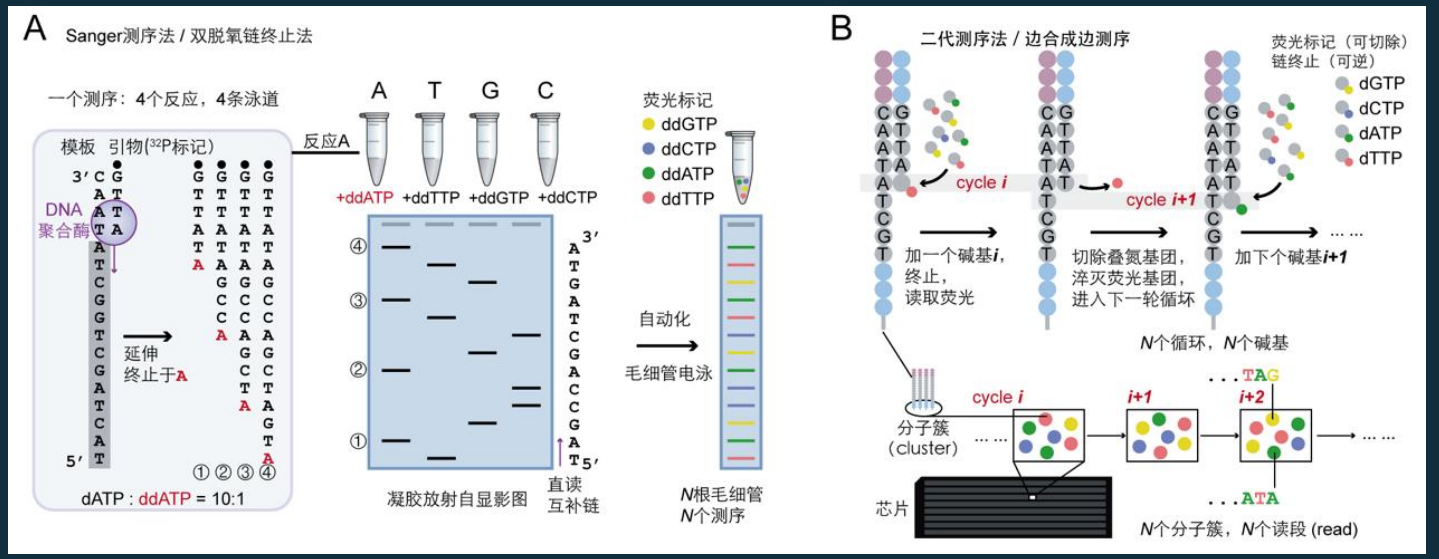




测序原理及方法



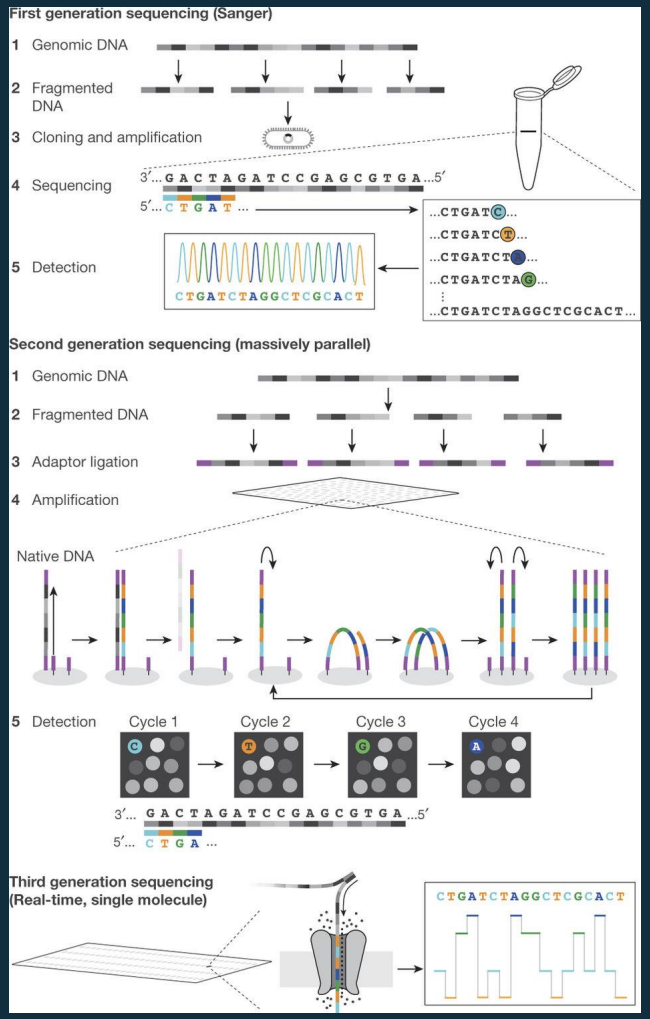
Frederick Sanger (1918-2013)
Nobel Prize in Chemistry (1958, 1980)

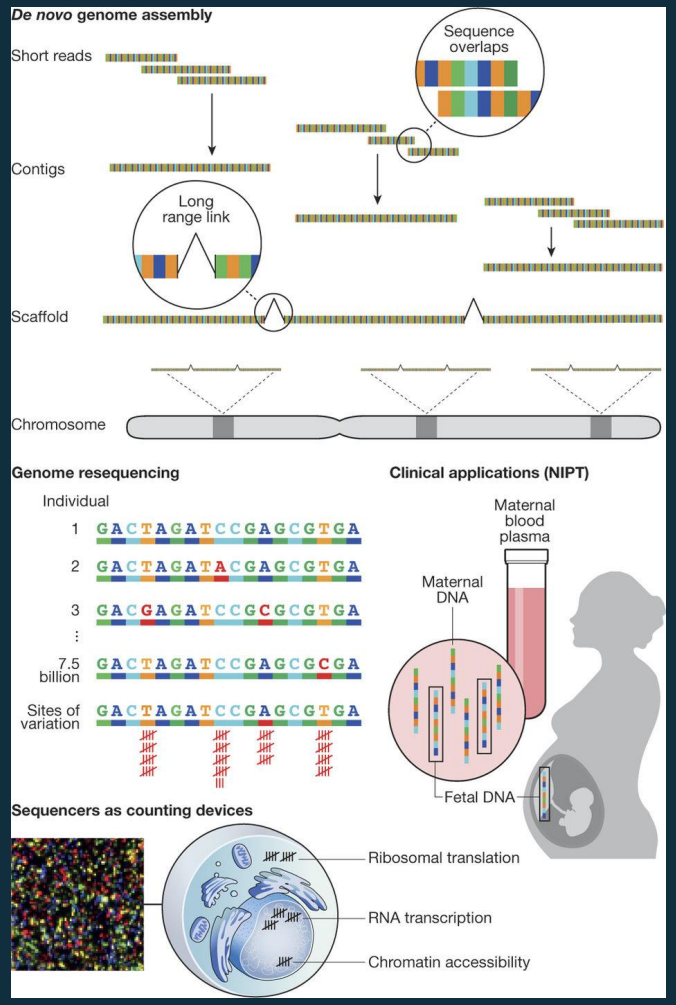


(A)Sanger测序法基本原理及其毛细管电泳自动化。(B)二代高通量并行测序原理。

Sequencing applications

DNA Sequencing

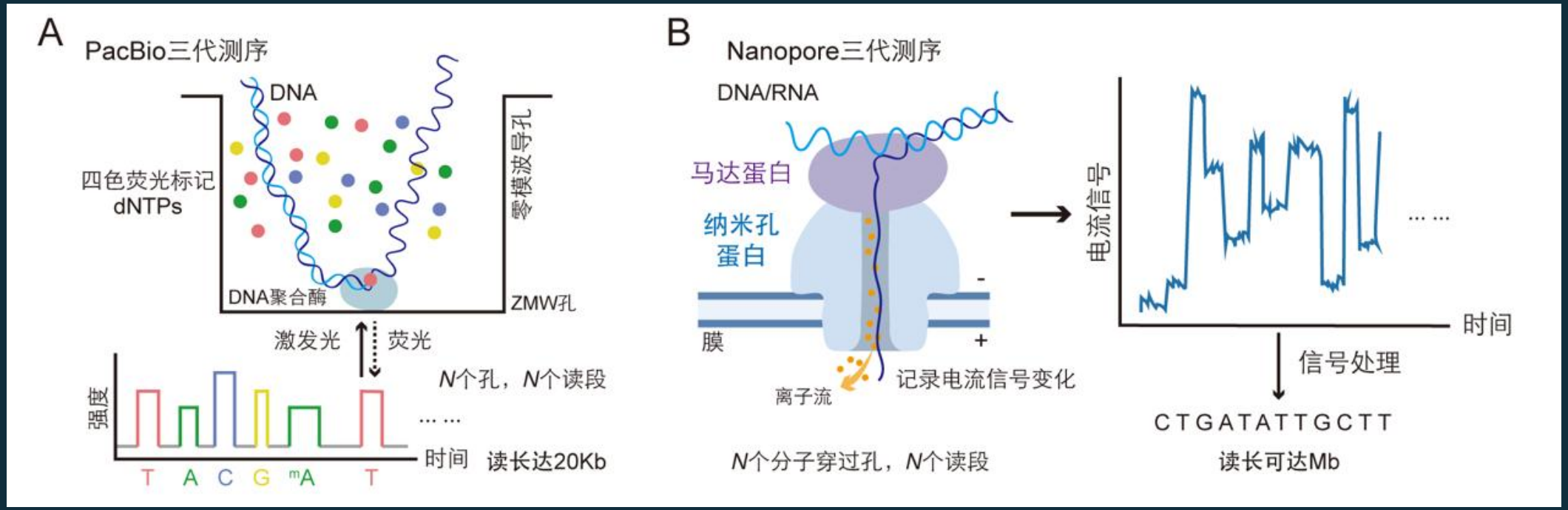




测序原理及方法

	仪器成本（$）	测序成本 ($/Mbase)	读长（bp）	通量	测序质量（Q30）	特点
HiSeq (NovaSeq XPlus)	~100万	~0.02	2x150	每日～8Tb	≥85%	通量高、单Read成本低
BGISEQ -T20	~100万	~0.01	2x150	每小时900Gb	≥80%	成本效益高

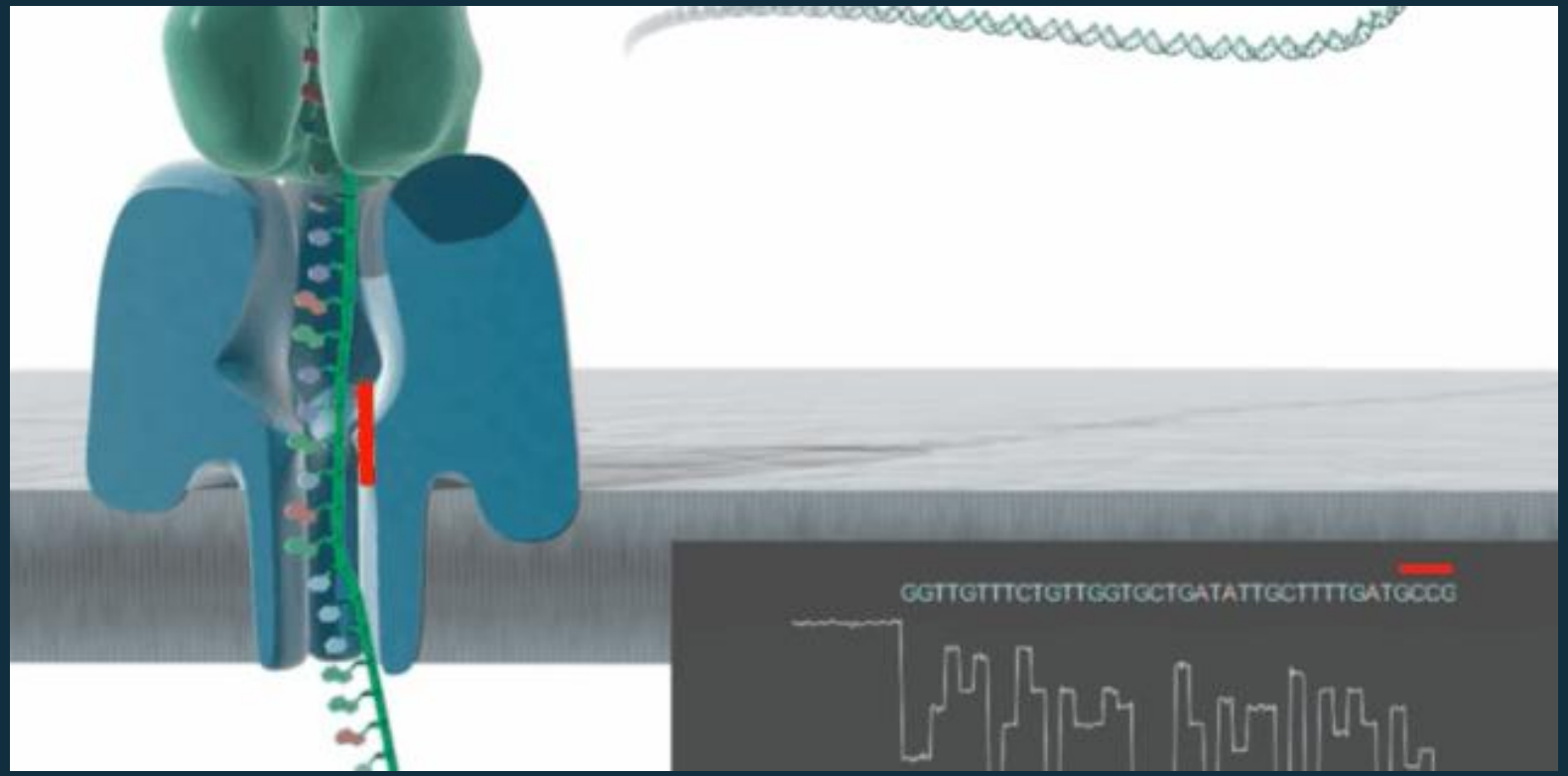
测序原理及方法



三代测序技术原理示意图。(A)PacBio长读长测序法基本原理。(B)Nanopore长读长测序基本原理。

生物学视角

3GS DNA Sequencing



测序原理及方法

一代、二代、三代PacBio测序技术的共同点在于基于在DNA复制中对掺入的A/C/G/T引入标记信号，通过不同的方法读出DNA序列：

一代测序技术通过按片段大小依次读出末端终止碱基；

二代测序技术并行、循环可逆地边合成边读取信号（掺入终止碱基/读取信号/去除终止基团和信号）；

三代测序技术并行、以单分子实时读取掺入的碱基信号。

它们的主要区别在于：

一代测序技术通量最低、读长稍长于二代测序技术；

二代测序技术读长最短，但通量最大；

三代测序技术读长最长，但通量低于二代测序技术。

生物学视角

在今天，DNA测序技术已经在诸多方面达到了临床应用的具体要求。

科学家估计，全世界每年大约有四百万到六百万孕妇正在通过外周血游离DNA对胎儿的21三体综合征进行诊断，而十年之内，这个数字将超过1500万。

在高收入国家，基因组测序已经广泛用于多种疾病的产前诊断，可以揭示大约30%的出生缺陷，同时，这一数字也正随着数据解读能力的成熟而逐渐上升。

在肿瘤学领域，液体活检在最近几年已经成为了肿瘤相关学术，产业以及投资界的新宠。基于DNA测序的液体活检被认为正逐步发展为癌症诊断与预后评估的标准方法，能够在可知的时间内逐步补充甚至取代传统的创伤性癌症诊断技术。

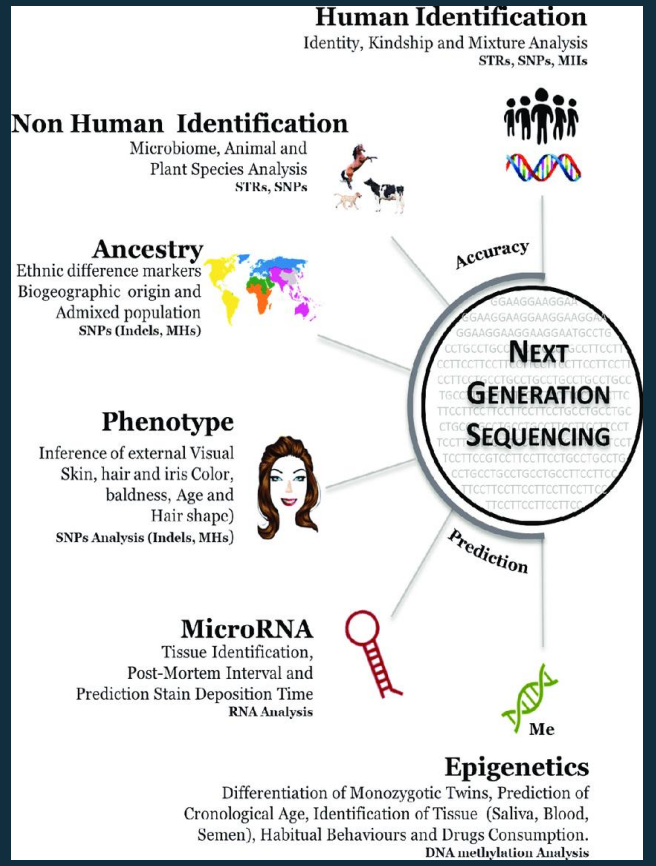
同样，手持DNA测序仪等设备的开发也使得流行病学家甚至能够在最为偏远的地区高效完成对人类样本，动物以及昆虫病原载体甚至是空气，水，食物的基因检测。

流行病学家和公共卫生专家也开始讨论如何通过对城市垃圾中微生物的DNA测序辅助传染性疾病的预防与控制。

海洋生物学家也正在通过宏基因组学技术来对海洋的生态健康进行监测与研究。

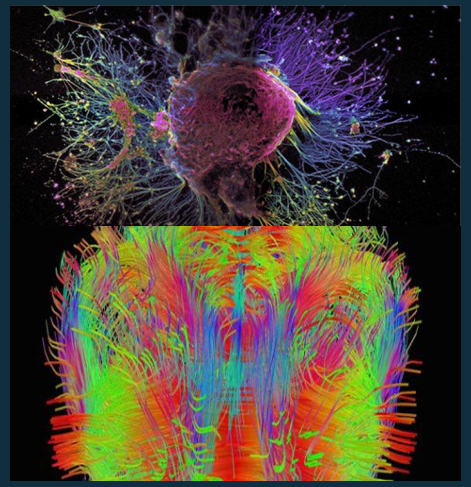
在法医领域，便携式DNA测序仪可以将DNA分析带出法医实验室，使其成为一线警务工作人员的随身工具。帮助警方即使通过DNA监测确定嫌疑人，发展成为诸如酒精探测器一类的便捷工具。

在人们的家里，DNA测序设备或许也可以成为下一个“智能”或“连接”设备，一些评论者甚至认定厕所是通过实时DNA测序监测家庭成员健康的理想场所。

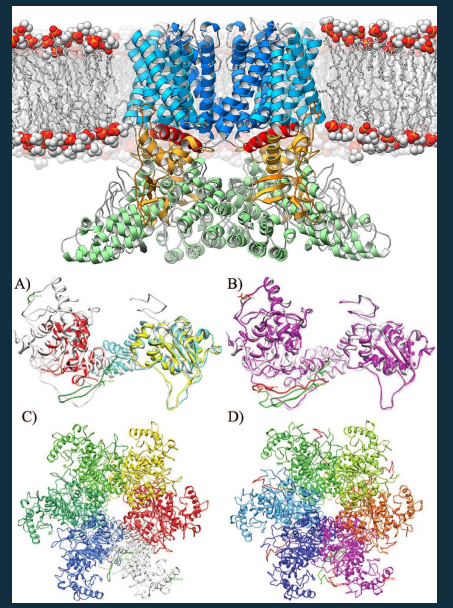


生物信息学 @HUST

生命不只是序列的，但是生命始终是数字的！



- 结构生物学（Structure biology）
- 生物图像（Bio-imaging）
- 。。。



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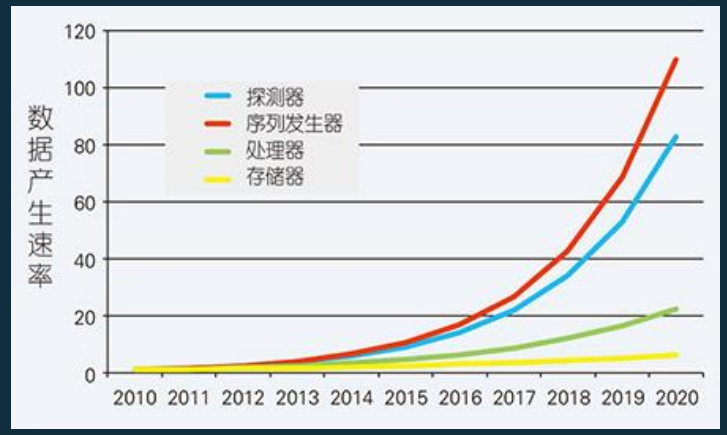
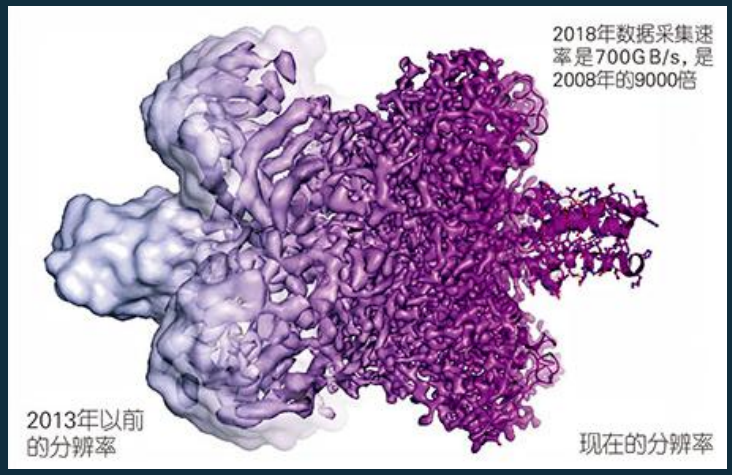
生命不只是序列的，但是生命始终是数字的！





Microbiome and big-data…

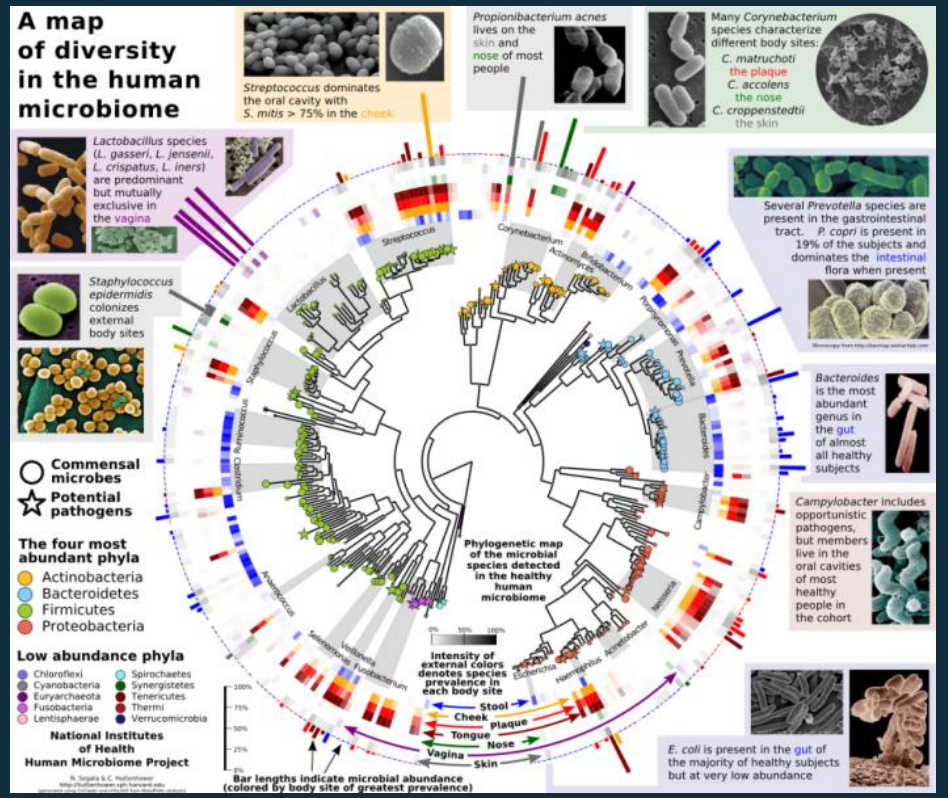
- A cell is already very complex
- 一个细胞已经非常复杂了

- A microbial community is much more complex than a cell
- 一个微生物群落就更为复杂了

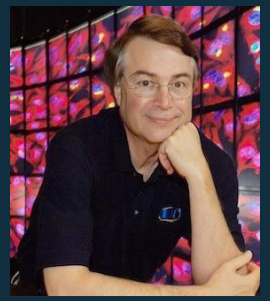
- But much more big-data
- 但是也代表了更多的数据

Microbiome and big-data…

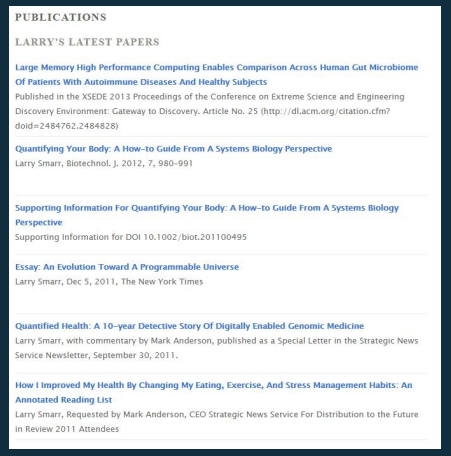
在生物信息眼里，这全是大数据。。。



Microbiome and big-data…



Larry Smarr
Founding Director of the California Institute for Telecommunications and Information Technology (Calit2)



PUBLICATIONS

LARRY'S LATEST PAPERS

Large Memory High Performance Computing Enables Comparison Across Human Gut Microbiome Of Patients With Autoimmune Diseases And Healthy Subjects
Published in the XSEDE 2013 Proceedings of the Conference on Extreme Science and Engineering Discovery Environment: Gateway to Discovery. Article No. 25 (http://dl.acm.org/citation.cfm?doid=2484762.2484828)

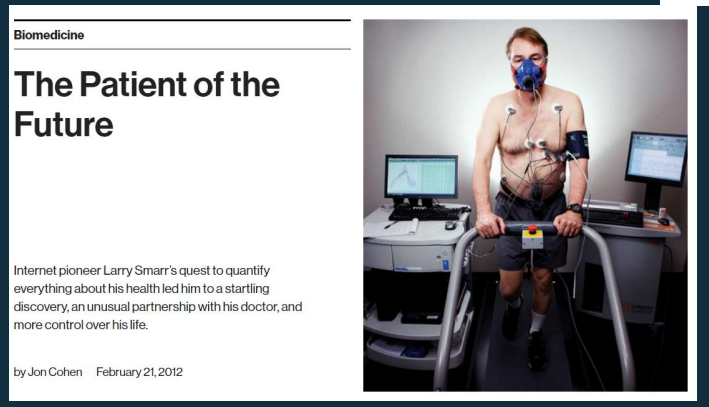
Quantifying Your Body: A How-to Guide From A Systems Biology Perspective
Larry Smarr, Biotechnol. J. 2012, 7, 980–991

Supporting Information For Quantifying Your Body: A How-to Guide From A Systems Biology Perspective
Supporting Information for DOI 10.1002/biot.201100495

Essay: An Evolution Toward A Programmable Universe
Larry Smarr, Dec 5, 2011, The New York Times

Quantified Health: A 10-year Detective Story Of Digitally Enabled Genomic Medicine
Larry Smarr, with commentary by Mark Anderson, published as a Special Letter in the Strategic News Service Newsletter, September 30, 2011.

How I Improved My Health By Changing My Eating, Exercise, And Stress Management Habits: An Annotated Reading List
Larry Smarr, Requested by Mark Anderson, CEO Strategic News Service For Distribution to the Future in Review 2011 Attendees



Biomedicine

The Patient of the Future

Internet pioneer Larry Smarr's quest to quantify everything about his health led him to a startling discovery, an unusual partnership with his doctor, and more control over his life.

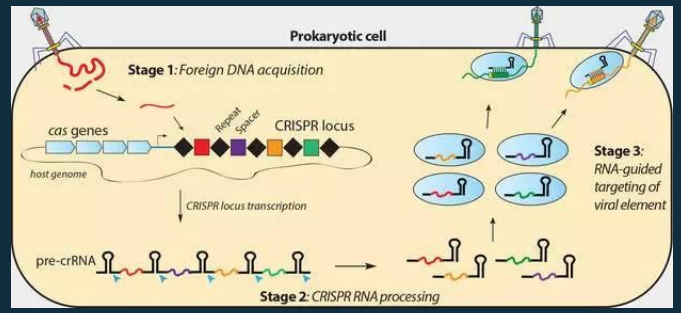
by Jon Cohen February 21, 2012

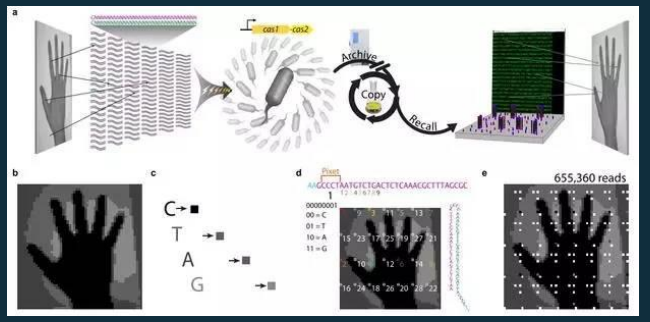
Biomedical big-data…

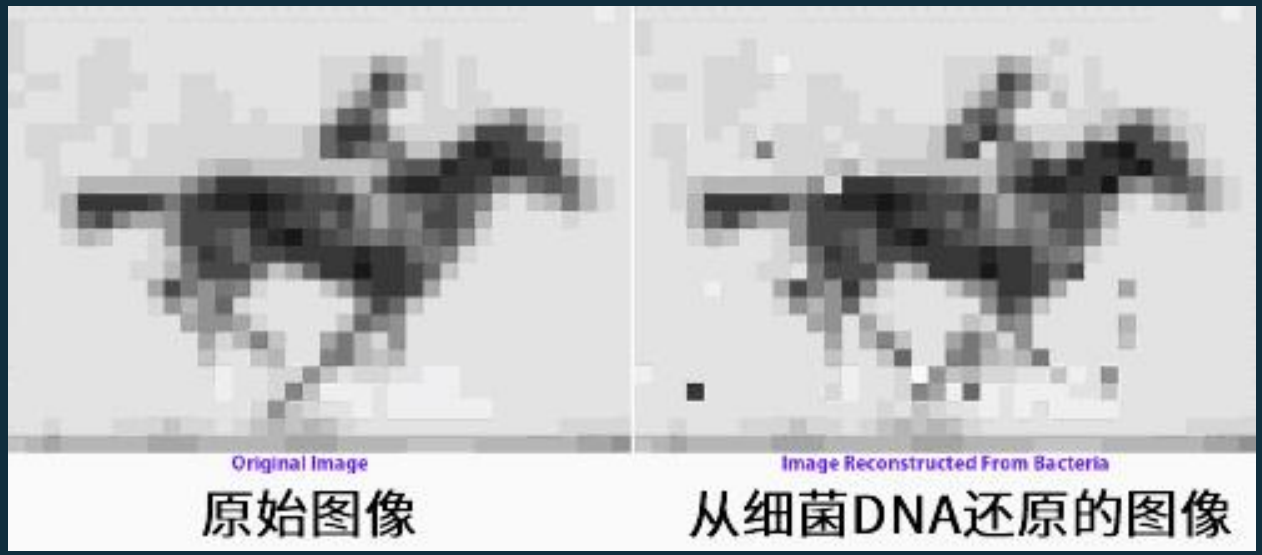
“Have you ever figured how information-rich your stool is?,” Larry asks me with a wide smile, his gray-green eyes intent behind rimless glasses. “There are about 100 billion bacteria per gram. Each bacterium has DNA whose length is typically one to 10 megabases—call it 1 million bytes of information. This means human stool has a data capacity of 100,000 terabytes of information stored per gram. That’s many orders of magnitude more information density than, say, in a chip in your smartphone or your personal computer. So your stool is far more interesting than a computer.”

-- Larry Smarr

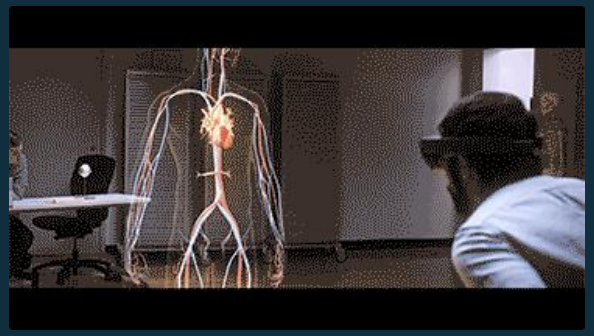
Understand it, create it!



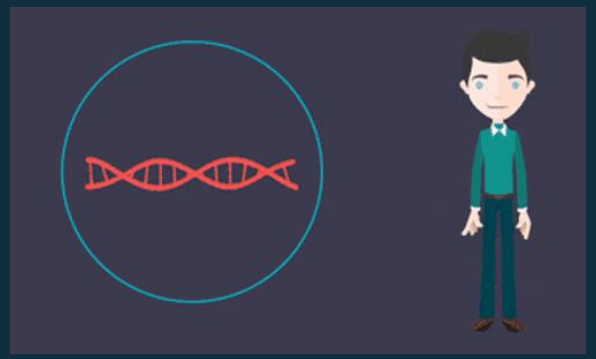




See it!



Understand it!



Create it!

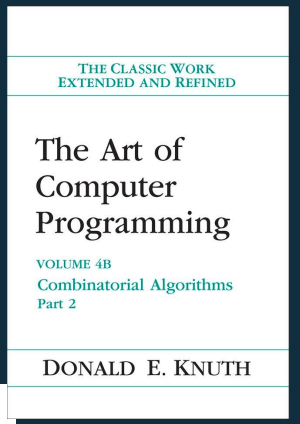
LECTURE 01

生物统计学：计算科学视角

Donald Knuth（高德纳）

Donald Knuth, the "father of the analysis of algorithms.”

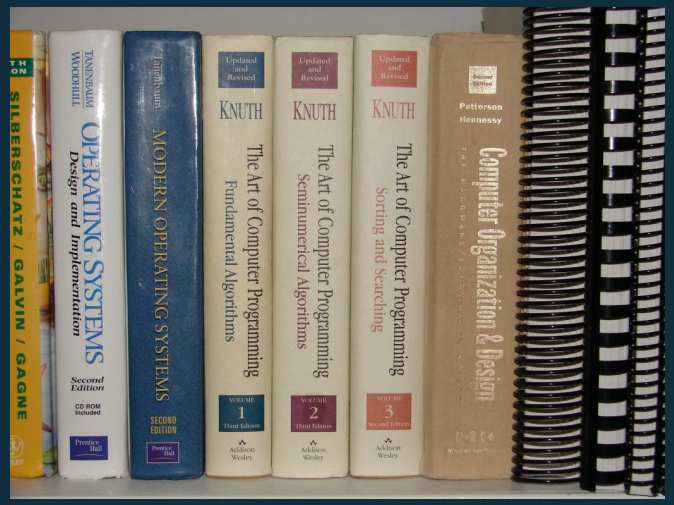
The Art of Computer Programming（计算机程序设计艺术）



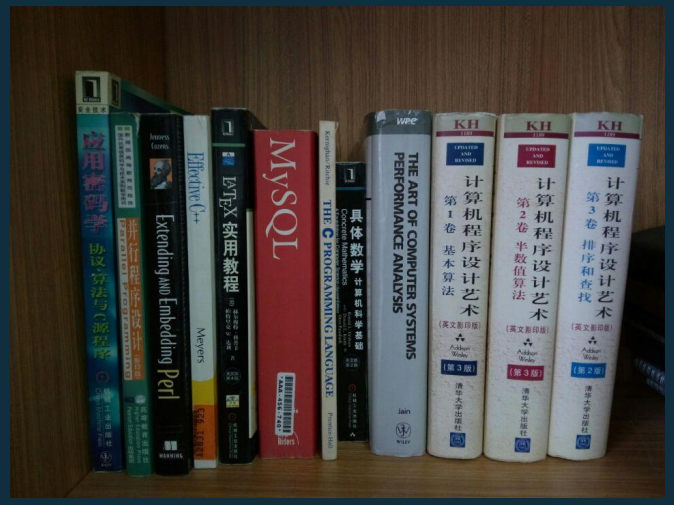
Markup	Renders as
`The quadratic formula is $-b \pm \sqrt{b^2 - 4ac} \over 2a$ \bye`	The quadratic formula is $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

“生物信息学为算法研究提供了500年的问题”– Don Knuth

计算科学视角

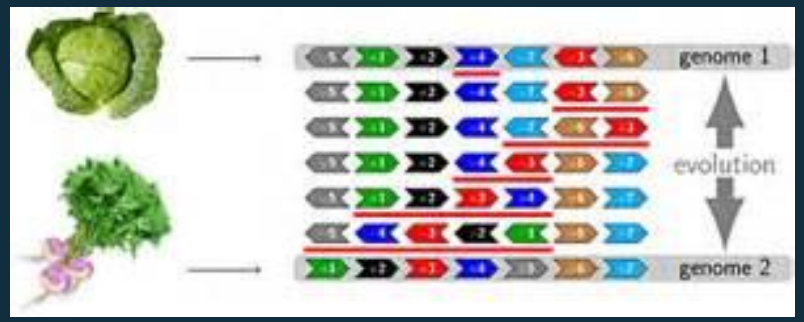


“definitely send me a résumé if you finish this fiendishly difficult book” – Bill Gates



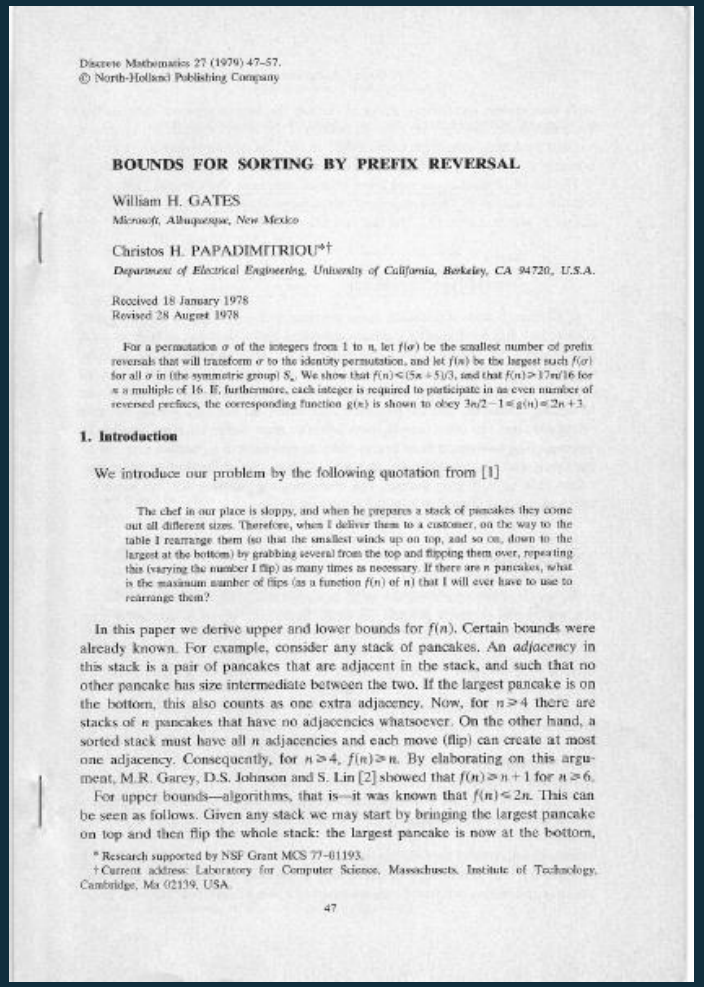
“definitely come to talk about algorithm if you read half of this book” – Kang Ning

Bill Gates（比尔盖茨）



比尔盖茨:下个世界首富出自基因检测领域

Sorting by reversal problem



Discrete Mathematics 27 (1979) 47–57.
© North-Holland Publishing Company

BOUNDS FOR SORTING BY PREFIX REVERSAL

William H. GATES
Microsoft, Albuquerque, New Mexico

Christos H. PAPADIMITRIOU*†
Department of Electrical Engineering, University of California, Berkeley, CA 94720, U.S.A.

Received 18 January 1978
Revised 28 August 1978

For a permutation σ of the integers from 1 to n, let $f(\sigma)$ be the smallest number of prefix reversals that will transform σ to the identity permutation, and let $f(n)$ be the largest such $f(\sigma)$ for all σ in (the symmetric group) S_n. We show that $f(n) \leq (5n+5)/3$, and that $f(n) \geq 17n/16$ for n a multiple of 16. If, furthermore, each integer is required to participate in an even number of reversed prefixes, the corresponding function $g(n)$ is shown to obey $3n/2 - 1 \leq g(n) \leq 2n+3$.

1. Introduction

We introduce our problem by the following quotation from [1]

> The chef in our place is sloppy, and when he prepares a stack of pancakes they come out all different sizes. Therefore, when I deliver them to a customer, on the way to the table I rearrange them (so that the smallest winds up on top, and so on, down to the largest at the bottom) by grabbing several from the top and flipping them over, repeating this (varying the number I flip) as many times as necessary. If there are n pancakes, what is the maximum number of flips (as a function $f(n)$ of n) that I will ever have to use to rearrange them?

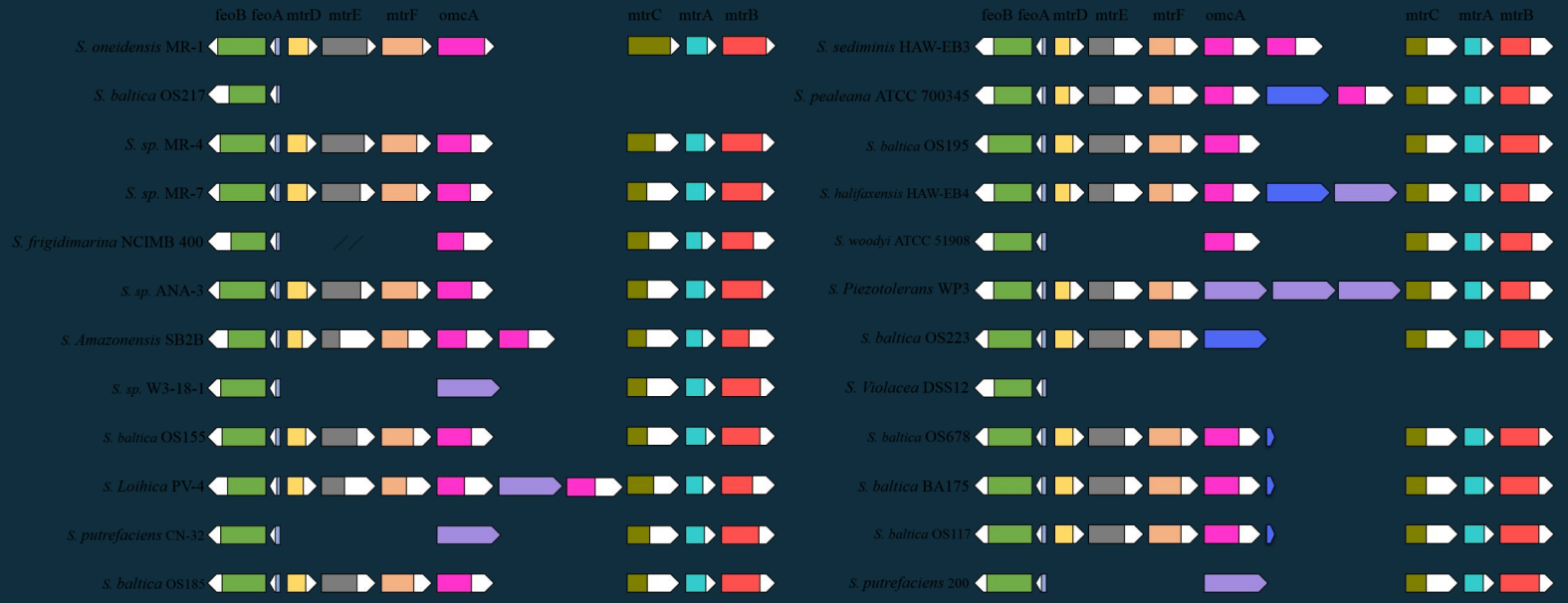
In this paper we derive upper and lower bounds for $f(n)$. Certain bounds were already known. For example, consider any stack of pancakes. An *adjacency* in this stack is a pair of pancakes that are adjacent in the stack, and such that no other pancake has size intermediate between the two. If the largest pancake is on the bottom, this also counts as one extra adjacency. Now, for $n \geq 4$ there are stacks of n pancakes that have no adjacencies whatsoever. On the other hand, a sorted stack must have all n adjacencies and each move (flip) can create at most one adjacency. Consequently, for $n \geq 4$, $f(n) \geq n$. By elaborating on this argument, M.R. Garey, D.S. Johnson and S. Lin [2] showed that $f(n) \geq n+1$ for $n \geq 6$.

For upper bounds—algorithms, that is—it was known that $f(n) \leq 2n$. This can be seen as follows. Given any stack we may start by bringing the largest pancake on top and then flip the whole stack: the largest pancake is now at the bottom,

* Research supported by NSF Grant MCS 77-01193.
†Current address: Laboratory for Computer Science, Massachusetts Institute of Technology, Cambridge, Ma 02139, USA.

计算科学视角

How many reversal steps for this REAL case?

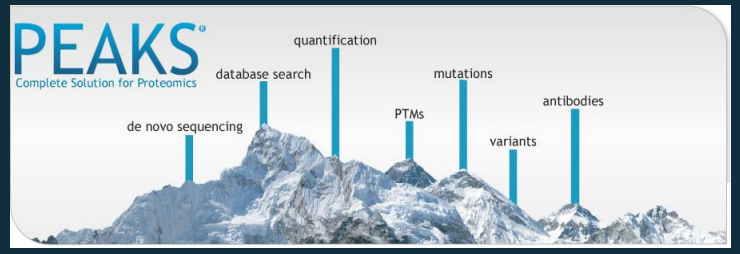


Ming Li（李明）

滑铁卢大学是微软招聘毕业生最多的学校之一

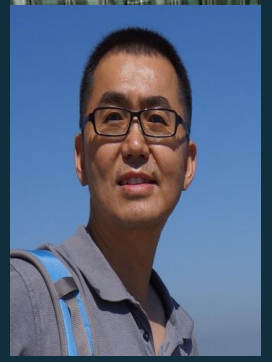
University of Waterloo

PatternHunter, ExonHunter, …



理论、生物信息、应用都重要！

Ming Li（李明）& Tao Jiang（姜涛）



SIAM J. COMPUT.
Vol. 24, No. 5, pp. 1122–1139, October 1995

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012

ON THE APPROXIMATION OF SHORTEST COMMON SUPERSEQUENCES AND LONGEST COMMON SUBSEQUENCES*

TAO JIANG† AND MING LI‡

Abstract. The problems of finding shortest common supersequences (SCS) and longest common subsequences (LCS) are two well-known **NP**-hard problems that have applications in many areas, including computational molecular biology, data compression, robot motion planning, and scheduling, text editing, etc. A lot of fruitless effort has been spent in searching for good approximation algorithms for these problems. In this paper, we show that these problems are inherently hard to approximate in the worst case. In particular, we prove that (i) SCS does not have a polynomial-time linear approximation algorithm unless $\mathbf{P} = \mathbf{NP}$; (ii) There exists a constant $\delta > 0$ such that, if SCS has a polynomial-time approximation algorithm with ratio $\log^{\delta} n$, where n is the number of input sequences, then **NP** is contained in $\mathbf{DTIME}(2^{\text{polylog } n})$; (iii) There exists a constant $\delta > 0$ such that, if LCS has a polynomial-time approximation algorithm with performance ratio n^{δ}, then $\mathbf{P} = \mathbf{NP}$. The proofs utilize the recent results of Arora et al. [*Proc. 23rd IEEE Symposium on Foundations of Computer Science*, 1992, pp. 14–23] on the complexity of approximation problems.

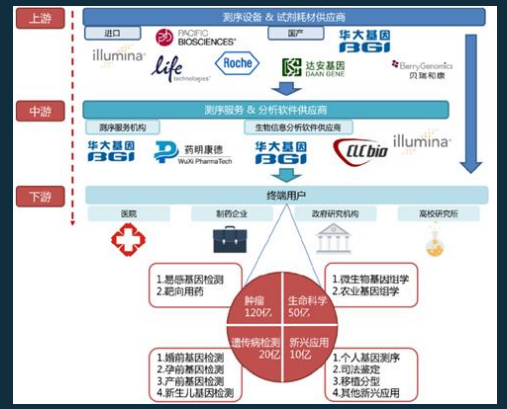
In the second part of the paper, we introduce a new method for analyzing the average-case performance of algorithms for sequences, based on Kolmogorov complexity. Despite the above nonapproximability results, we show that near optimal solutions for both SCS and LCS can be found on the average. More precisely, consider a fixed alphabet Σ and suppose that the input sequences are generated randomly according to the uniform probability distribution and are of the same length n. Moreover, assume that the number of input sequences is polynomial in n. Then, there are simple greedy algorithms which approximate SCS and LCS with expected additive errors $O(n^{0.707})$ and $O(n^{1/2+\epsilon})$ for any $\epsilon > 0$, respectively.

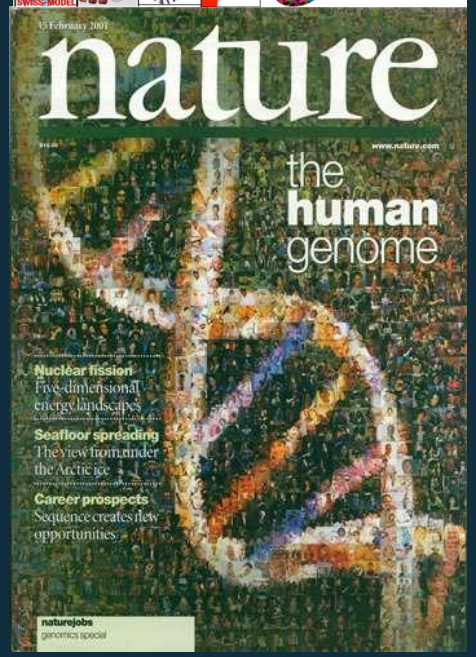
Incidentally, our analyses also provide tight upper and lower bounds on the expected LCS and SCS lengths for a set of random sequences solving a generalization of another well-known open question on the expected LCS length for two random sequences [K. Alexander, *The rate of convergence of the mean length of the longest common subsequence*, 1992, manuscript], [V. Chvatal and D. Sankoff, *J. Appl. Probab.*, 12 (1975), pp. 306–315], [D. Sankoff and J. Kruskall, eds., *Time Warps, String Edits, and Macromolecules: The Theory and Practice of Sequence Comparison*, Addison–Wesley, Reading, MA, 1983].

Key words. shortest common supersequence, longest common subsequence, approximation algorithm, **NP**-hardness, average-case analysis, random sequence

Current status（现今态势）

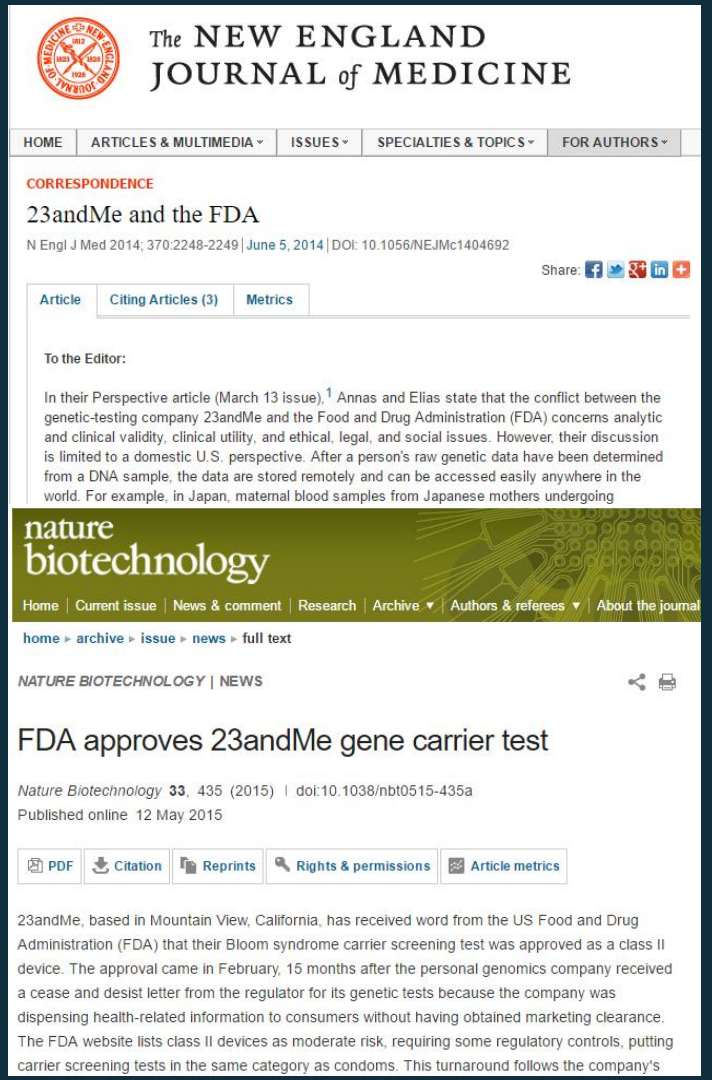
很难找到
与生物信息学和生物统计学
没有关系的
生物学与生物工程
研究和应用领域了。。。





Alphabet（谷歌）

Google 的基因组学梦想



The NEW ENGLAND JOURNAL of MEDICINE

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CORRESPONDENCE

23andMe and the FDA

N Engl J Med 2014; 370:2248-2249 | June 5, 2014 | DOI: 10.1056/NEJMc1404692

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To the Editor:

In their Perspective article (March 13 issue),[1] Annas and Elias state that the conflict between the genetic-testing company 23andMe and the Food and Drug Administration (FDA) concerns analytic and clinical validity, clinical utility, and ethical, legal, and social issues. However, their discussion is limited to a domestic U.S. perspective. After a person's raw genetic data have been determined from a DNA sample, the data are stored remotely and can be accessed easily anywhere in the world. For example, in Japan, maternal blood samples from Japanese mothers undergoing

nature biotechnology

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NATURE BIOTECHNOLOGY | NEWS

FDA approves 23andMe gene carrier test

Nature Biotechnology **33**, 435 (2015) | doi:10.1038/nbt0515-435a

Published online 12 May 2015

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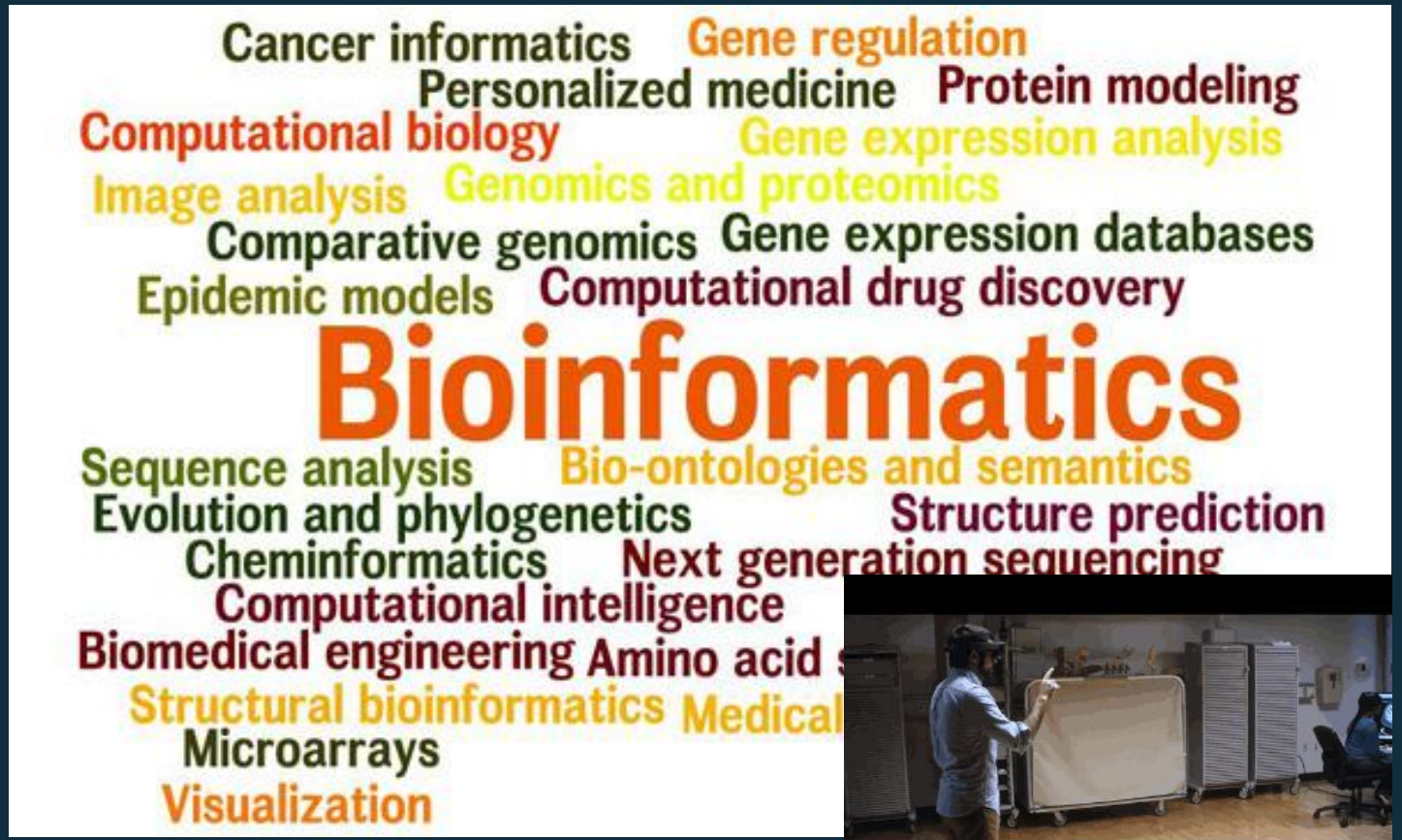
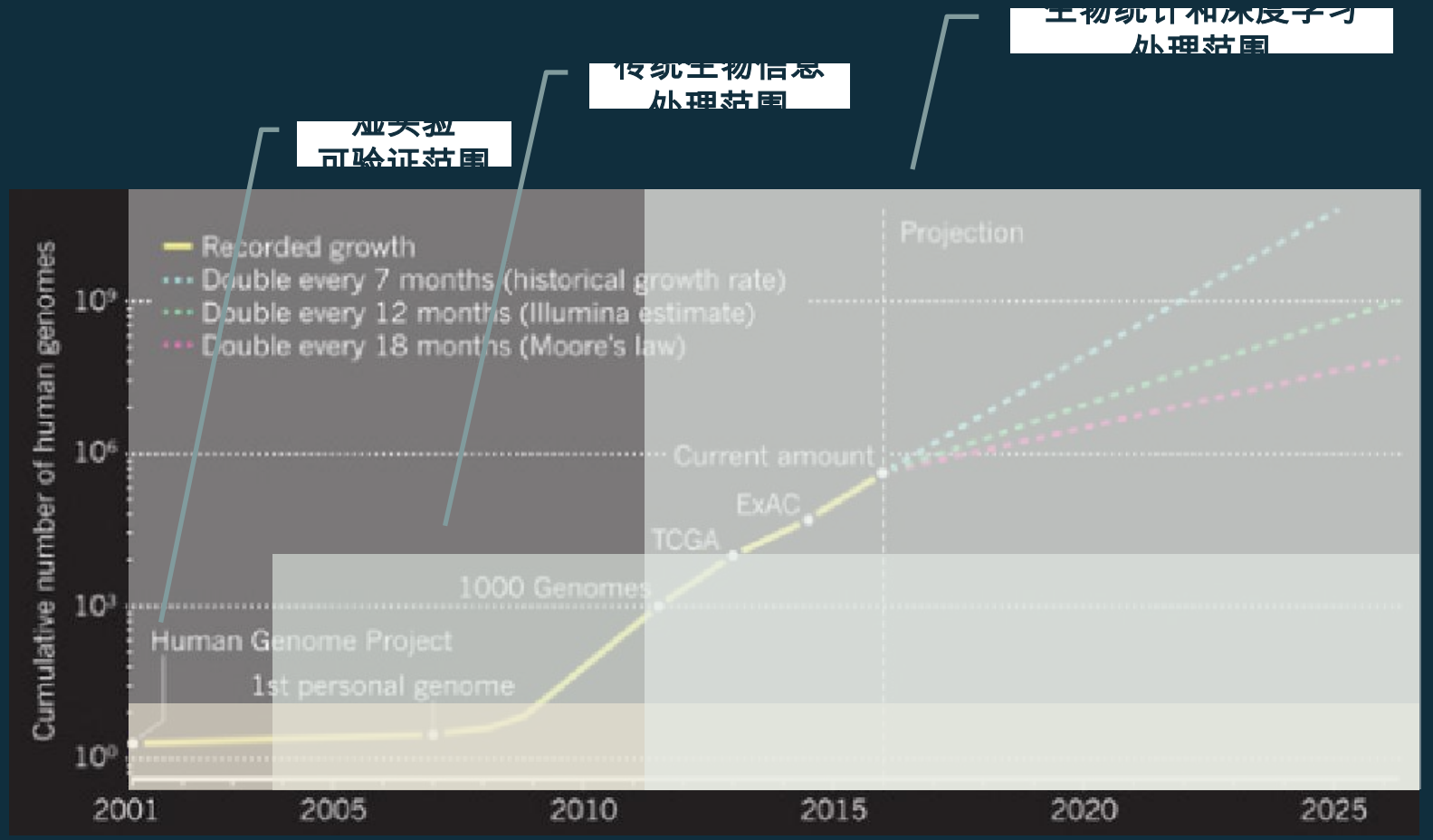
23andMe, based in Mountain View, California, has received word from the US Food and Drug Administration (FDA) that their Bloom syndrome carrier screening test was approved as a class II device. The approval came in February, 15 months after the personal genomics company received a cease and desist letter from the regulator for its genetic tests because the company was dispensing health-related information to consumers without having obtained marketing clearance. The FDA website lists class II devices as moderate risk, requiring some regulatory controls, putting carrier screening tests in the same category as condoms. This turnaround follows the company's

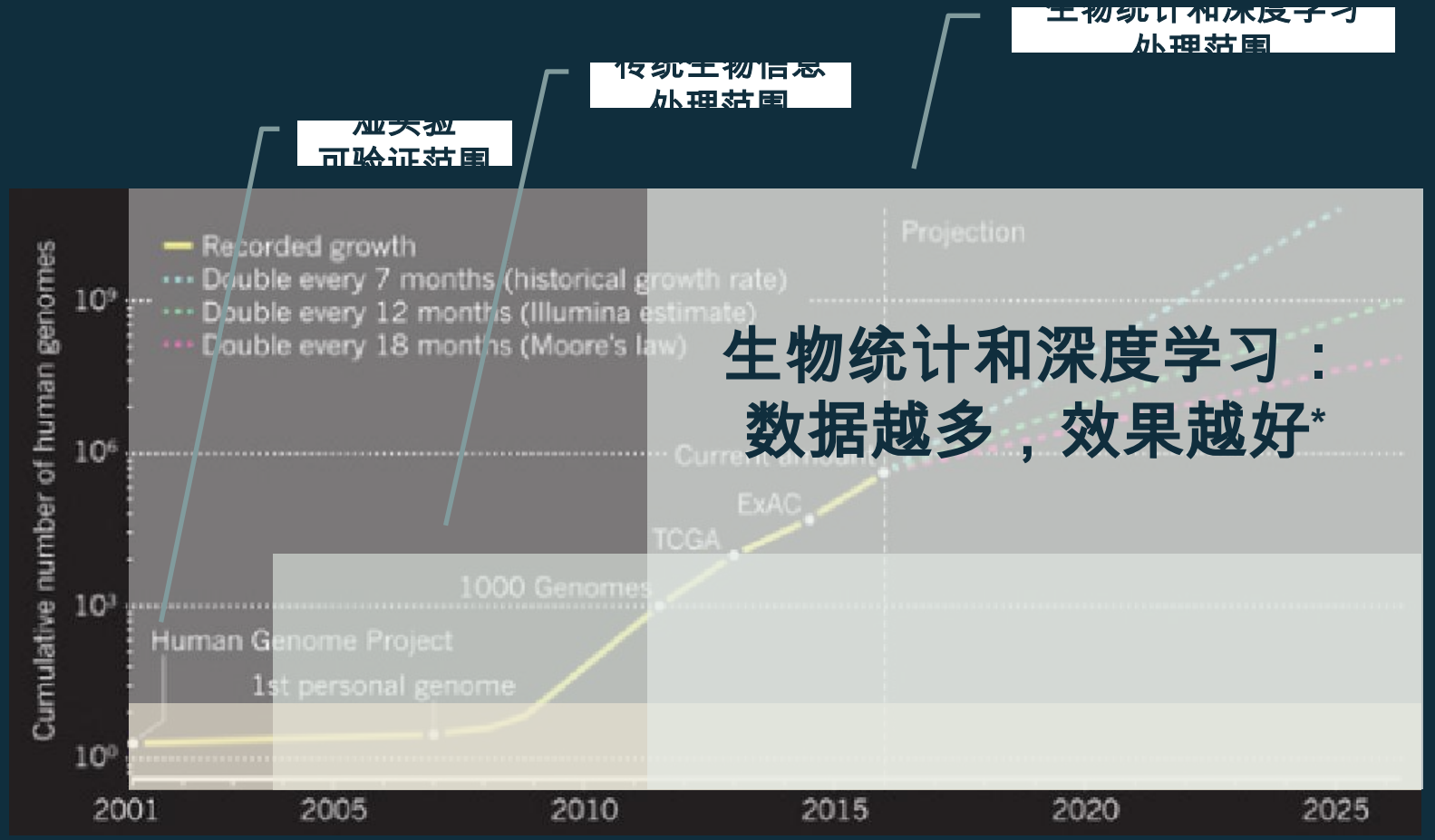
Future（未来）


Cancer informatics
Gene regulation
Personalized medicine
Protein modeling
Computational biology
Gene expression analysis
Image analysis
Genomics and proteomics
Comparative genomics
Gene expression databases
Epidemic models
Computational drug discovery
Bioinformatics
Sequence analysis
Bio-ontologies and semantics
Evolution and phylogenetics
Structure prediction
Cheminformatics
Next generation sequencing
Computational intelligence
Biomedical engineering
Amino acid
Structural bioinformatics
Medical
Microarrays
Visualization

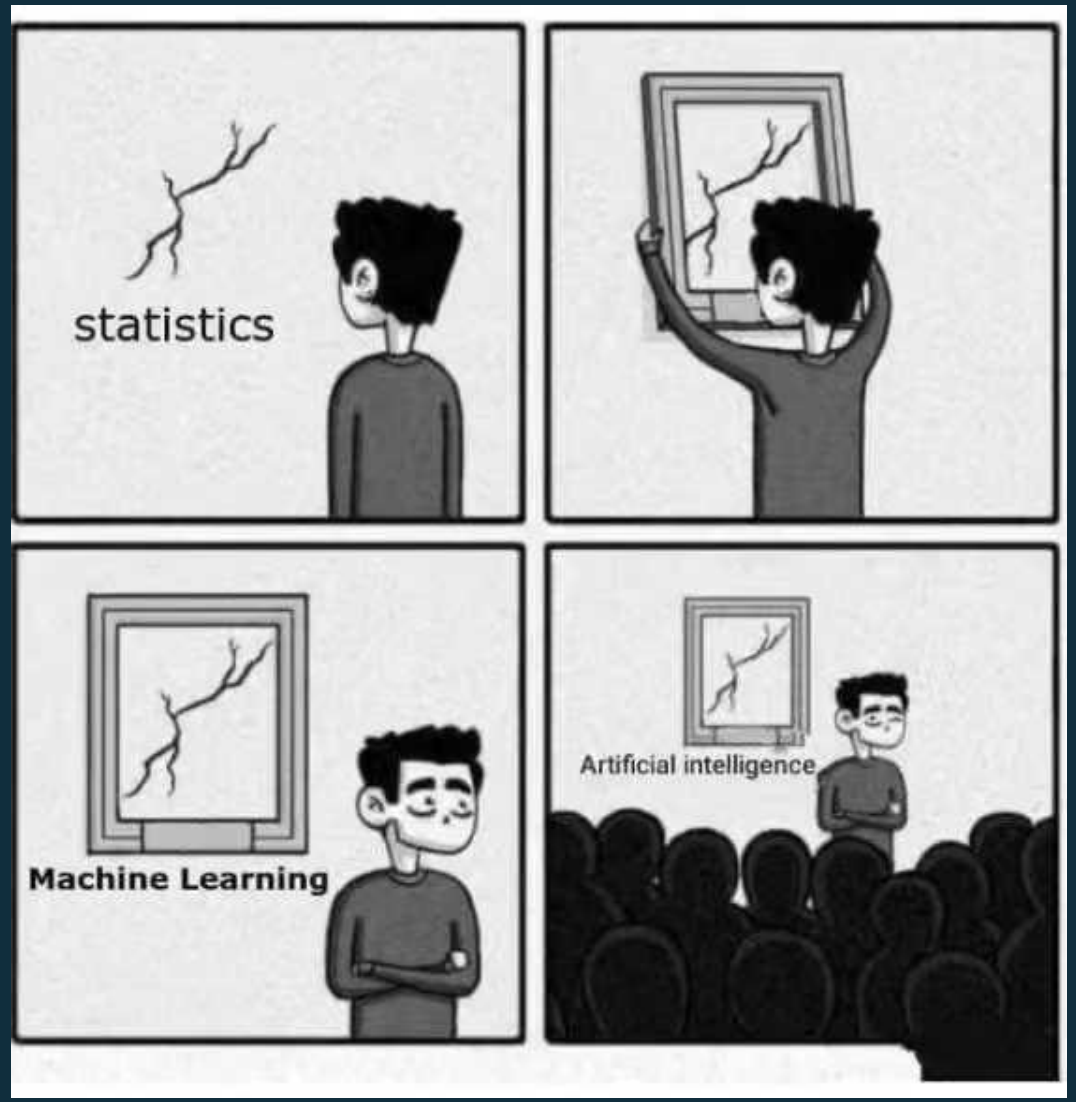
Biostatistics



Biostatistics

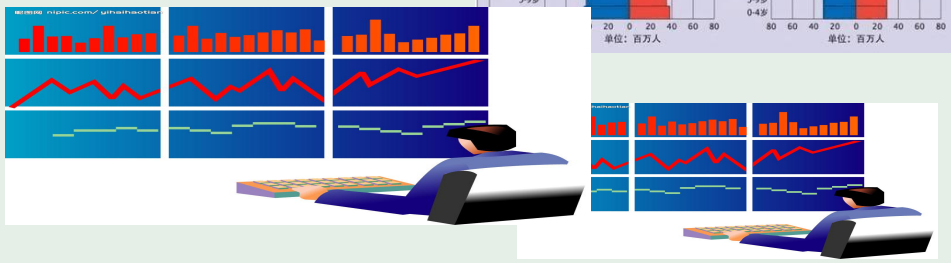


Biostatistics



统计学：让数字说话！

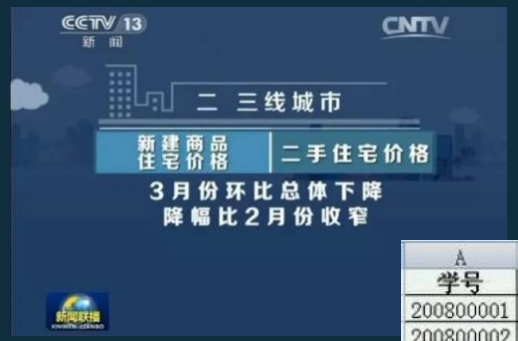
人口数量和结构可以预测吗？



股市可以预测吗？

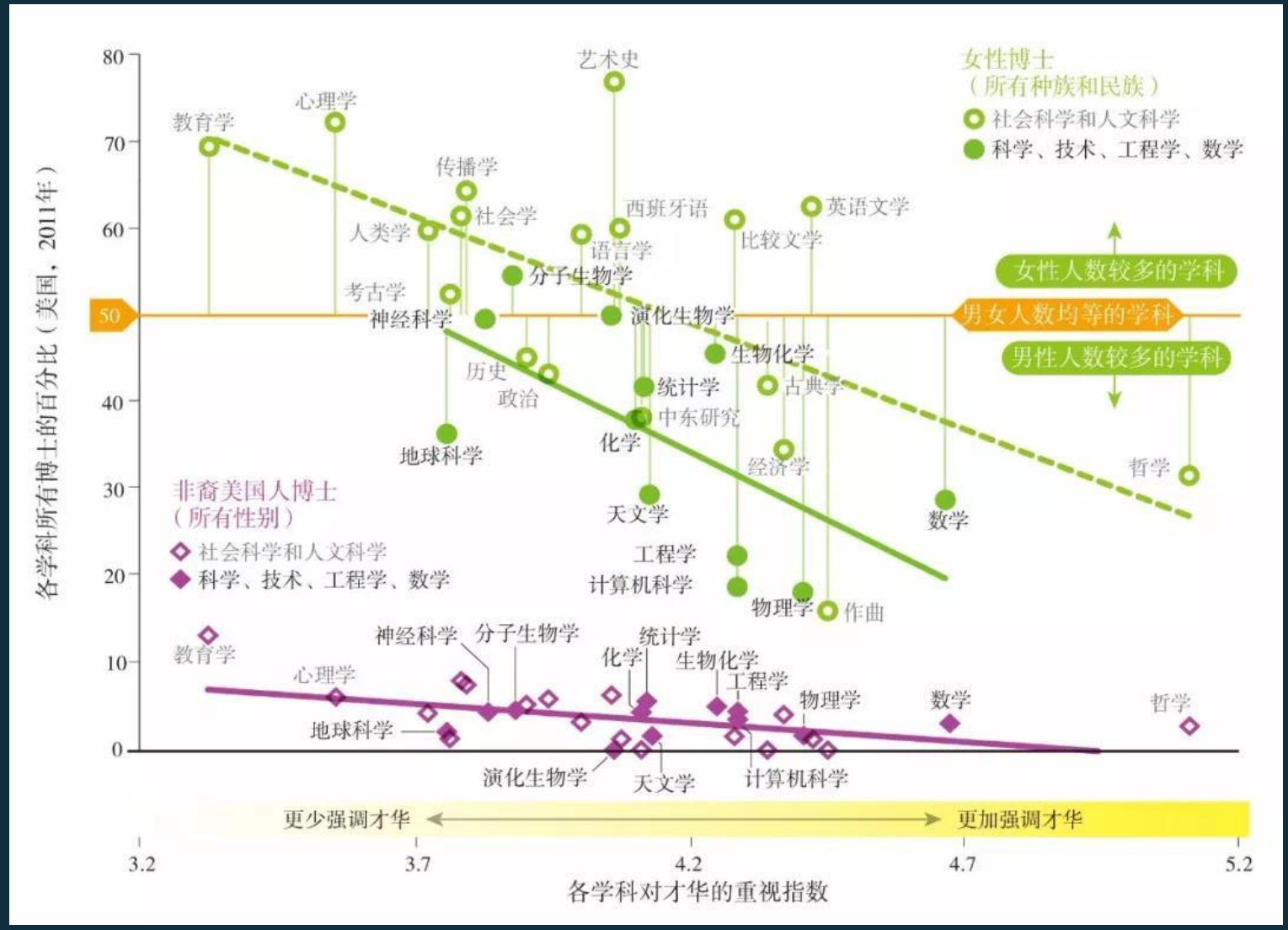
吸烟可以致癌？新药测试？

统计学：无处不在！



A	B	C	D	E	F	G
学号	姓名	语文	数学	英语	平均分	名次
200800001	狮子王	89	65	56	70.00	13
200800002	米老鼠	96	80	98	91.33	1
200800003	花木兰	83	82	83	82.67	9
200800004	唐老鸭	91	90	87	89.33	3
200800005	阿童木	93	92	88	91.00	2
200800006	灰姑娘	90	86	80	85.33	6
200800007	大力士	94	79	91	88.00	4
200800008	跳跳虎	83	43	94	73.33	10
200800009	小蚂蚁	89	91	82	87.33	5
200800010	皮诺曹	48	95	76	73.00	11
200800011	爱丽丝	87	83	85	85.00	7
200800012	睡美人	82	79.5	90	83.83	8
200800013	加菲猫	81	54	80	71.67	12

统 计 学：无处不在！


女性博士
（所有种族和民族）
社会科学和人文科学
科学、技术、工程学、数学
女性人数较多的学科
男女人数均等的学科
男性人数较多的学科
非裔美国人博士
（所有性别）
社会科学和人文科学
科学、技术、工程学、数学
各学科所有博士的百分比（美国，2011年）
各学科对才华的重视指数
更少强调才华
更加强调才华
艺术史
心理学
教育学
传播学
社会学
人类学
西班牙语
英语文学
比较文学
语言学
分子生物学
考古学
神经科学
演化生物学
生物化学
历史
政治
统计学
古典学
中东研究
地球科学
化学
经济学
哲学
天文学
数学
工程学
计算机科学
物理学
作曲
神经科学
分子生物学
统计学
教育学
化学
生物化学
心理学
工程学
物理学
数学
哲学
地球科学
演化生物学
天文学
计算机科学

统 计 学：无处不在！

习近平说：中国是世界第二大经济体，有13亿多人口的大市场，有960多万平方公里的国土，中国经济是一片大海，而不是一个小池塘。大海有风平浪静之时，也有风狂雨骤之时。没有风狂雨骤，那就不是大海了。狂风骤雨可以掀翻小池塘，但不能掀翻大海。经历了无数次狂风骤雨，大海依旧在那儿！经历了5000多年的艰难困苦，中国依旧在这儿！面向未来，中国将永远在这儿！

概率：

$P(掀翻 \cap 狂风骤雨 | 小池塘)=high$

$P(掀翻 \cap 小雨 | 小池塘)=low$

$P(掀翻 \cap 狂风骤雨 | 大海)=low$

$P(掀翻 \cap 小雨 | 大海)=low$

$P(掀翻 | 大海)=P(掀翻 \cap 狂风骤雨 | 大海)+P(掀翻 \cap 小雨 | 大海)$

贝叶斯推断：

$P(掀翻 | 大海)=P(大海 | 掀翻)*P(掀翻)/P(大海)=low$

vs.

$P(掀翻 | 小池塘)=P(小池塘 | 掀翻)*P(掀翻)/P(小池塘)=high$

统 计 学：产生价值！

基于大数据统计分析的防控平台
（社会效益）

基于大数据统计分析的决策平台
（经济效益）

统 计 学：产生价值！

在高校学科设置上，我特别支持你们重视统计学。计算机科学不仅仅是技术，还应该以统计学为基础。大数据需要统计学，信息科学需要统计学，生命科学也需要统计学。国家要搞人工智能，更要重视统计学。统计学不是一个纯粹的学科，而是每一个学科都要以统计学为基础。

[公司文件] 加强与国内大学合作，吸纳全球优秀人才，共同推动中国基础研究——任总与中国科学技术大学包信和校长座谈的讲话 荐

2018-12-13 17:59 6799 113 只看楼主

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电邮讲话【2018】128号 签发人：任正非

加强与国内大学合作，吸纳全球优秀人才，共同推动中国基础研究

——任总与中国科学技术大学包信和校长座谈的讲话

2018年11月19日

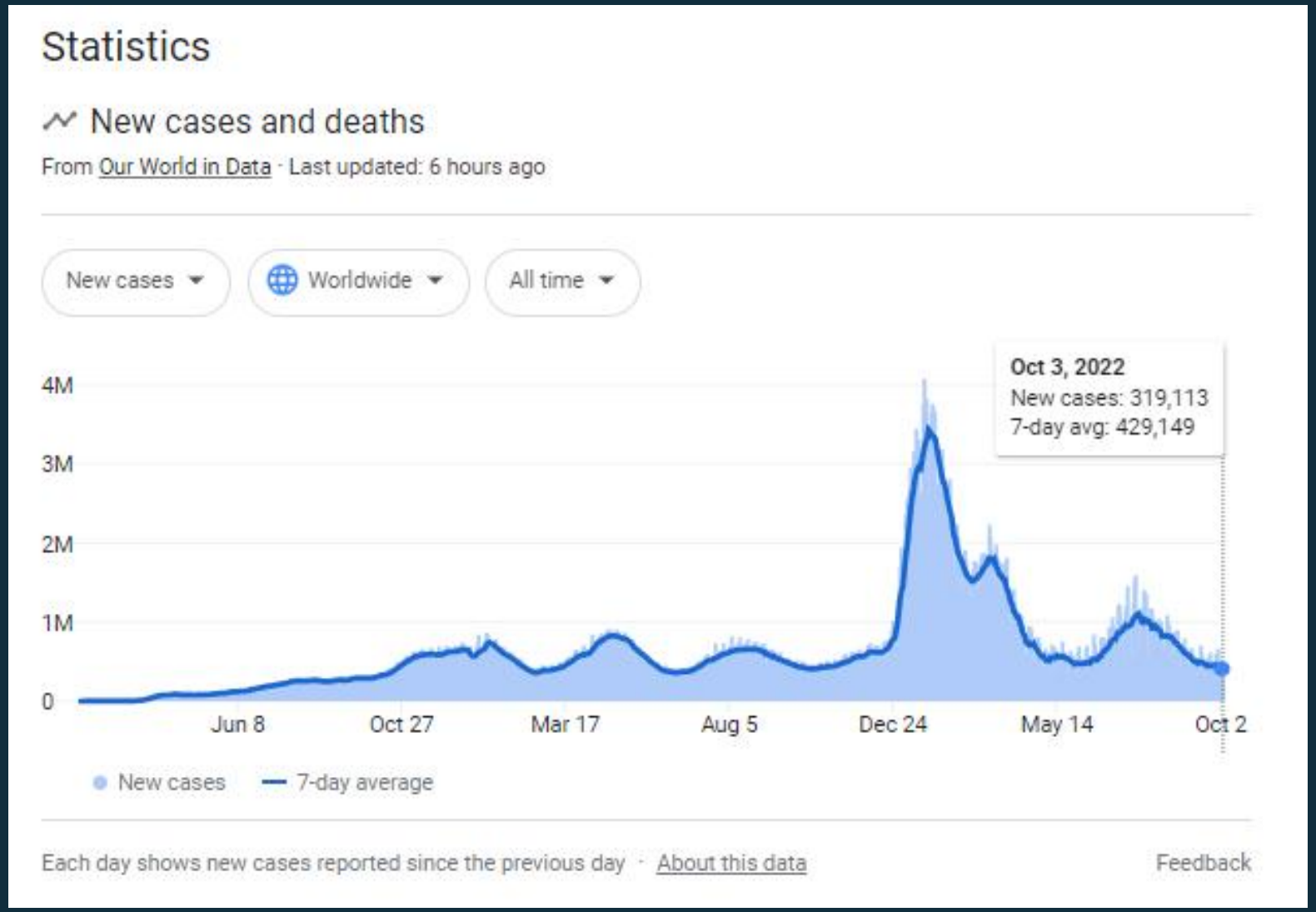
统 计 学：概率不等于事实！

盖洛普民意测验与美国总统大选关联度一览表（1936—2000）

年代	候选人	盖洛普最后民意测验结果（%）	总统选举真实结果（%）	盖洛普误差（%）
2000	布什	48.0	47.9	+0.1
1996	克林顿	52.0	49.2	+2.8
1992	克林顿	49.0	43.3	+5.7
1988	老布什	56.0	53.9	+2.1
1984	里根	59.0	59.2	-0.2
1980	里根	47.0	50.8	-3.8
1976	卡特	48.0	50.1	-2.1
1972	尼克松	62.0	61.8	+0.2
1968	尼克松	43.0	43.5	-0.5
1964	约翰逊	64.0	61.3	+2.7
1960	肯尼迪	51.0	50.1	+0.9
1956	艾森豪威尔	59.5	57.8	+1.7
1952	艾森豪威尔	51.0	55.4	-4.4
1948	杜鲁门	44.5	49.5	-5.0
1944	罗斯福	51.5	53.8	-2.3
1940	罗斯福	52.0	55.0	-3.0
1936	罗斯福	55.7	62.5	-6.8

盖洛普民意测验创始人乔治·盖洛普

统计学：性命攸关！


Statistics
New cases and deaths
From Our World in Data · Last updated: 6 hours ago
New cases
Worldwide
All time
Oct 3, 2022
New cases: 319,113
7-day avg: 429,149
4M
3M
2M
1M
0
Jun 8
Oct 27
Mar 17
Aug 5
Dec 24
May 14
Oct 2
New cases
7-day average
Each day shows new cases reported since the previous day · About this data
Feedback

统计学：历史和地位！

人口普查

F检验

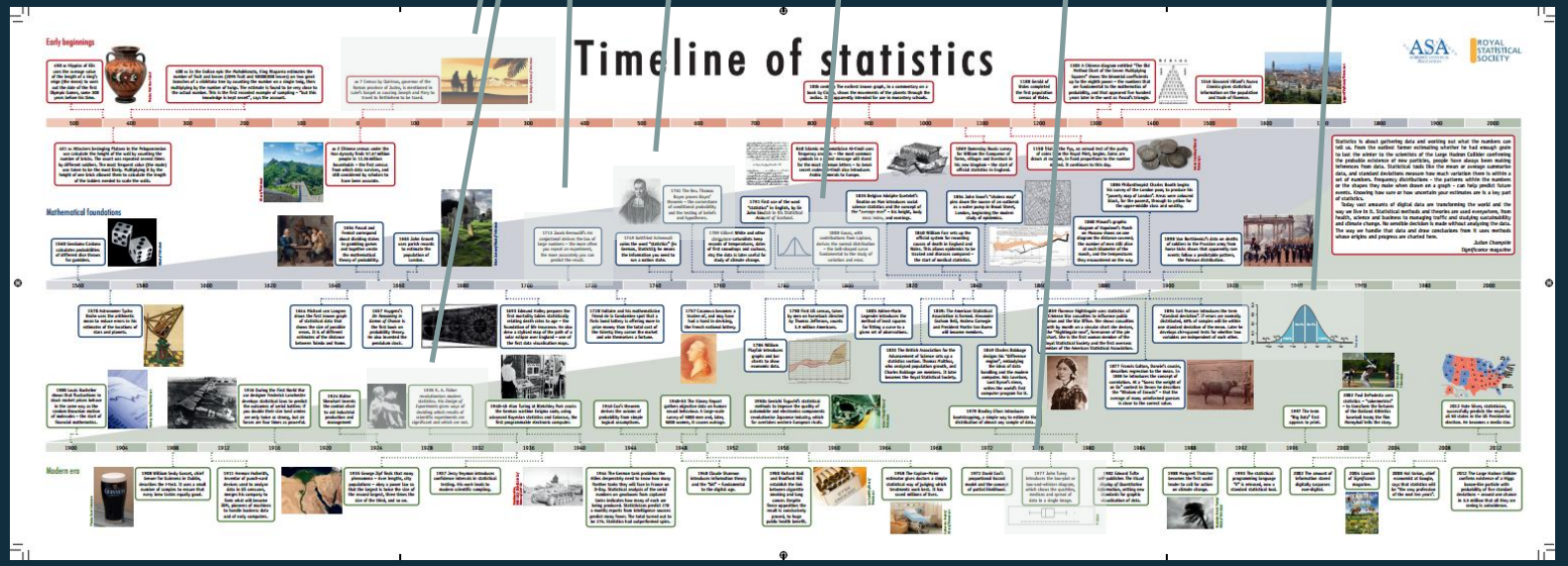
大数定理

贝叶斯推断

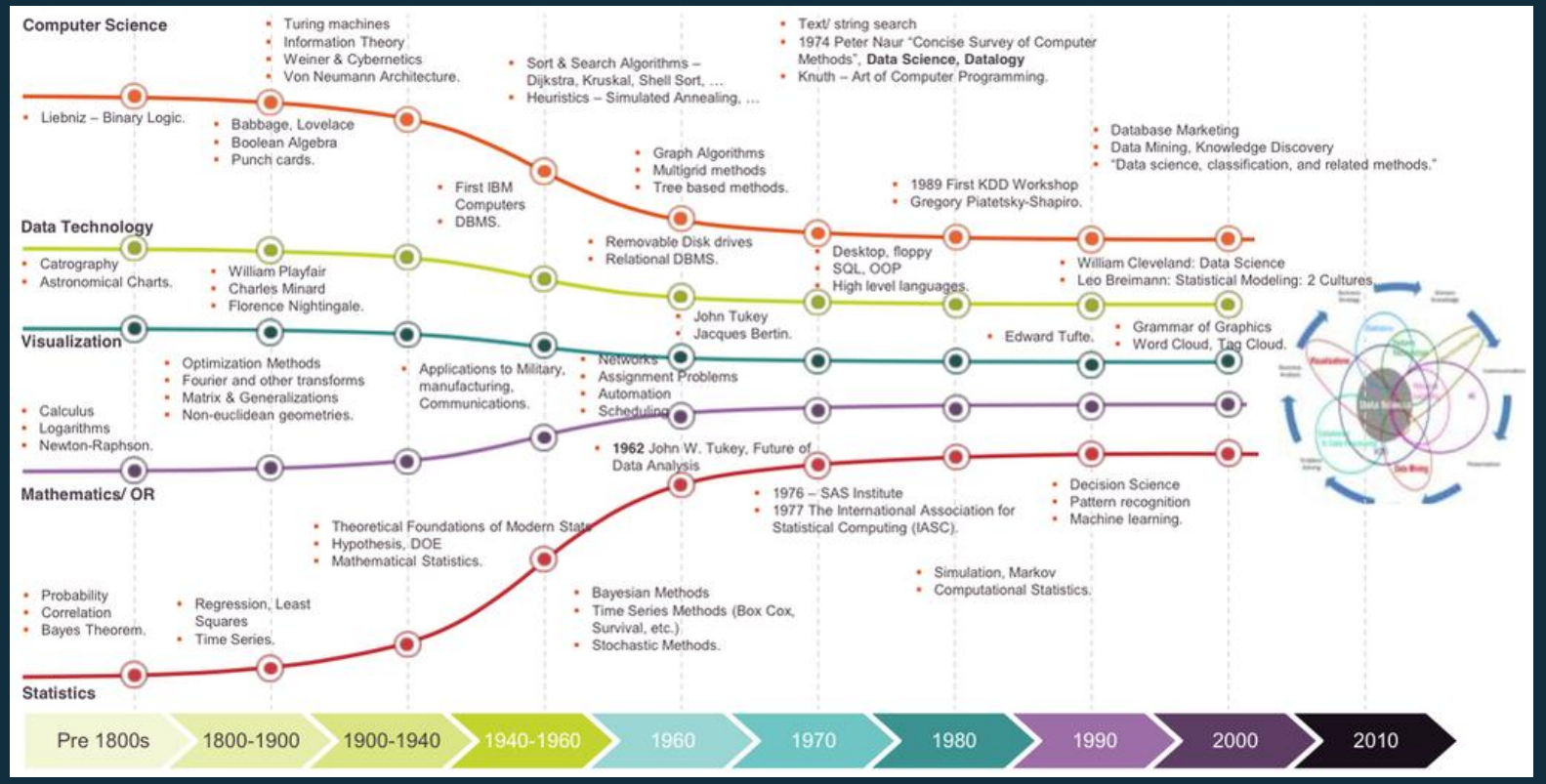
正态分布

箱式图

方差

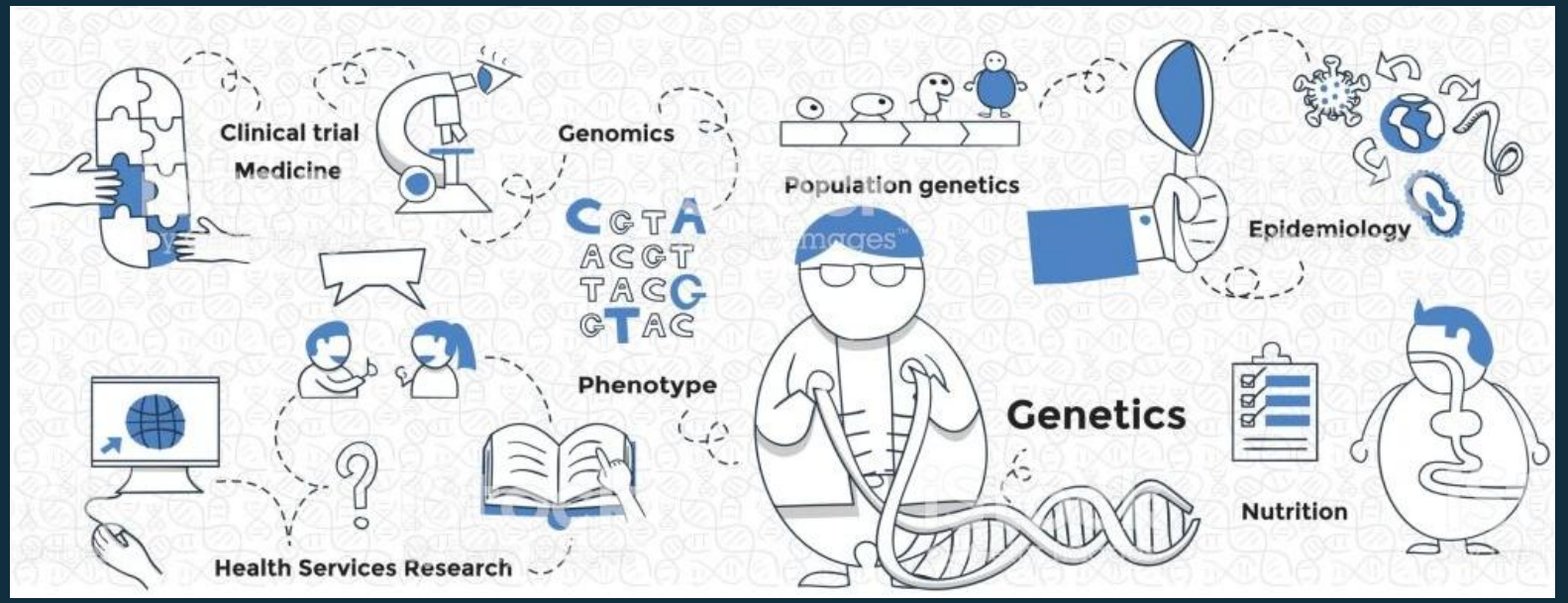


统计学：历史和地位！


Computer Science
Liebniz – Binary Logic.
Turing machines
Information Theory
Weiner & Cybernetics
Von Neumann Architecture.
Babbage, Lovelace
Boolean Algebra
Punch cards.
Sort & Search Algorithms – Dijkstra, Kruskal, Shell Sort, ...
Heuristics – Simulated Annealing, ...
First IBM Computers
DBMS.
Graph Algorithms
Multigrid methods
Tree based methods.
Text/ string search
1974 Peter Naur "Concise Survey of Computer Methods", Data Science, Datalogy
Knuth – Art of Computer Programming.
1989 First KDD Workshop
Gregory Piatetsky-Shapiro.
Database Marketing
Data Mining, Knowledge Discovery
"Data science, classification, and related methods."
Data Technology
Catrography
Astronomical Charts.
William Playfair
Charles Minard
Florence Nightingale.
Removable Disk drives
Relational DBMS.
Desktop, floppy
SQL, OOP
High level languages.
William Cleveland: Data Science
Leo Breimann: Statistical Modeling: 2 Cultures
Visualization
John Tukey
Jacques Bertin.
Edward Tufte.
Grammar of Graphics
Word Cloud, Tag Cloud.
Optimization Methods
Fourier and other transforms
Matrix & Generalizations
Non-euclidean geometries.
Applications to Military, manufacturing, Communications.
Networks
Assignment Problems
Automation
Scheduling
Calculus
Logarithms
Newton-Raphson.
1962 John W. Tukey, Future of Data Analysis
Mathematics/ OR
1976 – SAS Institute
1977 The International Association for Statistical Computing (IASC).
Decision Science
Pattern recognition
Machine learning.
Theoretical Foundations of Modern Stats
Hypothesis, DOE
Mathematical Statistics.
Simulation, Markov
Computational Statistics.
Probability
Correlation
Bayes Theorem.
Regression, Least Squares
Time Series.
Bayesian Methods
Time Series Methods (Box Cox, Survival, etc.)
Stochastic Methods.
Statistics
Pre 1800s
1800-1900
1900-1940
1940-1960
1960
1970
1980
1990
2000
2010

统 计 学 : 生 物统计学 !

Bridging the sciences
BIOSTATISTICS


Clinical trial
Medicine
Genomics
Population genetics
Epidemiology
Phenotype
Genetics
Nutrition
Health Services Research

两品种小麦的对比

为什么要学习生物统计学

一株转基因小麦各项指标都优于另一株非转基因小麦，是否可以确定转基因小麦产量提高？

实际情况可能是：

所选转基因小麦是实验田中长得最好的，而所选非转基因小麦是麦田中非常普通的一株。

怎样才能确认转基因小麦更优呢？

选取所有的小麦进行比较？

如 何 选 取

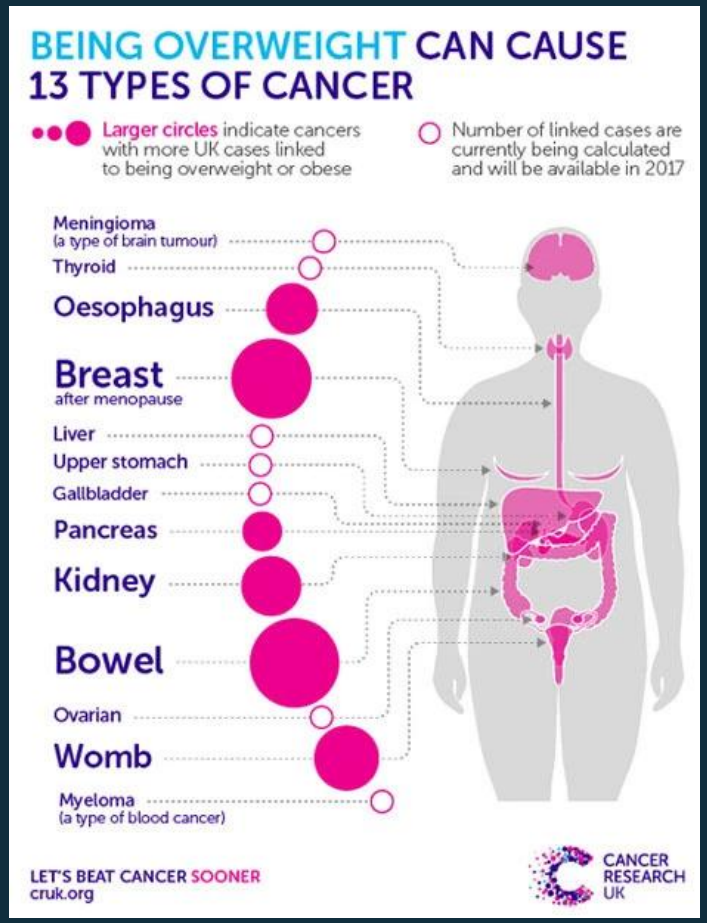
摘取多少小麦才能更好地保证对比结果的准确性呢？

为什么要学习生物统计学

阿司匹林对抗癌的疗效

口服阿司匹林，可能有助于治疗好多种癌症。

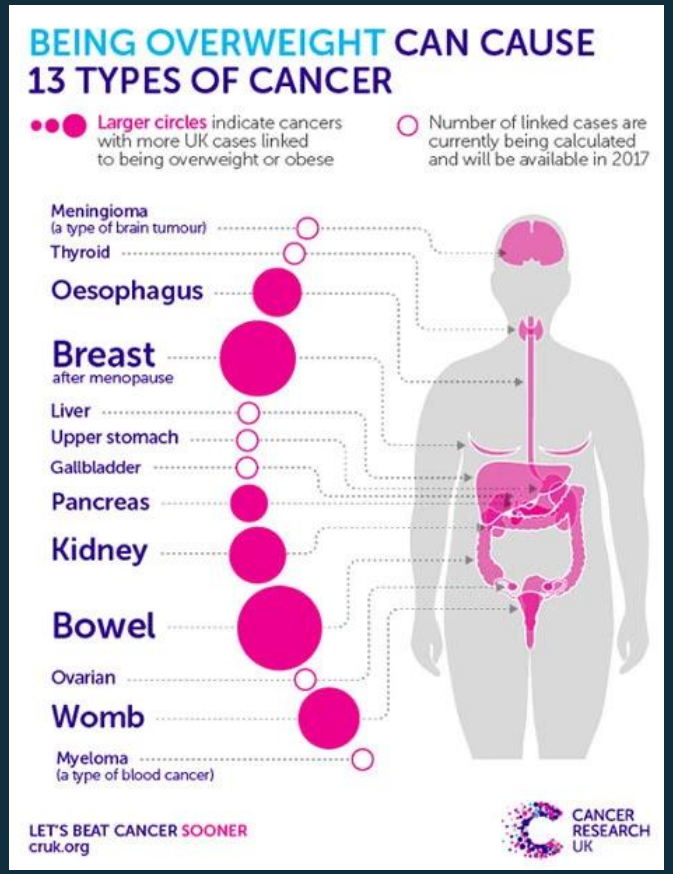
是真的吗？



为什么学习生物统计学

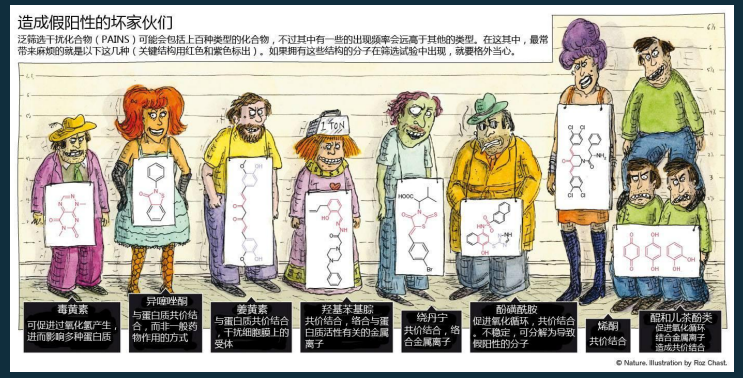
实际情况可能是：

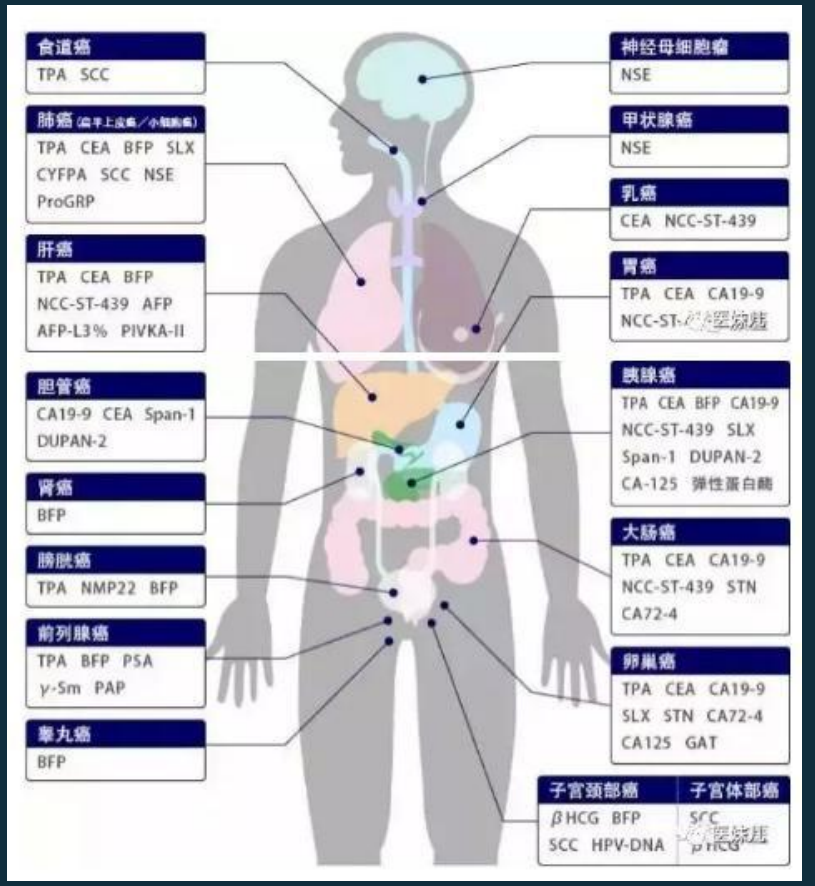
口服阿司匹林的同时，可能也口服了其它抗癌药物。。。



如何选取

要评估口服阿司匹林的效果，必须做好实验设计，控制单变量，评估其影响。





如何确立可靠的关联性

为什么要学习生物统计学

生物统计学是生物科学研究的基本工具

生物现象的特点：

变异性：个体之间存在差异

不确定性（随机性）：变异不能准确推算

复杂性：影响因素众多，有些是未知的

常规的数学方法不能解决问题

为什么要学习生物统计学

必须利用生物统计学才能回答的问题

疾病已经进入哪个阶段了？

哪些基因在疾病发生发展中起到关键作用？

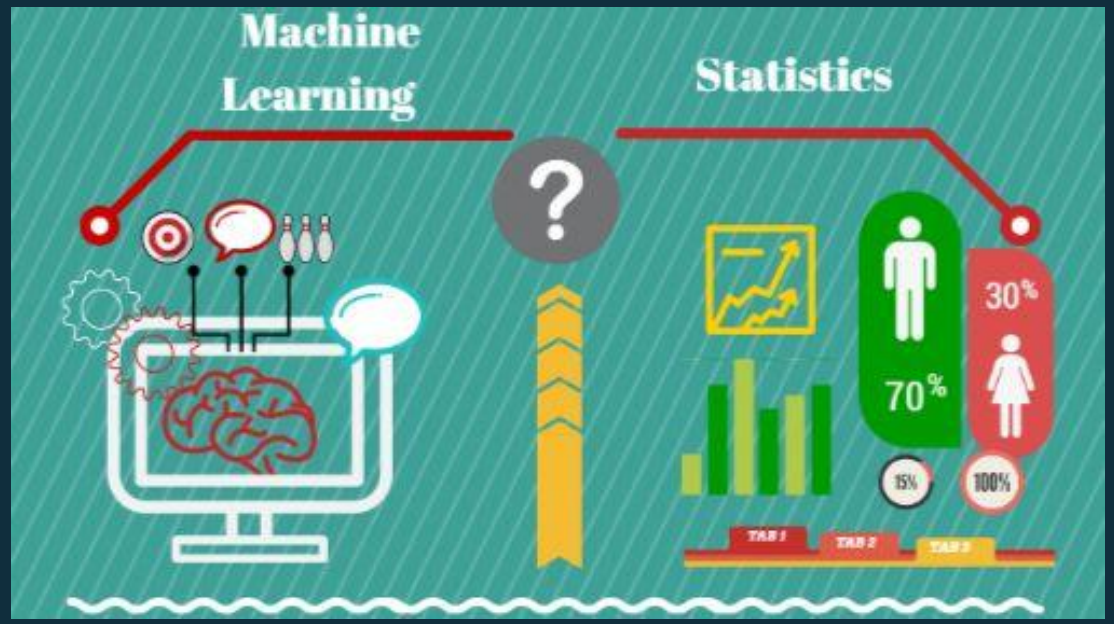
基因和环境是否有关？

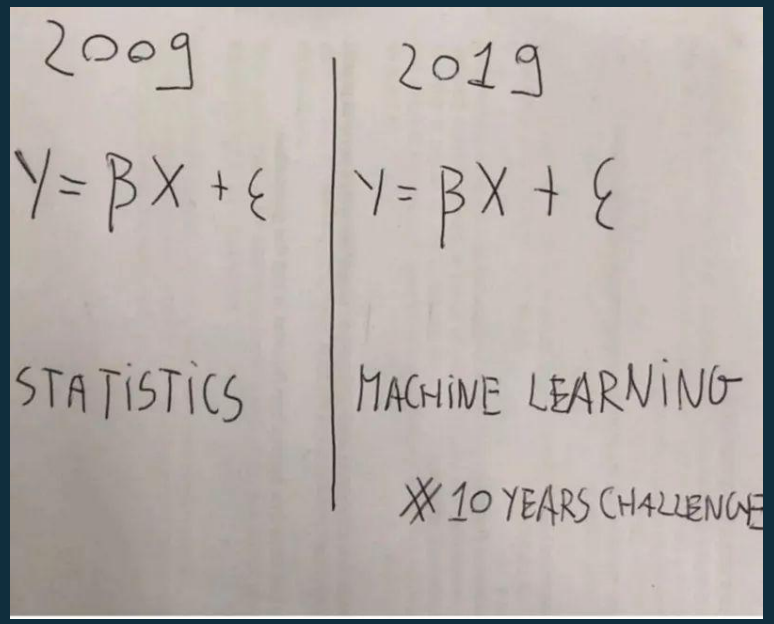
新药物是否更有效？

遗传与环境哪个更重要？

.....

为什么要学习生物统计学





https://mp.weixin.qq.com/s/xCJBowXS89UlHA07R8WNuw

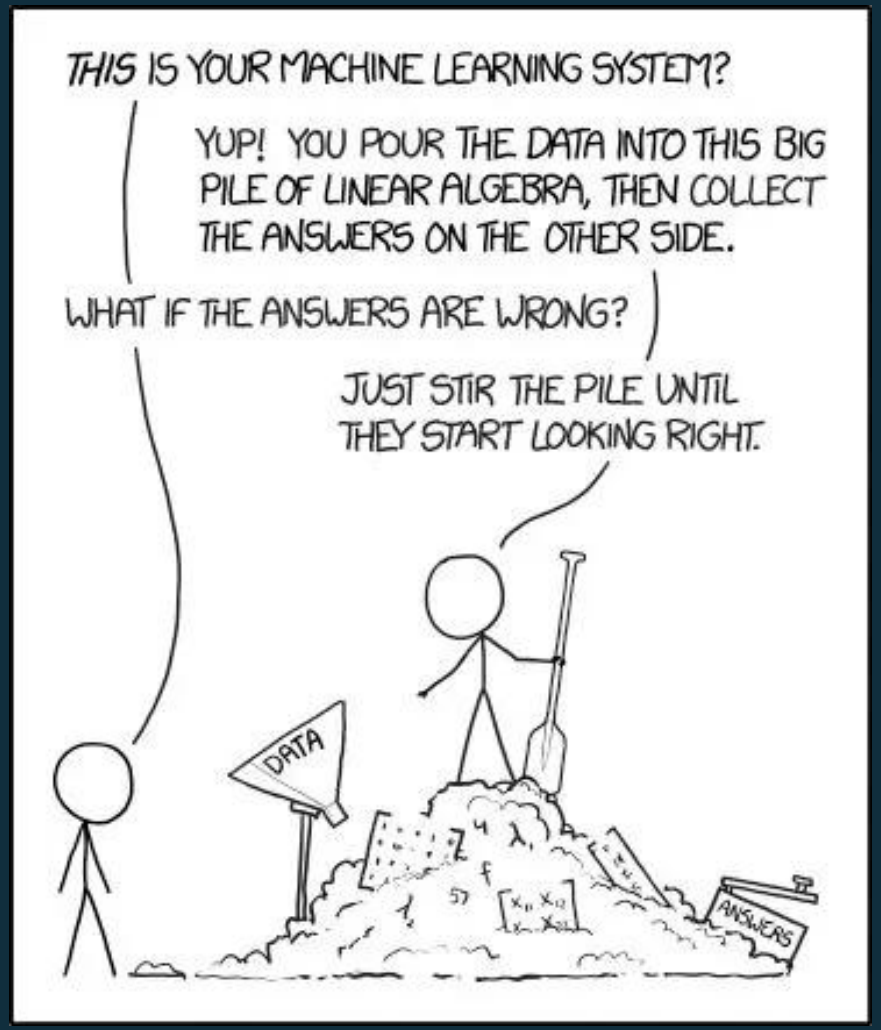
https://towardsdatascience.com/the-actual-difference-between-statistics-and-machine-learning-64b49f07ea3?gi=412e8f93e22e

为什么要学习生物统计学

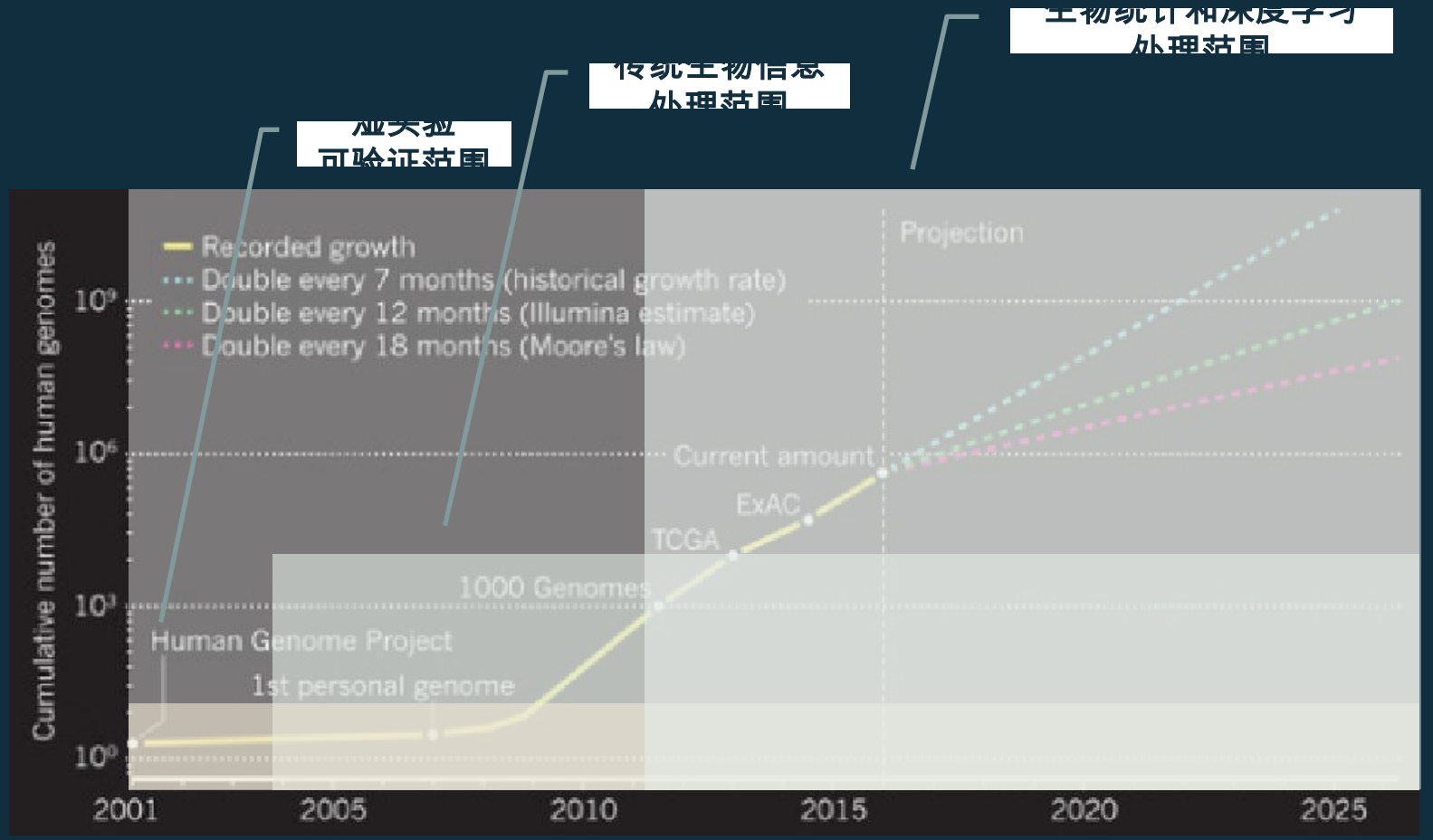
https://towardsdatascience.com/the-actual-difference-between-statistics-and-machine-learning-64b49f07ea3?gi=412e8f93e22e

如果你只是想从数据中找出哪类人更容易得某种疾病，机器学习可能是更好的选择。

如果你希望找出变量之间的关系或从数据中得出推论，选择统计模型会更好。



为什么要学习生物统计学



学习方法与要求

要弄懂统计的基本原理和基本公式；

要认真做好习题作业，加深对公式及统计步骤的理解，达到能熟练地应用统计方法；

注意培养科学的统计思维方法 ，理论联系实际，结合专业，了解统计方法的实际应用。

教材及参考书目

教学参考书：

《生物统计学：生物大数据的概率统计模型与机器学习方法》. 华中科技大学出版社. 2023年出版. 宁康等编著

《生物统计学》（第5版），普通高等教育十二五国家级规划教材). 科学出版社. 2013年6月出版. 李春喜，姜丽娜，邵云，张黛静编著.

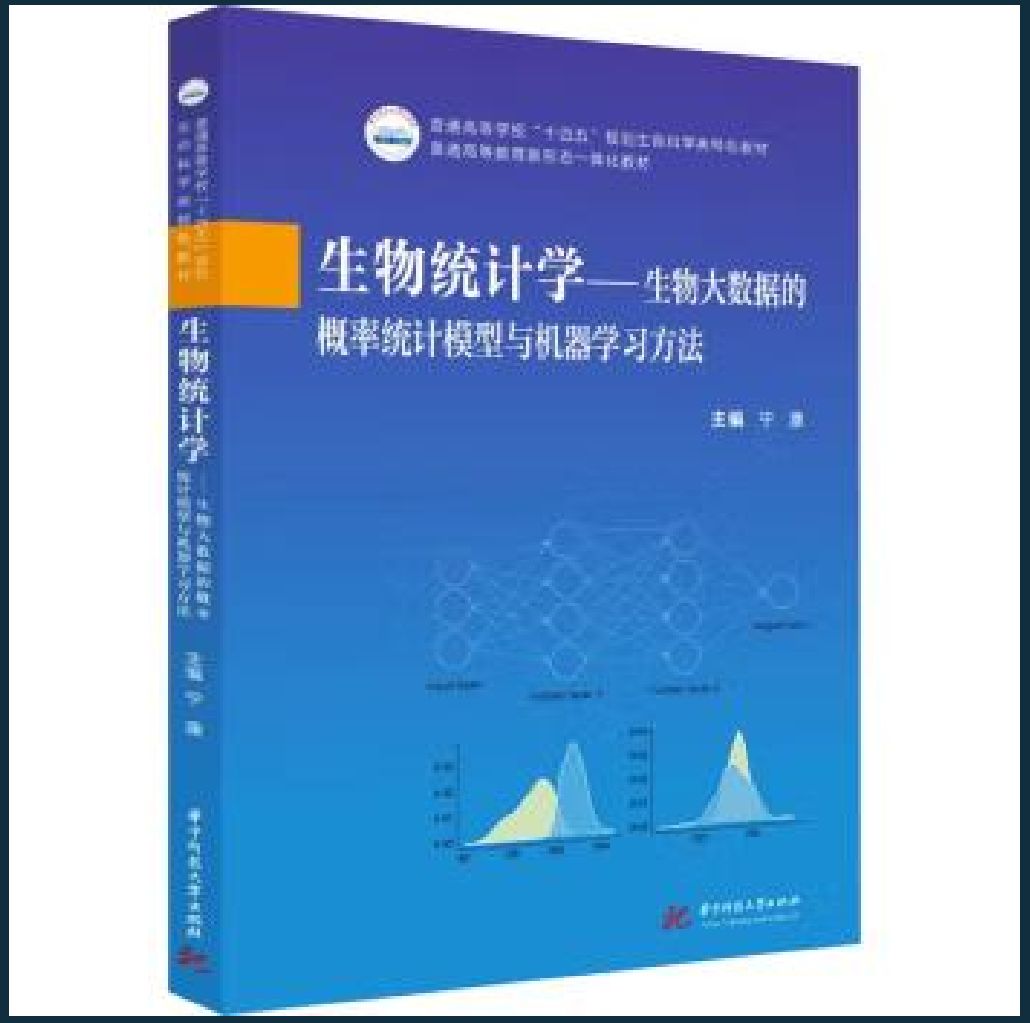
《生物序列分析》（第1版）.科学出版社. 2010年8月出版. R. Durbin等编著，王俊等主译.

课外参考书：

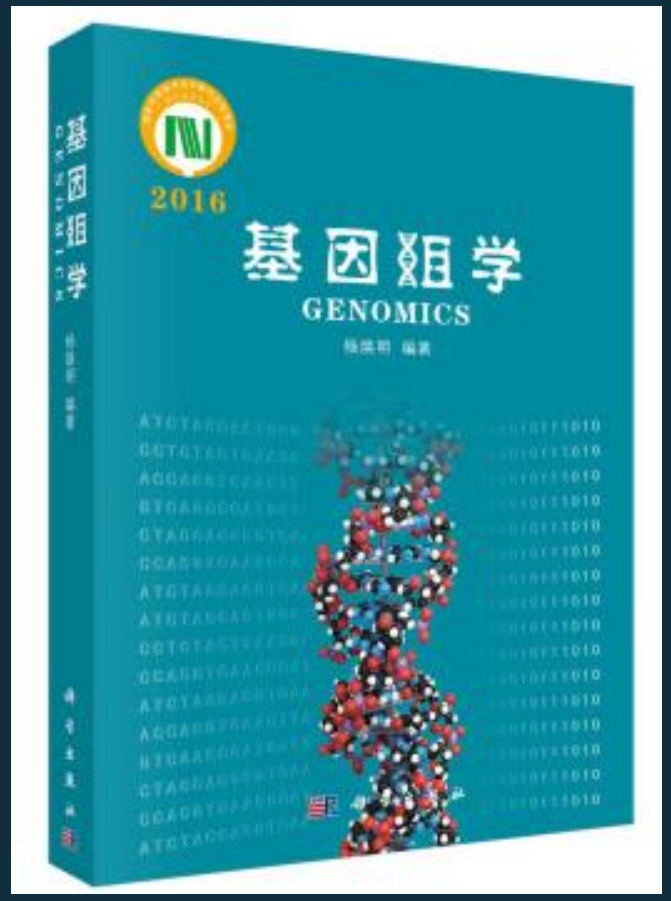
《生物统计学习题与详解》. 华中科技大学出版社. 2022年出版. 宁康等编著

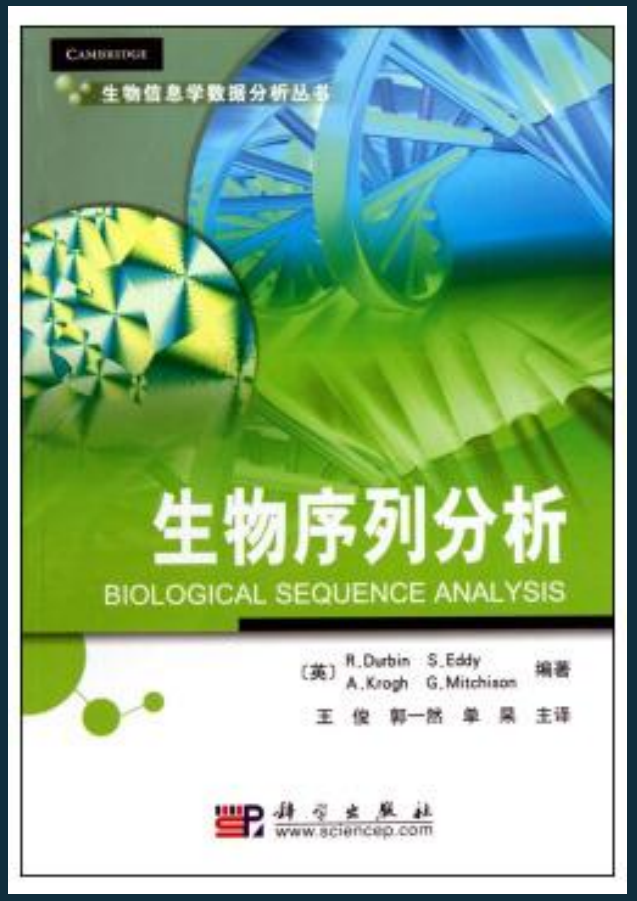
《生物信息学》（101系列教材）.高等教育出报社. 2025年出版. 陈铭、吕辉、宁康、张勇、张子丁等。

References



References


2016
基因组学
GENOMICS


CAMBRIDGE
生物信息学数据分析丛书
生物序列分析
BIOLOGICAL SEQUENCE ANALYSIS
R.Durbin S.Eddy
A.Krogh G.Mitchison
王俊 郭一然 单昊 主译
www.sciencep.com

生物统计学
(第五版)

References

智能时代 深度学习 · 文津图书奖 · 中华优秀出版物提名奖
吴军
著
JUST PUB
数学之美
第三版
中国工信出版集团
人民邮电出版社
POSTS & TELECOM PRESS

课程范围

生物统计学的范围

一切和生物相关数据的分析有关的统计

面向生物信息和大数据挖掘的生物统计学特点

兼容并包、同时注重方法和应用

生物统计学的应用

精准医学的应用

课程结构

生物统计学基础；

生物信息中的算法设计与概率统计模型；

生物大数据和深度学习。

LECTURE 01

生物统计学：生物学背景知识

Biostatistics

Biostatistics is the application of statistics to a wide range of topics in biology.

The science of biostatistics encompasses the design of biological experiments, especially in medicine, pharmacy, agriculture and fishery; the collection, summarization, and analysis of data from those experiments; and the interpretation of, and inference from, the results.

A major branch of this is medical biostatistics, which is exclusively concerned with medicine and health.

生物信息学

Bioinformatics is the research, development, or application of computational tools and approaches for expanding the use of biological, medical, behavioral or health data, including those to acquire, store, organize, archive, analyze, or visualize such data;

Computational biology is the development and application of data-analytical and theoretical methods, mathematical modeling and computational simulation techniques to the study of biological, behavioral, and social systems.

生物信息学

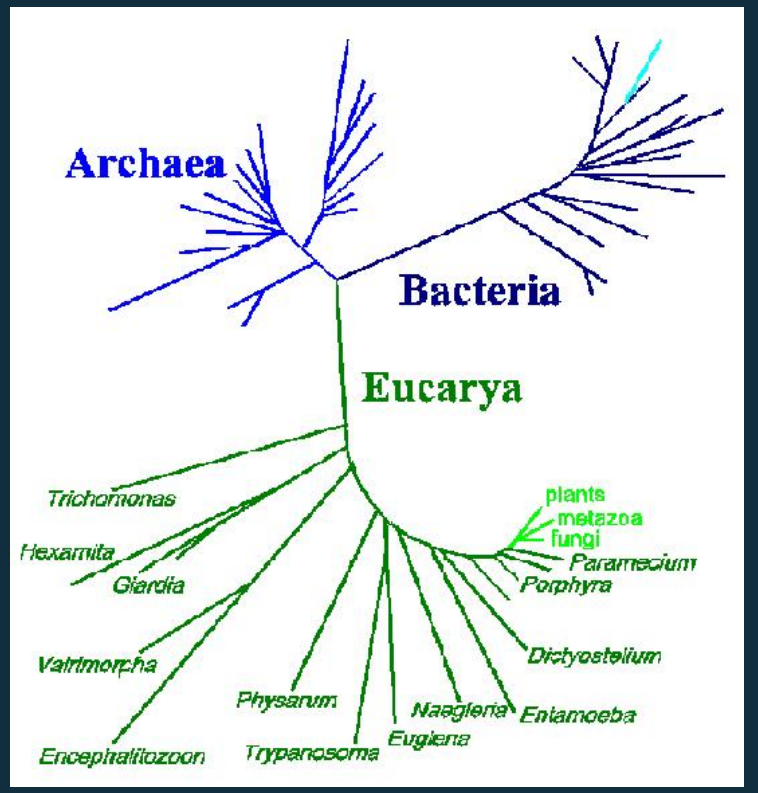
Overlay of Biology, Computer science and Statistics.

Topics:

- Sequence alignment
- Protein folding
- Gene finding
- Functional annotation
- Network inference

研究对象: 生物序列, 进化树, 生物网络, 基因表达...

Tree of Life



modified from N.R. Pace, ASM News 62:464, 1996

Molecules of Life

DNA

RNA

Protein

DNA

Deoxyribonucleic acid(脱氧核糖核酸)

Consist of four nucleotides

A Adenine(腺嘌呤)

C Cytosine(胞嘧啶)

G Guanine(鸟嘌呤)

T Thymine(胸腺嘧啶)

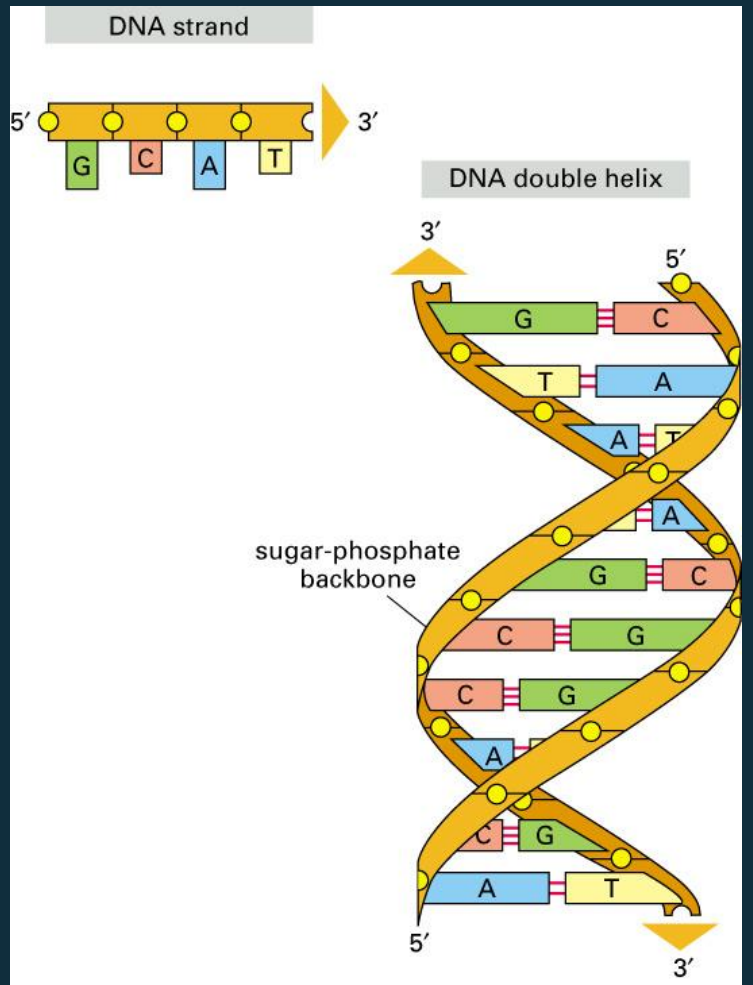


Figure 4–3 part 2 of 2. Molecular Biology of the Cell, 4th Edition.

RNA

Ribonucleic acid (核糖核酸)

mRNA : Messenger RNAs, code for proteins

rRNA: Ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis.

tRNA: Transfer RNAs, central to protein synthesis as adaptors between mRNA and amino acids.

Proteins

Main building blocks and functional molecules, take up ~20% of eukaryotic cell’s weight.

Structural proteins

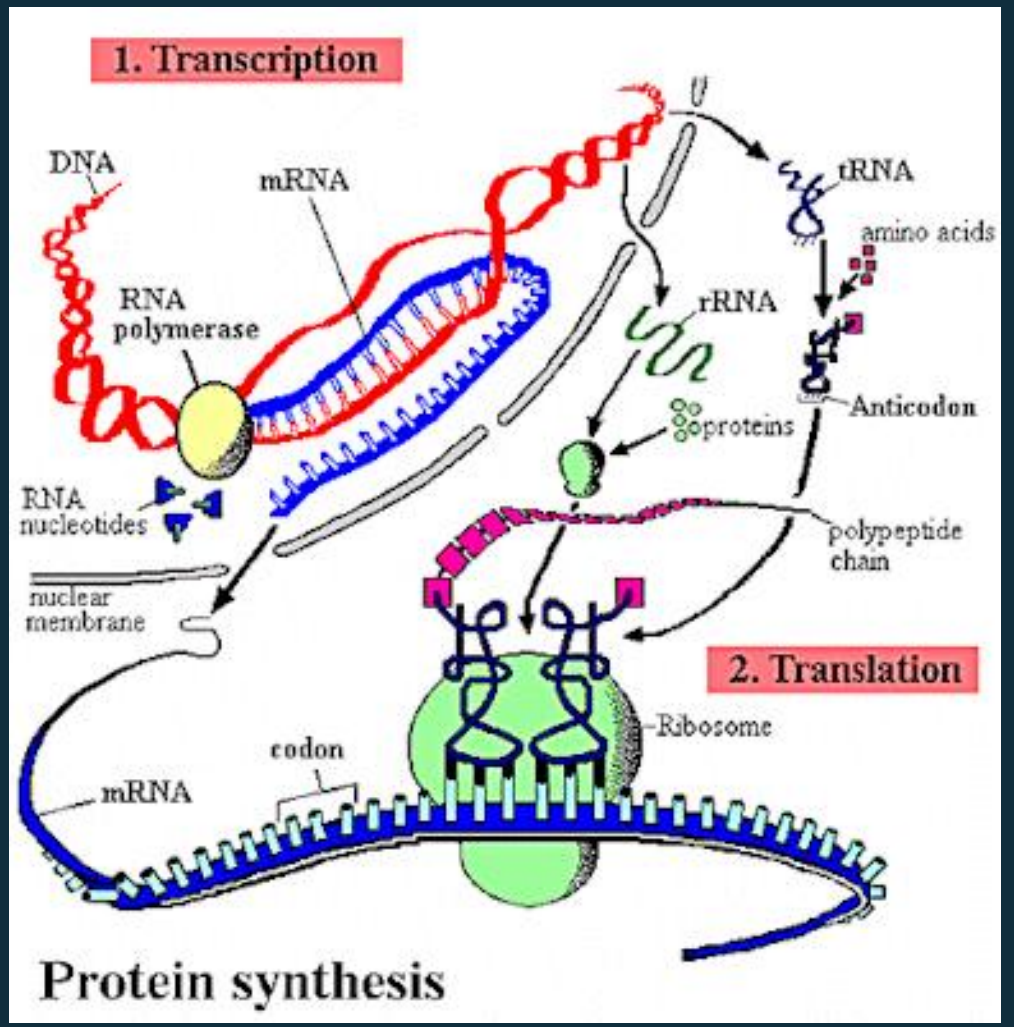
Enzymes

Antibodies

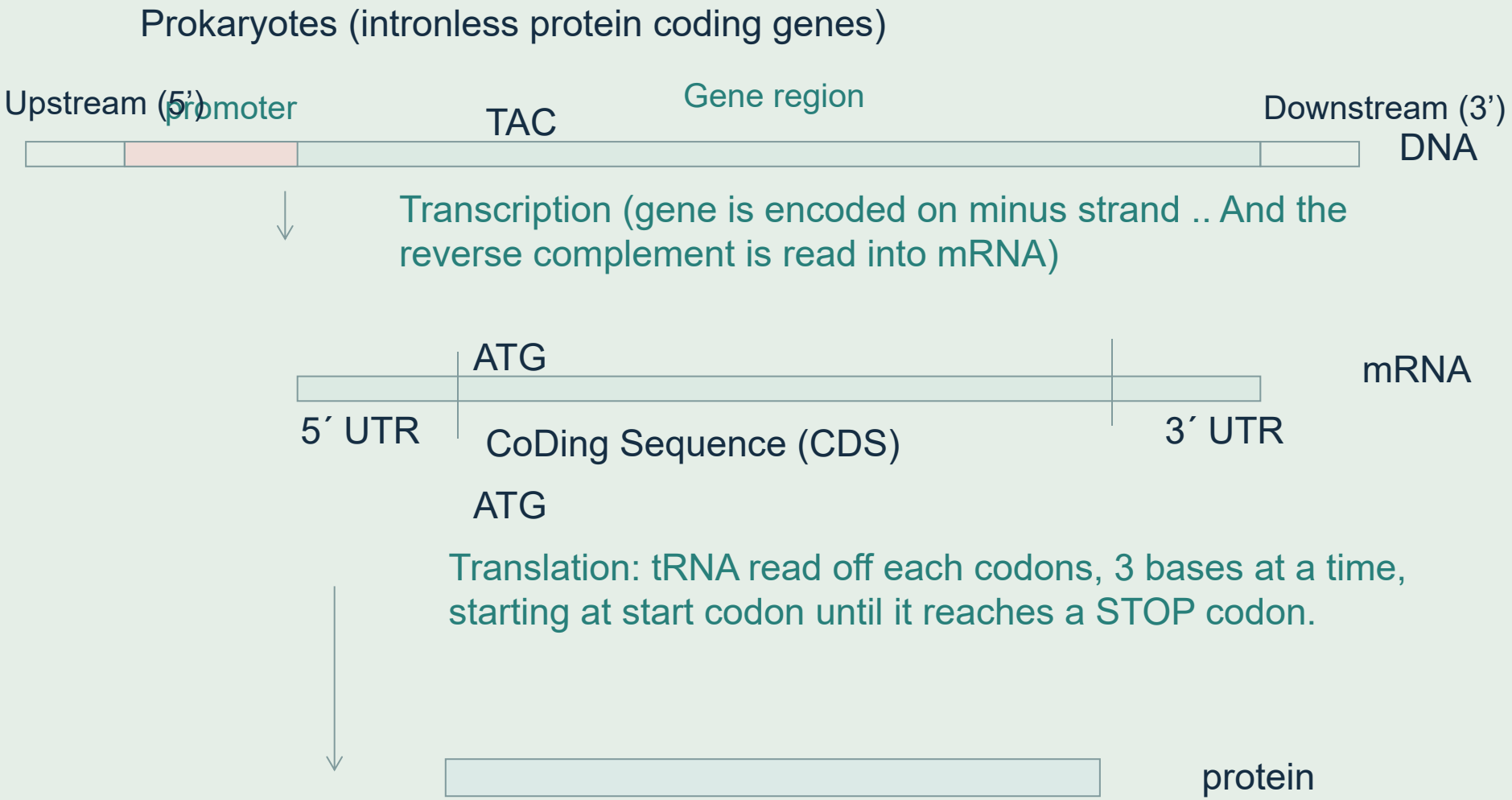
Transmembrane proteins

The Central Dogma

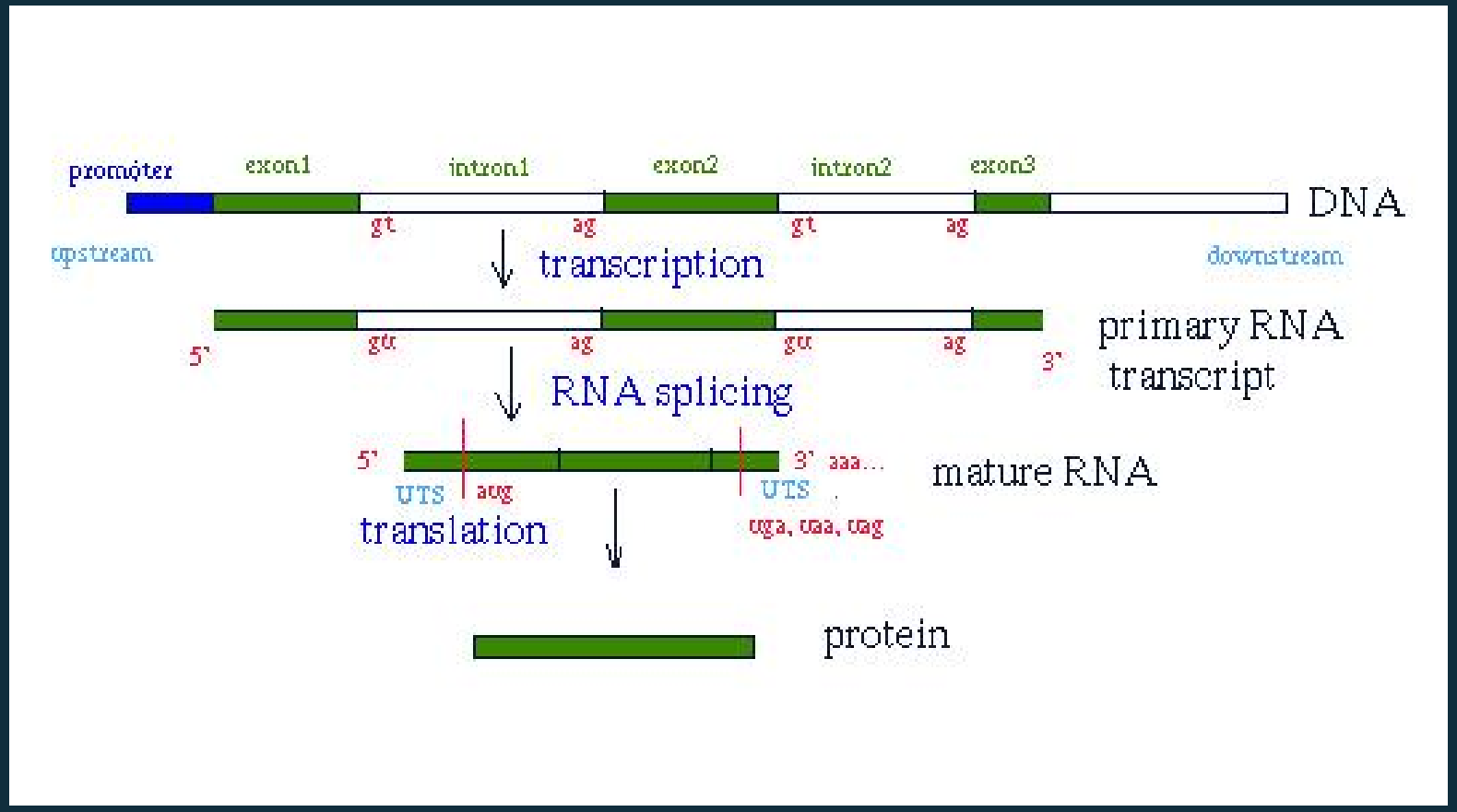
DNA→RNA→Protein



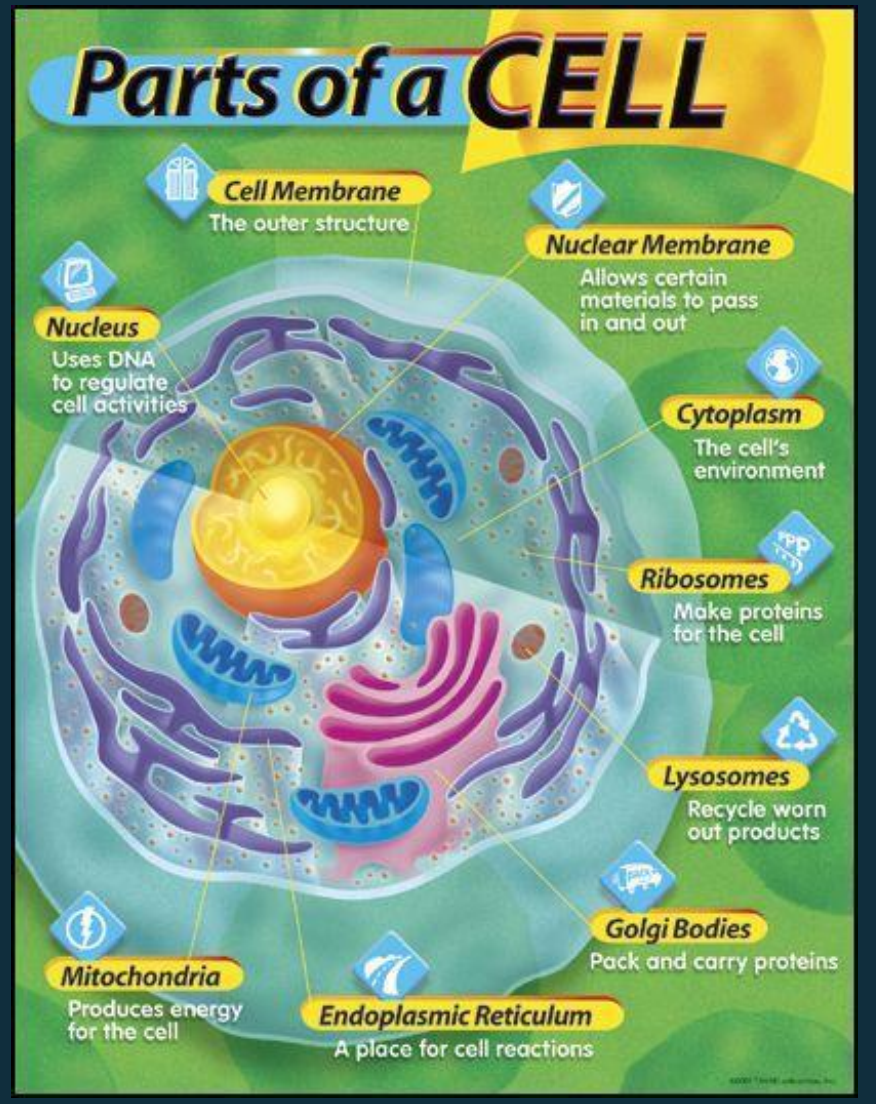
Prokaryotic Genes



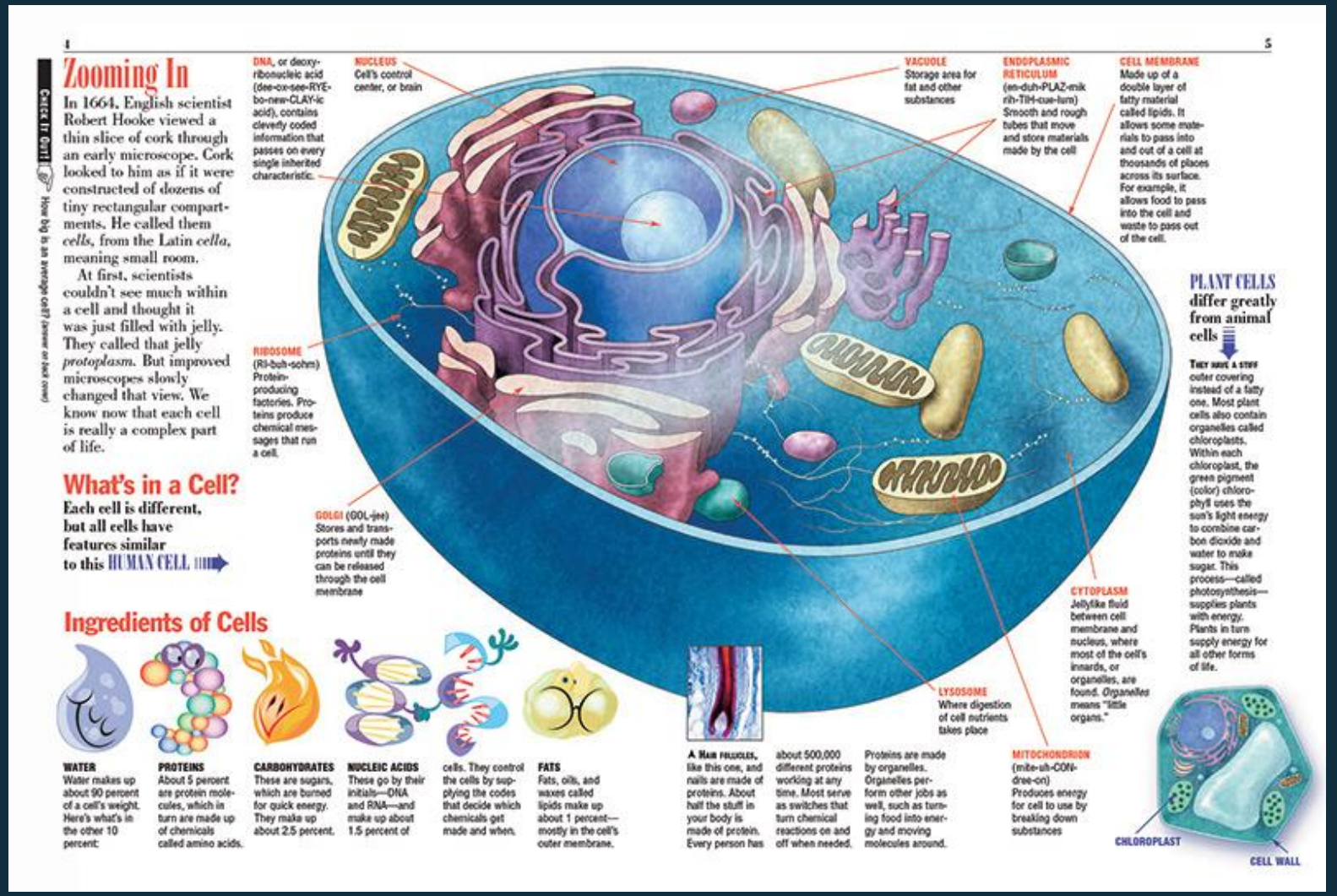
Eukaryotic Genes



Cell structure



Cell structure



Biostatistics

差异表达基因分析

基因共表达分析

基因表达数据的聚类和分类

基因集分析

基因调控网络

Case 1：基因表达数据分析

Biostatistics

通过测序，找到了一批“interesting” 的基因

差异表达基因

不做差异基因鉴定，直接做基因集分析

生物学功能上是否存在关联？

某种功能是否显著？

计算分析方法

基因本体 (Gene Ontology)

KEGG (Kyoto Encyclopedia of Genes and Genomes)

超几何分布

Case 2：基因集合分析

Biostatistics

物种的富集分析

功能的富集分析

MWAS分析

Case 3：微生物组生态数据分析

LECTURE 01

生物统计学：统计建模和人工智能背景知识

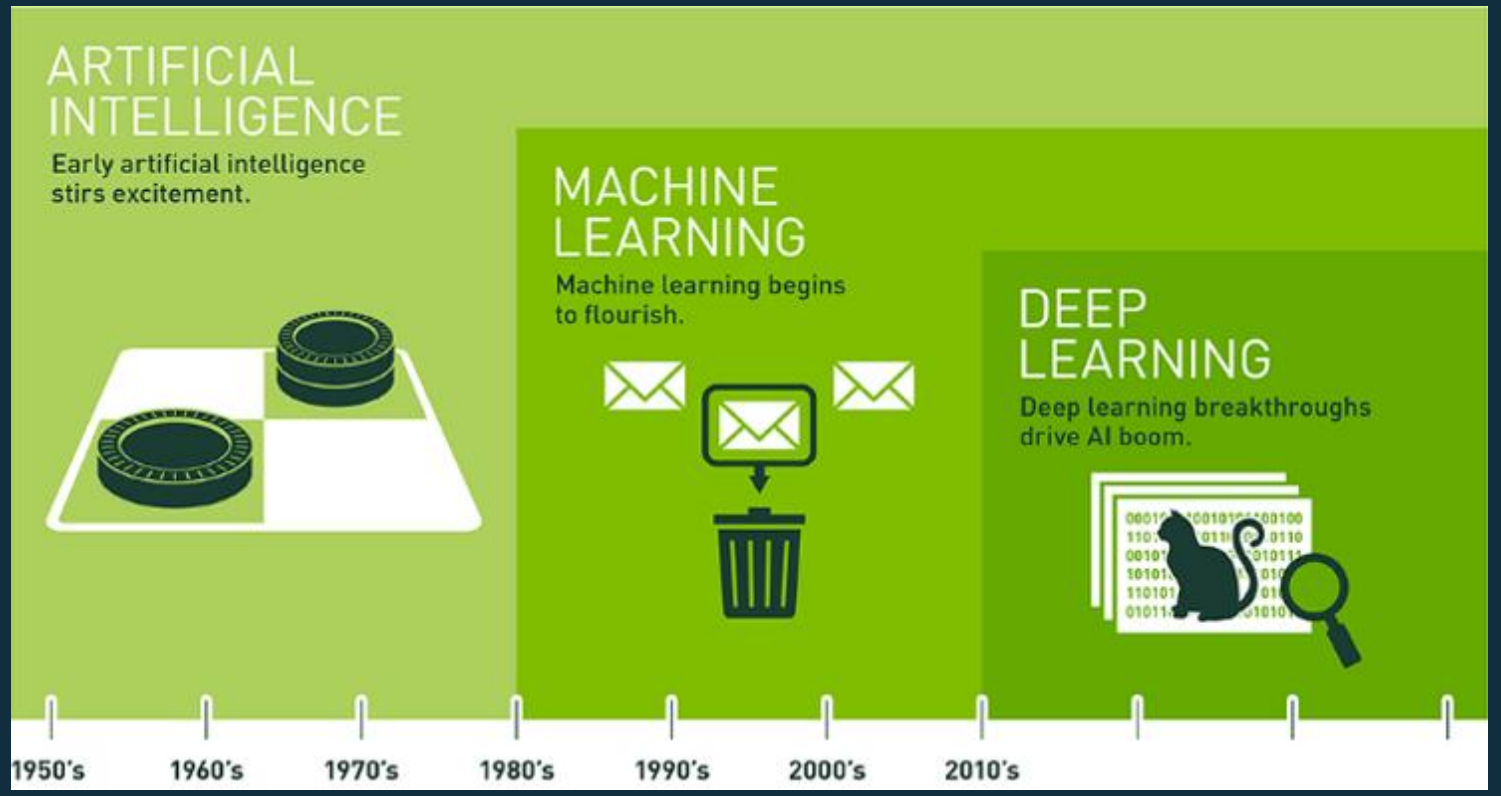
人工智能与机器学习、深度学习的关系

Introduction

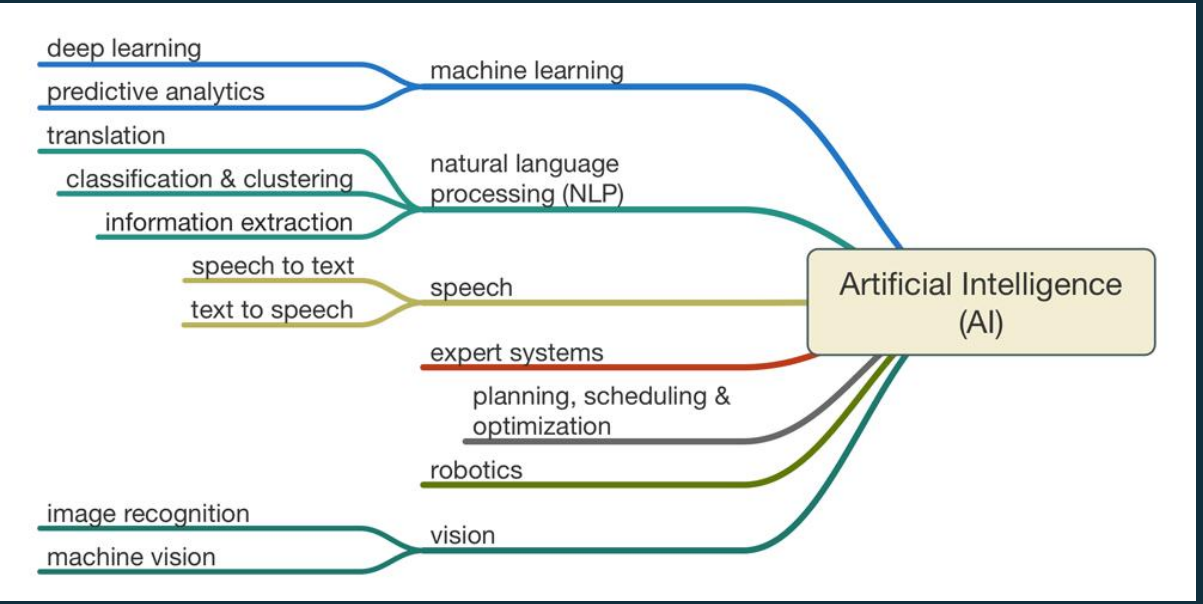
Artificial Intelligence

Machine learning

“Deep” learning

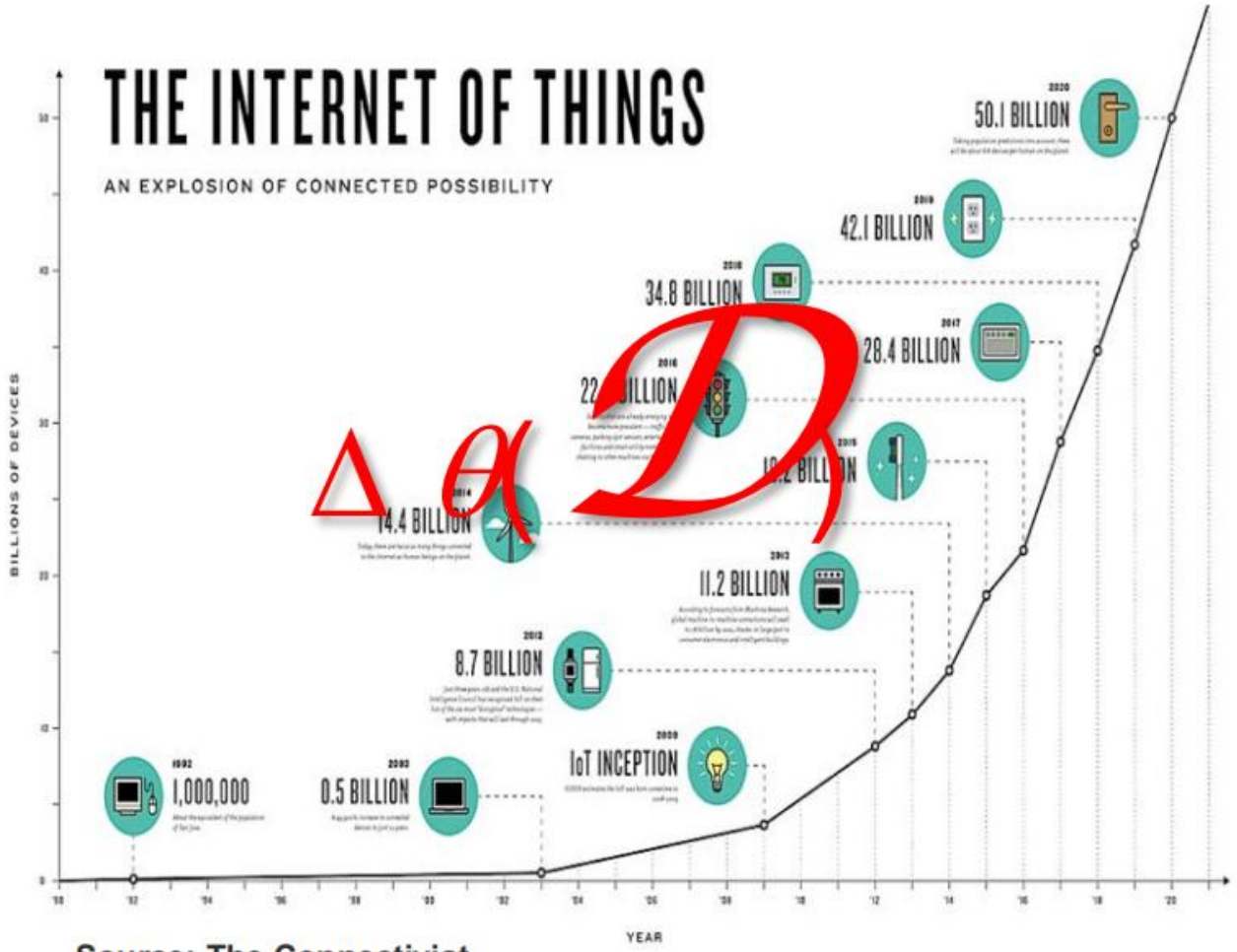


Artificial Intelligence

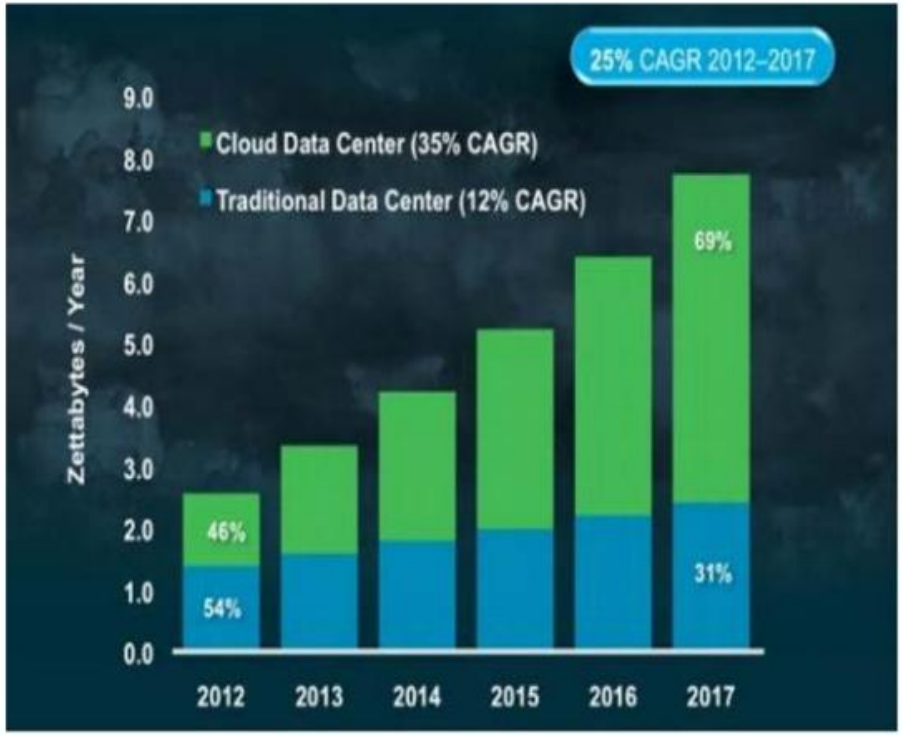


Hatley, L. (2016). Presentation, New Designs for Learning: Games and Gamification
https://en.wikipedia.org/wiki/Applications_of_artificial_intelligence

Big-data

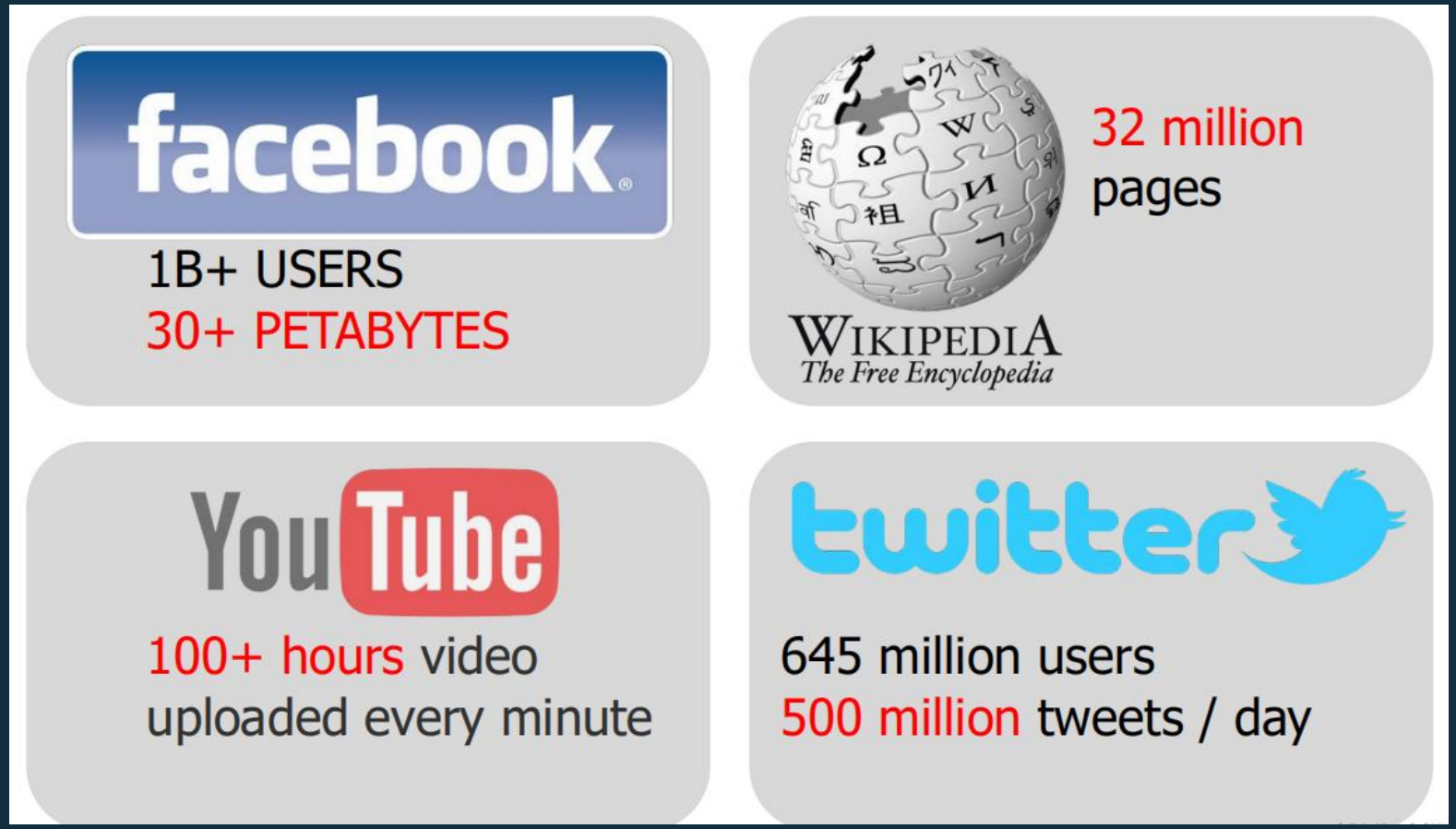


Source: The Connectivist

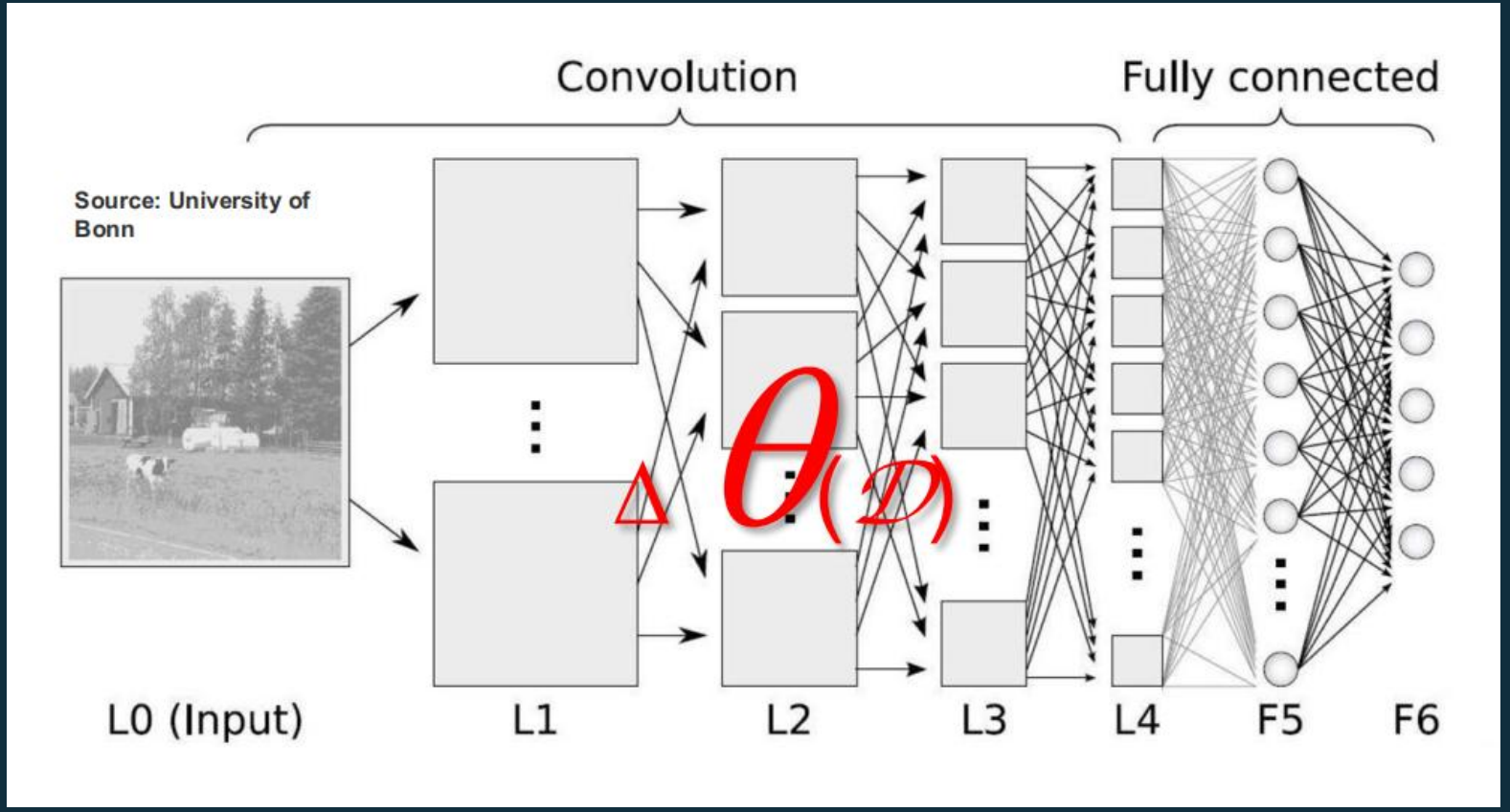


Source: Cisco Global Cloud Index

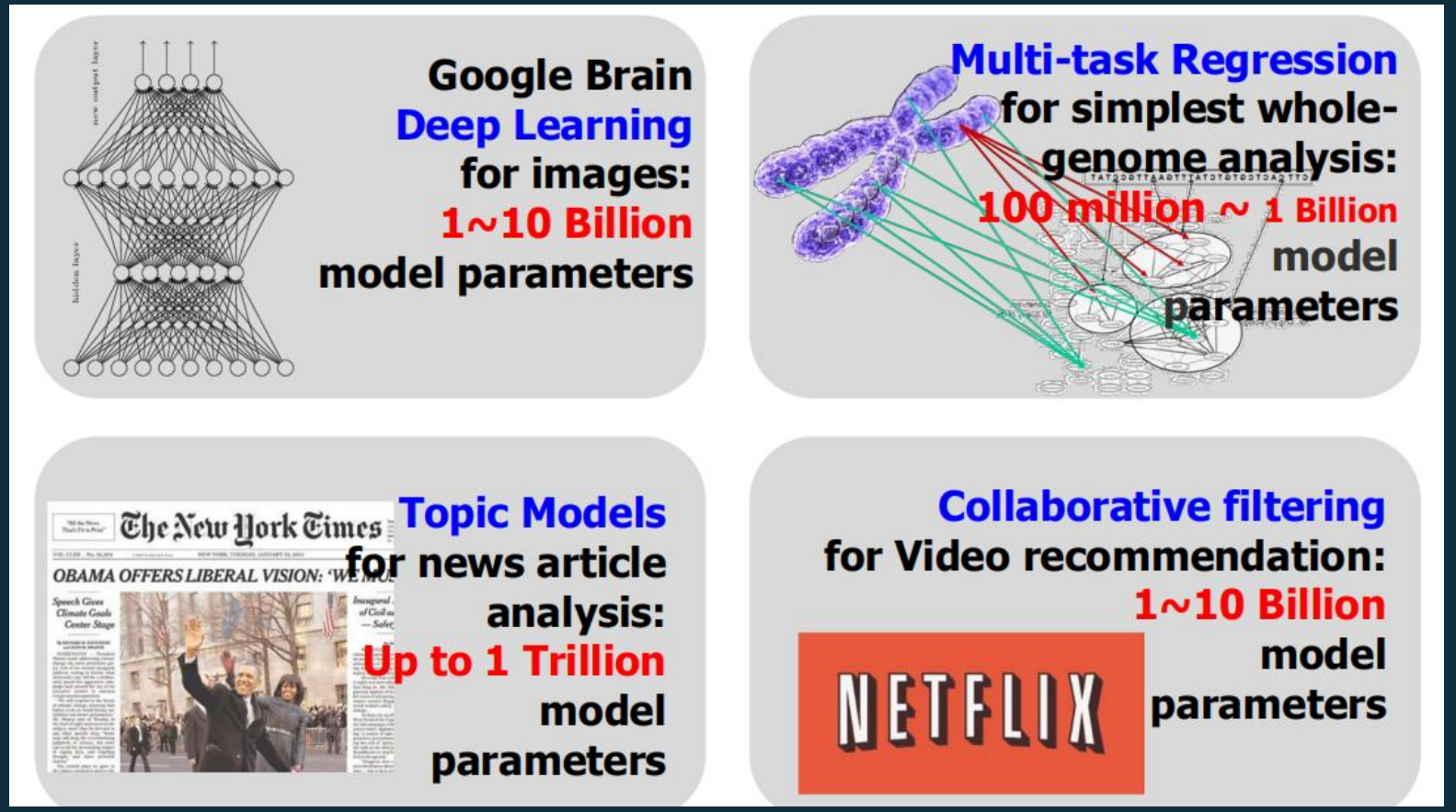
Big-data


facebook
1B+ USERS
30+ PETABYTES
32 million
pages
WIKIPEDIA
The Free Encyclopedia
YouTube
100+ hours video
uploaded every minute
twitter
645 million users
500 million tweets / day

Big-model


Convolution
Fully connected
Source: University of Bonn
$\Delta\theta(\mathcal{D})$
L0 (Input)
L1
L2
L3
L4
F5
F6

Big-model


Google Brain
Deep Learning
for images:
1~10 Billion
model parameters
Multi-task Regression
for simplest whole-genome analysis:
100 million ~ 1 Billion
model parameters
The New York Times
OBAMA OFFERS LIBERAL VISION:
Topic Models
for news article analysis:
Up to 1 Trillion
model parameters
Collaborative filtering
for Video recommendation:
1~10 Billion
model parameters
NETFLIX

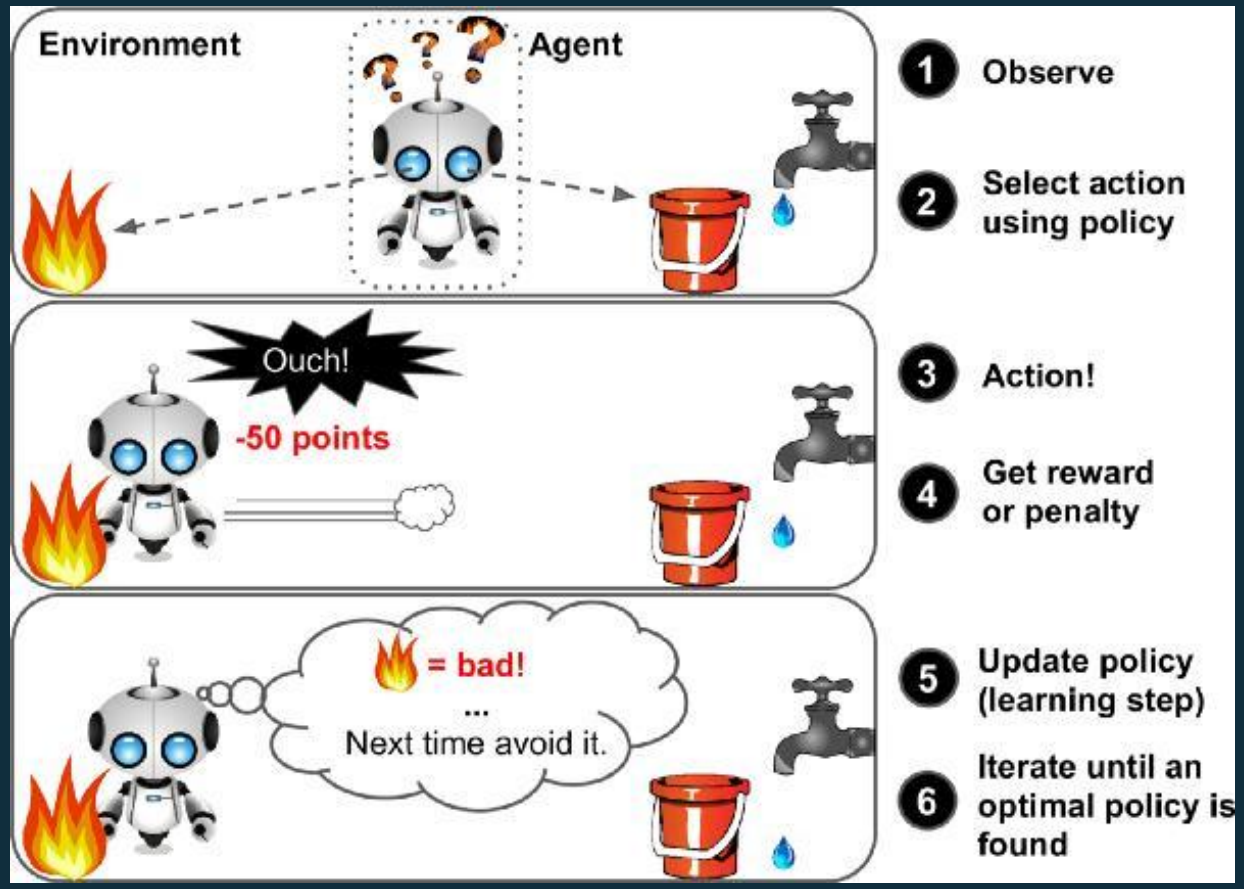
Big-data + Deep learning

Big-data is a term for data sets that are so large or complex that traditional data processing application software is inadequate to deal with them. Challenges include capture, storage, analysis, data curation, search, sharing, transfer, visualization, querying, updating and information privacy.

Deep learning is part of a broader family of machine learning methods based on learning representations of data. Research in this area attempts to make better representations and create models to learn these representations from large-scale unlabeled data.

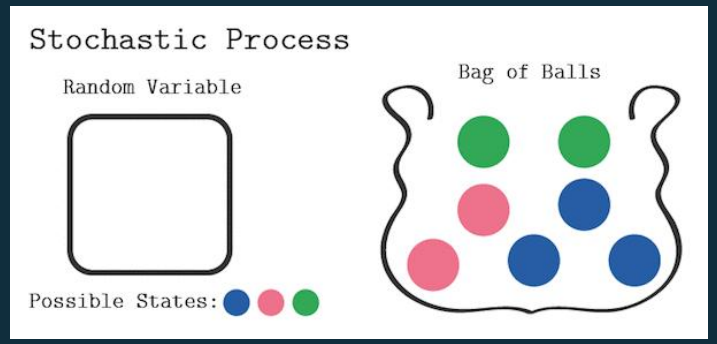
Big-data + Deep learning

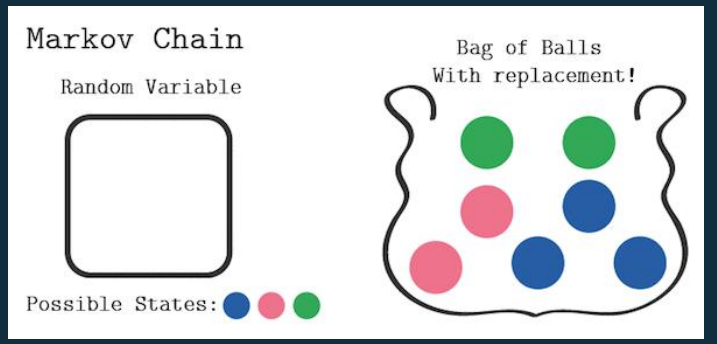
Reinforcement Learning



Big-data + Deep learning

Markov chain





“The future is independent of the past given the present!”

Big-data + Deep learning

IMAGENET Large Scale Visual Recognition Challenge

Year 2010
NEC-UIUC

Dense descriptor grid:
HOG, LBP

Coding: local coordinate, super-vector

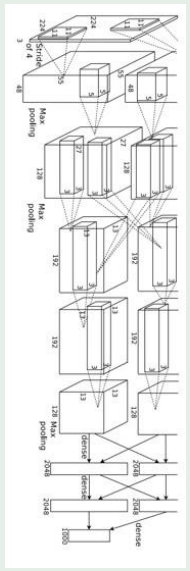
Pooling,
SPM

Linear SVM

[Lin CVPR 2011]

Lion image by Swissfrog is licensed under CC BY 3.0

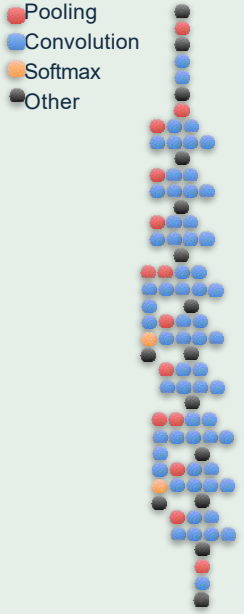
Year 2012
SuperVision



[Krizhevsky NIPS 2012]

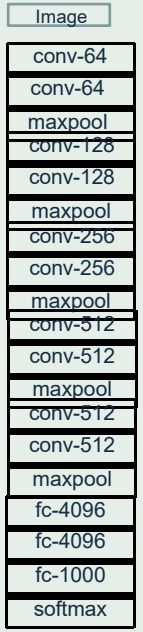
Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

Year 2014
GoogLeNet



[Szegedy arxiv 2014]

VGG

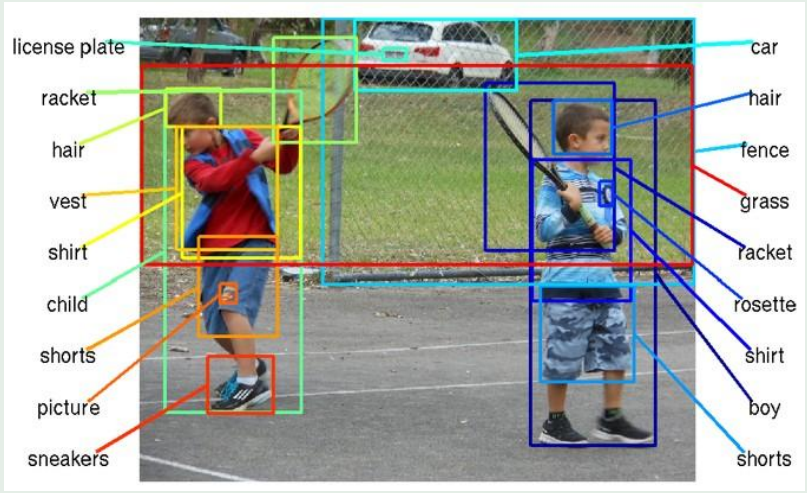


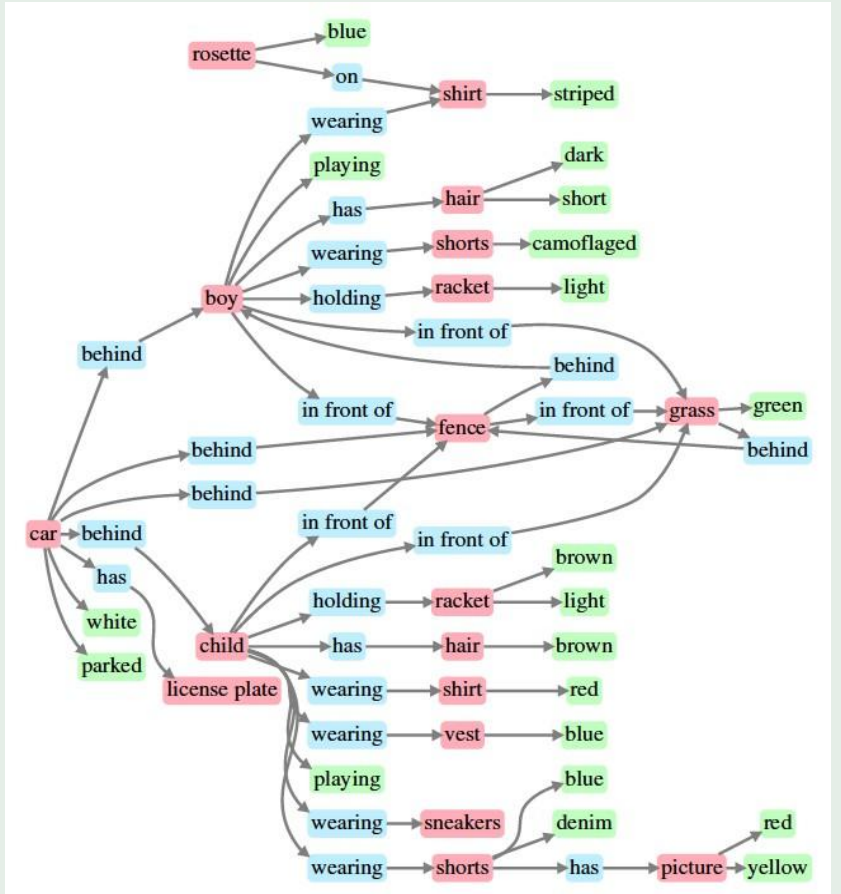
[Simonyan arxiv 2014]

Year 2015
MSRA

[He ICCV 2015]

Big-data + Deep learning





Johnson *et al.*, “Image Retrieval using Scene Graphs”, CVPR 2015

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Big-data + Deep learning

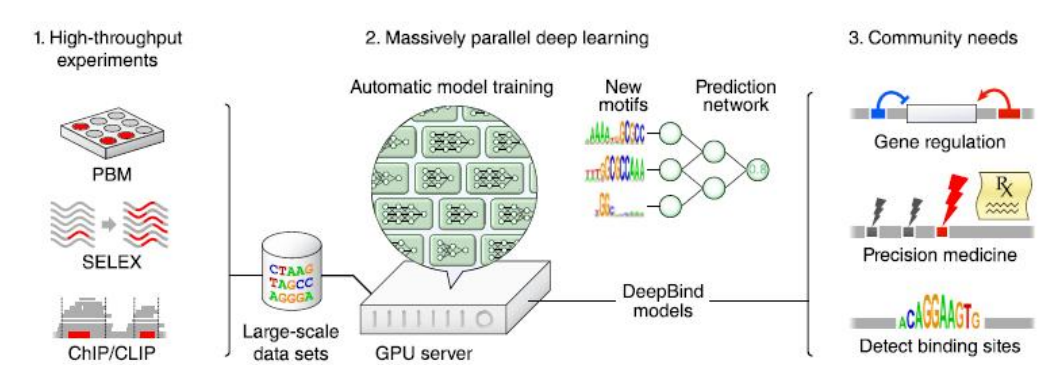


Figure 1 DeepBind's input data, training procedure and applications. 1. The sequence specificities of DNA- and RNA-binding proteins can now be measured by several types of high-throughput assay, including PBM, SELEX, and ChIP- and CLIP-seq techniques. 2. DeepBind captures these binding specificities from raw sequence data by jointly discovering new sequence motifs along with rules for combining them into a predictive binding score. Graphics processing units (GPUs) are used to automatically train high-quality models, with expert tuning allowed but not required. 3. The resulting DeepBind models can then be used to identify binding sites in test sequences and to score the effects of novel mutations.

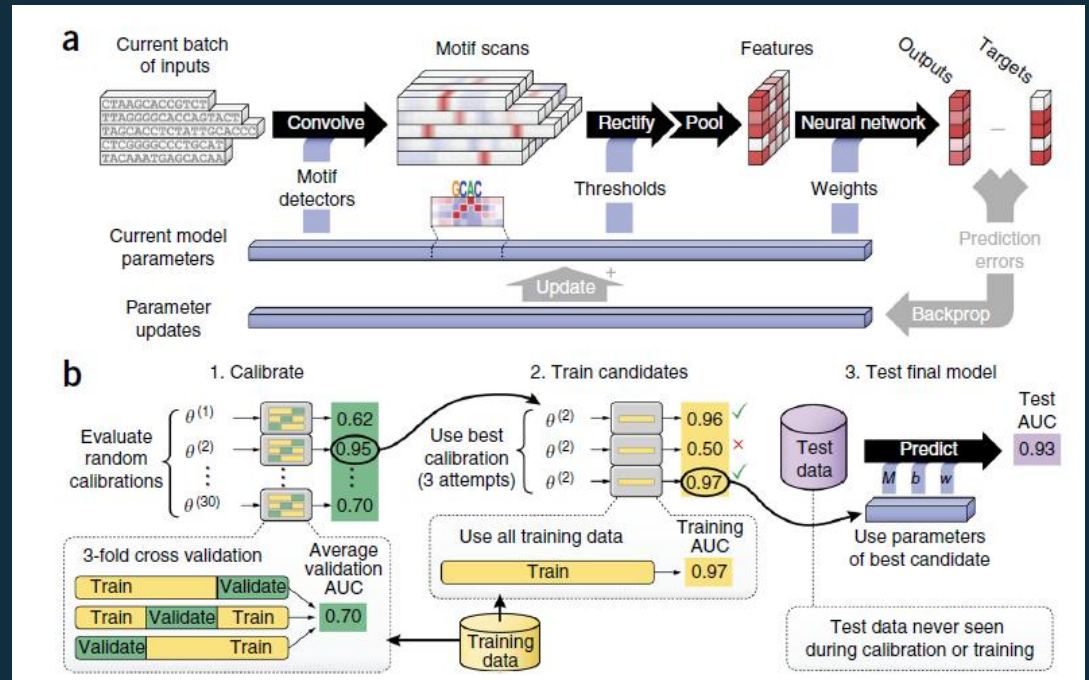
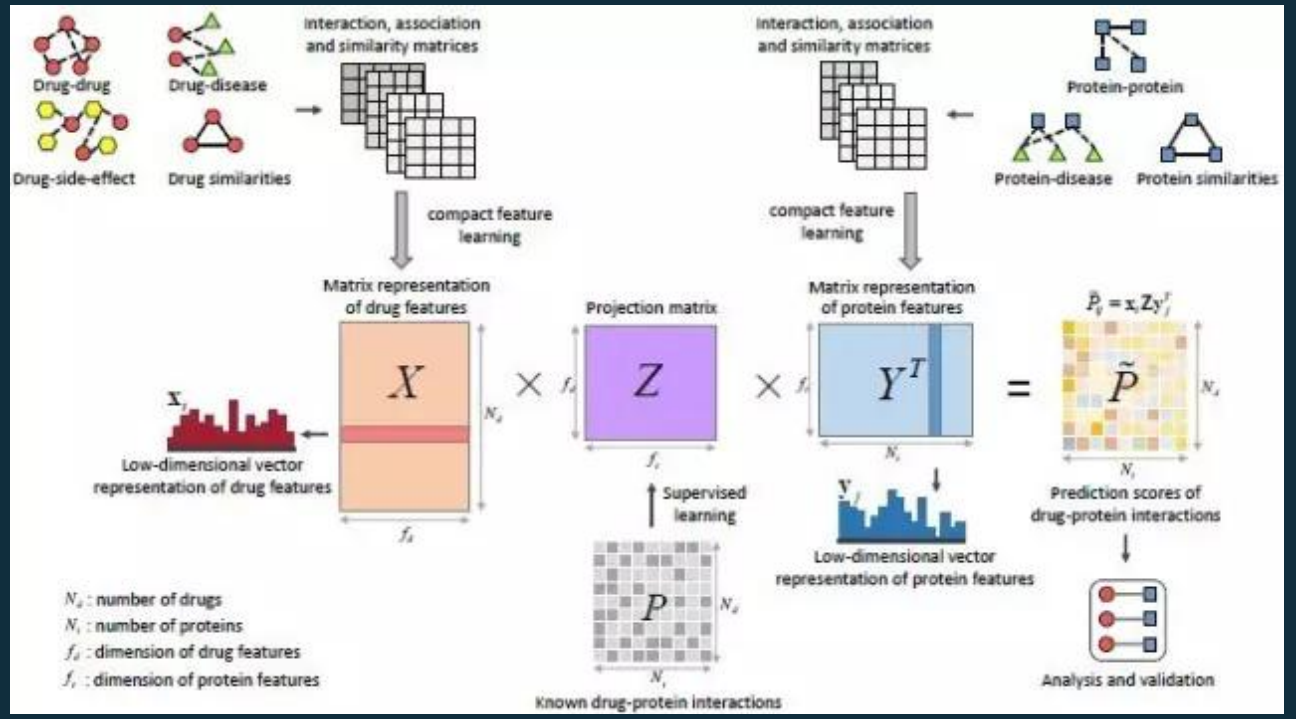


Figure 2 Details of inner workings of DeepBind and its training procedure. (**a**) Five independent sequences being processed in parallel by a single DeepBind model. The convolve, rectify, pool and neural network stages predict a separate score for each sequence using the current model parameters (**Supplementary Notes**, sec. 1). During the training phase, the backprop and update stages simultaneously update all motifs, thresholds and network weights of the model to improve prediction accuracy. (**b**) The calibration, training and testing procedure used throughout (**Supplementary Notes**, sec. 2).

Brendan Frey *et al.*, Nature Biotechnology, 2015

Big-data + Deep learning



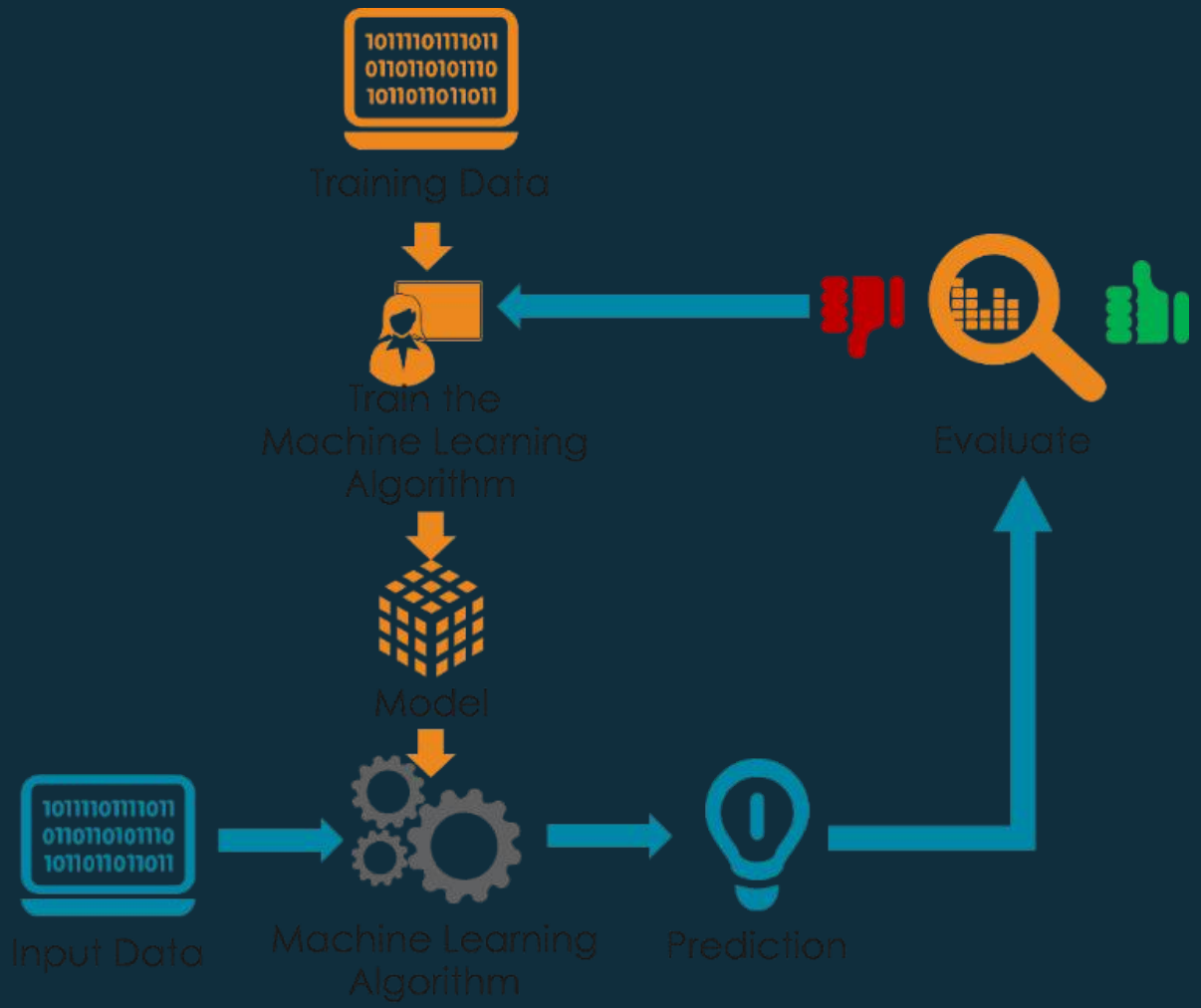
Jianyang Zeng *et al.*, Nature Communications, 2017

Machine Learning

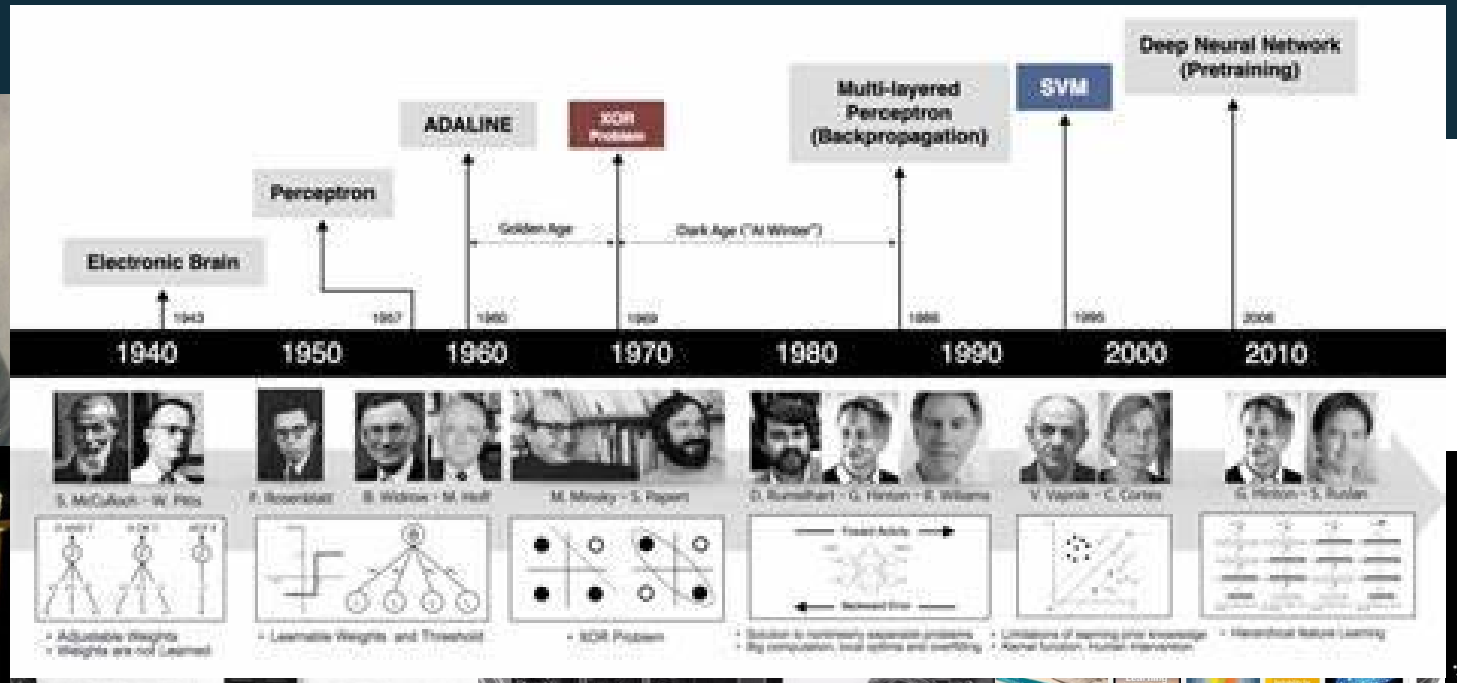
Machine learning is the science of getting computers to act without being explicitly programmed.

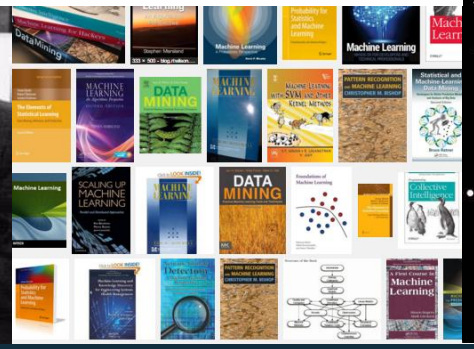
General workflow of Machine Learning

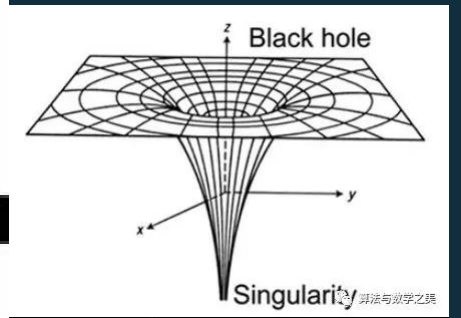
Pat Alvarado et al,2016,Building the Machine Learning Infrastructure

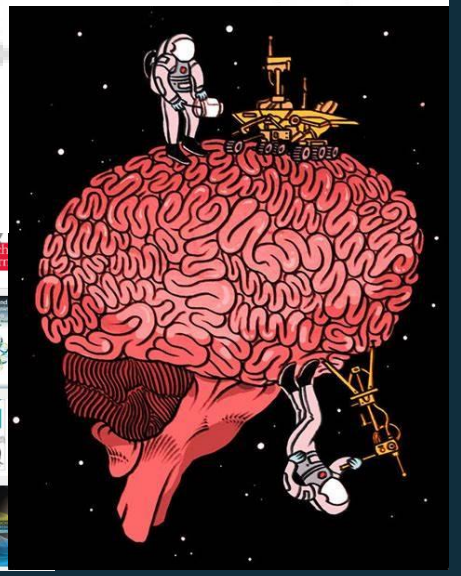


Machine Learning

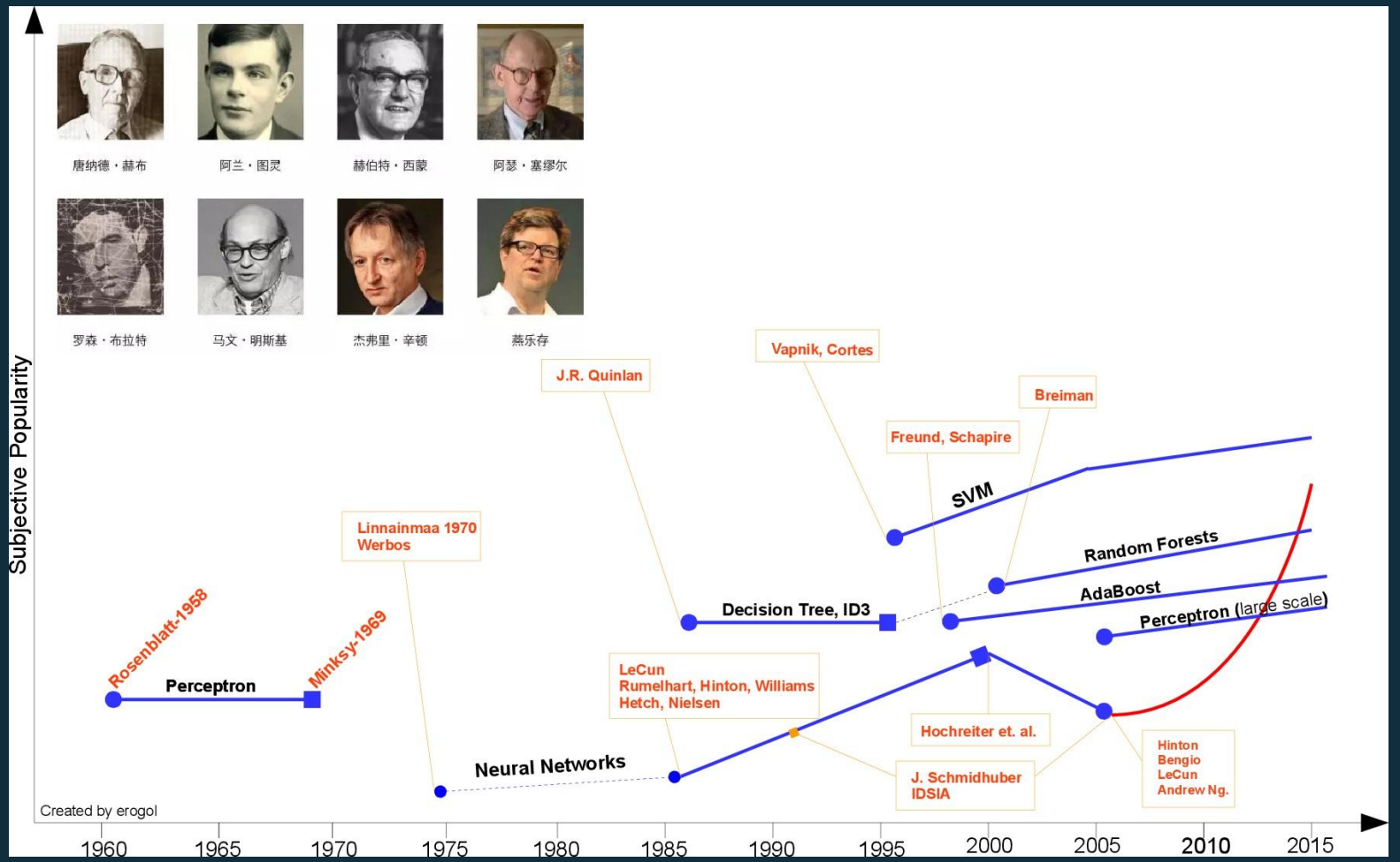




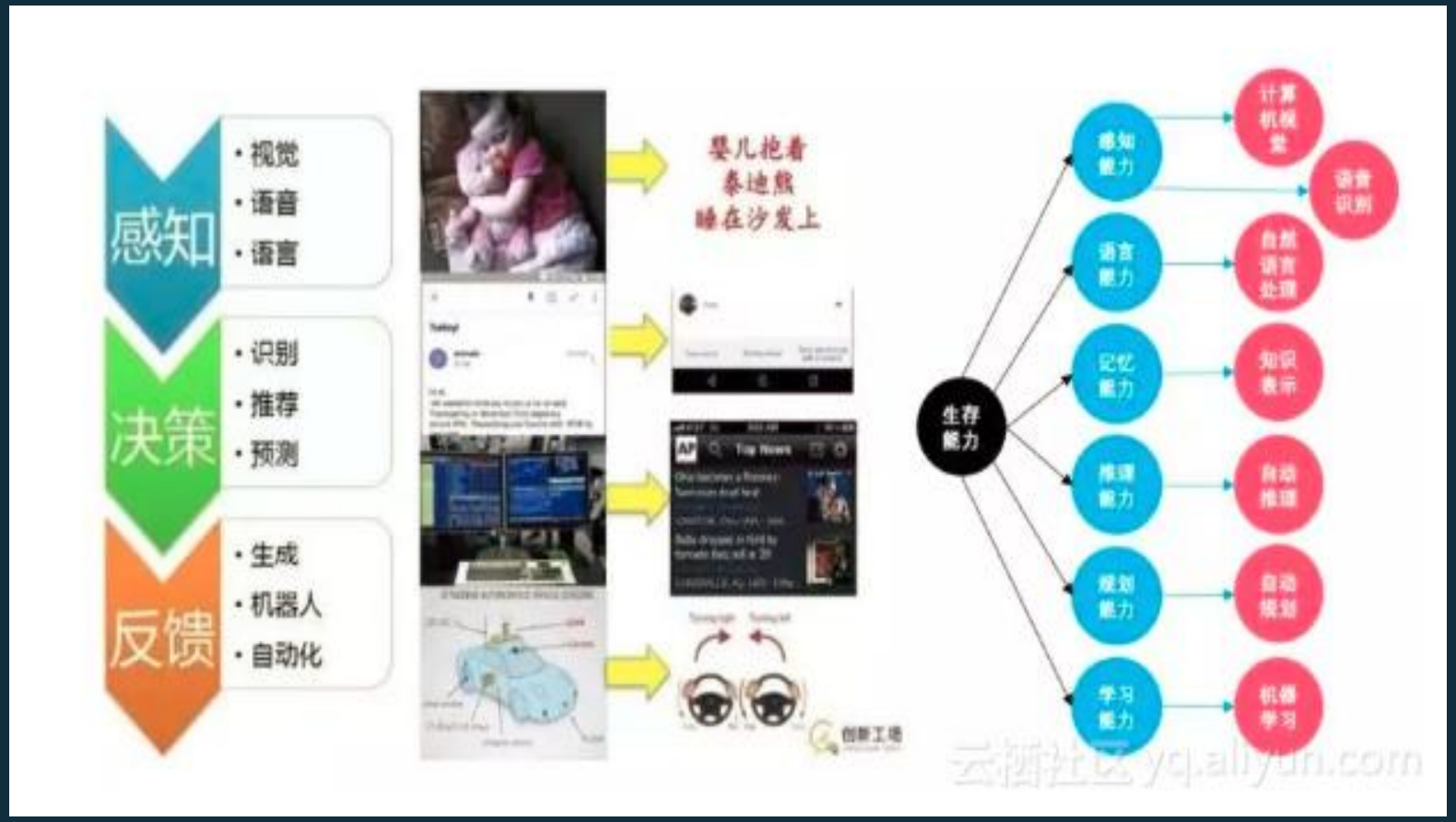




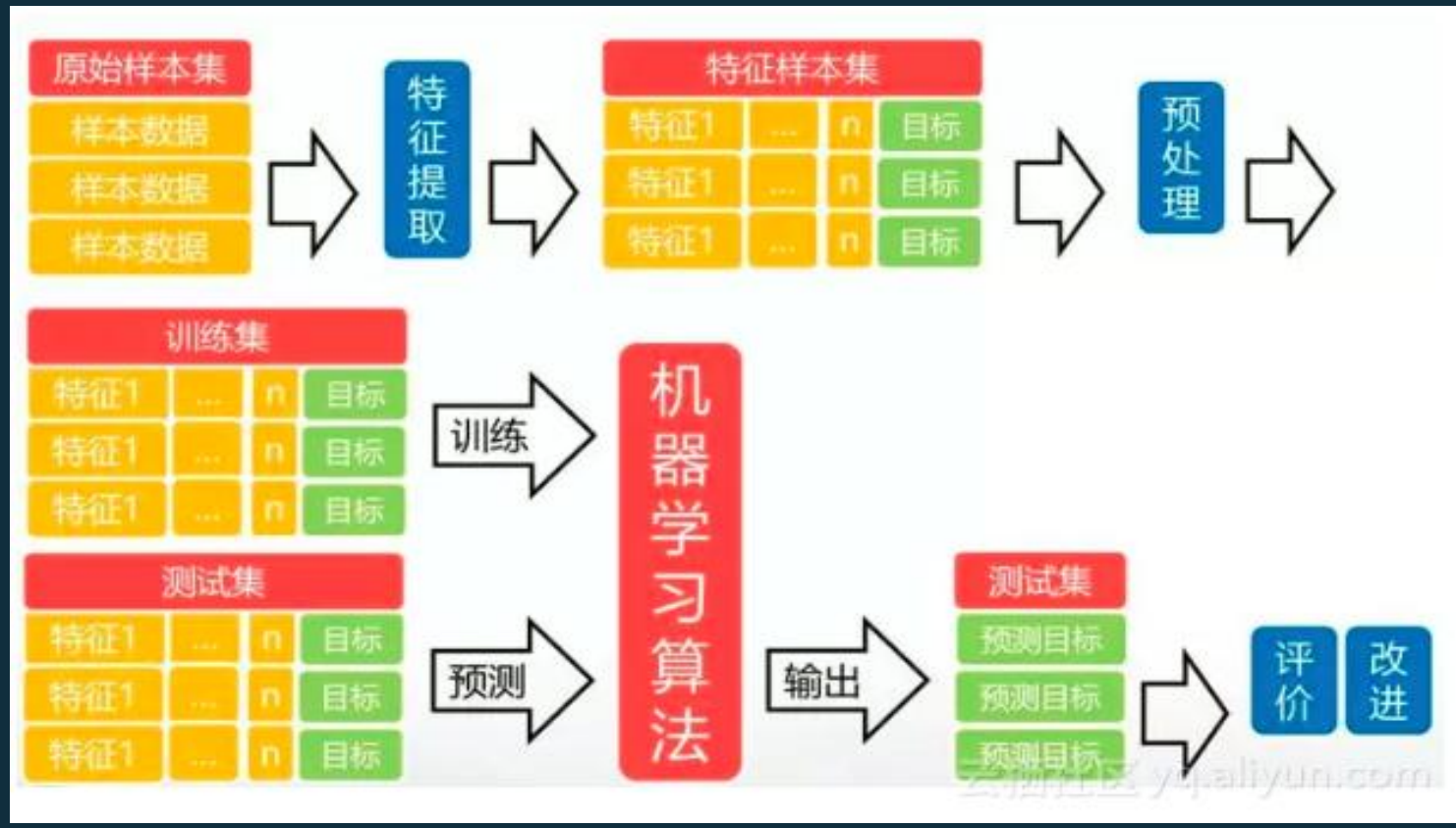
Machine Learning



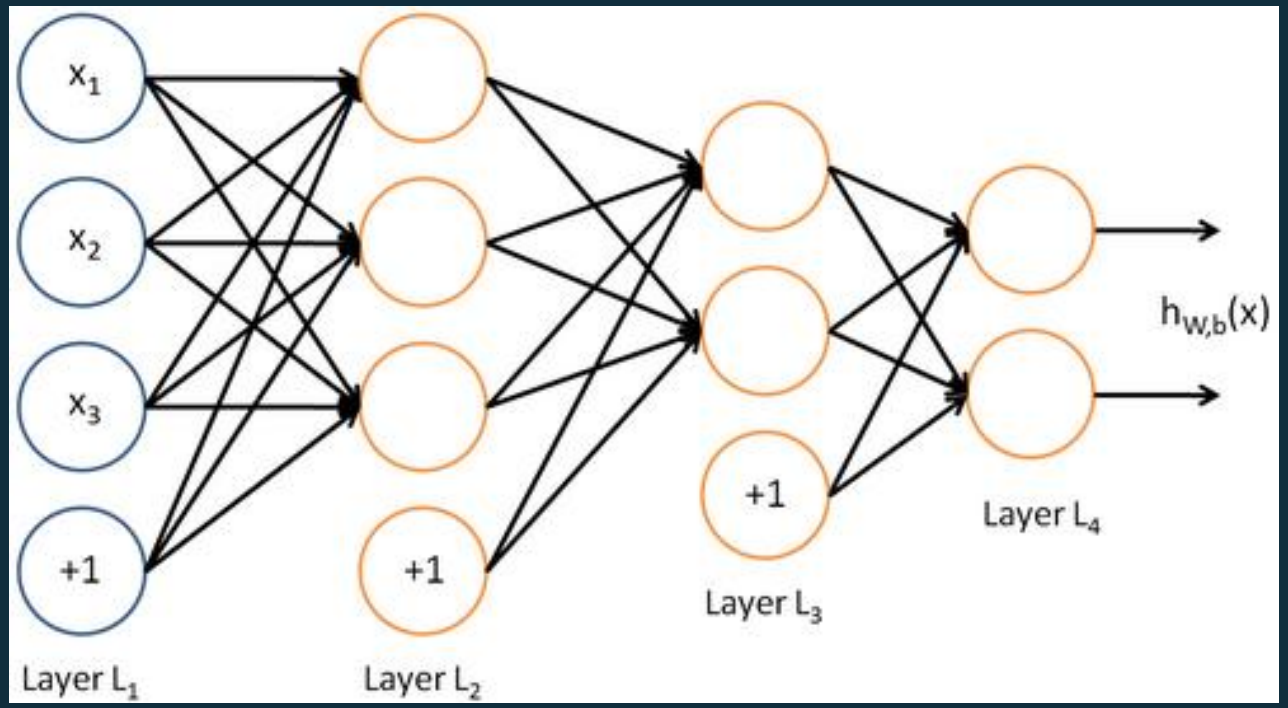
拥抱人工智能从机器学习开始



统计建模与人工智能


原始样本集
样本数据
样本数据
样本数据
特征提取
特征样本集
特征1 ... n 目标
特征1 ... n 目标
特征1 ... n 目标
预处理
训练集
特征1 ... n 目标
特征1 ... n 目标
特征1 ... n 目标
测试集
特征1 ... n 目标
特征1 ... n 目标
特征1 ... n 目标
训练
预测
机器学习算法
输出
测试集
预测目标
预测目标
预测目标
评价
改进

机器学习算法是使计算机具有智能的关键



机器学习算法分类

机器学习算法
回归
线性回归
分类
逻辑回归、K-近邻、朴素贝叶斯
决策树、支持向量机SVM
聚类
K-Means
关联分析
F-P Growth
降维
主成分分析PCA
人工神经网络
深度学习
传统机器学习算法
云栖社区 yq.aliyun.com

1. 线性回归：找到一条直线来预测目标值

House Size sq.ft (X)	1400	1600	1700	1875	1100	1550	2350	2450	1425	17(
House Price$ (Y)	245,000	312,000	279,000	308,000	199,000	219,000	405,000	324,000	319,000	255,

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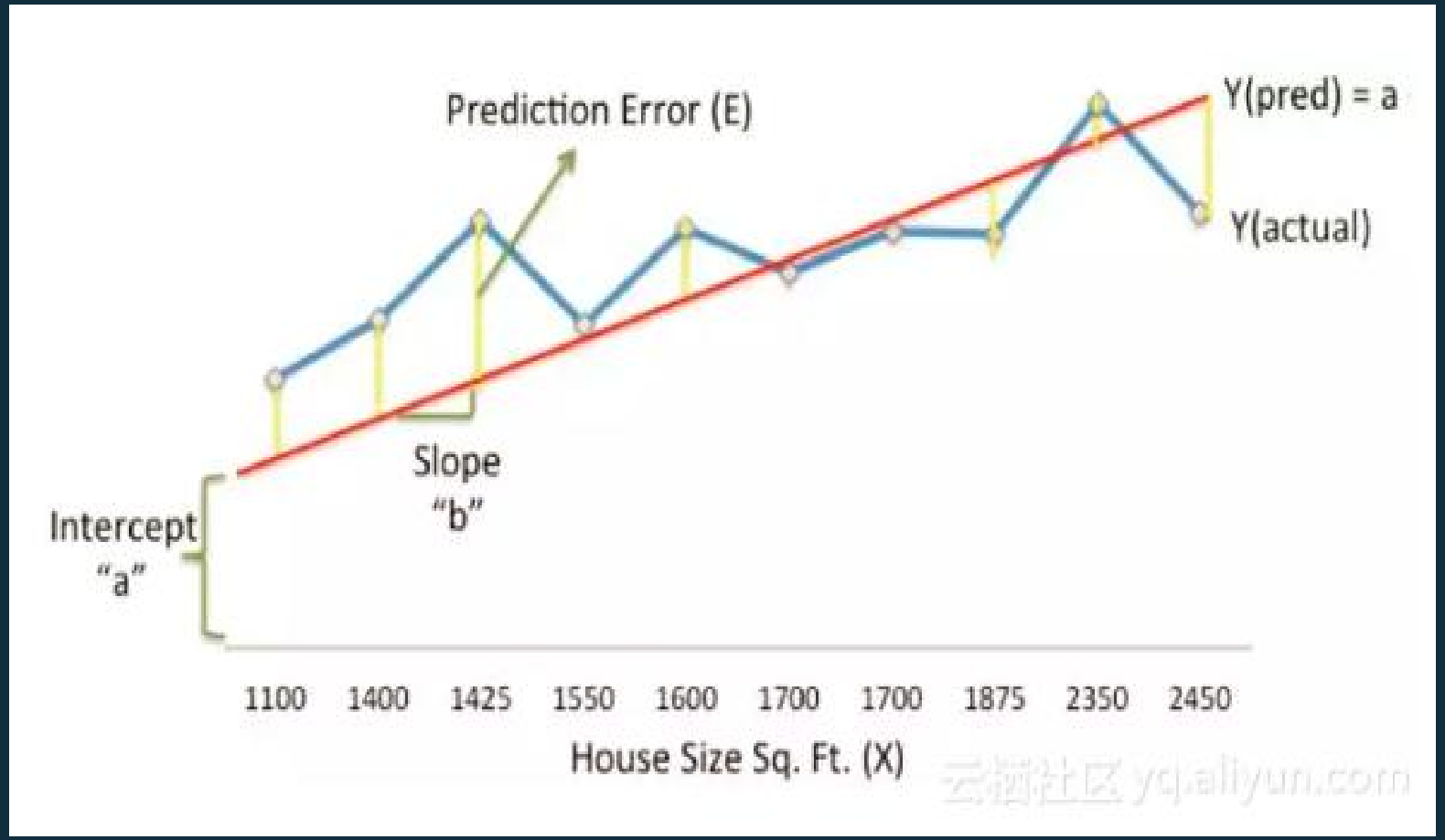
统计建模与人工智能


Prediction Error (E)
Y(pred) = a
Y(actual)
Slope
"b"
Intercept
"a"
1100
1400
1425
1550
1600
1700
1700
1875
2350
2450
House Size Sq. Ft. (X)

线性回归的应用

预测客户终生价值： 基于老客户历史数据与客户生命周期的关联关系，建立线性回归模型，预测新客户的终生价值，进而开展针对性的活动。

机场客流量分布预测： 以海量机场WiFi数据及安检登机值机数据，通过数据算法实现机场航站楼客流分析与预测。

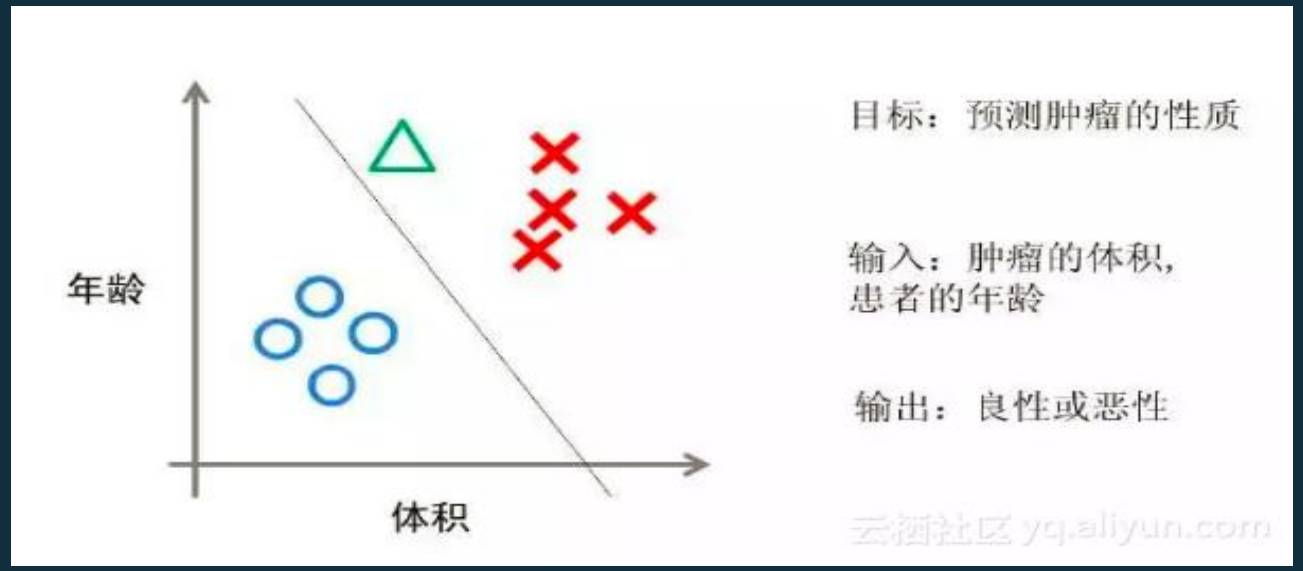
货币基金资金流入流出预测： 通过用户基本信息数据、用户申购赎回数据、收益率表和银行间拆借利率等信息，对用户的申购赎回数据的把握，精准预测未来每日的资金流入流出情况。

电影票房预测： 依据历史票房数据、影评数据、舆情数据等互联网公众数据，对电影票房进行预测。

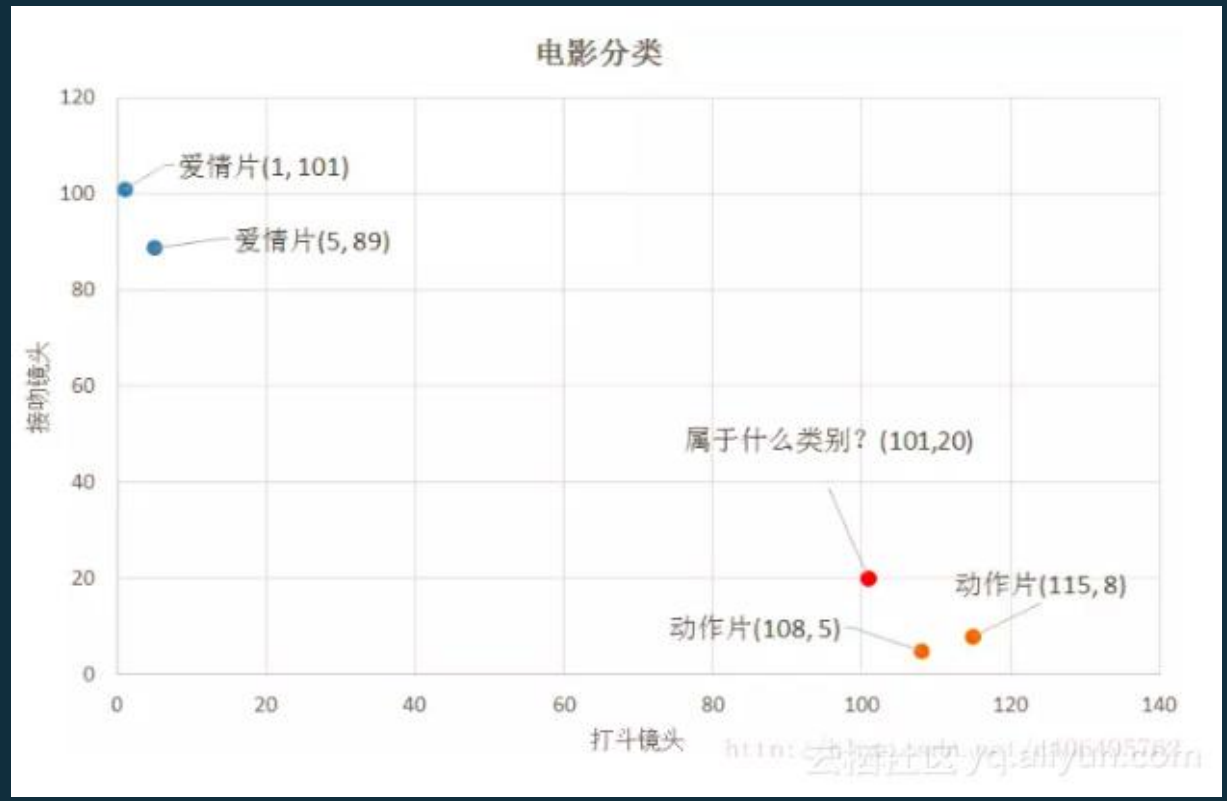
2. 逻辑回归：找到一条直线来分类数据

逻辑回归虽然名字叫回归，却是属于分类算法，是通过Sigmoid函数将线性函数的结果映射到Sigmoid函数中，预估事件出现的概率并分类。

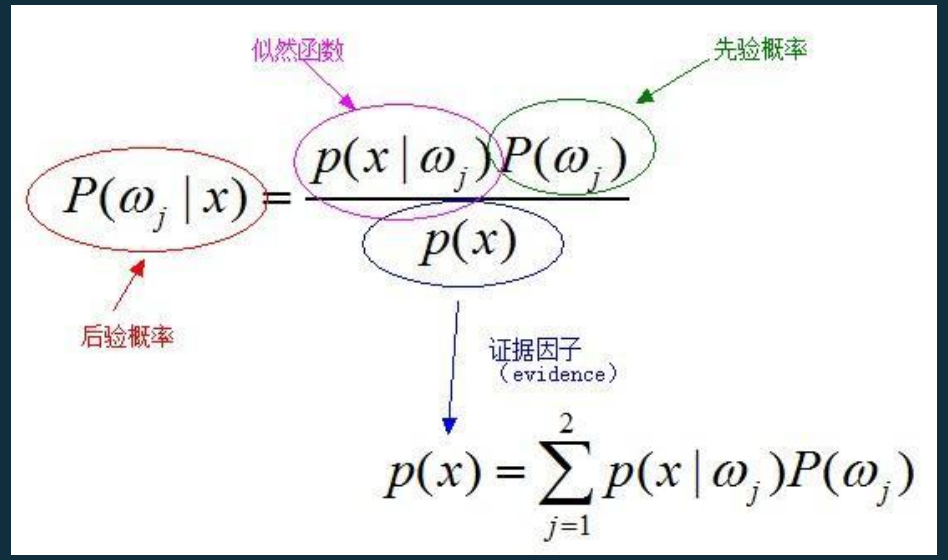
逻辑回归从直观上来说是画出了一条分类线。位于分类线一侧的数据，概率>0.5,属于分类A；位于分类线另一侧的数据，概率<0.5,属于分类B。



3. K-近邻：用距离度量最相邻的分类标签



4. 朴素贝叶斯：选择后验概率最大的类为分类标签



统计建模与人工智能

例如上面的例子中： P(X): 水果糖的概率为5/8
P(X|C1): 一号碗中水果糖的概率为3/4
P(X|C2): 二号碗中水果糖的概率为2/4
P(C1)=P(C2): 两个碗被选中的概率相同，为1/2

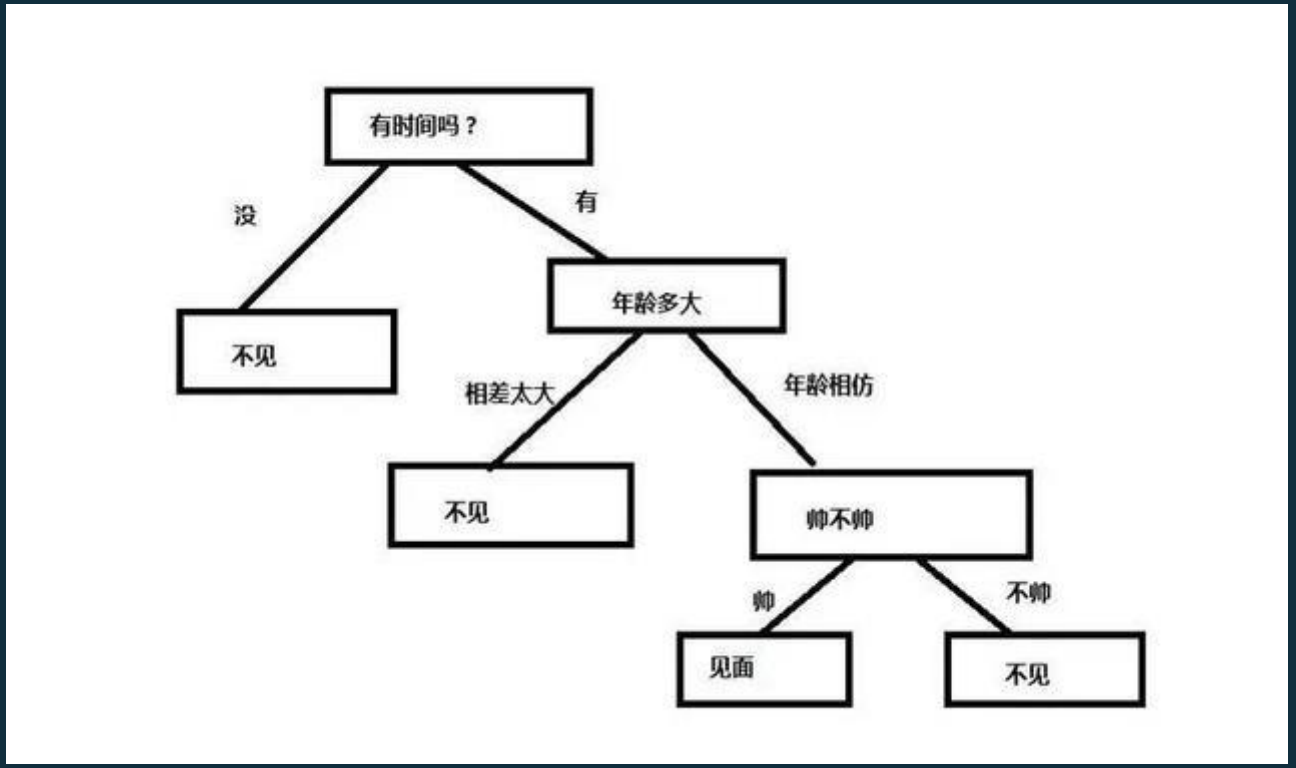
则水果糖来自一号碗的概率为:
$P(C1|X)=P(X|C1)P(C1)/P(X)=(3/4)(1/2)/(5/8)=3/5
水果糖来自二号碗的概率为:
P(C2|X)=P(X|C2)P(C2)/P(X)=(2/4)(1/2)/(5/8)=2/5
P(C1|X) > P(C2|X)
因此这颗糖最有可能来自一号碗。

朴素贝叶斯的主要应用有文本分类、垃圾文本过滤，情感判别，多分类实时预测等。

5. 决策树：构造熵值下降最快的分类树

一个简单的场景：
相亲时，可能首先检测相亲对方是否有时间。如果有，则考虑进一步接触，再观察其是否有上进心，如果没有，直接Say Goodbye。如果有，则在看帅不帅，帅的可以列入候选名单。

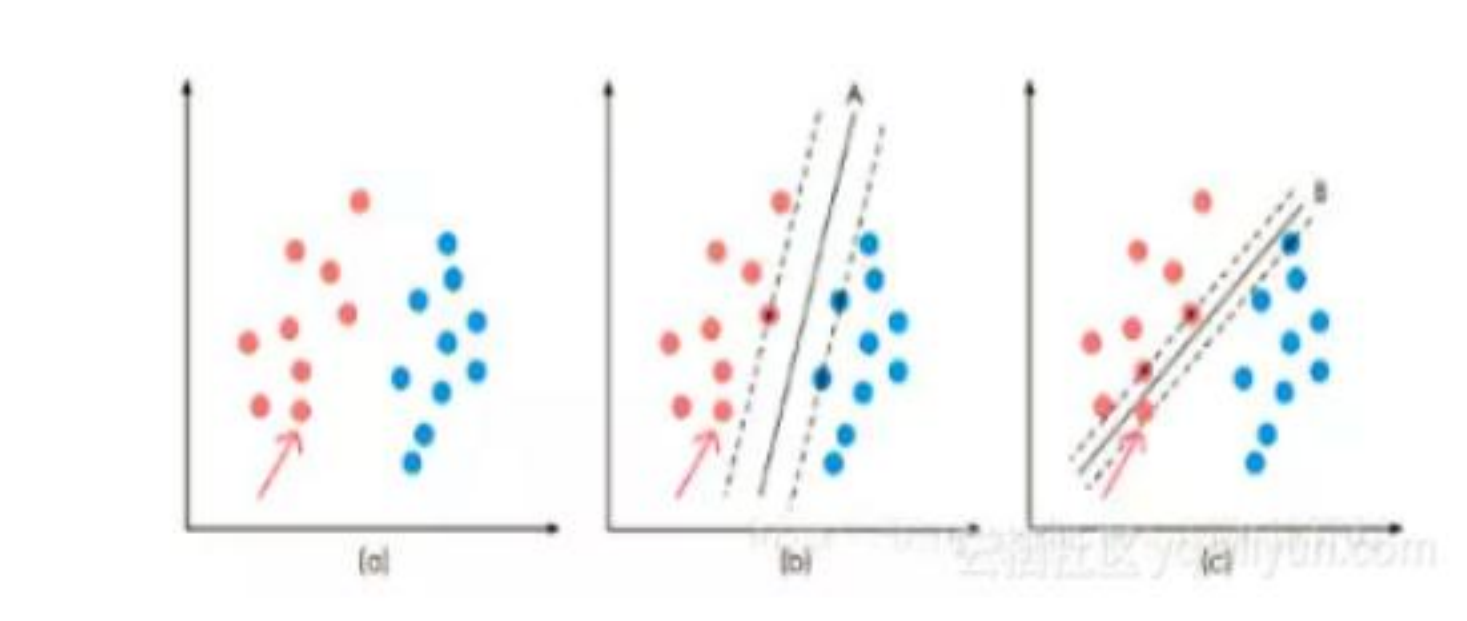
决策树可以应于：用户分级评估、贷款风险评估、选股、投标决策等。



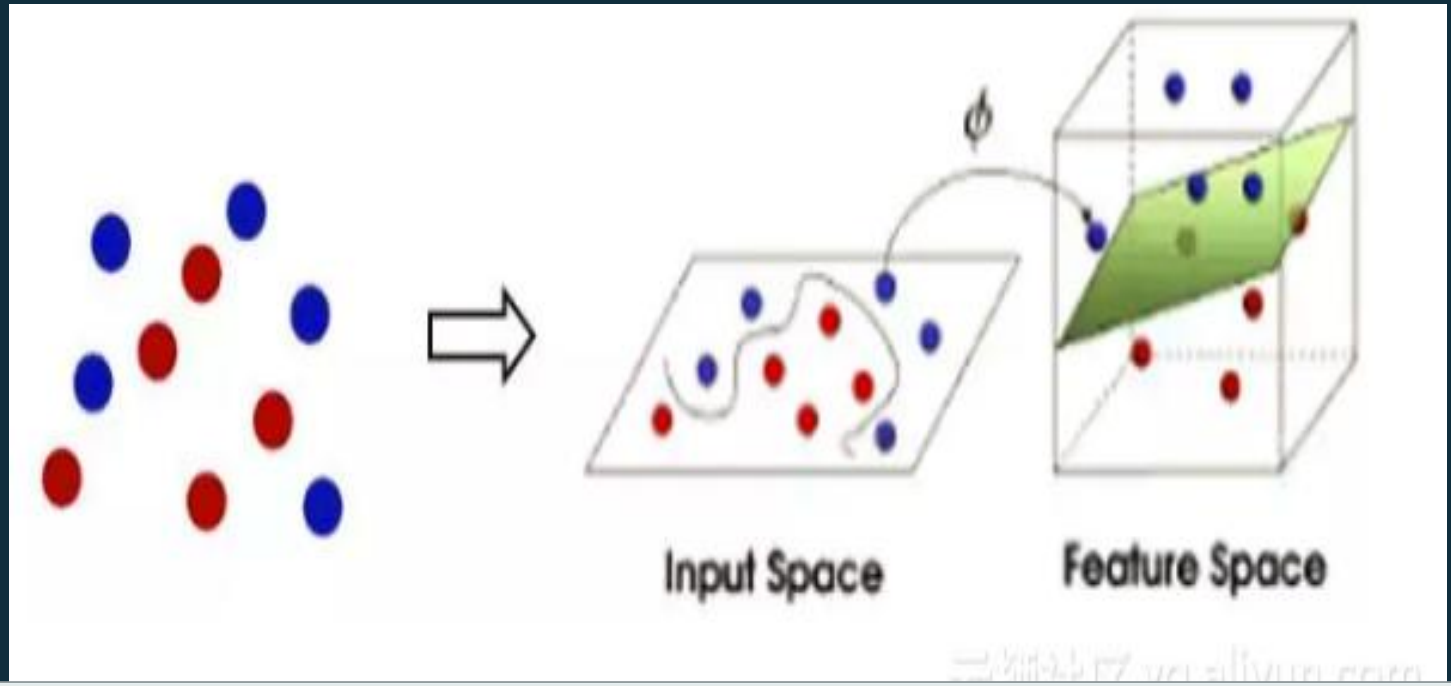
6. 支持向量机（SVM）：构造超平面，分类非线性数据

一个简单的场景：

要求用一根线将不同颜色的球分开，要求尽量在放更多球之后，仍然适用。A、B两条线都可以满足条件。再继续增加球，线A仍可以将球很好的分开，而线B则不可以。



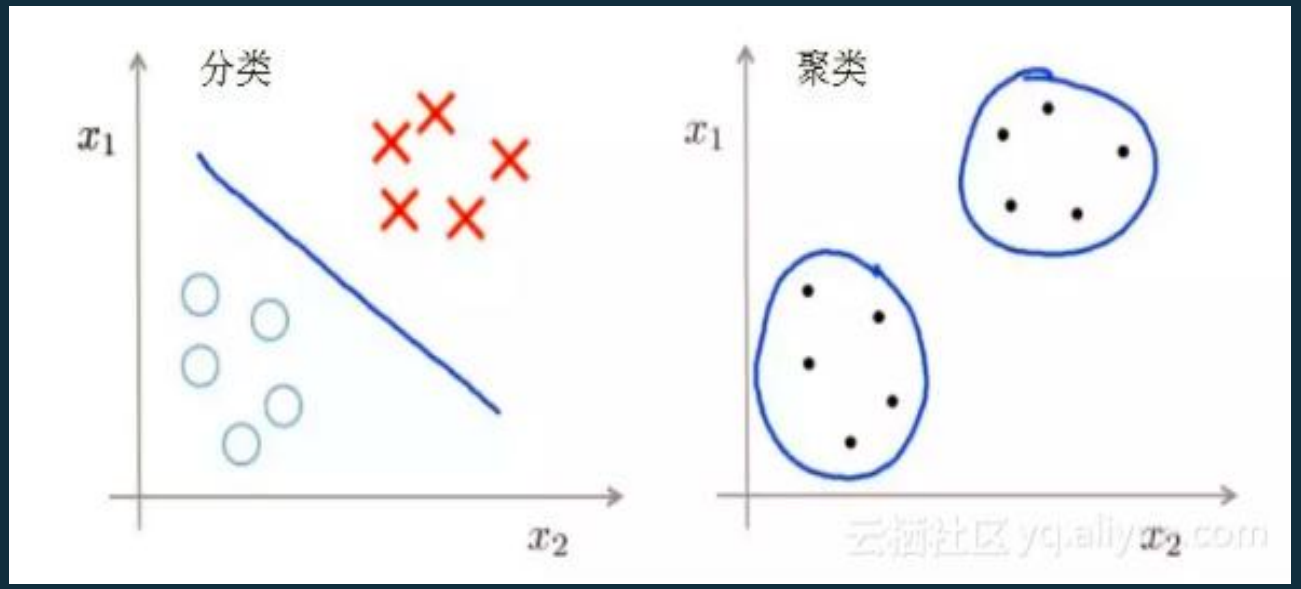
统计建模与人工智能



SVM 可应用于垃圾邮件识别、手写识别、文本分类、选股等。

7. K-means：计算质心，聚类无标签数据

在上面介绍的分类算法中，需要被分类的数据集已经有标记，例如数据集已经标记为○或者×，通过学习出假设函数对这两类数据进行划分。而对于没有标记的数据集，希望能有一种算法能够自动的将相同元素分为紧密关系的子集或簇，这就是聚类算法。



8. 关联分析：挖掘啤酒与尿布（频繁项集）的关联规则

算法中几个相关的概念：

频繁项集：在数据库中大量频繁出现的数据集合。例如购物单数据中{'啤酒'}、{'尿布'}、{'啤酒', '尿布'}出现的次数都比较多。

关联规则：由集合 A，可以在某置信度下推出集合 B。即如果 A 发生了，那么 B 也很有可能会发生。例如购买了{'尿布'}的人很可能会购买{'啤酒'}。

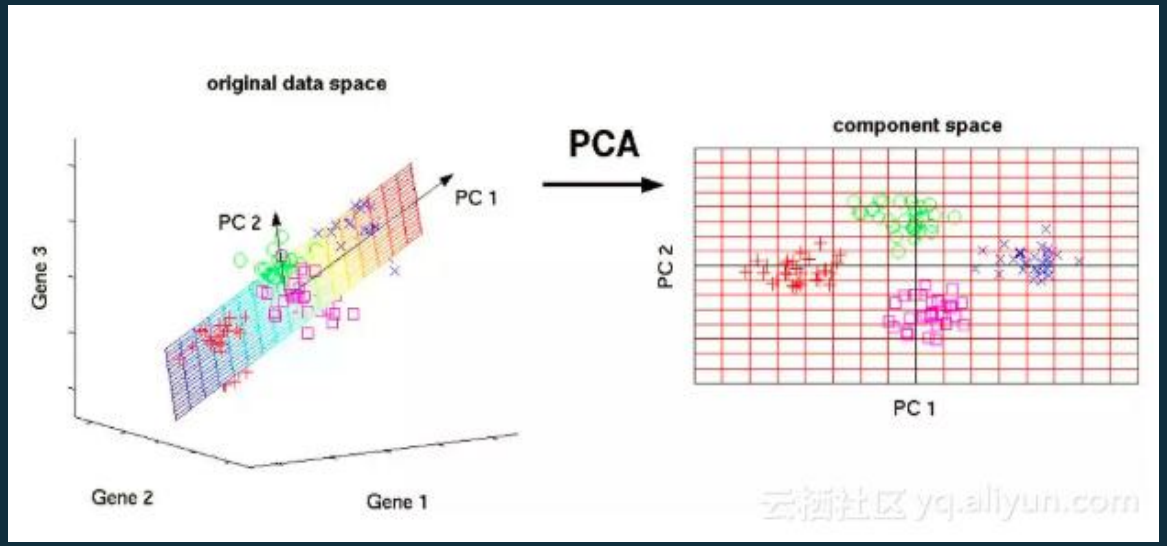
支持度：指某频繁项集在整个数据集中的比例。假设数据集有 10 条记录，包含{'啤酒', '尿布'}的有 5 条记录，那么{'啤酒', '尿布'}的支持度就是 5/10 = 0.5。

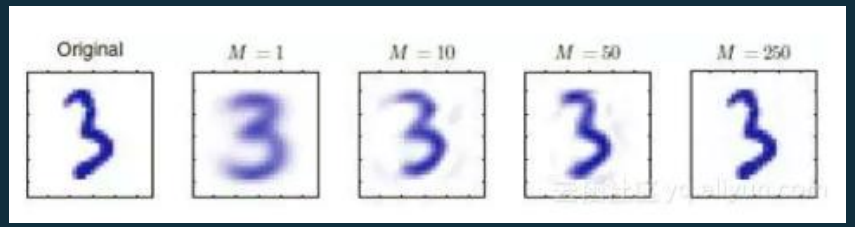
置信度：有关联规则如{'尿布'} -> {'啤酒'}，它的置信度为 {'尿布'} -> {'啤酒'}

假设{'尿布', '啤酒'}的支持度为 0.45，{'尿布'}的支持度为 0.5，则{'尿布'} -> {'啤酒'}的置信度为 0.45 / 0.5 = 0.9。

9. PCA降维：减少数据维度，降低数据复杂度

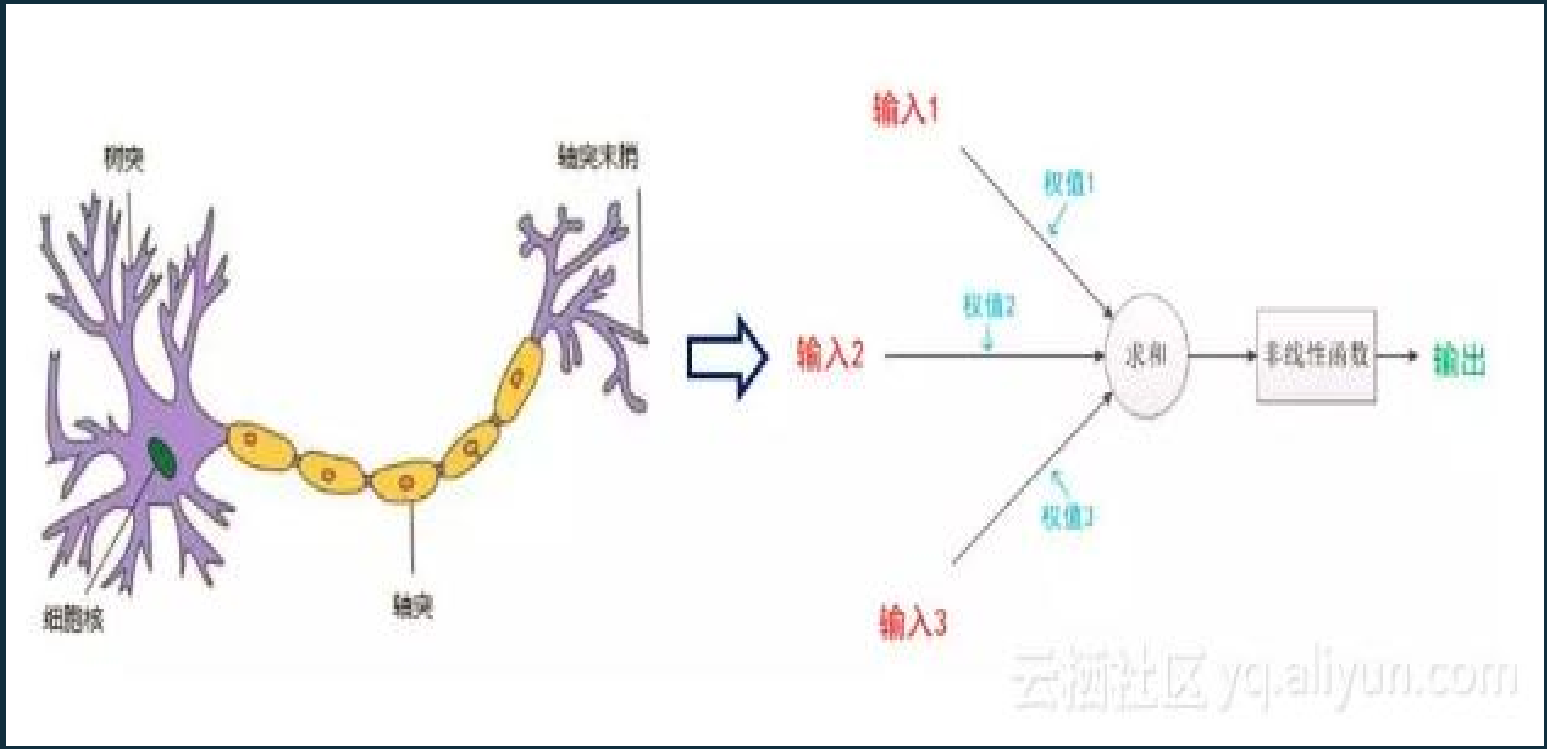
降维是指将原高维空间中的数据点映射到低维度的空间中。因为高维特征的数目巨大，距离计算困难，分类器的性能会随着特征数的增加而下降；减少高维的冗余信息所造成的误差,可以提高识别的精度。





10. 人工神经网络：逐层抽象，逼近任意函数

前面介绍了九种传统的机器学习算法，现在介绍一下深度学习的基础：人工神经网络。 它是模拟人脑神经网络而设计的模型，由多个节点（人工神经元）相互联结而成，可以用来对数据之间的复杂关系进行建模。



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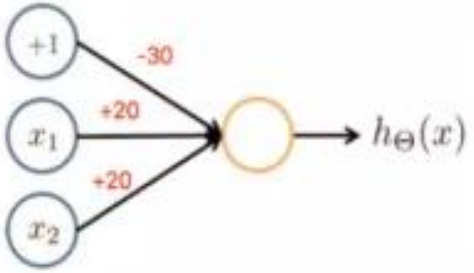
统计建模与人工智能

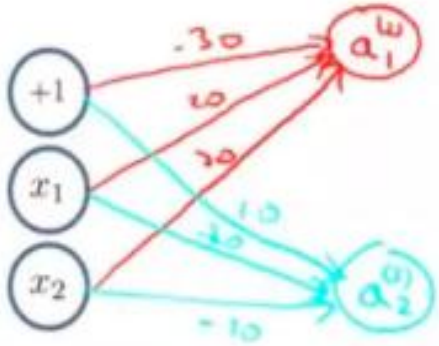
与门AND

$$h_\Theta(x) = g(-30 + 20x_1 + 20x_2)$$



x1	x2	hΘ(x)
0	0	g(−30)≈0
0	1	g(−10)≈0
1	0	g(−10)≈0
1	1	g(10)≈1

同或门XNOR



x1	x2	a1(2)	a2(2)	hΘ(x)
0	0	0	1	**1**
0	1	0	0	**0**
1	0	0	0	**0**
1	1	1	0	**1**

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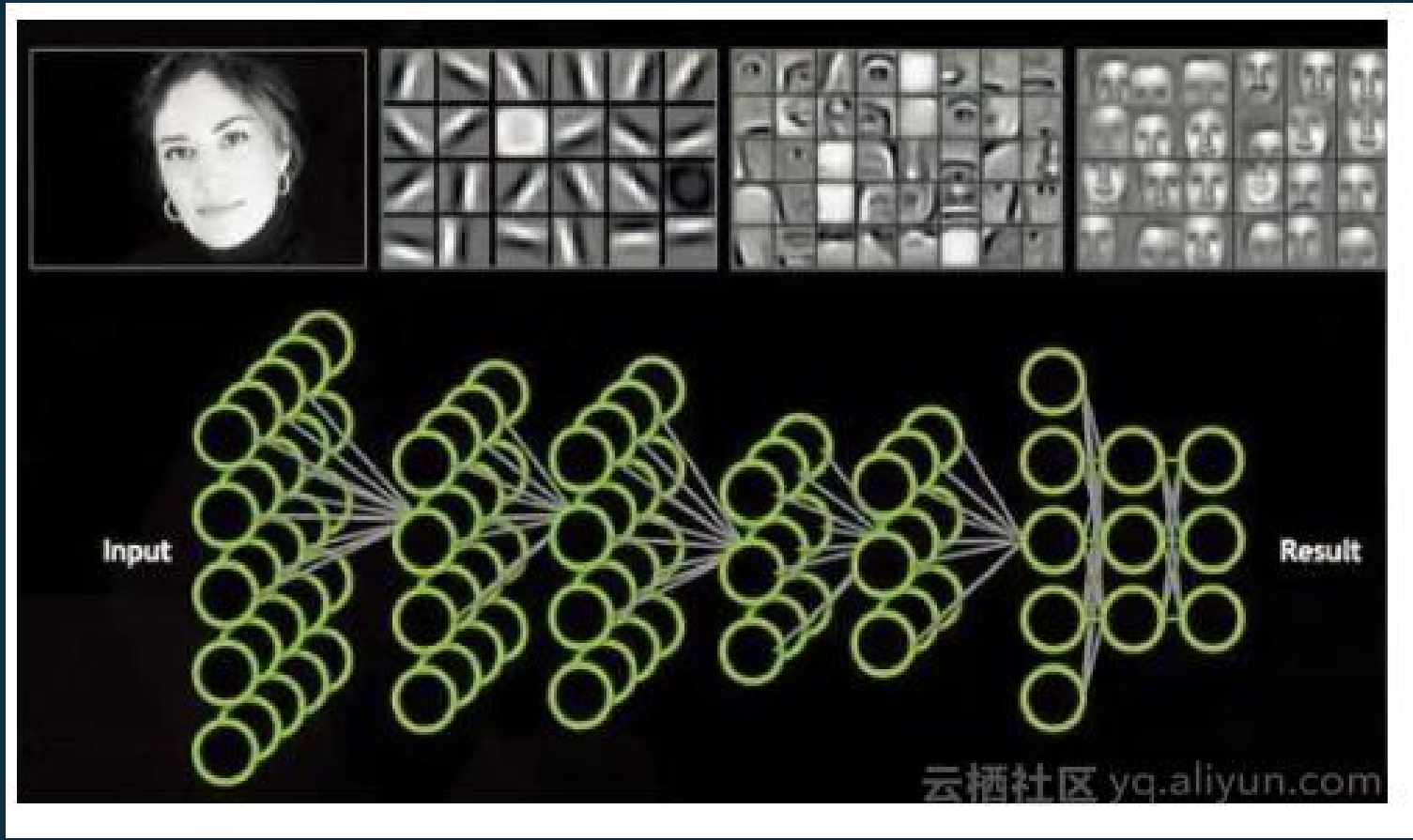
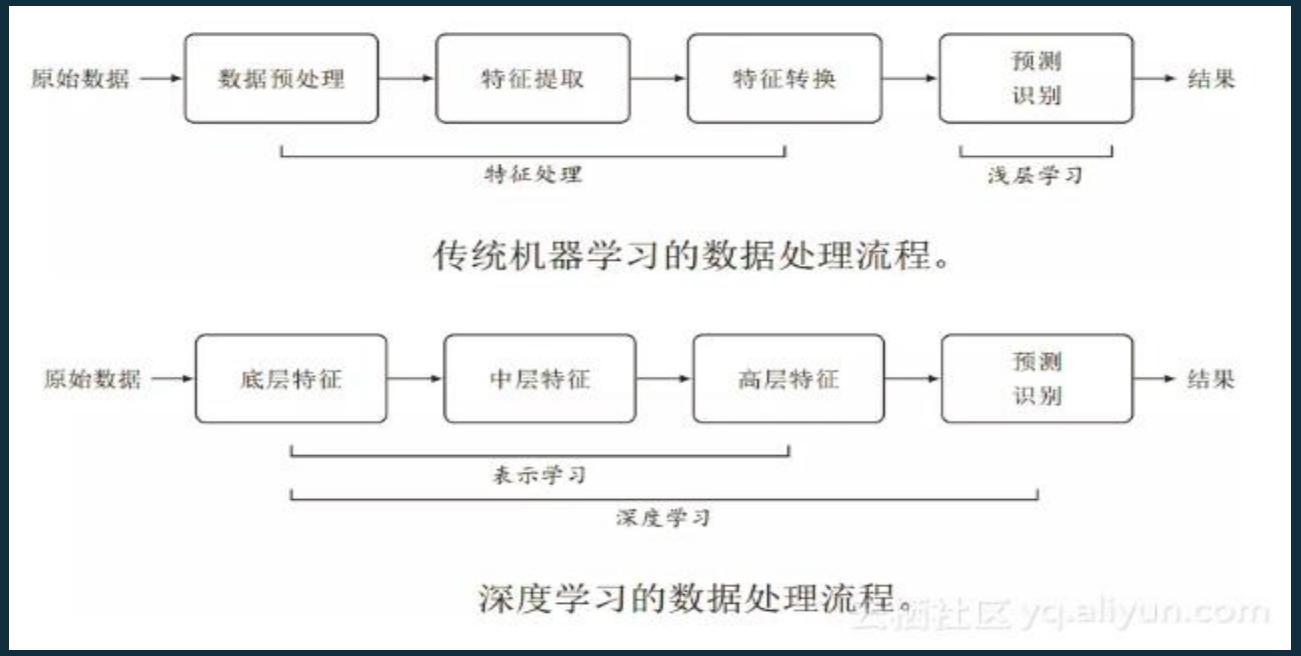
统计建模与人工智能

多层神经网络的每一层神经元学习到的是前一层神经元值的更抽象的表示，通过抽取更抽象的特征来对事物进行区分，从而获得更好的区分与分类能力。



11. 深度学习：赋予人工智能以璀璨的未来

深度学习就是一种基于对数据进行表征学习的方法，使用多层网络，能够学习抽象概念，同时融入自我学习，逐步从大量的样本中逐层抽象出相关的概念，然后做出理解，最终做出判断和决策。通过构建具有一定“深度”的模型，可以让模型来自动学习好的特征表示（从底层特征，到中层特征，再到高层特征），从而最终提升预测或识别的准确性。



统计建模与人工智能

深度学习的历史变迁：

深度学习经历了三次浪潮：

20世纪40年代～60年年代，深度学习的雏形出现在控制论中；

20世界80年代～90年代，深度学习表现为联结主义；

2006年以后，正式以深度学习之名复兴。

第一次浪潮：以感知机和线性模型为代表

不能解决与或问题

第二次浪潮：以多层感知机和BP模型为代表

以统计学为基础，应用核函数和图模型的支持向量机算法（SVM算法）等各种浅层有监督的机器学习模型广泛应用，且深度神经网络不可训练

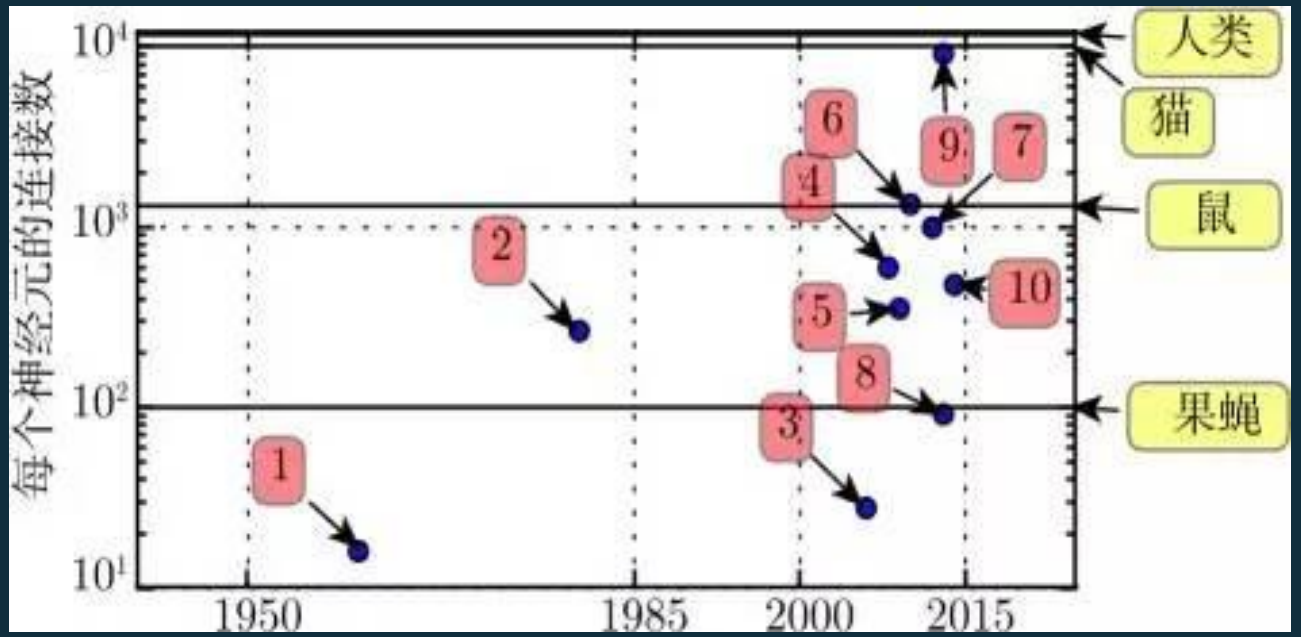
第三次浪潮：以无监督学习为代表。

解决了深层神经网络的计算能力问题；解决了深度神经网络后向误差反馈梯度消失的问题。

与日俱增的每个神经元的连接数

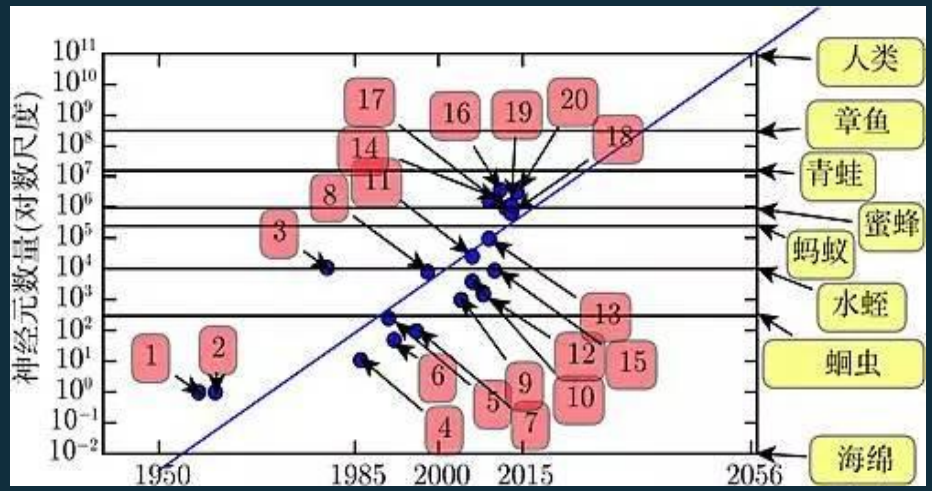
最初，人工神经网络中神经元之间的连接数受限于硬件能力。而现在，神经元之间的连接数大多是出于设计考虑。一些人工神经网络中每个神经元的连接数与猫一样多，并且对于其他神经网络来说，每个神经元的连接数与较小哺乳动物（如小鼠）一样多，这种情况是非常普遍的。甚至人类大脑每个神经元的连接数也没有过高的数量。

1. 自适应线性单元（Widrow and Hoff, 1960）；2. 神经认知机（Fukushima, 1980）；3. GPU- 加速卷积网络（Chellapilla et al., 2006）；4. 深度玻尔兹曼机（Salakhutdinov and Hinton, 2009a）；5. 无监督卷积网络（Jarrett et al., 2009b）；6. GPU- 加速多层感知机（Ciresan et al., 2010）；7. 分布式自编码器（Le et al., 2012）；8. Multi-GPU 卷积网络（Krizhevsky et al., 2012a）；9. COTS HPC 无监督卷积网络（Coates et al., 2013）；10. GoogLeNet（Szegedy et al., 2014a



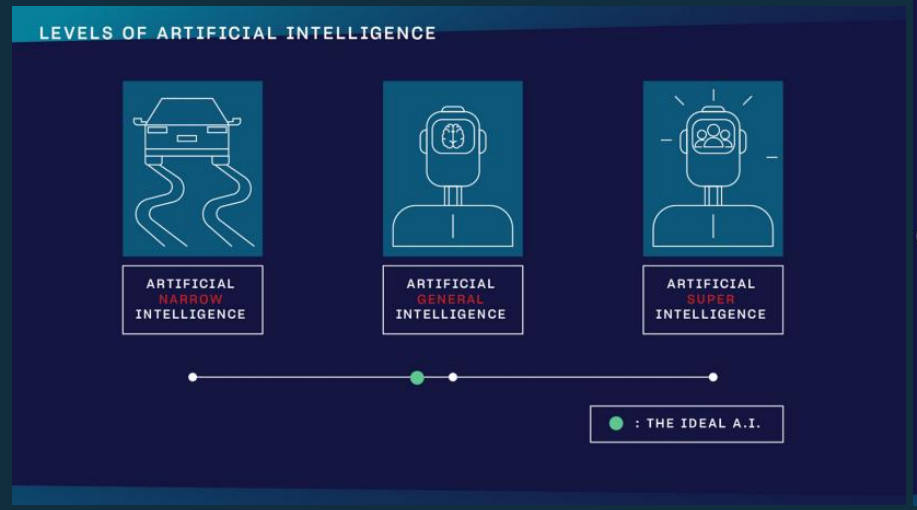
与日俱增的神经网络规模

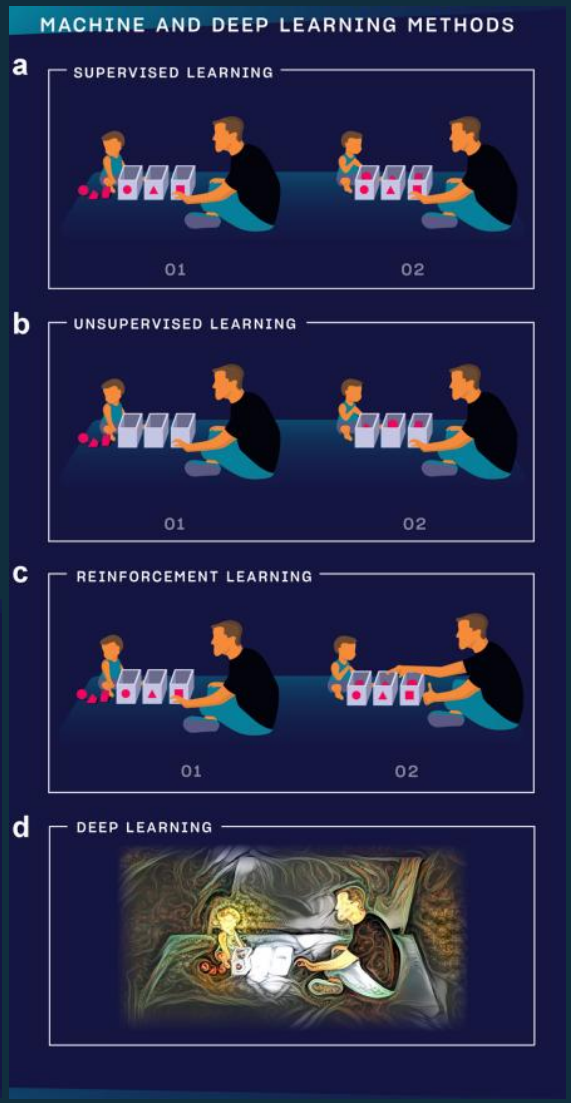
自从引入隐藏单元，人工神经网络的规模大约每 2.4 年翻一倍。



1. 感知机（ Rosenblatt, 1958, 1962 ）；2. 自适应线性单元（ Widrow and Hoff,1960 ）；3. 神经认知机（ Fukushima, 1980 ）；4. 早期后向传播网络（ Rumelhart et al., 1986b ）；5. 用于语音识别的循环神经网络（ Robinson and Fallside, 1991 ）；6. 用于语音识别的多层感知机（ Bengio et al., 1991 ）；7. 均匀场 sigmoid 信念网络（ Saul et al., 1996 ）；8. LeNet5（ LeCun et al., 1998c ）；9. 回声状态网络（ Jaeger and Haas, 2004 ）；10. 深度信念网络（ Hinton et al., 2006a ）；11. GPU- 加速卷积网络（ Chellapilla et al., 2006 ）；12. 深度玻尔兹曼机（ Salakhutdinov and Hinton, 2009a ）；13. GPU加速深度信念网络（ Raina et al., 2009a ）；14. 无监督卷积网络（ Jarrett et al., 2009b ）；15. GPU- 加速多层感知机（ Ciresan et al., 2010 ）；16. OMP-1 网络（ Coates and Ng, 2011 ）；17. 分布式自编码器（ Le et al., 2012 ）；18. MultiGPU 卷积网络（ Krizhevsky et al., 2012a ）；19. COTS HPC 无监督卷积网络（ Coates et al., 2013 ）；20. GoogLeNet（ Szegedy et al., 2014a ）

AI in a nutshell

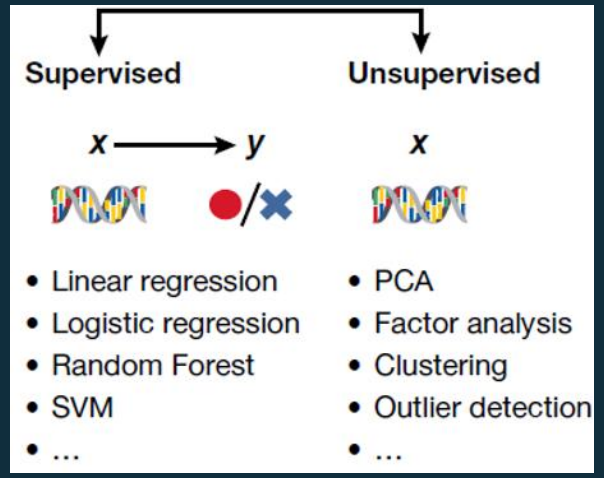


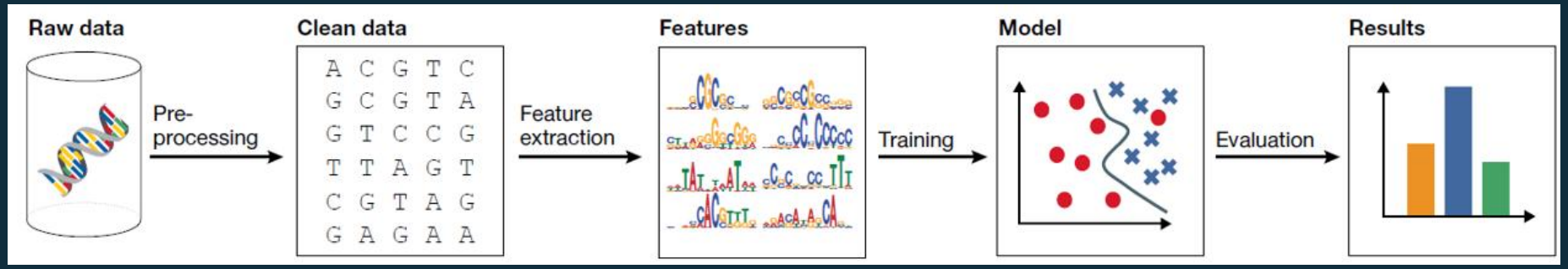


统计建模与人工智能

Siri
Google
Translate
Break through language barriers
Happy
Male
Eyes are open
Mustache
Finance
AAPL
GOOG
TSLA
Radiology
Malignant tumor
Benign tumor
云栖社区 yq.aliyun.com

Machine Learning in biology

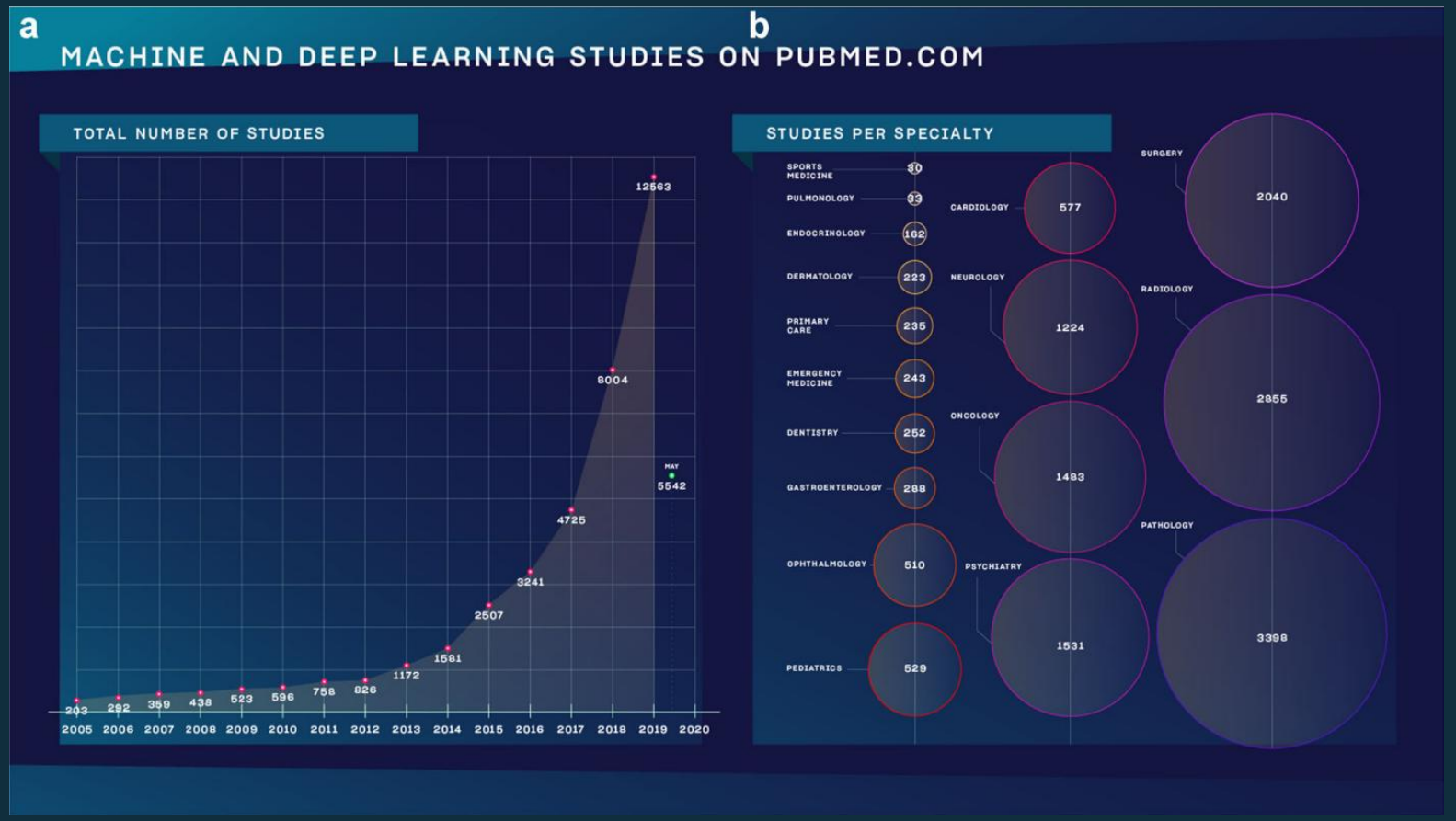




An exemplification of Machine learning in biology : classification model

Angermueller C, et al, 2016, Deep learning for computational biology

Machine Learning in biology



课程安排

生物背景和课程简介

传统生物统计学及其应用

生物统计学和生物大数据挖掘

- Hidden Markov Model (HMM)及其应用
  - Markov Chain
  - HMM理论
  - HMM和基因识别 (Topic I)
  - HMM和序列比对 (Topic II)
- 进化树的概率模型 (Topic III)
- Motif finding中的概率模型 (Topic IV)
  - EM algorithm
  - Markov Chain Monte Carlo (MCMC)
- 基因表达数据分析 (Topic V)
  - 聚类分析-Mixture model
  - Classification-Lasso Based variable selection
- 基因网络推断 (Topic VI)
  - Bayesian网络
  - Gaussian Graphical Model
- 基因网络分析 (Topic VII)
  - Network clustering
  - Network Motif
  - Markov random field (MRF)
- Dimension reduction及其应用 (Topic VIII)

面向生物大数据挖掘的深度学习

研究对象:生物序列,进化树,生物网络,基因表达…

方法：生物计算与生物统计

LECTURE 01

生物统计学：案例分析

Topic I: Sequence's Feature Detection

Problem I: CpG island finding

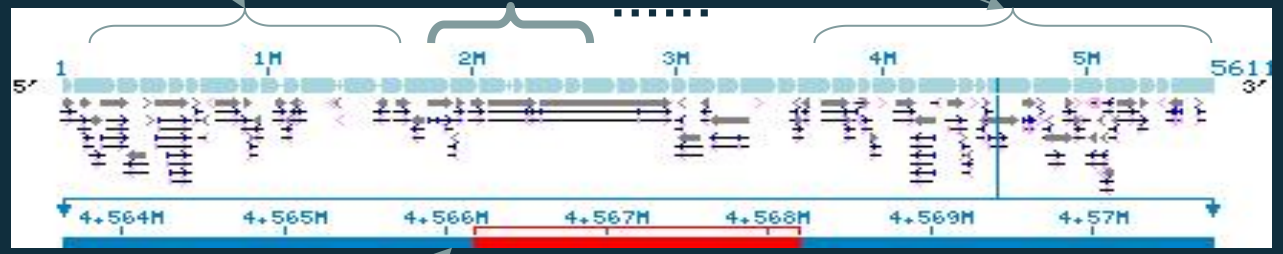
Problem II: Gene finding (promoter prediction, Splicing site prediction, Translation Initial Site Prediction etc.)

Hidden Markov Model is a powerful method for these problem

什么是CpG岛？

CG-poor regions:
P(CG) ~ 0.07!

CG-rich region:
P(CG) ~ 0.25



Gene coding region

Promoter region

CpG岛的生物学意义

CpG-rich regions are associated with genes which are *frequently transcribed*.

Helps to understand gene expression related to *location* in genome.

HMM对于CpG岛识别的意义

Q: Why an HMM?

It can answer the questions:

Short sequence: *does it come from a CpG island or not?*

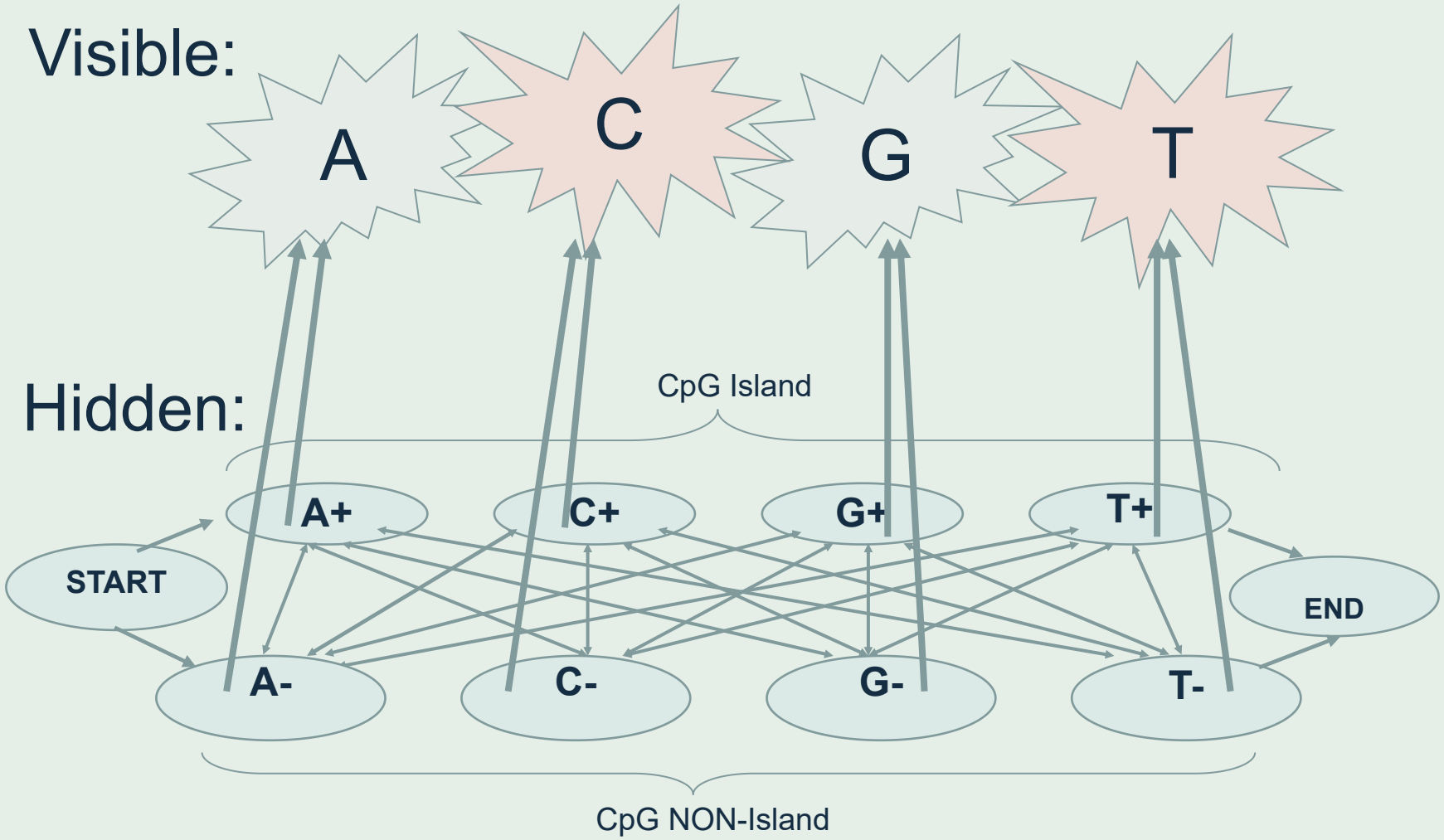
Long sequence: *where are the CpG islands?*

So, what's a good model?

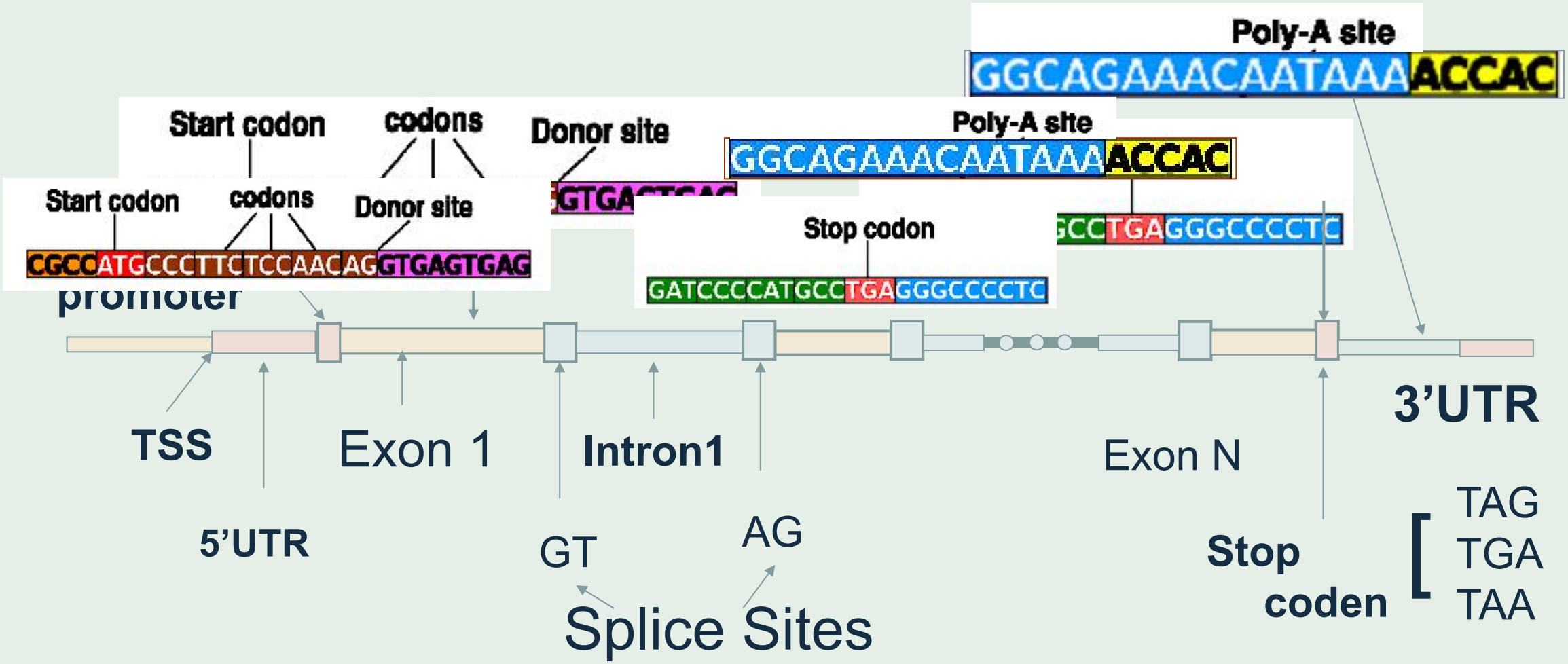
Well, we need states for ISLAND bases and

NON-ISLAND bases …

HMM示意


Visible:
A
C
G
T
Hidden:
CpG Island
A+
C+
G+
T+
START
END
A-
C-
G-
T-
CpG NON-Island

基因的结构


Poly-A site
GGCAGAAACAATAAAACCAC
Start codon
codons
Donor site
Poly-A site
GGCAGAAACAATAAA
ACCAC
Start codon
codons
Donor site
CGCCATGCCCTTCTCCAACAGGTGAGTGAG
Stop codon
GATCCCCATGCCTGAGGGCCCCTC
promoter
TSS
5'UTR
Exon 1
Intron1
Exon N
3'UTR
GT
AG
Splice Sites
Stop coden
TAG
TGA
TAA

HMMs and Gene Structure

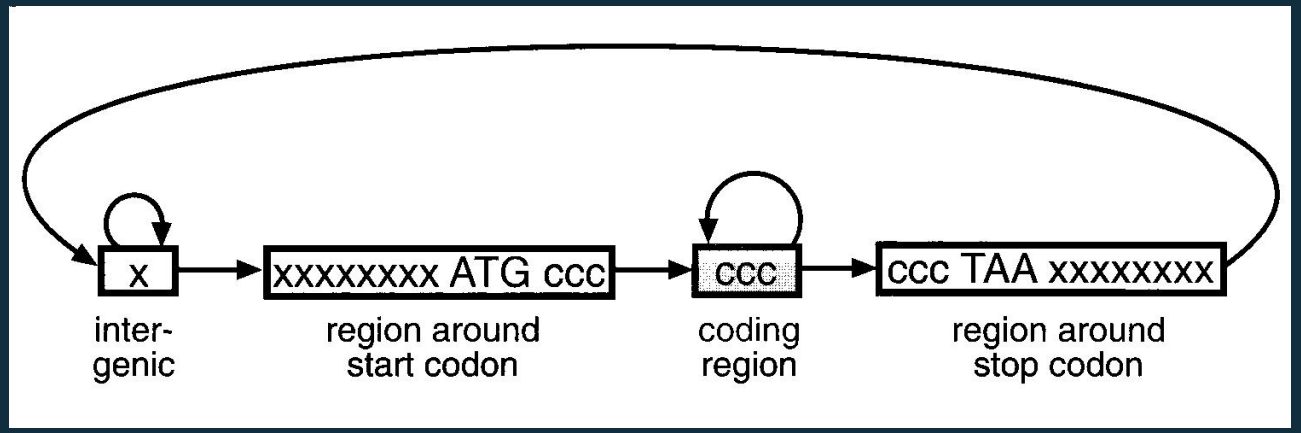
Nucleotides *{A,C,G,T}* are the observables

Different states generates generate nucleotides at different frequencies

A simple HMM for unspliced genes:

AAAGC ATG CAT TTA ACG AGA GCA CAA GGG CTC TAA TGCCG

The sequence of states is an annotation of the generated string – each nucleotide is generated in intergenic, start/stop, coding state



Genscan

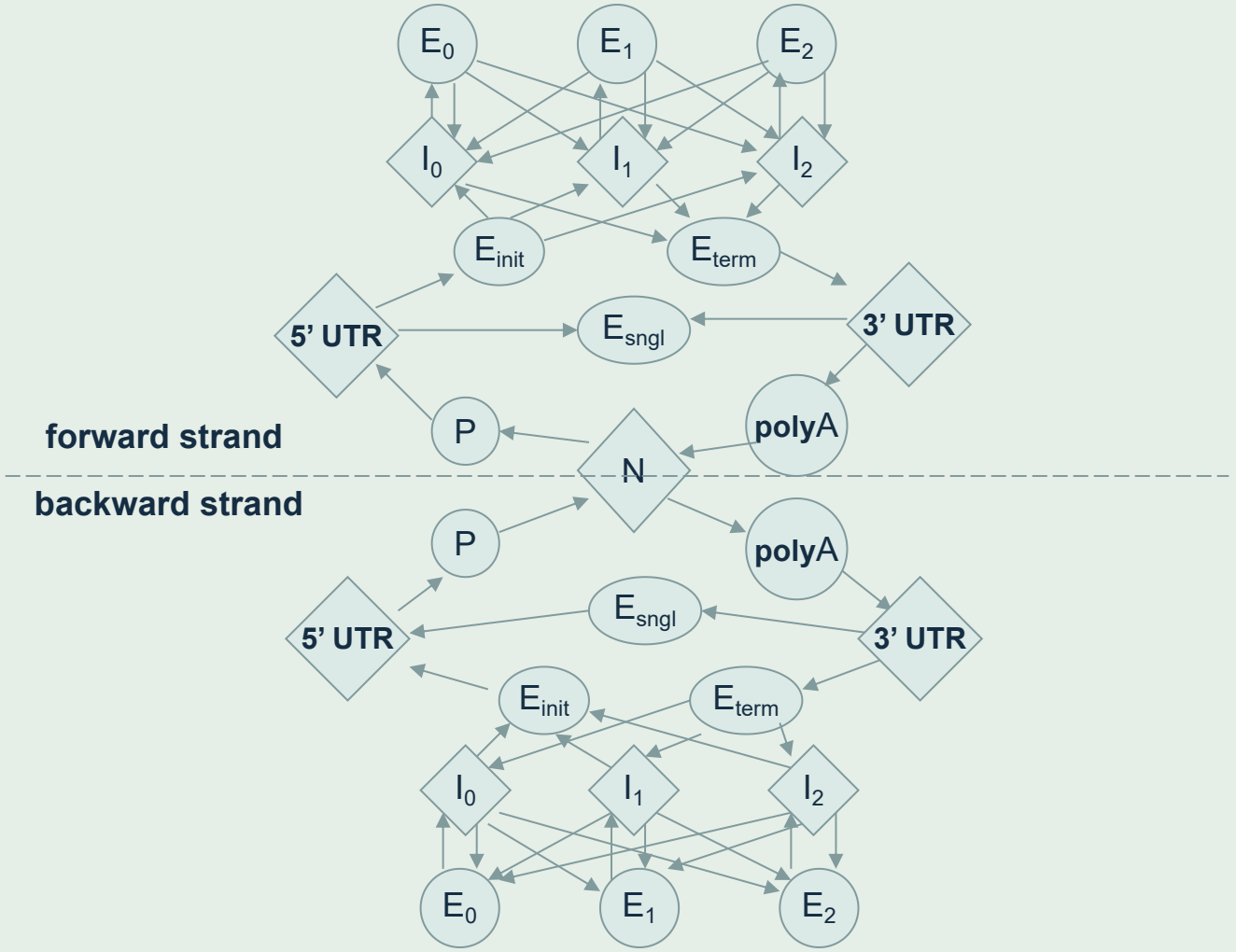
Developed by Chris Burge 1997

One of the most accurate *ab initio* programs

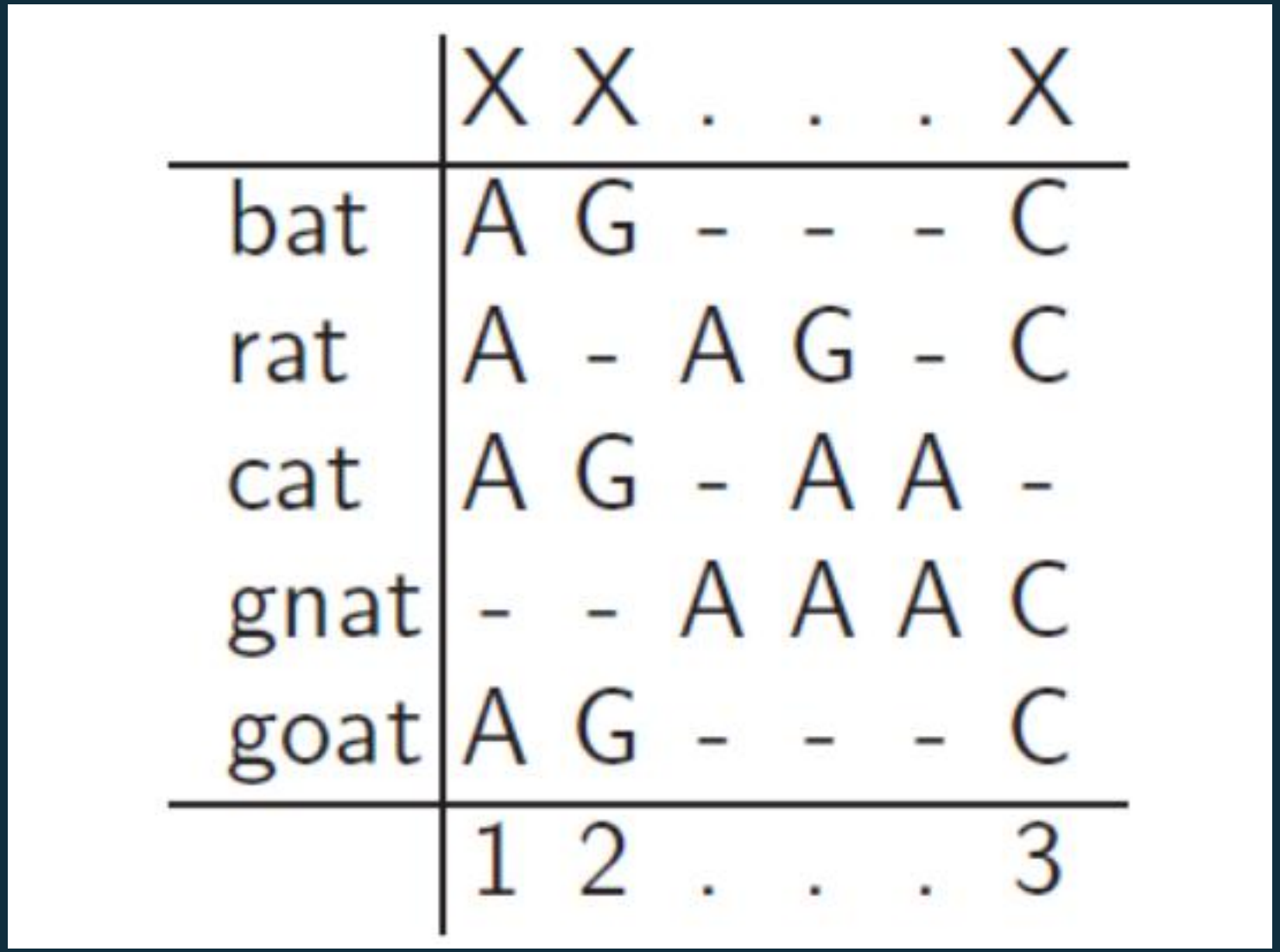
Uses explicit state duration HMM to model gene structure (different length distributions for exons)

Different model parameters for regions with different GC content

生物统计学案例

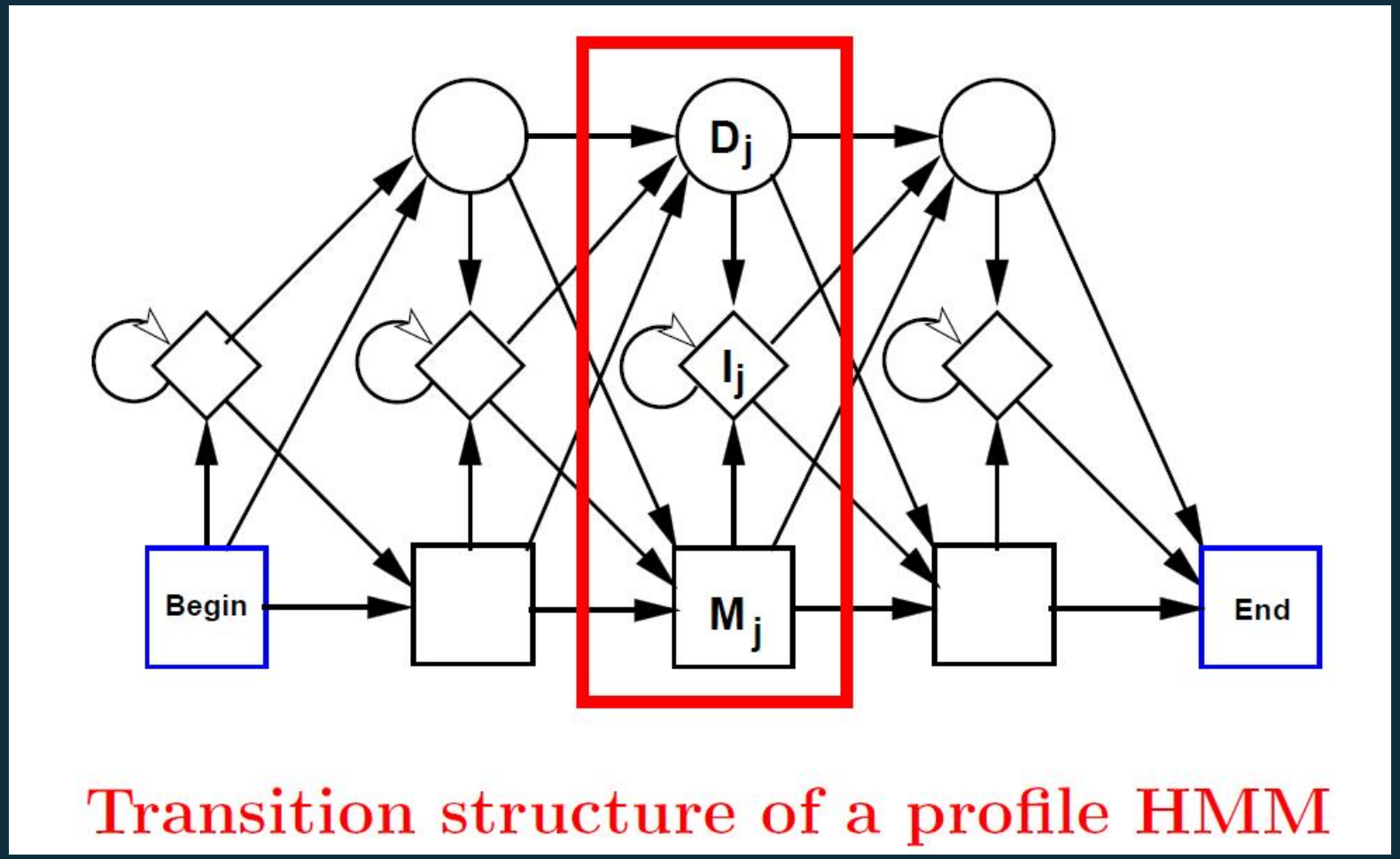

E0
E1
E2
I0
I1
I2
Einit
Eterm
5' UTR
Esngl
3' UTR
P
polyA
N
forward strand
backward strand
P
polyA
5' UTR
Esngl
3' UTR
Einit
Eterm
I0
I1
I2
E0
E1
E2

Topic II: Multiple Alignment



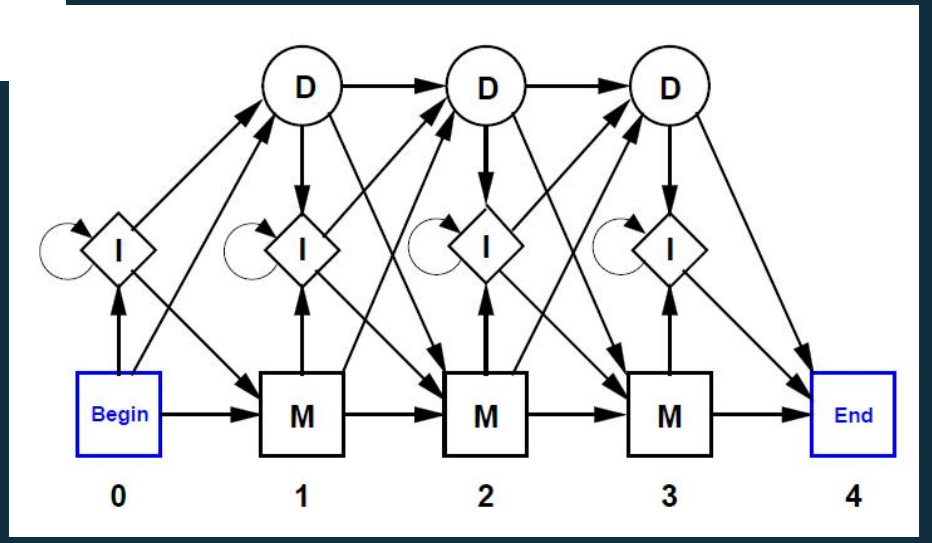
	X	X	.	.	.	X
bat	A	G	-	-	-	C
rat	A	-	A	G	-	C
cat	A	G	-	A	A	-
gnat	-	-	A	A	A	C
goat	A	G	-	-	-	C
	1	2	.	.	.	3

Profiled HMM

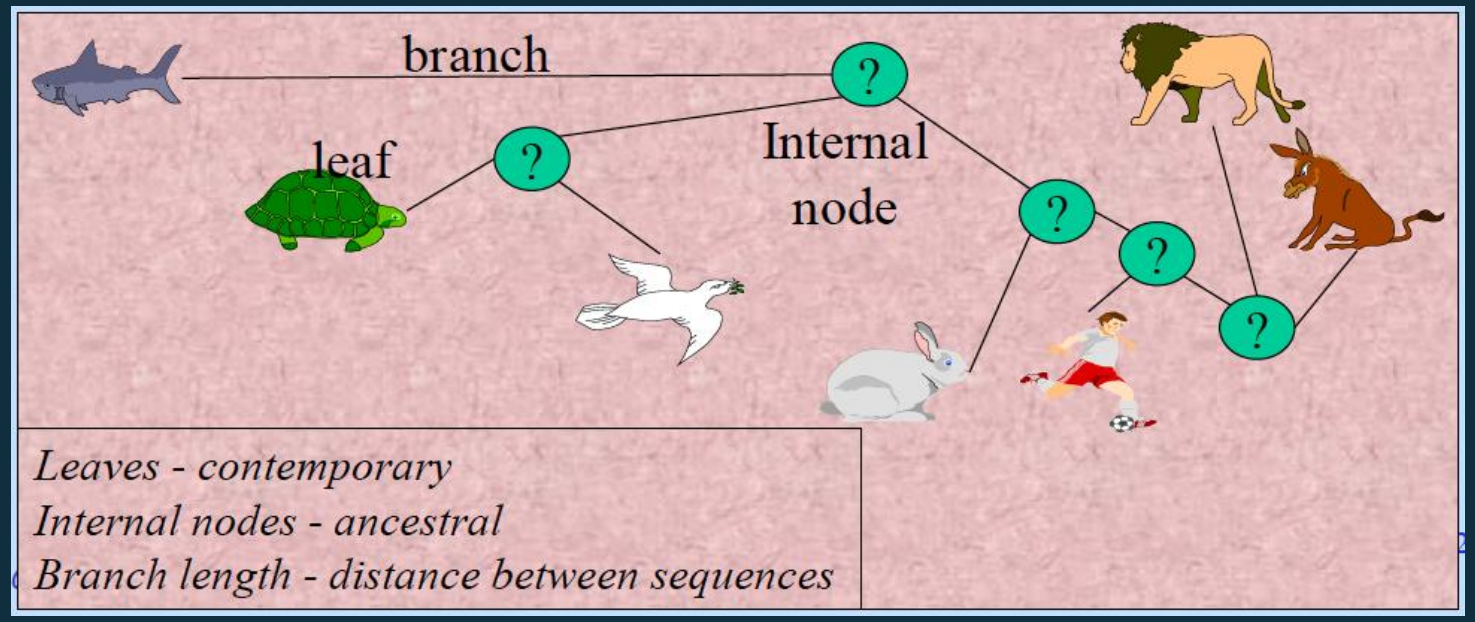


Example of Profile HMM

	X	X	.	.	.	X
bat	A	G	-	-	-	C
rat	A	-	A	G	-	C
cat	A	G	-	A	A	-
gnat	-	-	A	A	A	C
goat	A	G	-	-	-	C
	1	2	.	.	.	3

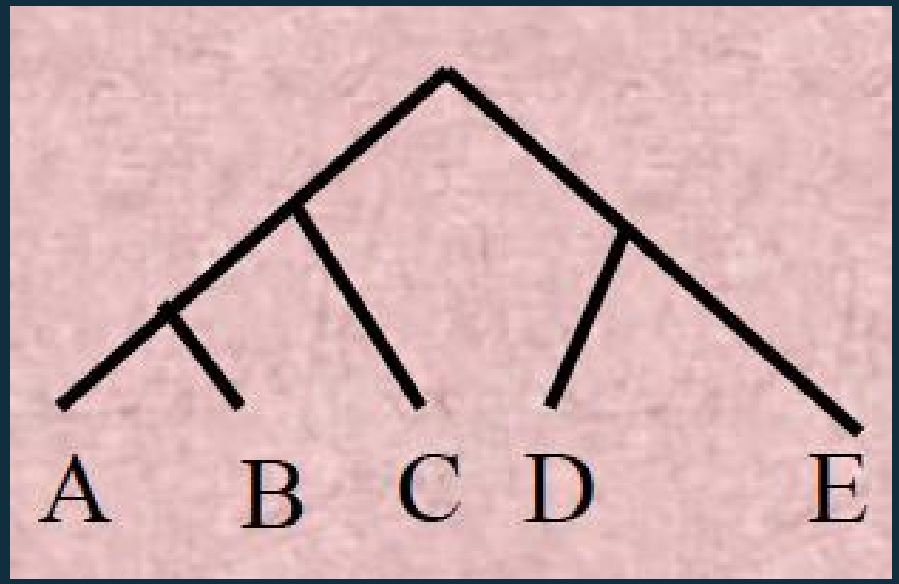


Topic III: Tree of life

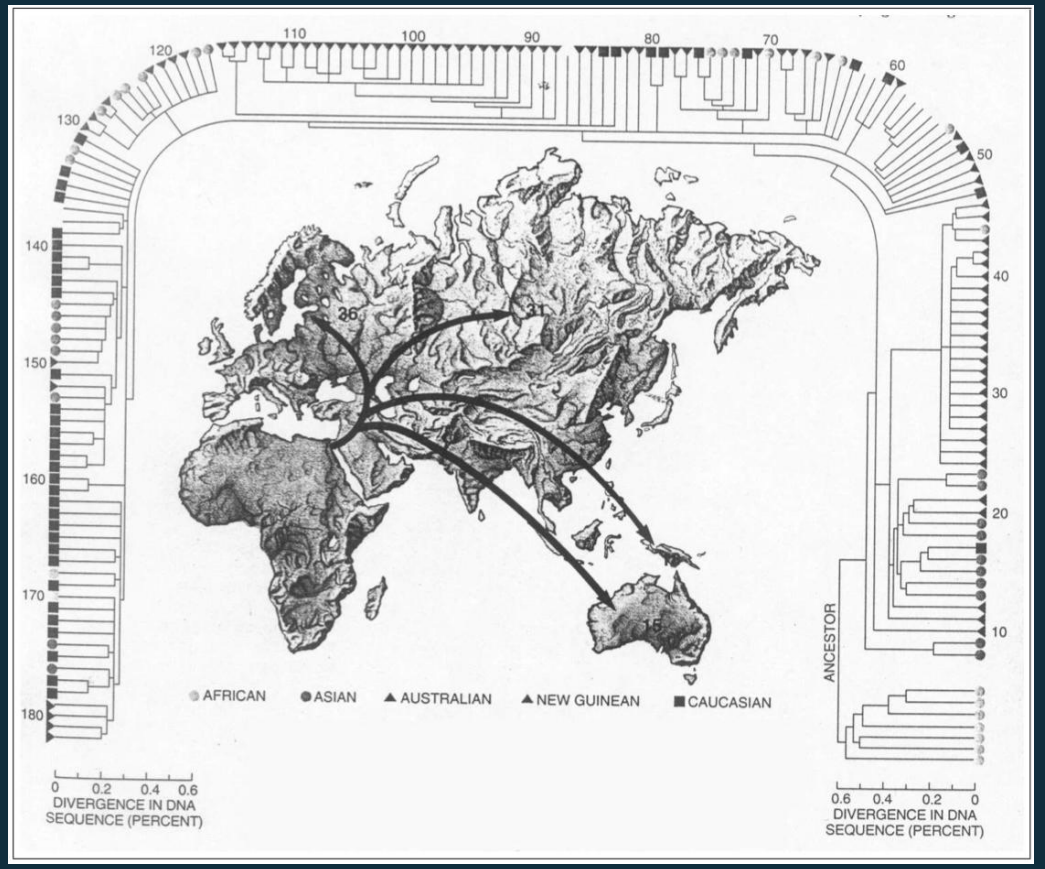


Inferring a phylogenetic tree

A: CAGGTA
B: CAGACA
C: CGGGTA
D: TGCACT
E: TGCGTA

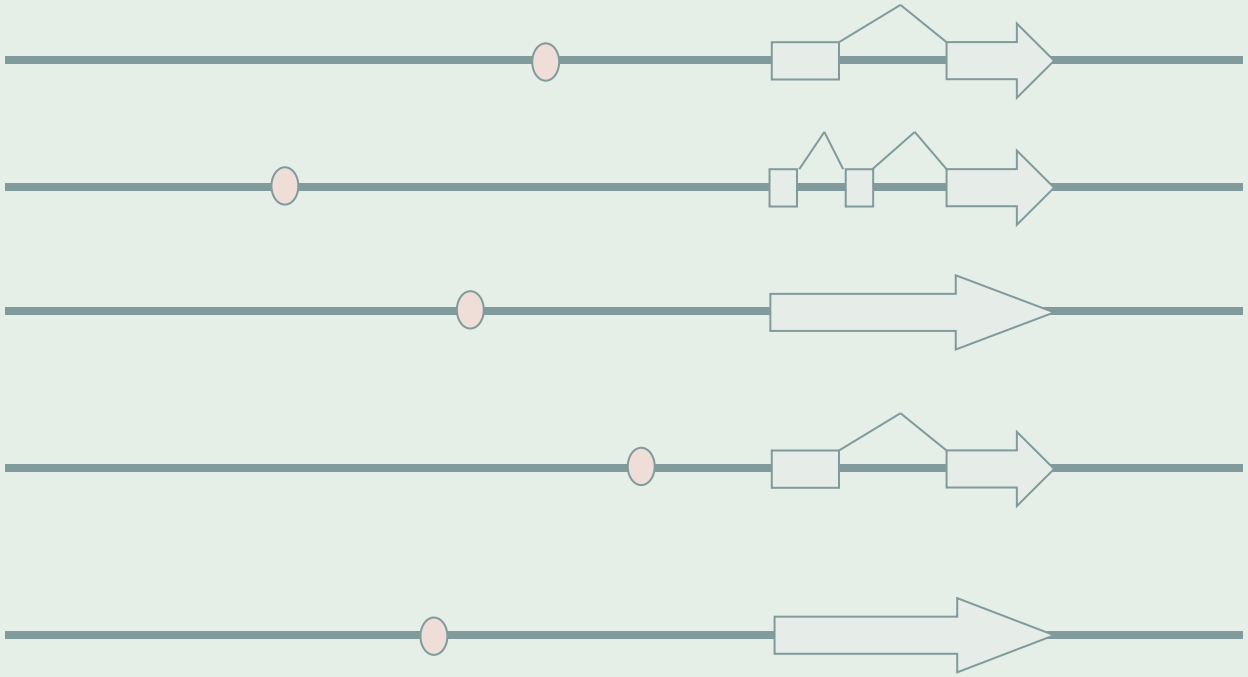


An example: Out of Africa



Topic IV: Motif Finding

Find promoter motifs associated with **co-regulated** or **functionally related** genes



转录调控

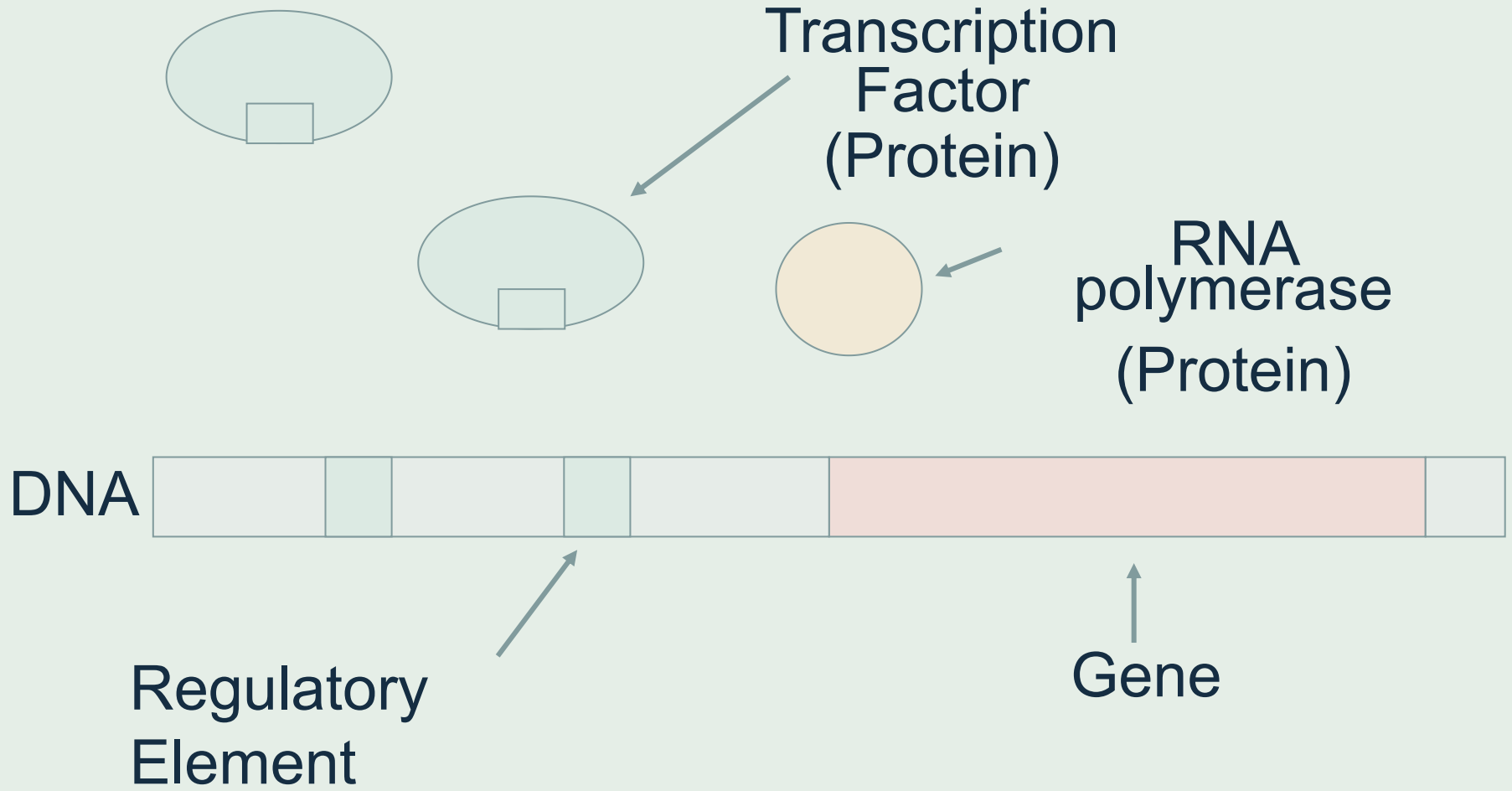
The transcription of each gene is controlled by a regulatory region of DNA relatively near the transcription start site (TSS).

two types of fundamental components

short DNA regulatory elements

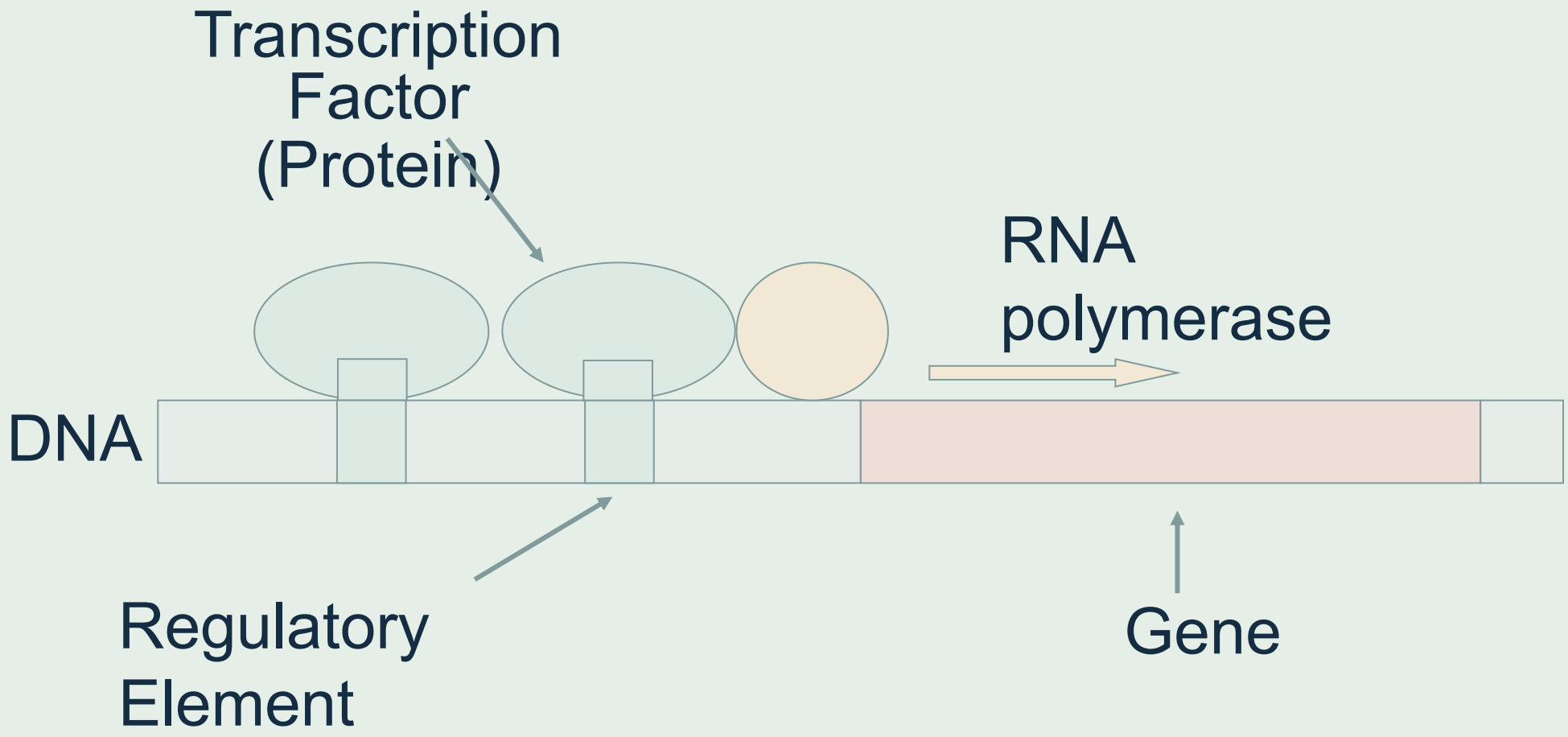
gene regulatory proteins that recognize and bind to them.

Regulation of Genes



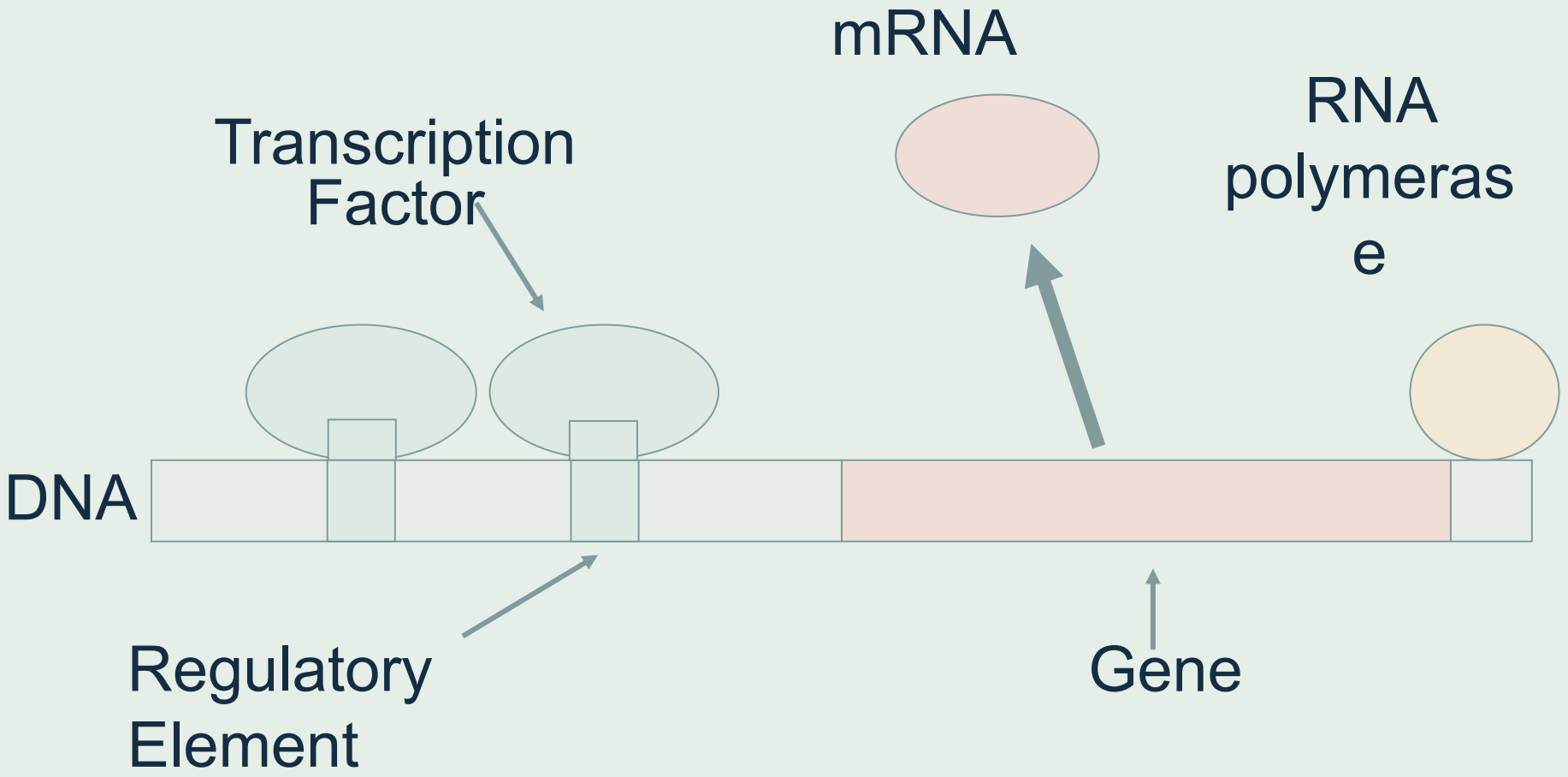
source: M. Tompa, U. of Washington

Regulation of Genes



source: M. Tompa, U. of Washington

Regulation of Genes



source: M. Tompa, U. of Washington

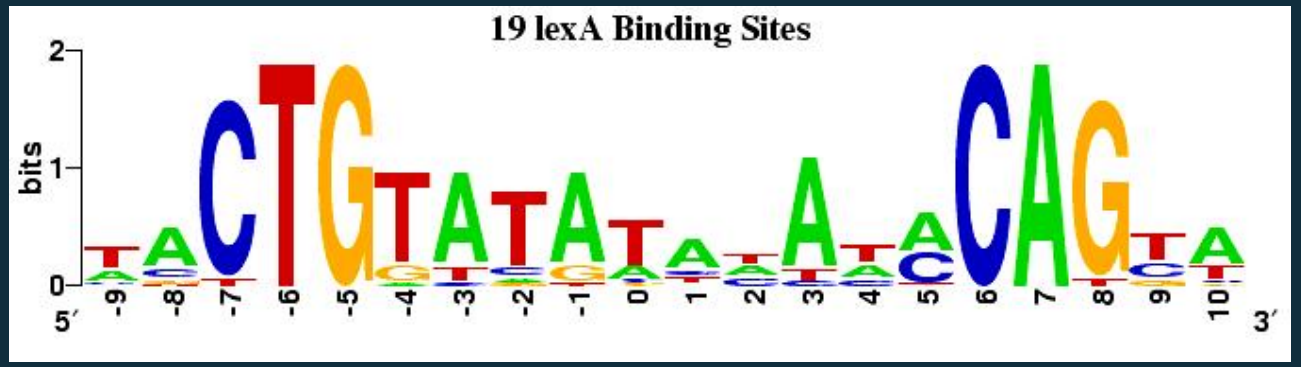
Motif Finding Problem

Characterizing the motif:
Positional weight matrix

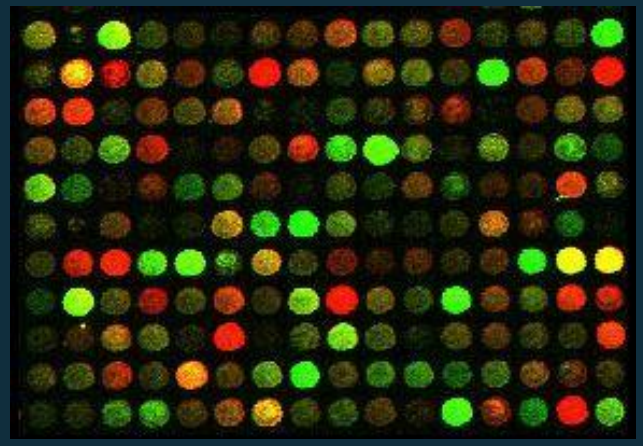
Finding the motif

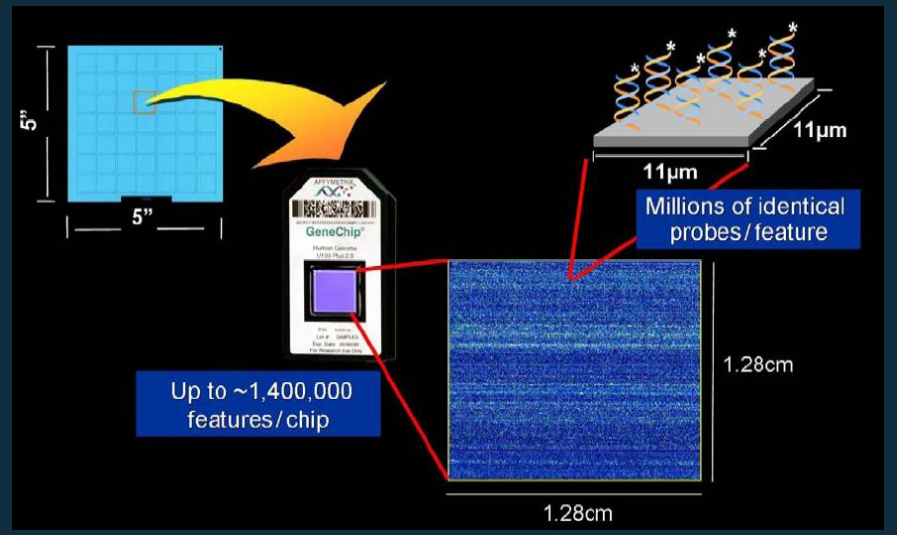
Gibbs Sampler (AlignACE)

EM algorithm (MEME)



Topic V: Gene Expression DataClustering and Biomarker Discovery

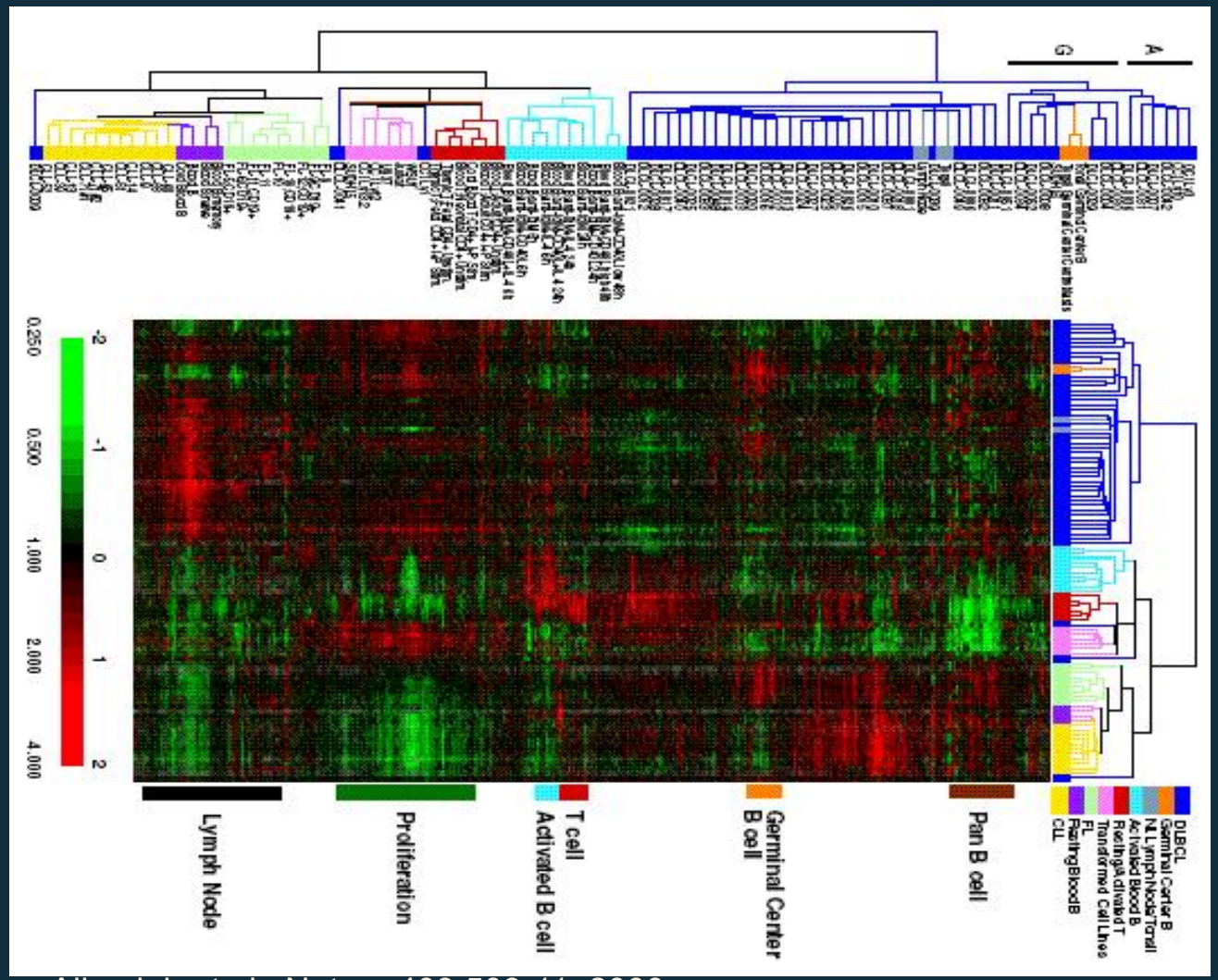




Microarrays

DNA microarray technology rely on the hybridization properties of nucleic acids to monitor DNA or RNA abundance on a genomic scale in different types of cells.

生物统计学案例



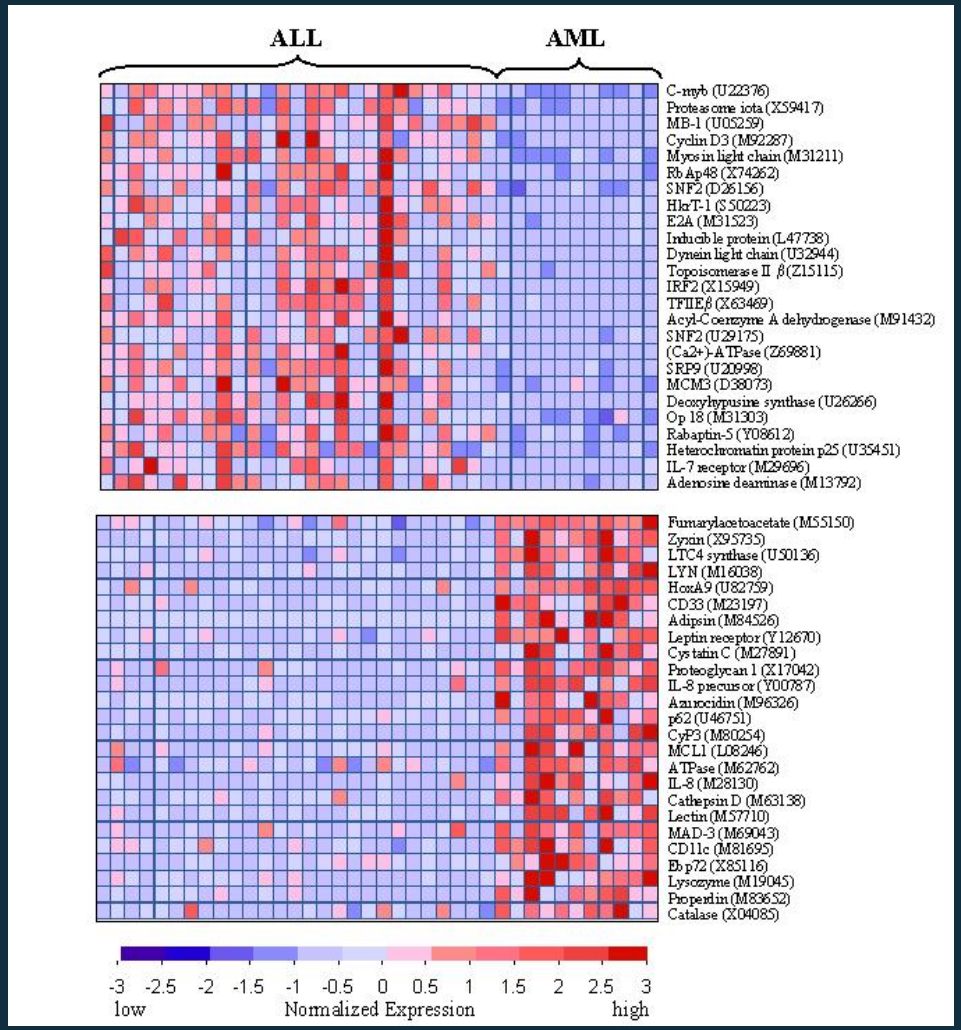
Alizadeh et al., Nature 403:503-11, 2000

Classification

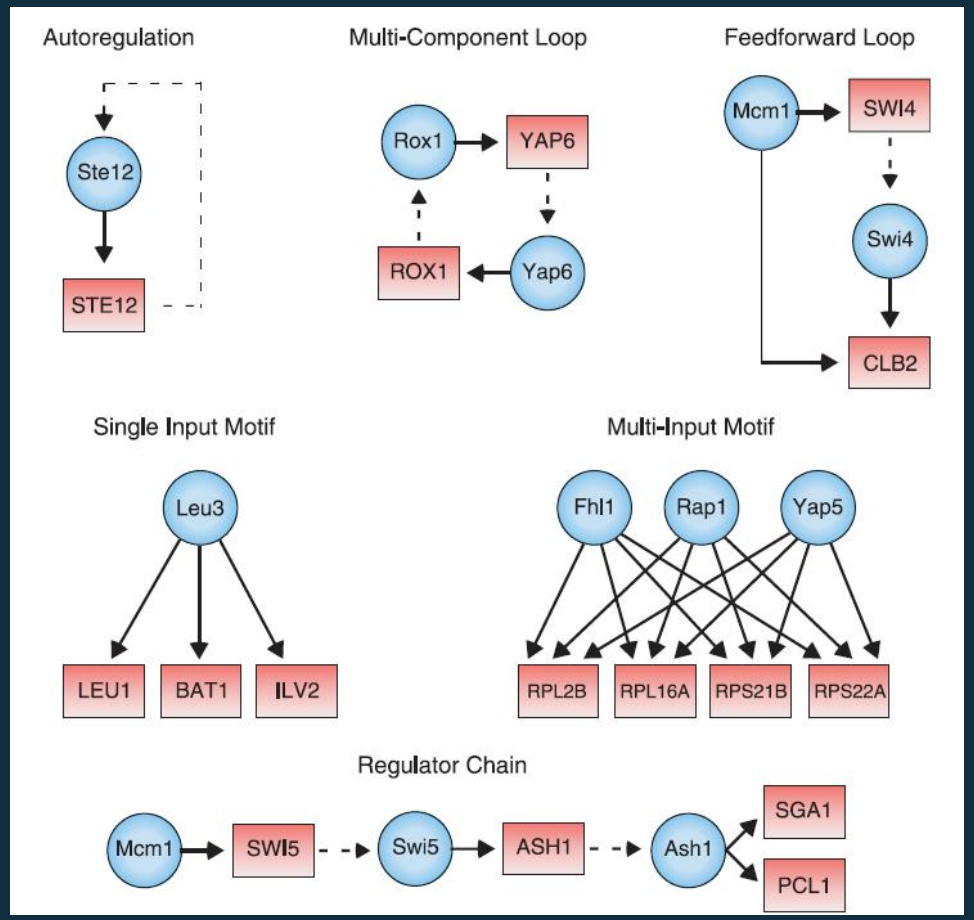
Given: the case, control gene expression data

Find: a set of genes (biomarker) can discriminate two classes.

Method: variable selection



Topic VI: Regulatory Network Inference from Gene Expression Data



Lee et al. Science 298: 799(2002)

Network Inference: Reverse Engineering

Given: a large set of gene expression observations

Find:

Wiring diagram

Transition rules

To fit the observation data

Methods

Bayesian Network

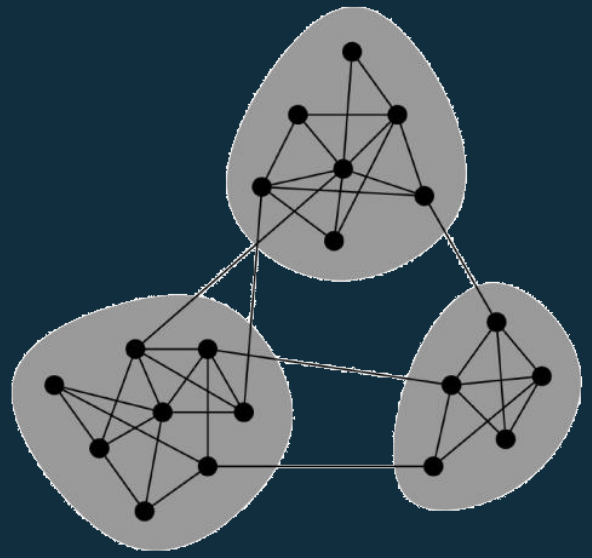
Gaussian graphical model

Dream Project

DREAM: Dialogue for Reverse Engineering Assessments and Methods

http://dreamchallenges.org/

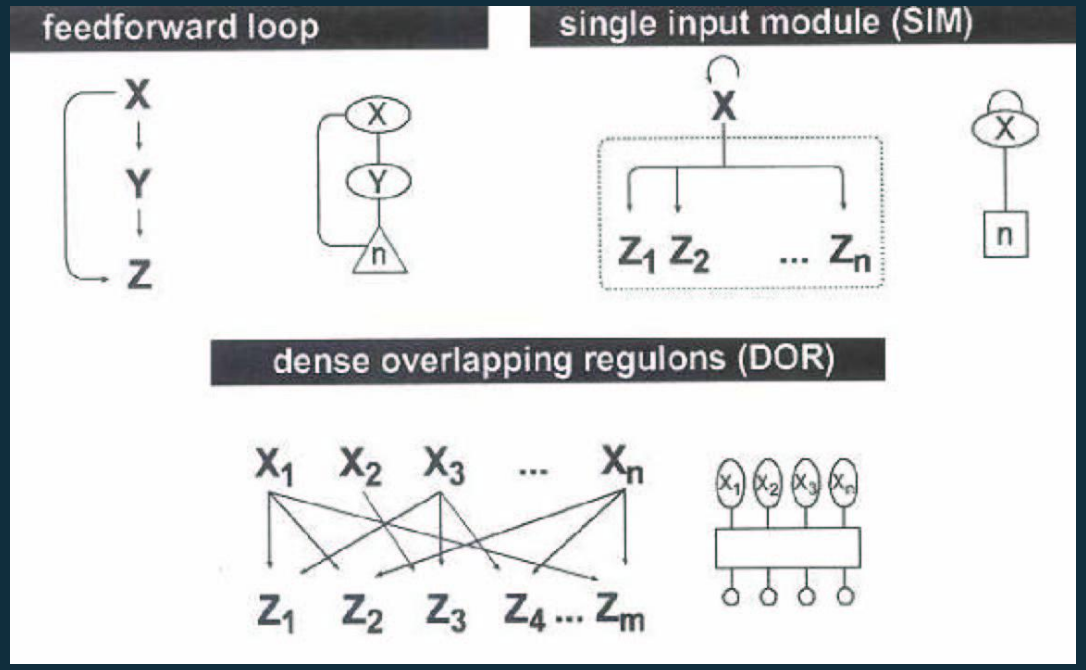
Topic VII: Network Analysis



Network Motif

Definition: Patterns of interconnections occurring in complex networks at numbers that are significantly higher than those in randomized networks (Milo, R., et. al. *Science* 298, 824–827)

Network Motifs



Milo et al. *Science,* **298**, 2002

Topic VIII: Dimension Reduction

Curse of dimensionality

Visualization in low dimension

Curse of Dimensionality

A major problem is *the curse of dimensionality.*

If the data x lies in high dimensional space, then an enormous amount of data is required to learn distributions or decision rules.

Example: 50 dimensions. Each dimension has 20 levels. This gives a total of 20^{50} cells. But the no. of data samples will be far less. There will not be enough data samples to learn.

Curse of Dimensionality

One way to deal with dimensionality is to assume that we know the form of the probability distribution.

For example, a Gaussian model in N dimensions has N + N(N-1)/2 parameters to estimate.

Requires $O(N^2)$ data to learn reliably. This $O(N^2)$ may be practical.

Dimension Reduction

One way to avoid the curse of dimensionality is by projecting the data onto a lower-dimensional space.

Techniques for dimension reduction:

Principal Component Analysis (PCA)

Singular value decomposition (SVD)

Multi-dimensional Scaling (MDS).

Recap

a breif summary...

Think again:

Why biostatistics

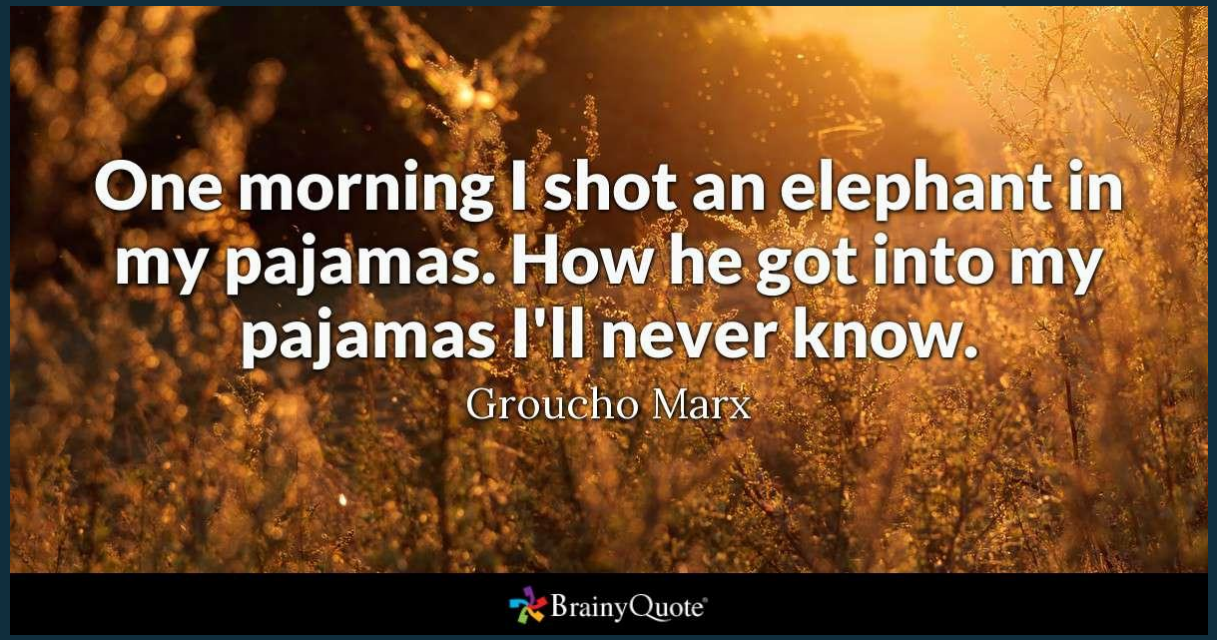
What is biostatistical modeling

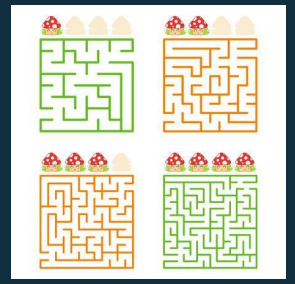
Where is the applications

Statistical modeling

数据无处不在，复杂性无处不在。。。

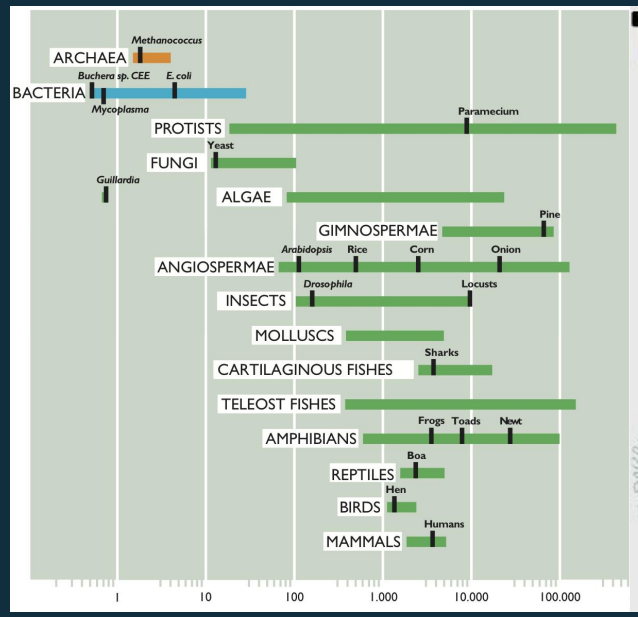
一天早上我在我的睡衣里打死一只大象，他怎么跑到我睡衣里来的，我就不知道了。

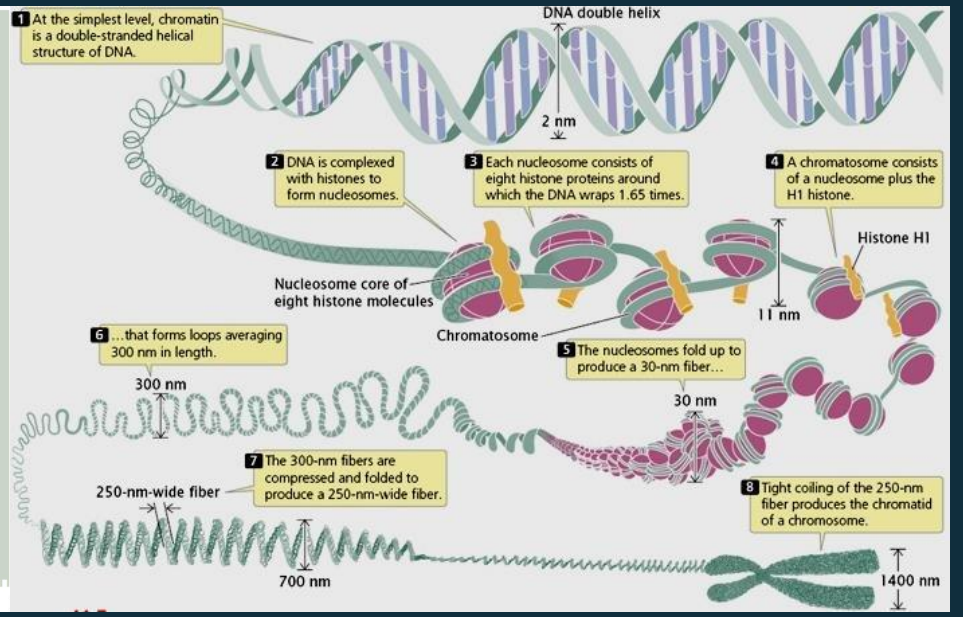




Statistical modeling

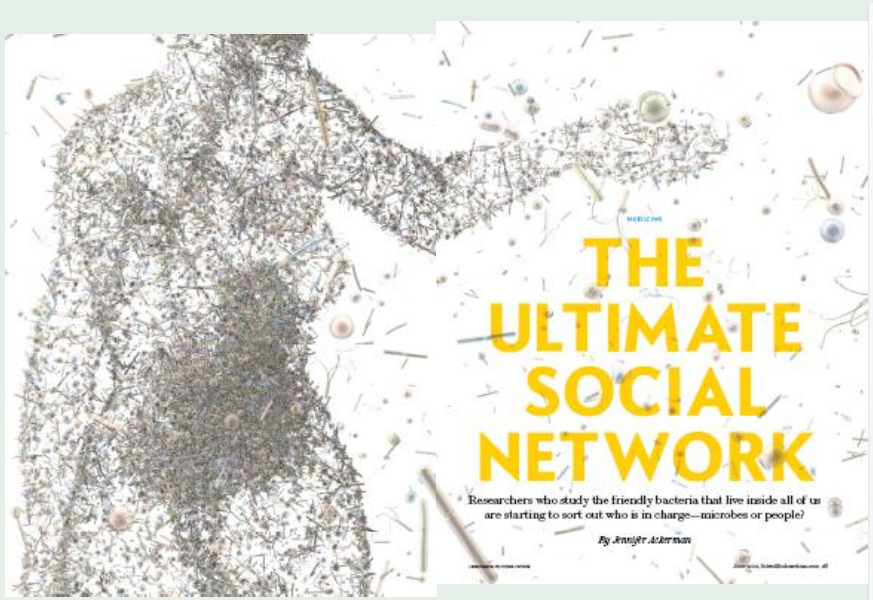
数据无处不在，复杂性无处不在。。。

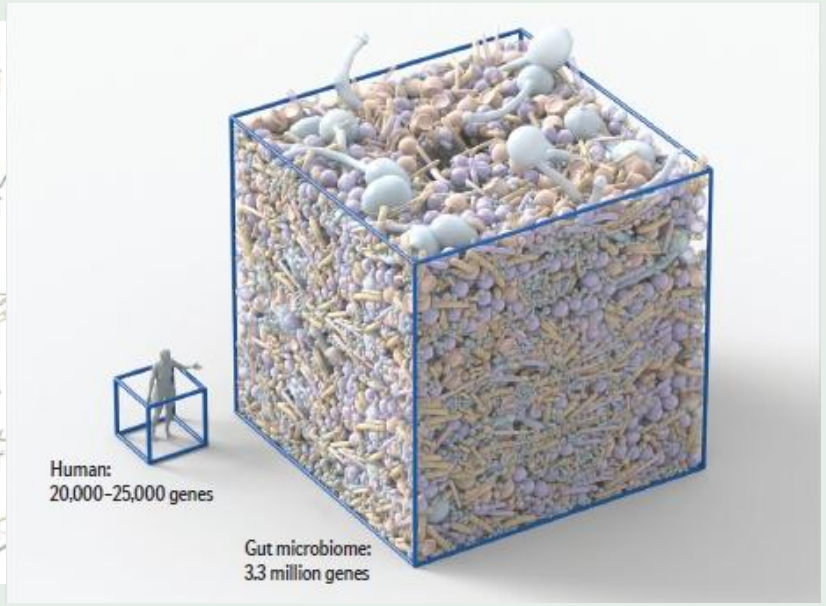




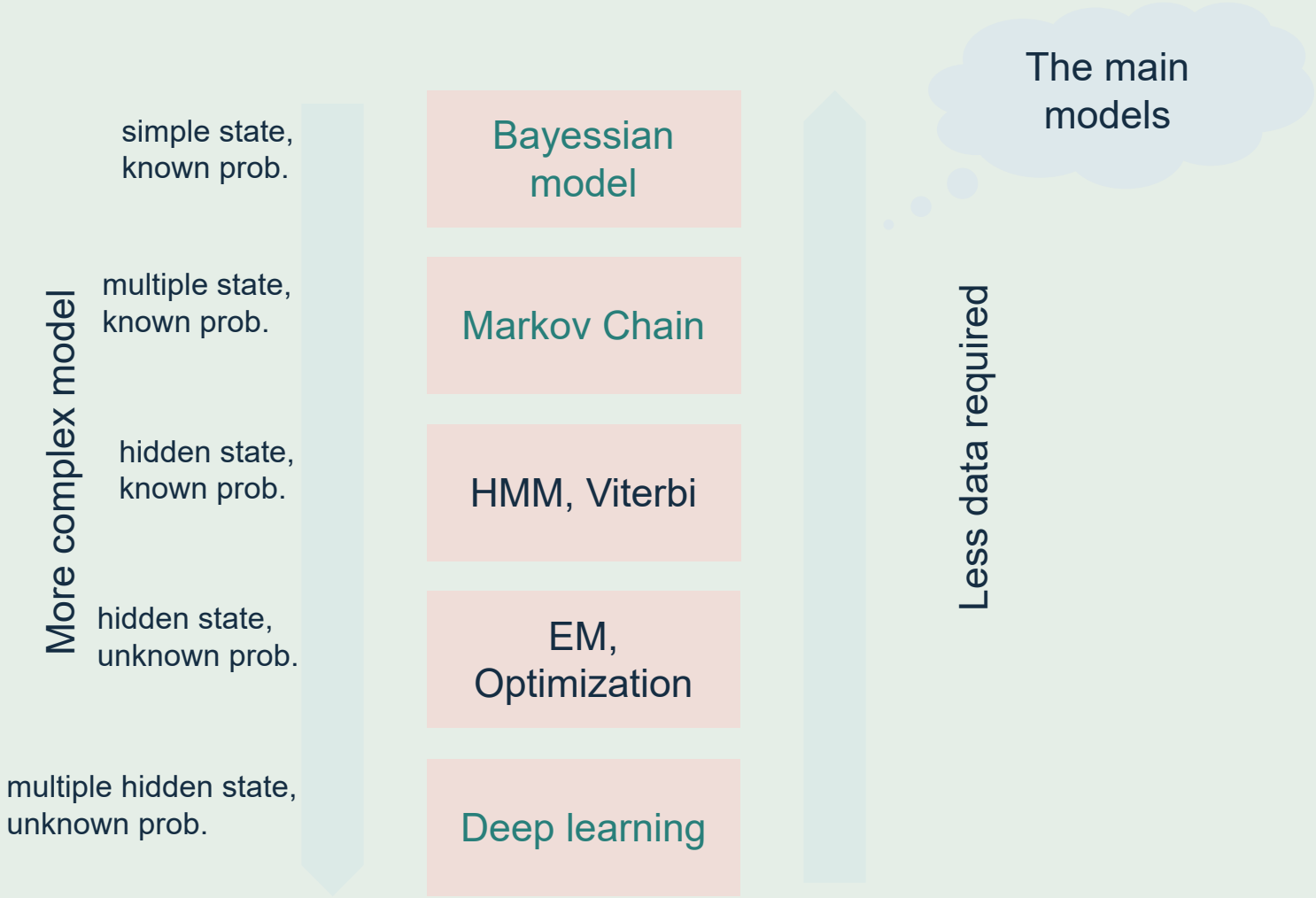
Statistical modeling

数据无处不在，复杂性无处不在。。。

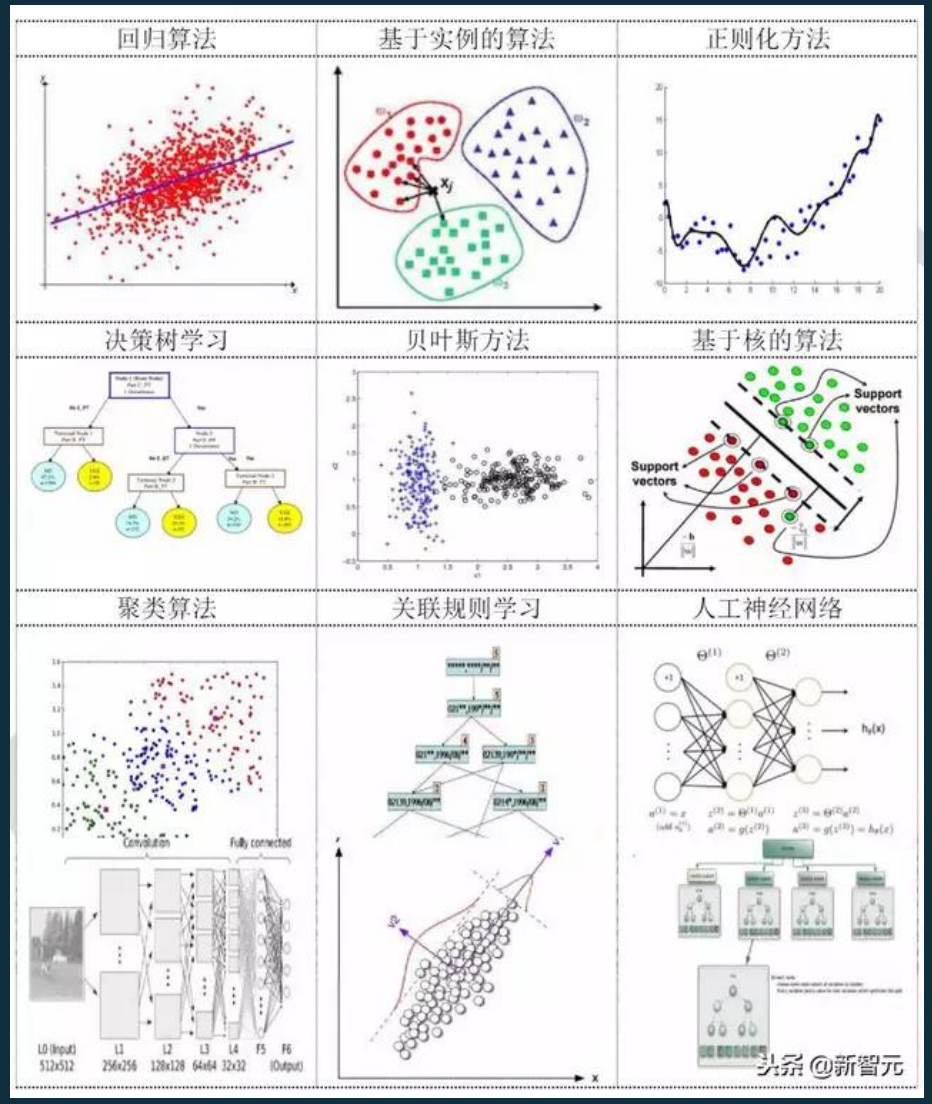




Statistical modeling



Statistical modeling



课程安排

生物背景和课程简介

传统生物统计学及其应用

生物统计学和生物大数据挖掘

- Hidden Markov Model (HMM)及其应用
  - Markov Chain
  - HMM理论
  - HMM和基因识别 (Topic I)
  - HMM和序列比对 (Topic II)
- 进化树的概率模型 (Topic III)
- Motif finding中的概率模型 (Topic IV)
  - EM algorithm
  - Markov Chain Monte Carlo (MCMC)
- 基因表达数据分析 (Topic V)
  - 聚类分析-Mixture model
  - Classification-Lasso Based variable selection
- 基因网络推断 (Topic VI)
  - Bayesian网络
  - Gaussian Graphical Model
- 基因网络分析 (Topic VII)
  - Network clustering
  - Network Motif
  - Markov random field (MRF)
- Dimension reduction及其应用 (Topic VIII)

面向生物大数据挖掘的深度学习

研究对象:生物序列,进化树,生物网络,基因表达…

方法：生物计算与生物统计

References

James D. Watson, Tania A. Baker, Stephen P. Bell. Molecular Biology of the Gene. Benjamin-Cummings Publishing Company. 2008.

Bruce Alberts. Molecular Biology of the Cell. Garland Publishing Inc. 2007.

Jocelyn E. Krebs, Stephen T. Kilpatrick, Elliott S. Goldstein. Lewin‘s Genes XI. Jones and Bartlett Publishers, Inc. 2012.

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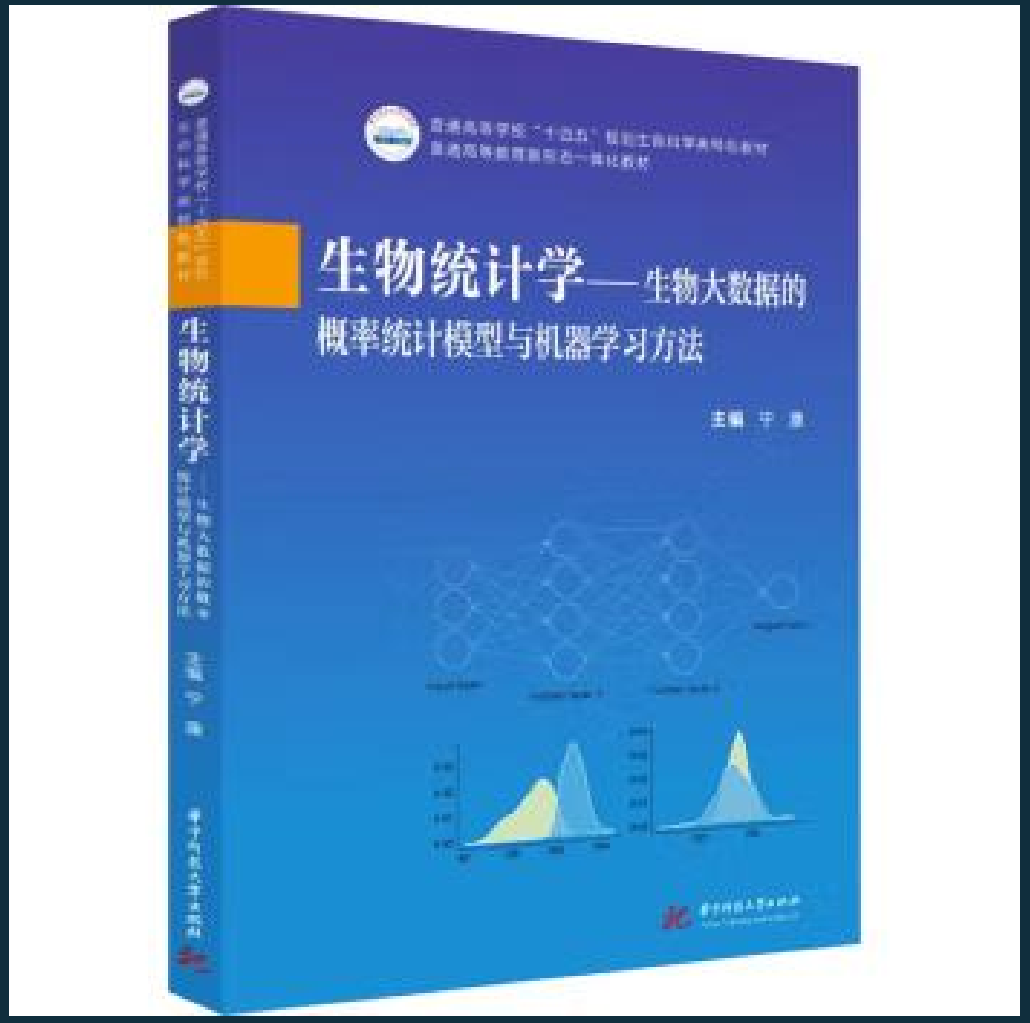

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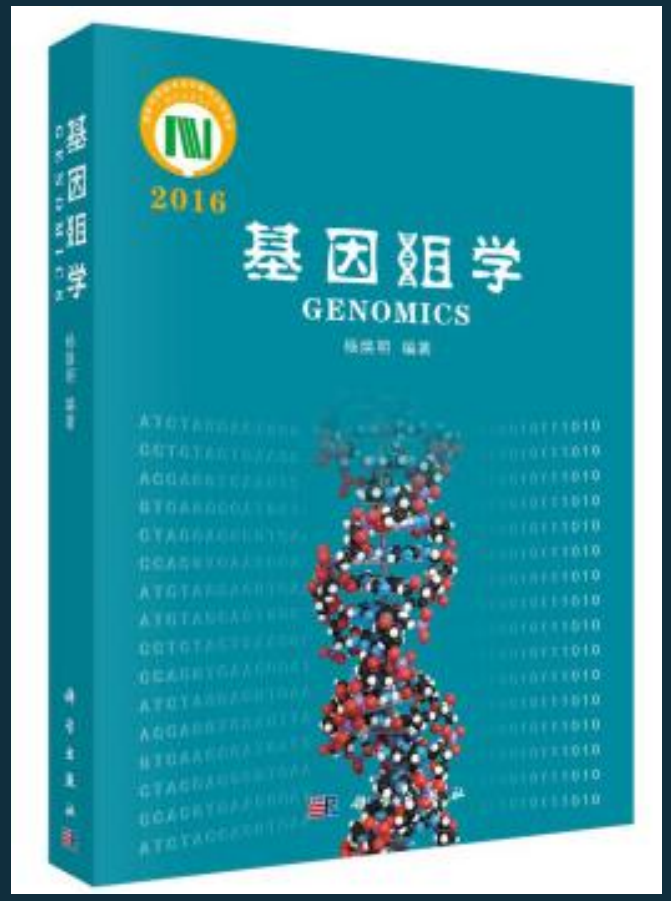
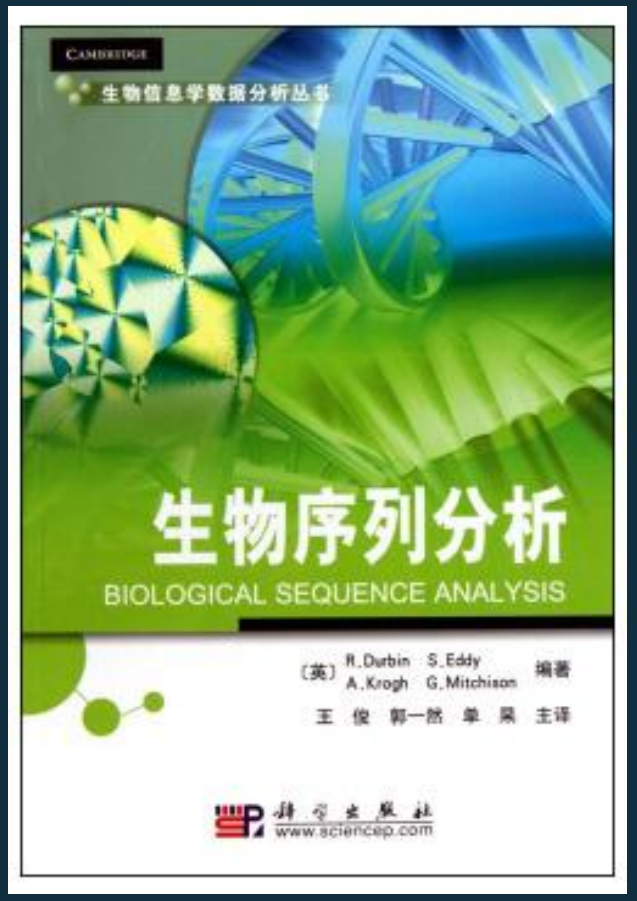

2016
基因组学
GENOMICS


CAMBRIDGE
生物信息学数据分析丛书
生物序列分析
BIOLOGICAL SEQUENCE ANALYSIS
R.Durbin S.Eddy
A.Krogh G.Mitchison
编著
王俊 郭一然 单昊 主译
www.sciencep.com

生物统计学
(第五版)

References

智能时代 深度学习 · 文津图书奖 · 中华优秀出版物提名奖
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计算之魂
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计 算 科 学 品 味 和 认 知 进 阶
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——生物统计学：卜东波@中国科学院计算技术研究所，邓明华@北京大学

——神经网络与深度学习: 邱锡鹏@复旦大学

——Introduction to Computational Biology and Bioinformatics: Xiaole Shirley Liu Lab@Havard University

——Combinatorial Methods in Computation Biology: Ken Sung Lab@NUS

——Deep Learning in the Life Sciences: MIT

——Probabilistic Graphical Models: Eric Xing@CMU

——Numerous other leading researchers and leading labs.....

LECTURE 01

补充知识

补充知识

生物信息学与生物统计学：

两种视角

四种技术

两类方法

生物信息学与生物统计学：

两种视角，四种技术，两类方法

两种视角

生物学视角
计算视角

四种技术

算法
数据
超算
深度学习

两类方法

经典软件
核心工具

LECTURE 01

生物信息学：生物学视角

我们是谁？我们从哪里来？我们到哪里去？

先看视屏！

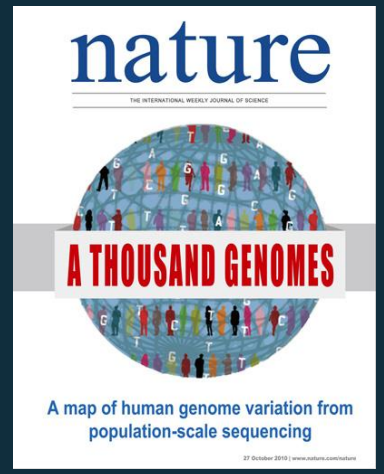
- 开讲啦：如何健康地活到100岁？（汪健@CCTV, http://www.biodiscover.com/talk/research/103880.html）
- 基因组学（Barry Schuler@TED, https://www.ted.com/talks/barry_schuler_genomics_101）
- 微生物组学（Rob Knight@TED, https://www.ted.com/talks/rob_knight_how_our_microbes_make_us_who_we_are）
- 一个IT人的故事（Larry Smarr@TED, http://www.tedmed.com/talks/show?id=18018）
- 大数据（Kenneth Cukier@TED, https://www.ted.com/talks/kenneth_cukier_big_data_is_better_data）

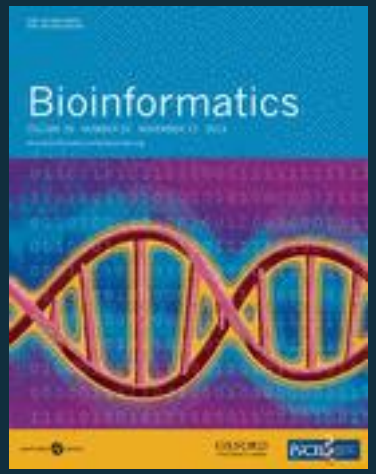
补充知识

生命是序列的，生命是数字的！

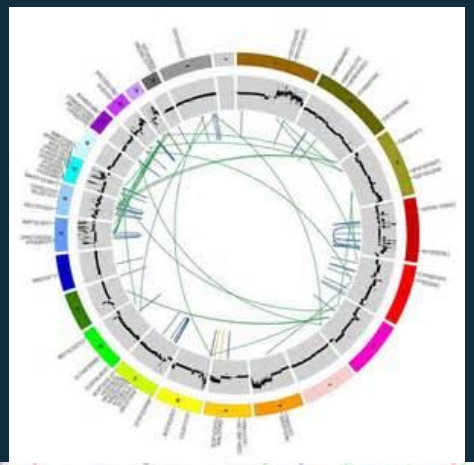
nature
Individual genomes from Africa and China
Acute myeloid leukaemia genome
Designer nucleases for gene therapy
Tracing gene flow across Europe
BREAK GLASS WHEN READY
YOUR LIFE IN YOUR HANDS
Instructions for the personal genome age
NATUREJOBS
Mentor awards

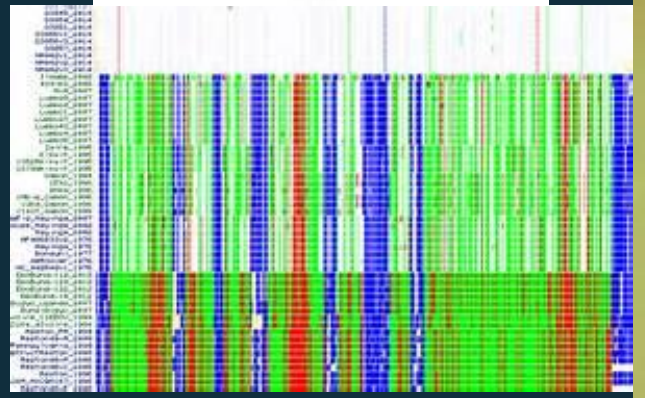
Science

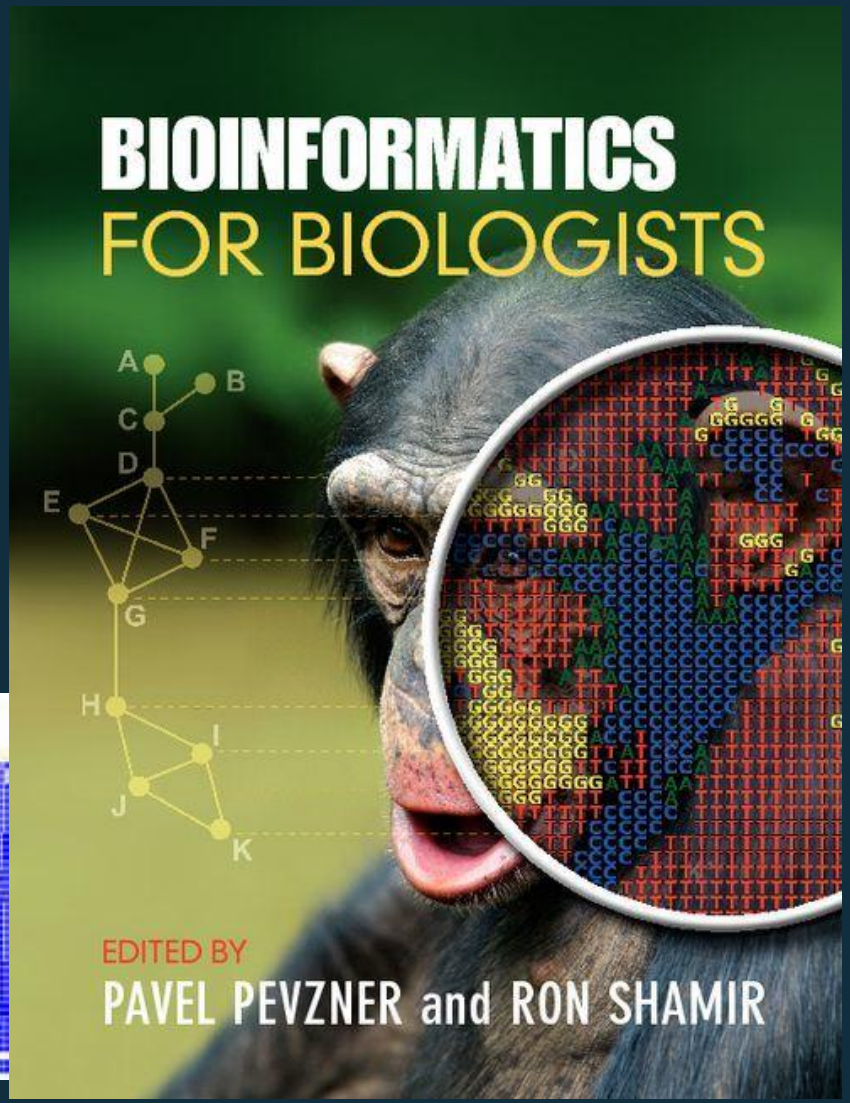

nature
A THOUSAND GENOMES
A map of human genome variation from population-scale sequencing


Bioinformatics

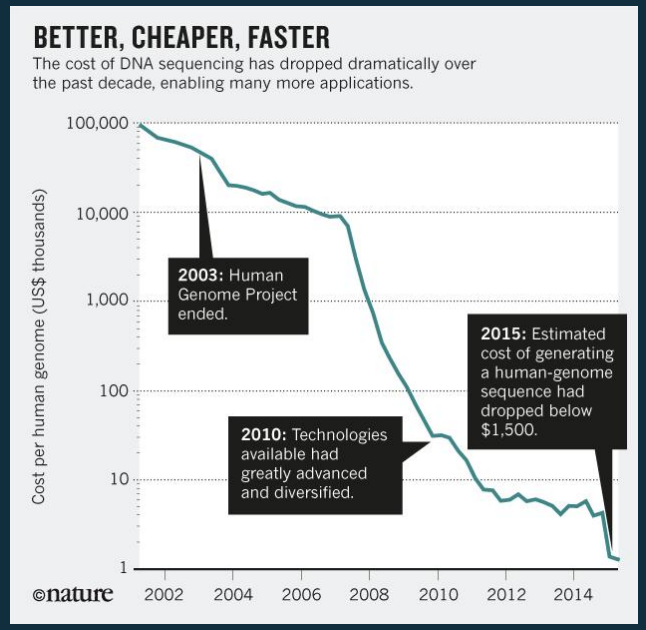
生命是序列的，生命是数字的！

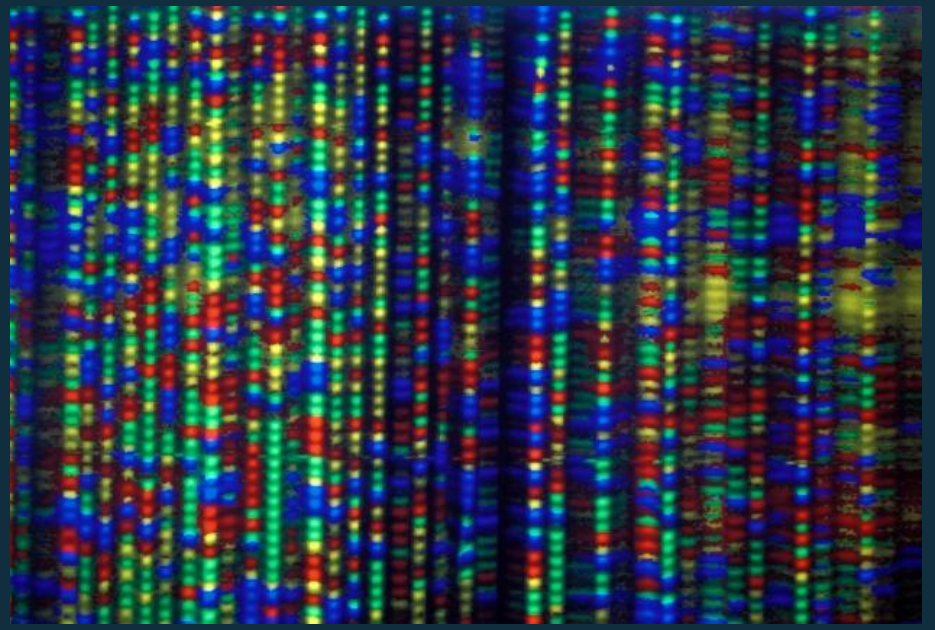






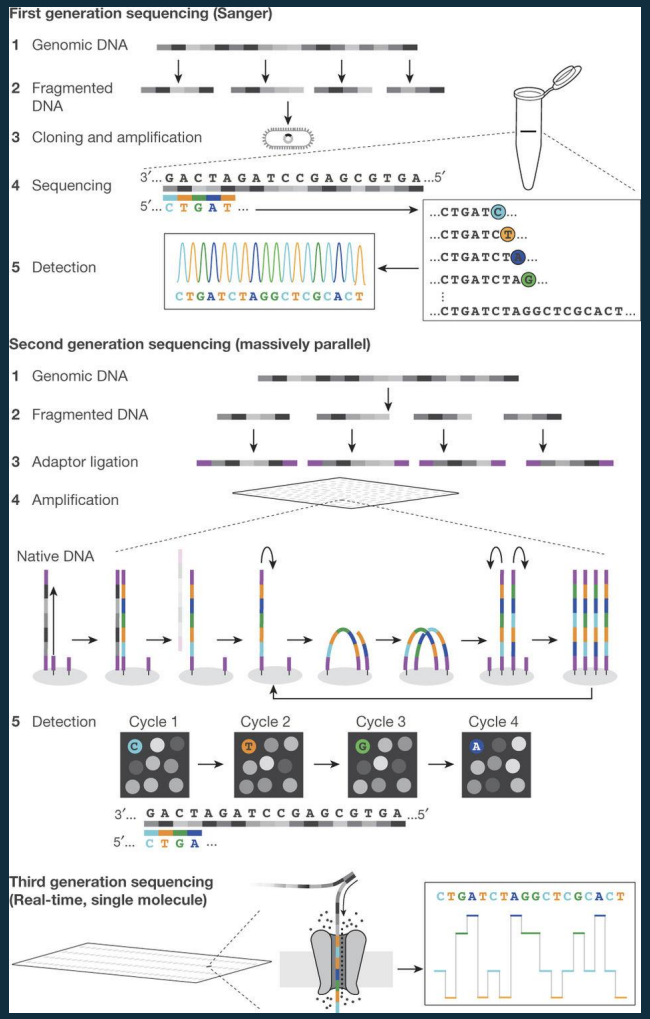
DNA测序与生物信息学

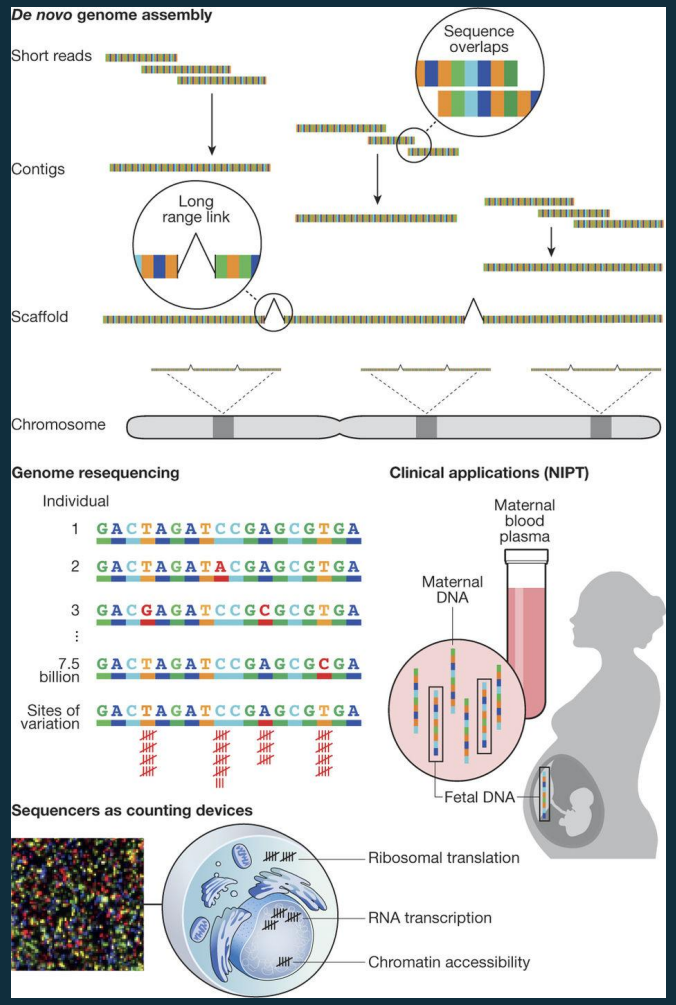




Sequencing applications

DNA Sequencing





补充知识

在今天，DNA测序技术已经在诸多方面达到了临床应用的具体要求。

科学家估计，全世界每年大约有四百万到六百万孕妇正在通过外周血游离DNA对胎儿的21三体综合征进行诊断，而十年之内，这个数字将超过1500万。

在高收入国家，基因组测序已经广泛用于多种疾病的产前诊断，可以揭示大约30%的出生缺陷，同时，这一数字也正随着数据解读能力的成熟而逐渐上升。

在肿瘤学领域，液体活检在最近几年已经成为了肿瘤相关学术，产业以及投资界的新宠。基于DNA测序的液体活检被认为正逐步发展为癌症诊断与预后评估的标准方法，能够在可知的时间内逐步补充甚至取代传统的创伤性癌症诊断技术。

同样，手持DNA测序仪等设备的开发也使得流行病学家甚至能够在最为偏远的地区高效完成对人类样本，动物以及昆虫病原载体甚至是空气，水，食物的基因检测。

流行病学家和公共卫生专家也开始讨论如何通过对城市垃圾中微生物的DNA测序辅助传染性疾病的预防与控制。

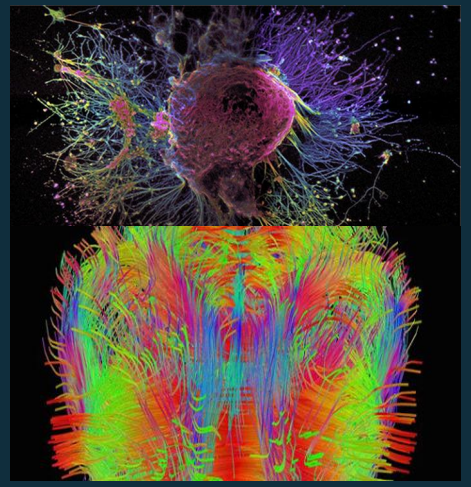
海洋生物学家也正在通过宏基因组学技术来对海洋的生态健康进行监测与研究。

在法医领域，便携式DNA测序仪可以将DNA分析带出法医实验室，使其成为一线警务工作人员的随身工具。帮助警方即使通过DNA监测确定嫌疑人，发展成为诸如酒精探测器一类的便捷工具。

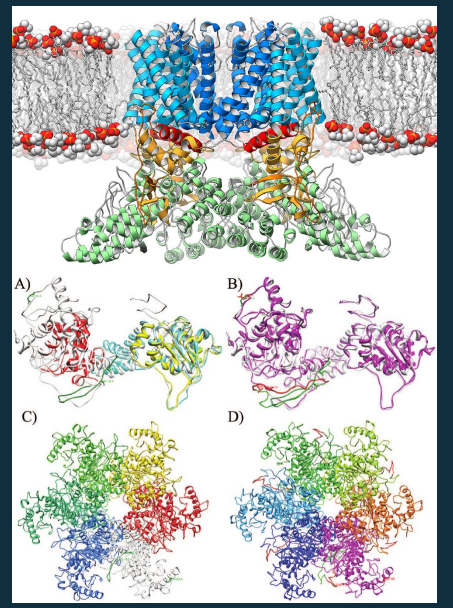
在人们的家里，DNA测序设备或许也可以成为下一个“智能”或“连接”设备，一些评论者甚至认定厕所是通过实时DNA测序监测家庭成员健康的理想场所。

生物信息学 @HUST

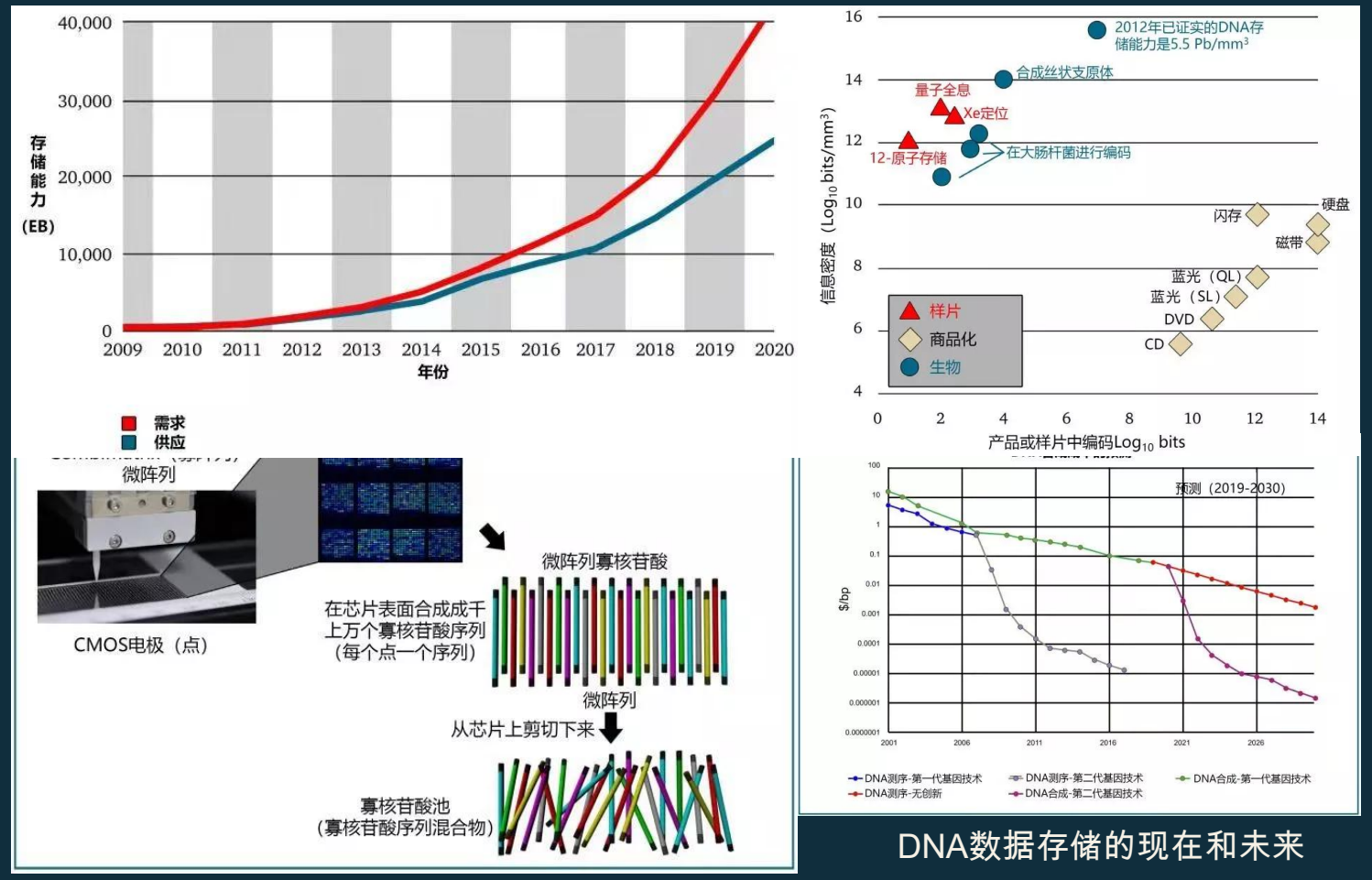
生命不只是序列的，但是生命始终是数字的！



- 结构生物学（Structure biology）
- 生物图像（Bio-imaging）
- 。。。

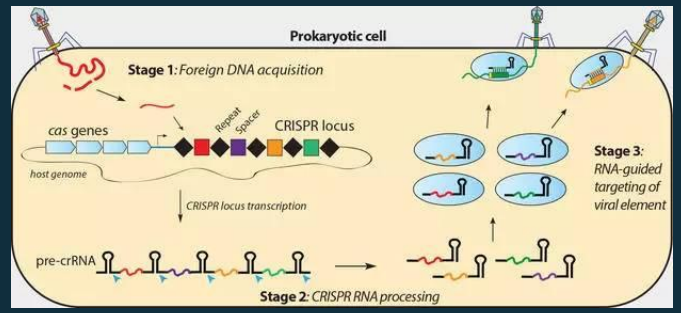


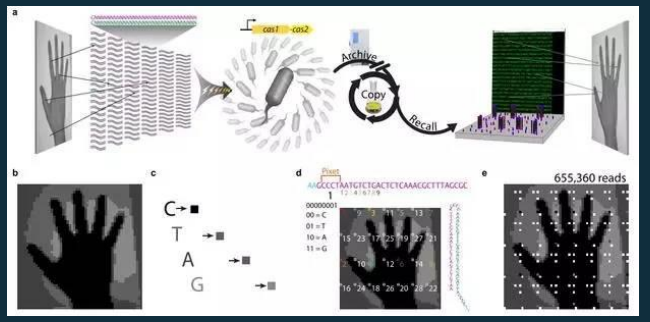
Understand it, create it!

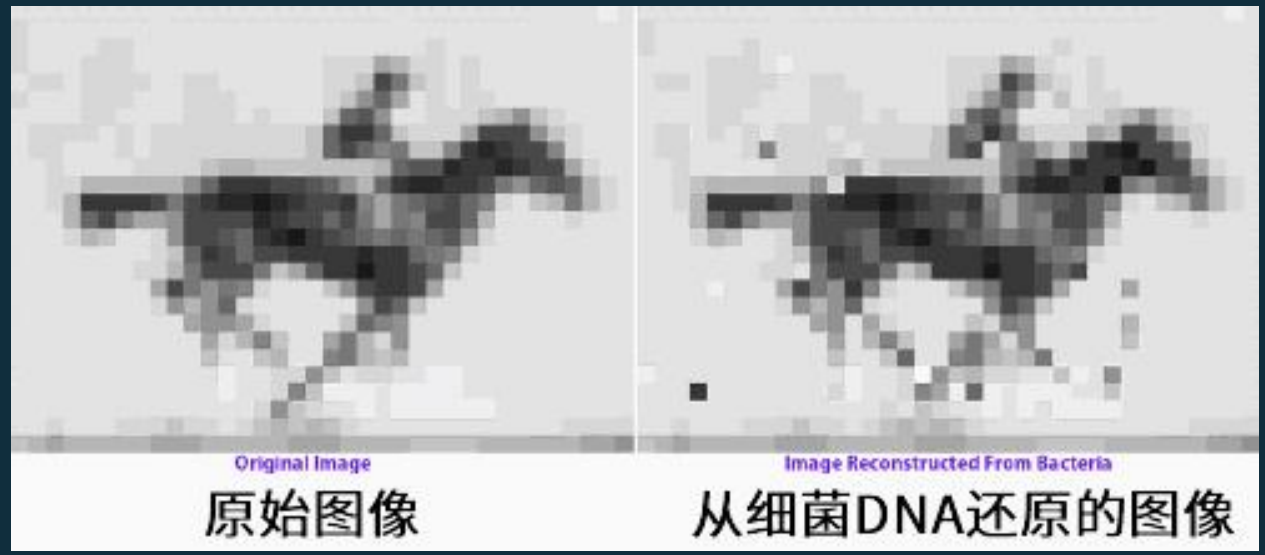


DNA数据存储的现在和未来

Understand it, create it!







Understand it, create it!

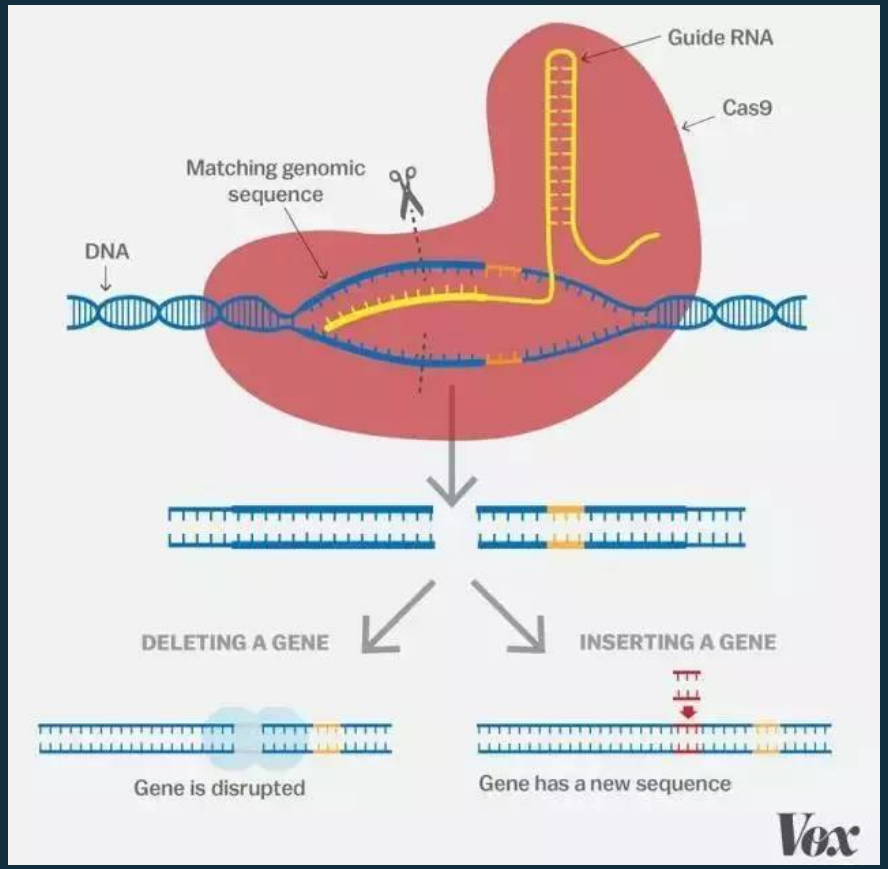
Our test on DNA storage: how can it tolerate mutations?

5% SNPs 2% SNPs 1% SNPs

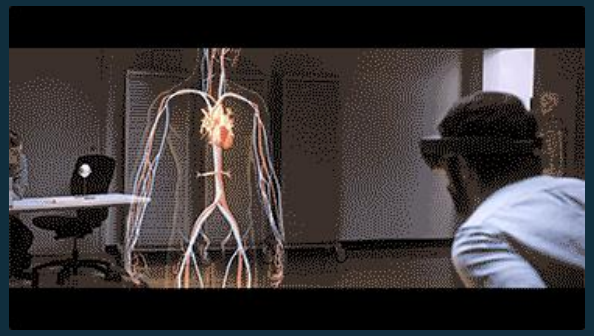
0.1% SNPs 0.05% SNPs

Understand it, create it!

CRISPR-Cas9

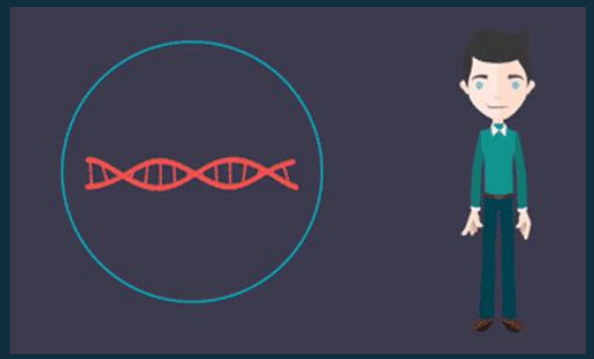


See it!



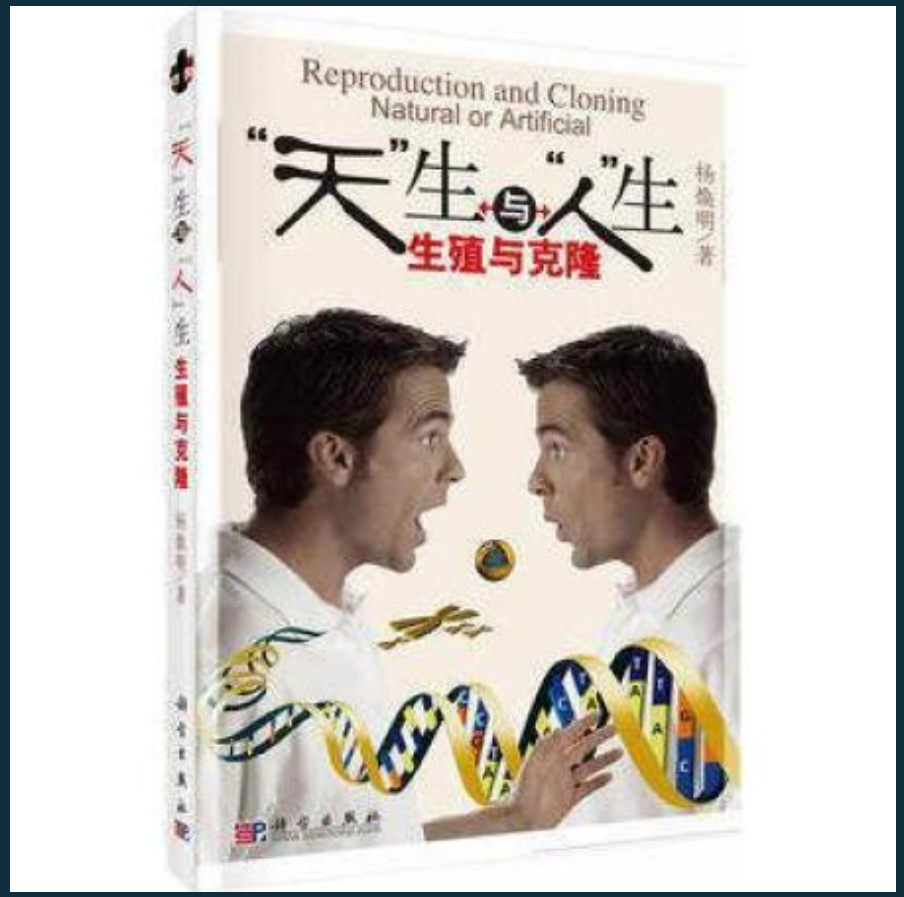
Understand it!

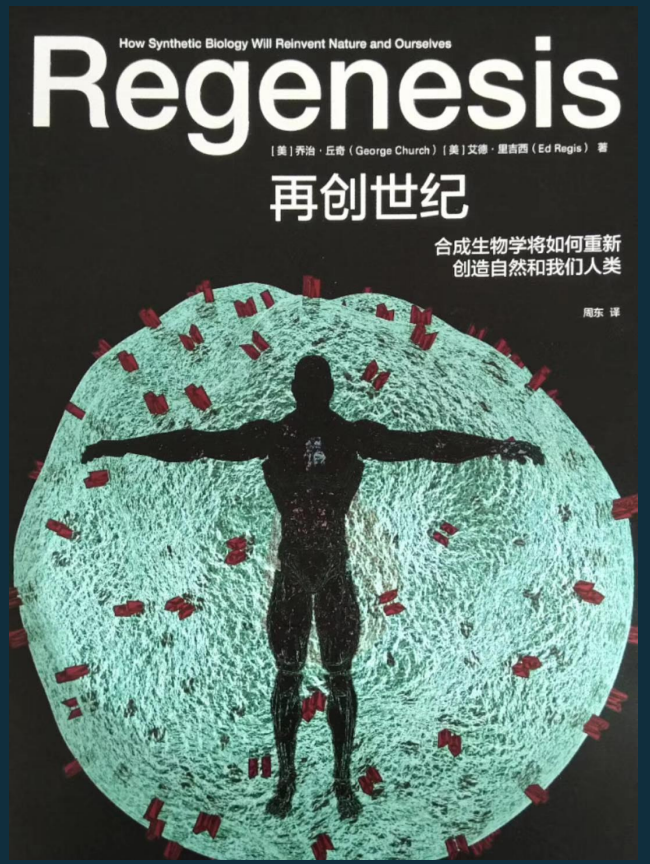
Create it!



Understand it, create it!

读物推荐

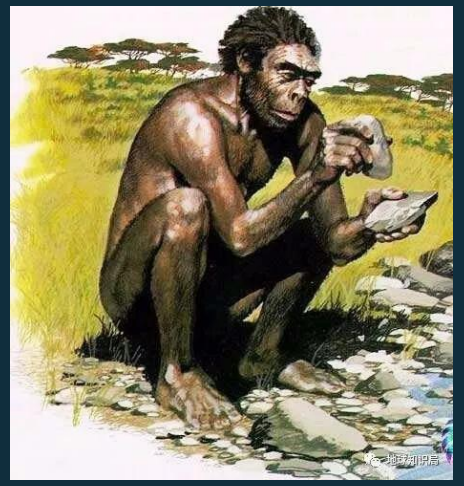




Understand it, create it!

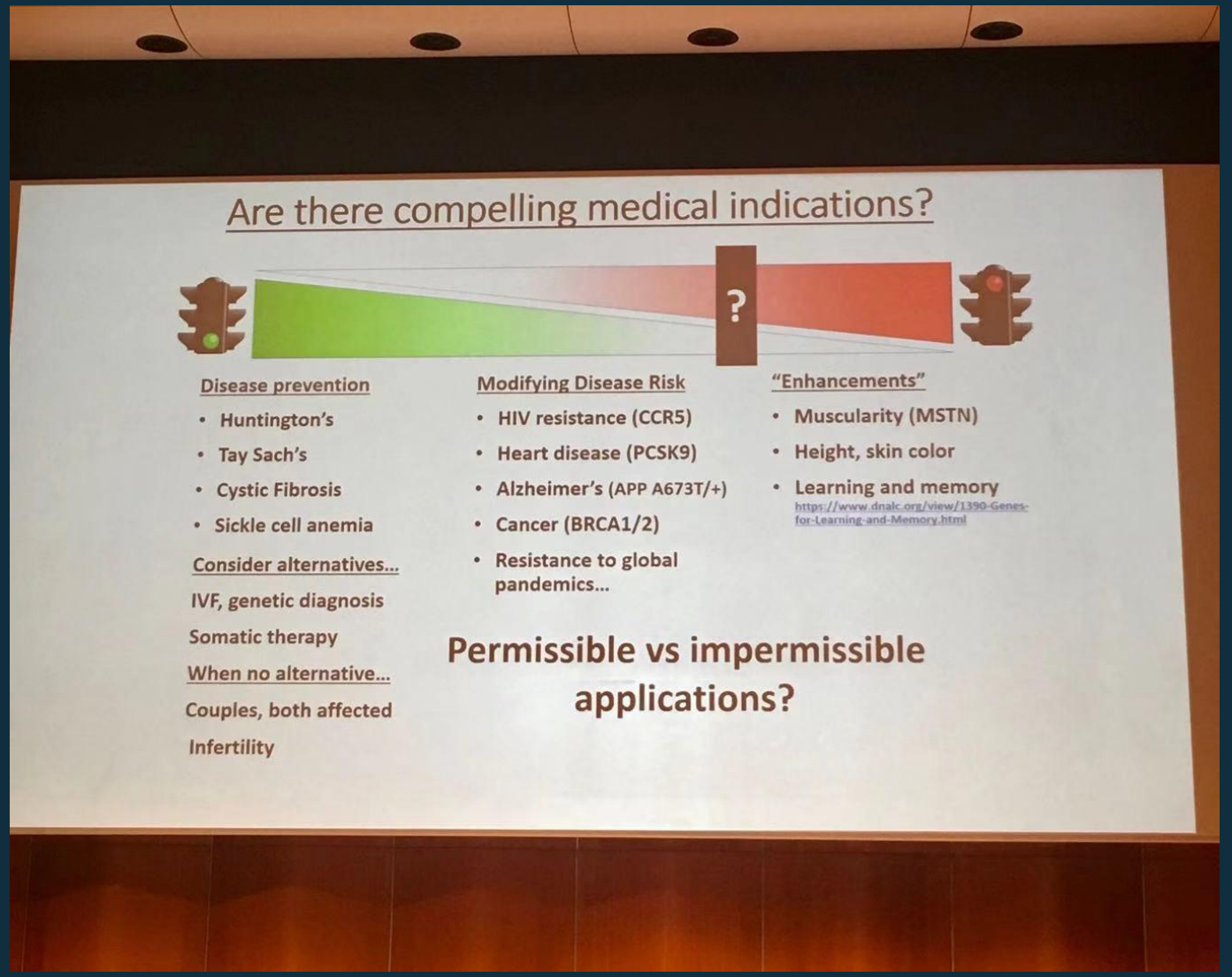
Understand it, create it!

CRISPR和基因编辑技术



OR

CRISPR和基因编辑技术

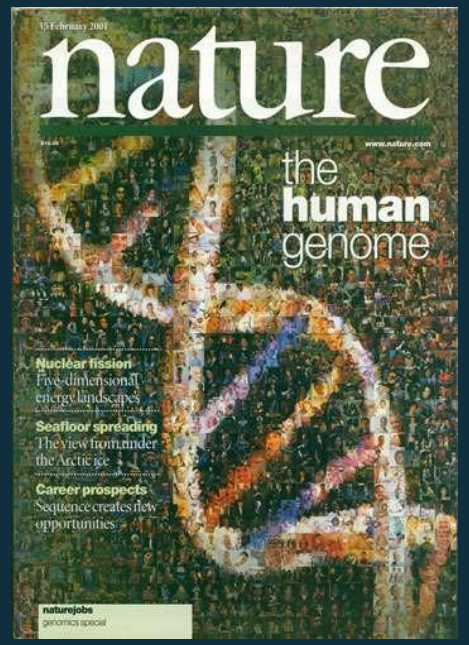


LECTURE 01

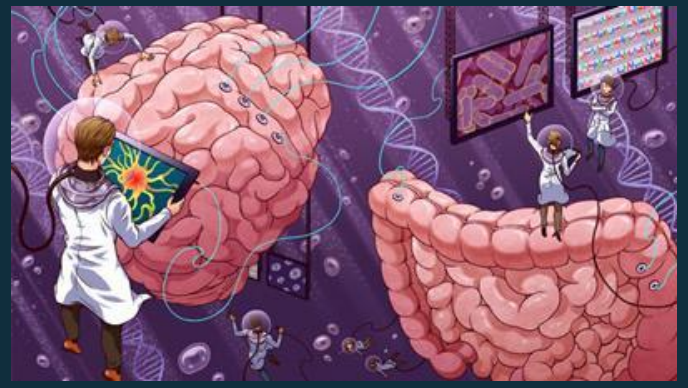
生物信息学：计算科学视角

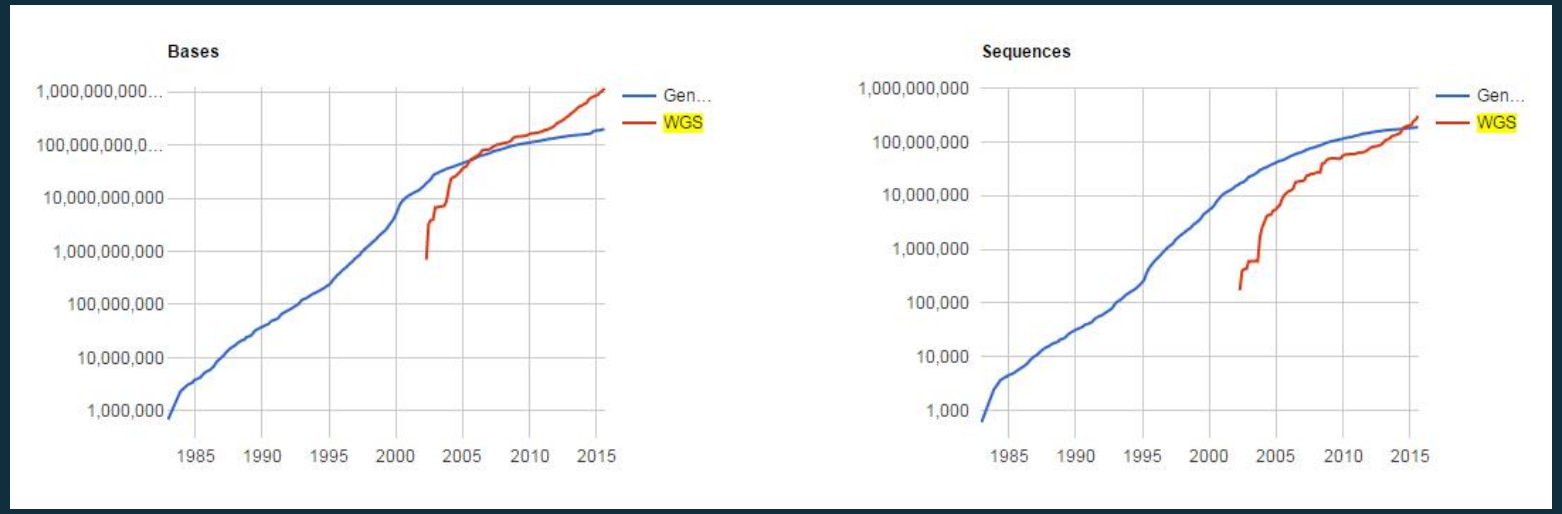
生物信息学 @HUST

我们是谁？我们从哪里来？我们到哪里去？



生命科学的前沿 – 生物大数据

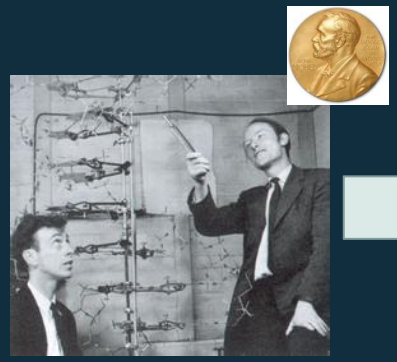




生物大数据 – 机遇与挑战

Next generation sequencing…

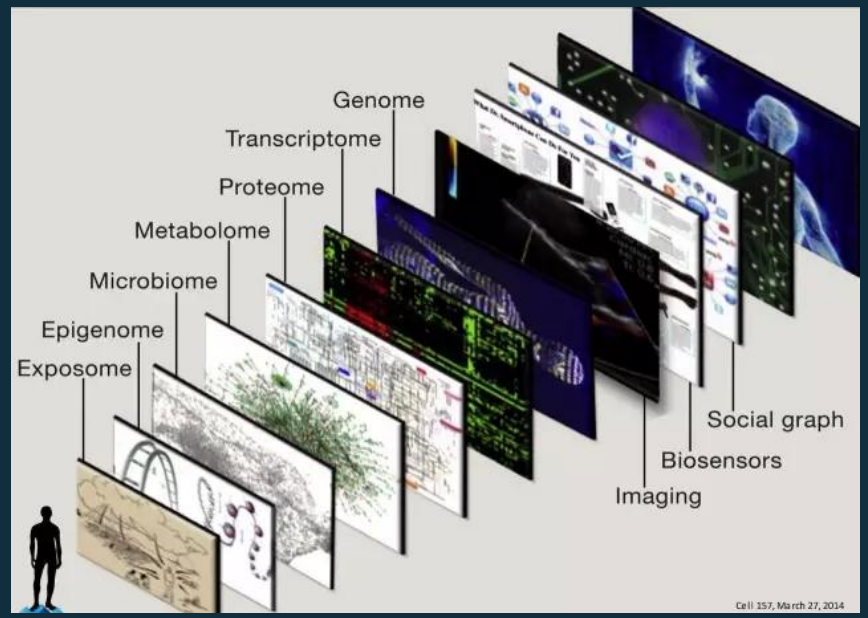
大数据的挑战 – 需要高通量的解决方案



1962

高通量测序

2003



2014

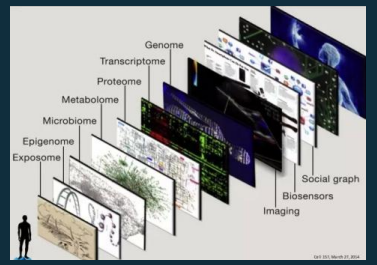
大数据的挑战 – 需要高通量的解决方案

1951

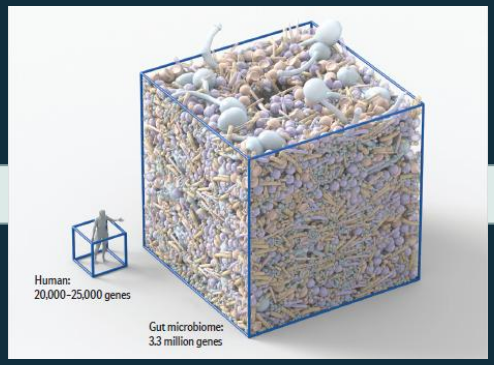
>100候选，>10000实验

1971

2015



2003

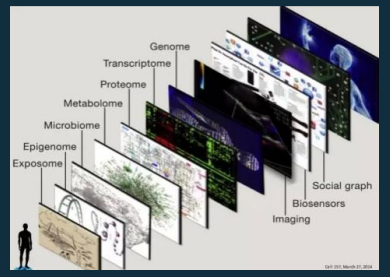


>>60亿研究对象，
>>EB(1000,000TB)数据量

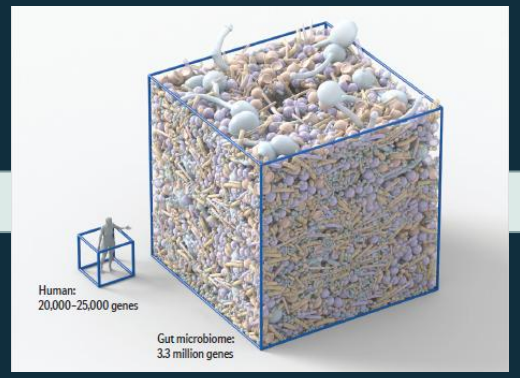
????

如果没有大数据分析技术…

大数据的挑战 – 需要高通量的解决方案



2003

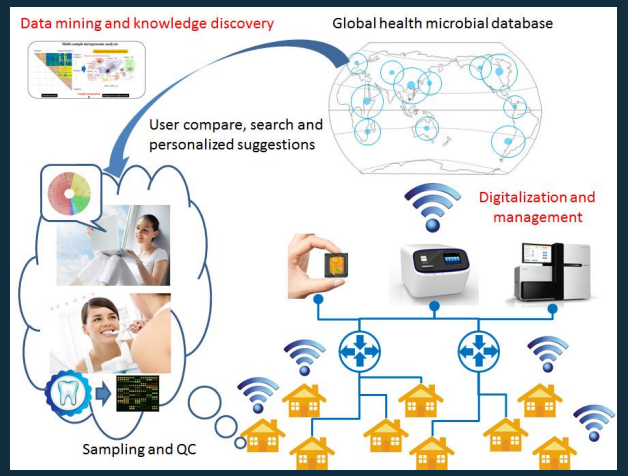


>60亿研究对象，>>EB数据量

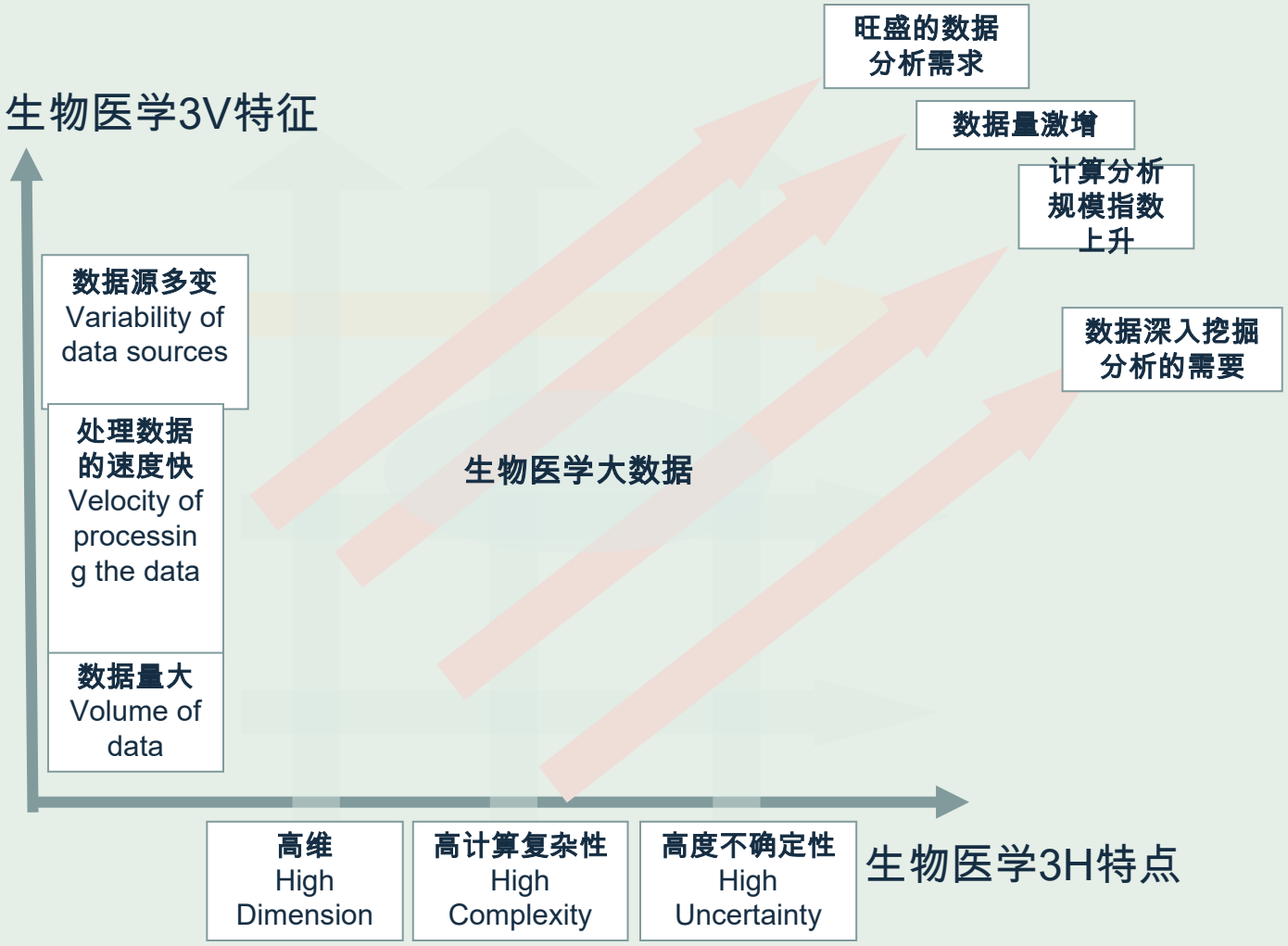
亟需高性能大数据分析技术...

!!!!

光有机器也枉然...



生物医学大数据3V特征



生物医学大数据涉及到广泛的领域

高通量测序

精度质谱仪 高

NCBI

大数据中心

UniProt

细胞生物学

合成生物学

分子生物学

海量生物数据

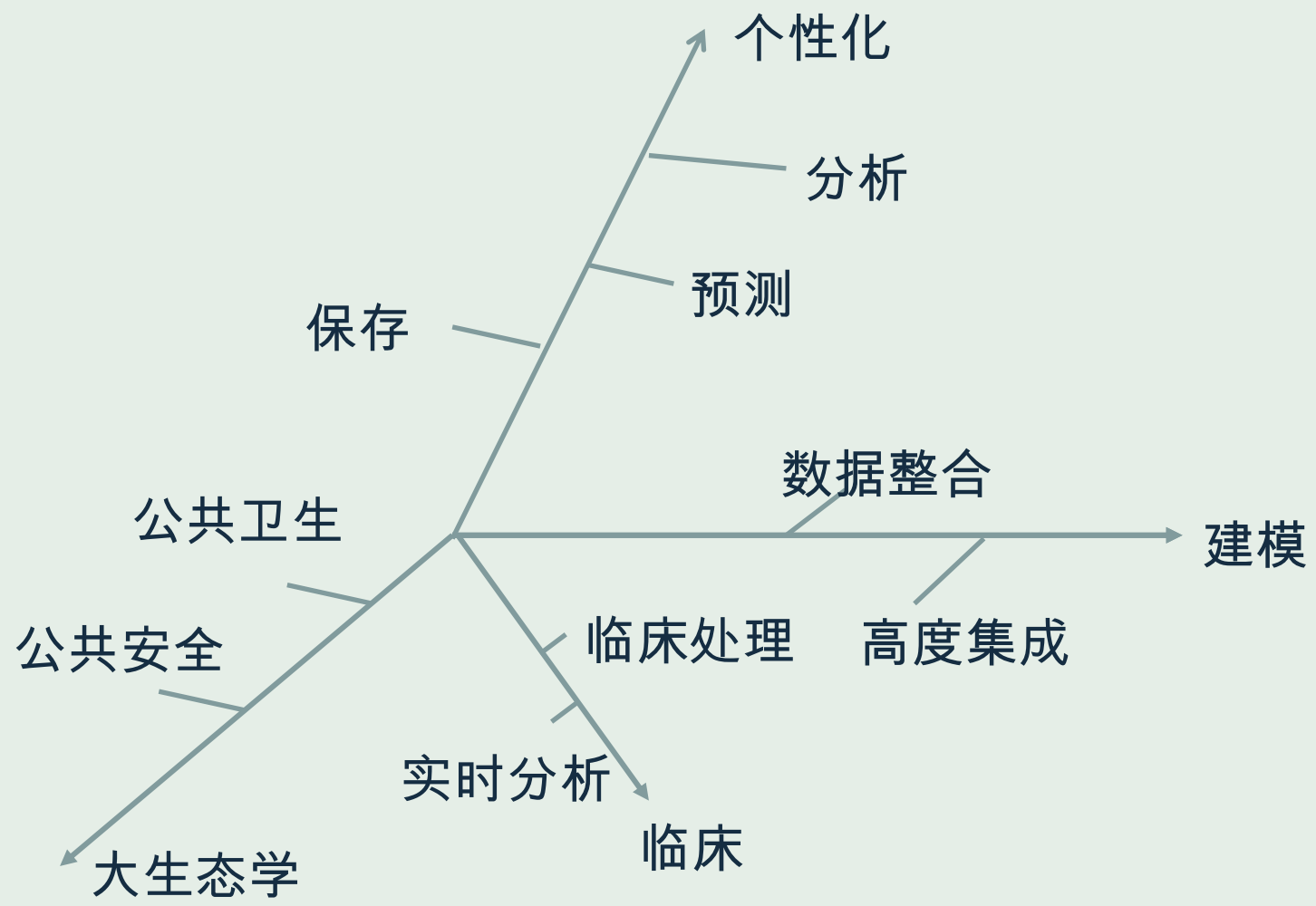
生命科学

生态学

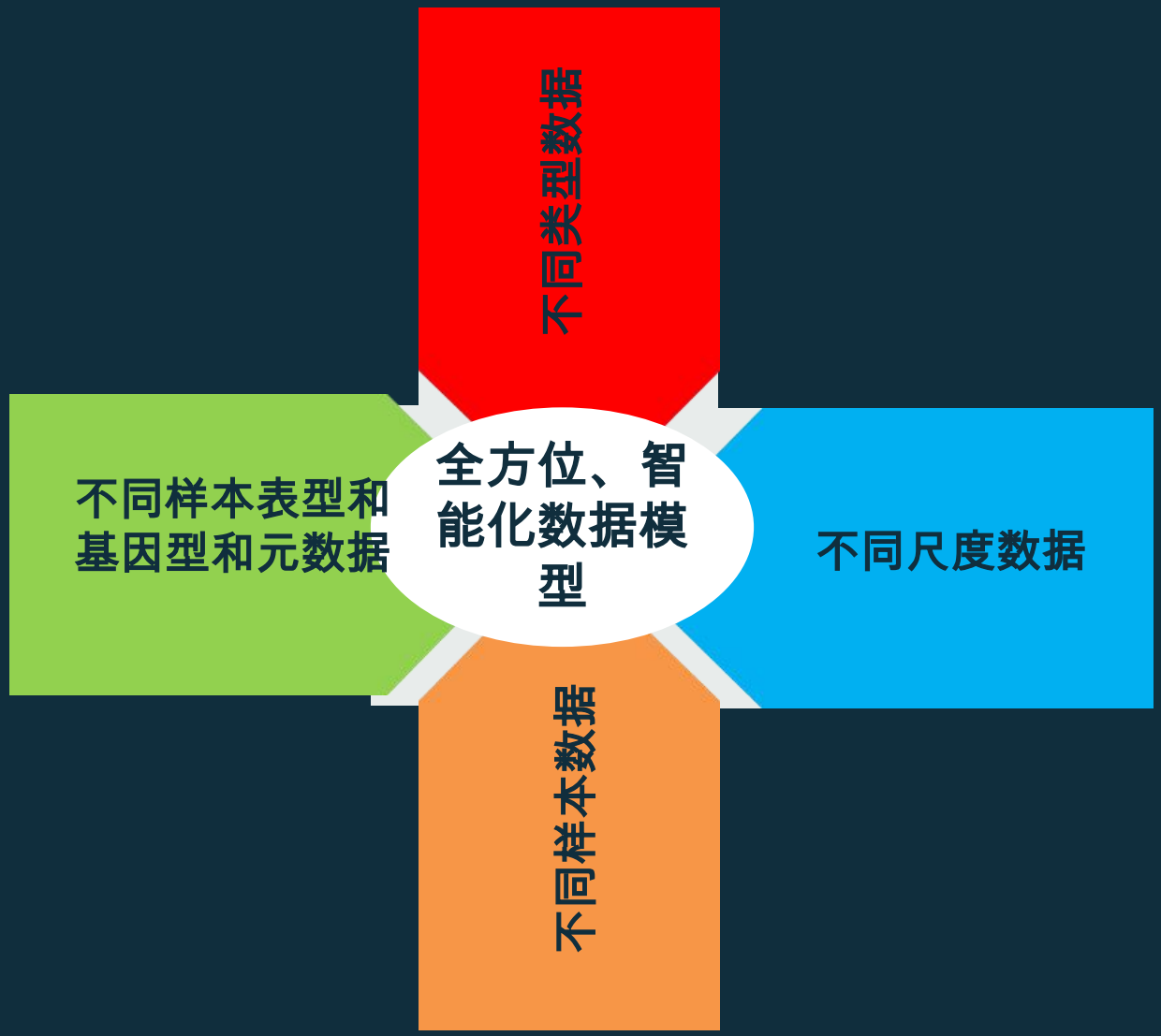
光学成像仪

…… ……

生物医学大数据涉及到完整研究链条

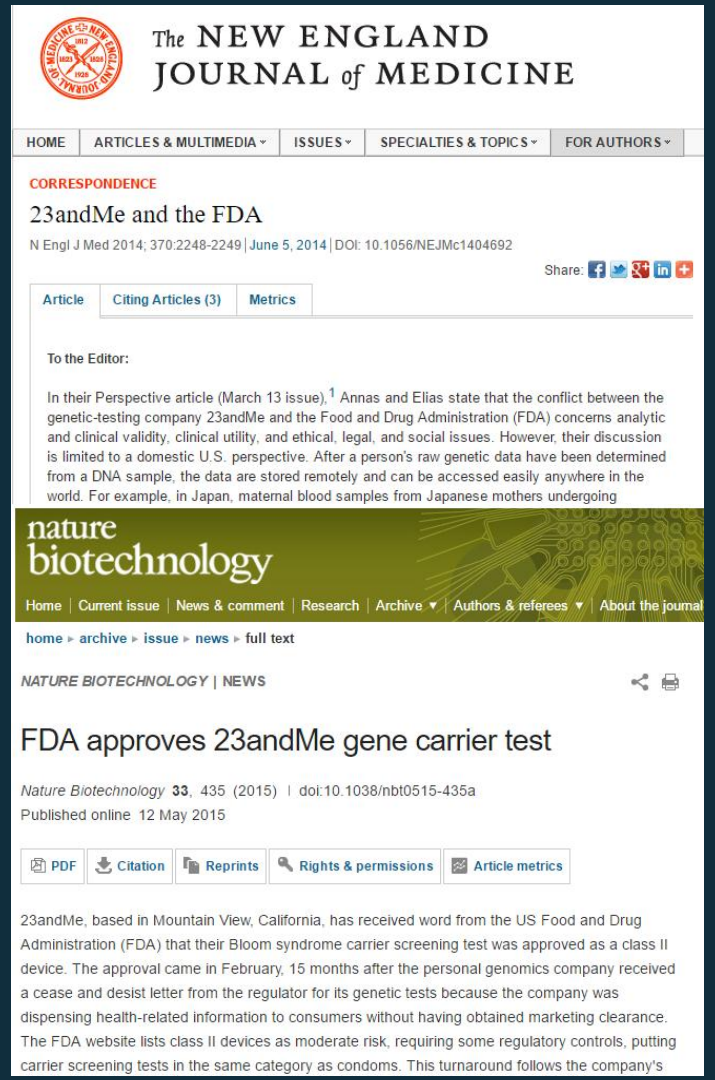


生物医学大数据依赖于大数据的分析方法



Alphabet（谷歌）

Google 的基因组学梦想



The NEW ENGLAND JOURNAL of MEDICINE

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CORRESPONDENCE

23andMe and the FDA

N Engl J Med 2014; 370:2248-2249 | June 5, 2014 | DOI: 10.1056/NEJMc1404692

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To the Editor:

In their Perspective article (March 13 issue),[1] Annas and Elias state that the conflict between the genetic-testing company 23andMe and the Food and Drug Administration (FDA) concerns analytic and clinical validity, clinical utility, and ethical, legal, and social issues. However, their discussion is limited to a domestic U.S. perspective. After a person's raw genetic data have been determined from a DNA sample, the data are stored remotely and can be accessed easily anywhere in the world. For example, in Japan, maternal blood samples from Japanese mothers undergoing

nature biotechnology

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NATURE BIOTECHNOLOGY | NEWS

FDA approves 23andMe gene carrier test

Nature Biotechnology **33**, 435 (2015) | doi:10.1038/nbt0515-435a

Published online 12 May 2015

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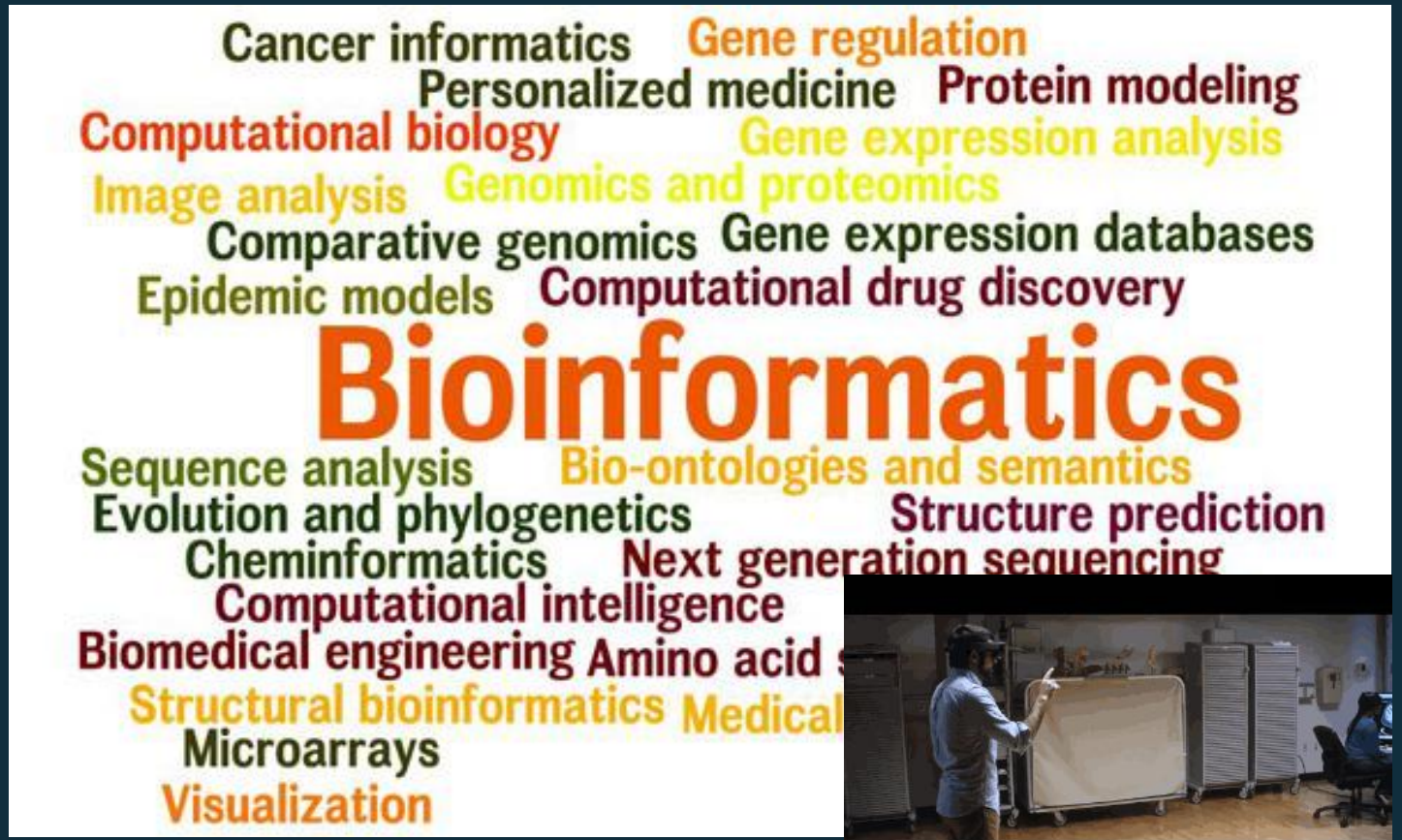
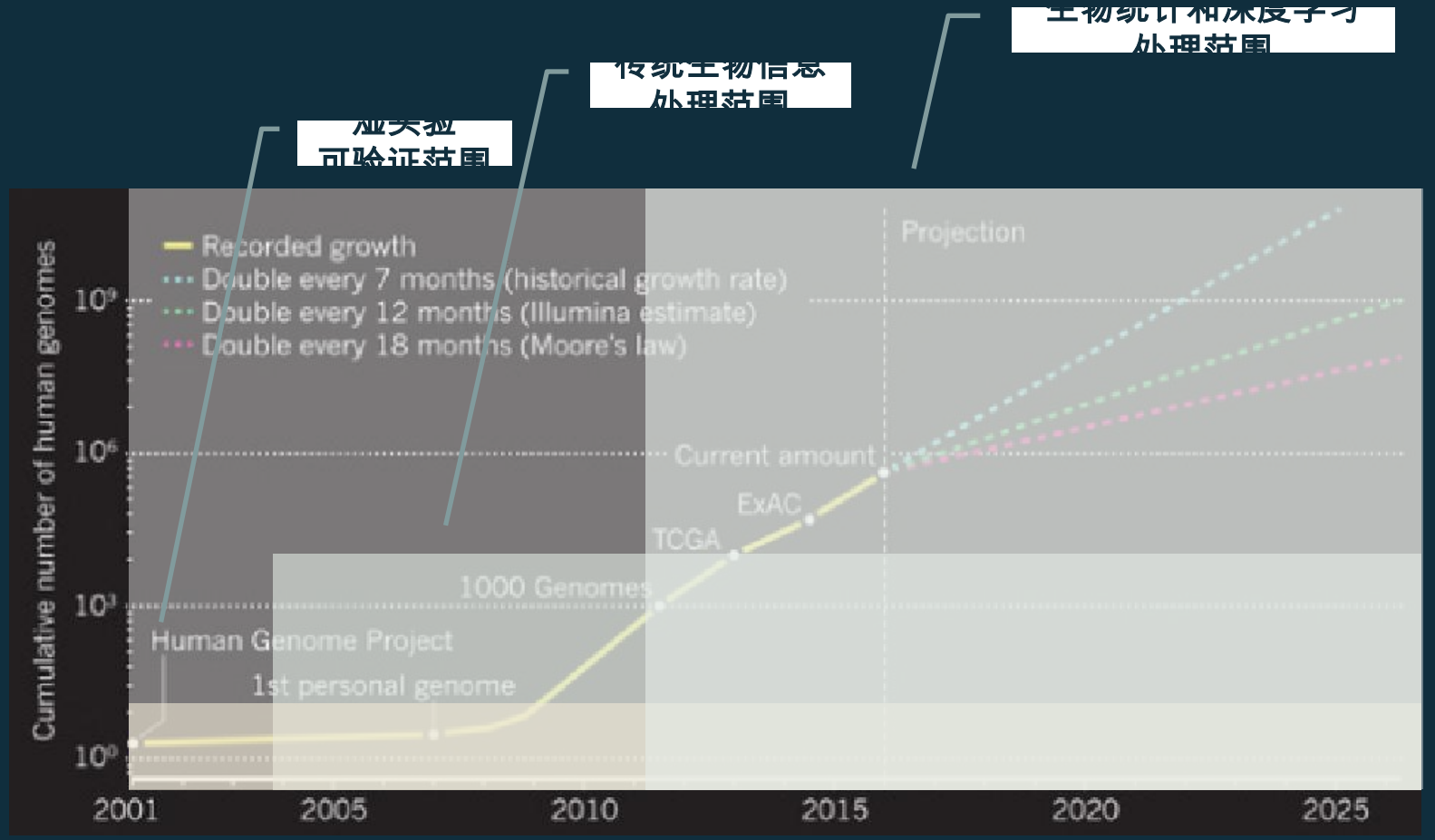
23andMe, based in Mountain View, California, has received word from the US Food and Drug Administration (FDA) that their Bloom syndrome carrier screening test was approved as a class II device. The approval came in February, 15 months after the personal genomics company received a cease and desist letter from the regulator for its genetic tests because the company was dispensing health-related information to consumers without having obtained marketing clearance. The FDA website lists class II devices as moderate risk, requiring some regulatory controls, putting carrier screening tests in the same category as condoms. This turnaround follows the company's

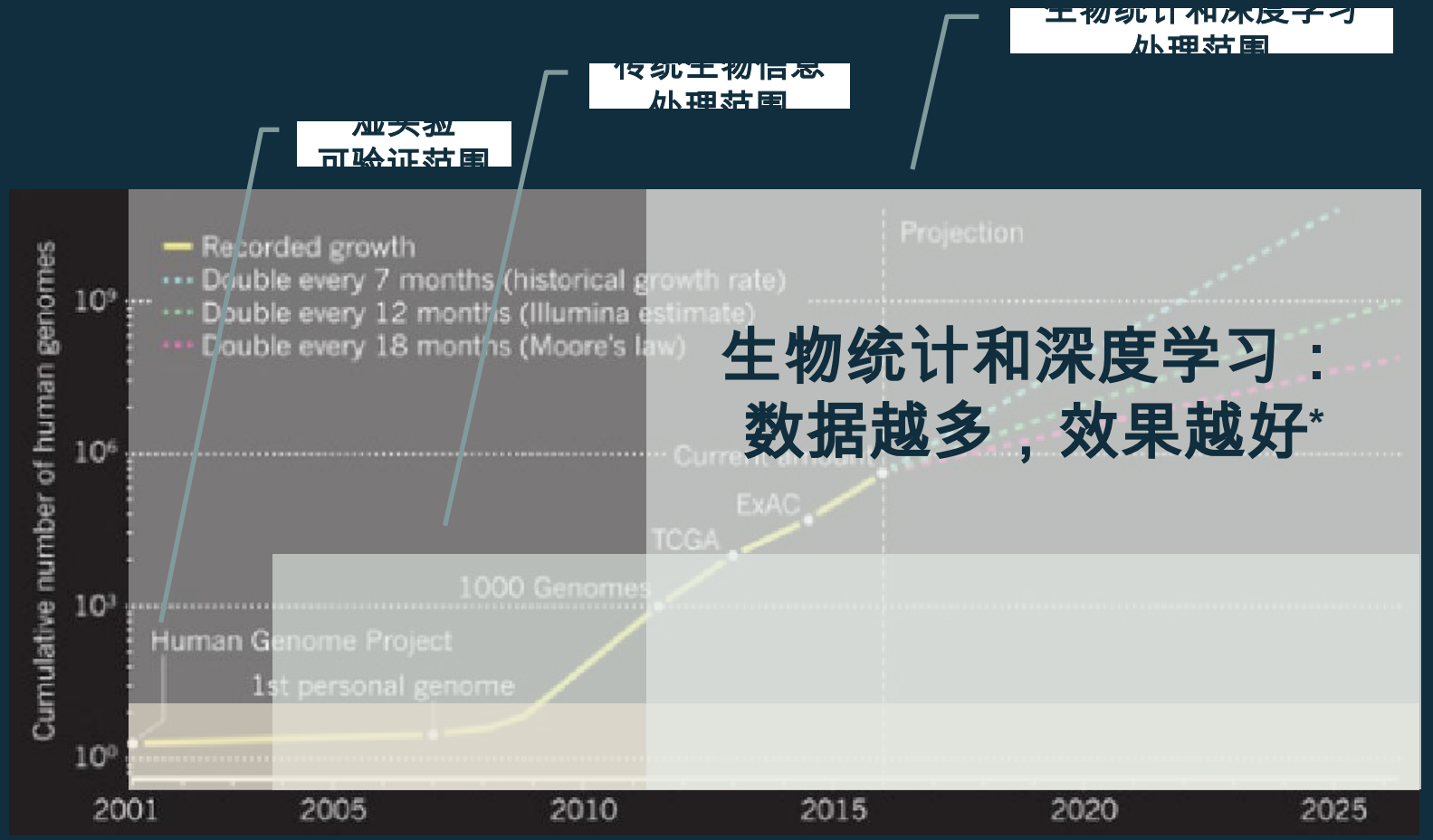
Future（未来）


Cancer informatics
Gene regulation
Personalized medicine
Protein modeling
Computational biology
Gene expression analysis
Image analysis
Genomics and proteomics
Comparative genomics
Gene expression databases
Epidemic models
Computational drug discovery
Bioinformatics
Sequence analysis
Bio-ontologies and semantics
Evolution and phylogenetics
Structure prediction
Cheminformatics
Next generation sequencing
Computational intelligence
Biomedical engineering
Amino acid
Structural bioinformatics
Medical
Microarrays
Visualization

Biostatistics



Biostatistics



生物信息学与生物统计学：

两种视角，四种技术，两类方法

两种视角

生物学视角
计算视角

四种技术

算法
数据
超算
深度学习

两类方法

经典软件
核心工具

生物信息学

Ontology and pathway (DAVID, GSEA)

Batch removal (COMBAT)

NGS for DNA-seq (FastQC, BWA, STAR)

RNA-seq (RSeQC, Salmon, DESeq2)

scRNA-seq (Seurat)

ChIP/DNase-seq (MACS, Cistrome)

scATAC-seq (Seurat / MAESTRO)

Genome-resequencing (Mutect)

CRISPR screens (MAGeCK)

Immunology (Polysolver, NetMHC, TIMER, TRUST, TIDE)

Do not be afraid to learn new algorithms

Bioinformatics Research

Level 0: modeling for modeling's sake

Published in low impact non-biological journals, read and cited only by non-biologists

Bioinformatics Research

Level 1: analyze unpublished data from a collaborator's project

Is it an important biological problem?

Are there existing algorithms to analyze the data?

Does the analyses and experiments inform each other?

Does the study have real biological findings?

Does the experimental group value the bioinformatics contribution?

Familiarity of bioinformatics algorithms and understanding of the biological problem are important

Can learn domain biological knowledge

and motivate methodology development

Avoid being a data janitor.

Bioinformatics Research

Level 2: algorithm development for mining biological data

Have a keen eye on cost-effective new techniques

Use sound statistics and develop user-friendly software

Continue to maintain and update algorithm with new data and new challenges

Stay in the biological domain, and provide complete solution to the domain's problem

User adoption is more important than publication journal IF

Time is the ultimate measure: Is the algorithm widely used and cited? Do people use it to make real findings?

Do well by doing good.

Bioinformatics Research

Level 3: make biological discoveries from public data integration and modeling

True big data mining!

Have unique biological angles, and not necessarily use sophisticated algorithms

Make real novel biological discoveries rather than “putting numbers on theories”

Is there experimental validation (by the informatics group or their experimental collaborators)?

Bioinformatics Research

Break

Level X: integrative papers from big science consortia

Sometimes have numbers on the paper title

Exhausting and time-consuming, political struggles of the consortium members

Published in high profile journals and highly cited, but contributions often dubious

Trainees might get some good papers / faculty positions, but may not do well later

Informatics group should caution not to sacrifice

Levels I, II, III for Level X-only research

生物信息学研究的三个层面

初级层面
中级层面
高级层面

生物信息学研究方法

初级层面

基于现有的生物信息数据库和资源，利用成熟的生物信息学工具（专业网站、软件）解决生物信息学问题

——生物信息数据库（NCBI、EBI等）

——基因组序列分析、序列比对软件（GCG、BLAST、CLUSTAL等）

——系统发育树构造软件（PHYLIP、PALM、MEGA等）

——分子动力学模拟软件（GROMACS、NAMD等）

——搜集、整理有特色的生物信息学数据集

生物信息学研究方法

中级层面

利用数值计算方法、数理统计方法和相关的工具，研究生物信息学问题

——概率、数理统计基础

——科学计算基础

——现有的数理统计和科学计算工具（EXCEL、SPSS、SAS、 MATLAB等）

——建立有特色的生物信息学数据库

生物信息学研究方法

高级层面

提出有重要意义的生物信息学问题；自主创新，发展新型方法，开发新型工具，引领生物信息学领域研究方向。

——面向生物学领域，解决生物学问题

——数学、物理、化学、计算科学等思想和方法

——建立模型，发展算法

——自行编程，开发软件，建立网页（Linux系统、C/C++、PERL、数据库技术）

生物信息学研究方法

从事生物信息学研究应具备多方面的科学基础：

（1）、一定的计算能力，包括相应的软、硬设备。要有各种数据库或者能与国际、国内的数据库系统进行有效的交流。要有发达、稳定的互联网络系统；

（2）、强有力的创新算法和软件。没有算法创新，生物信息学就无法获得持续的发展；

（3）、与实验科学，特别是与自动化的大规模高通量的生物学研究方法与平台技术建立广泛、紧密的联系。这些技术，既是产生生物信息数据的主要方法，又是验证生物信息学研究结果的关键手段。

从事生物信息学研究的人员必须具备多学科交叉的知识。

生物信息学的“降龙十八掌”

第一式 见龙在田

（1）

要掌握生物信息数据库及其查询搜索方法

(Database & searching)

——对分子生物信息数据库的种类以及某些具体数据库的掌握和了解

——从现有数据库中熟练获得需要的数据信息（尤其是二级数据库）

——能熟练地进行数据库查询和数据库搜索（数据库查询系统Entrez、SRS；搜索工具BLAST等）

——数据库技术、互联网技术

第二式 飞龙在天

（2）

要学会生物信息学软件和工具的应用

(Software & application)

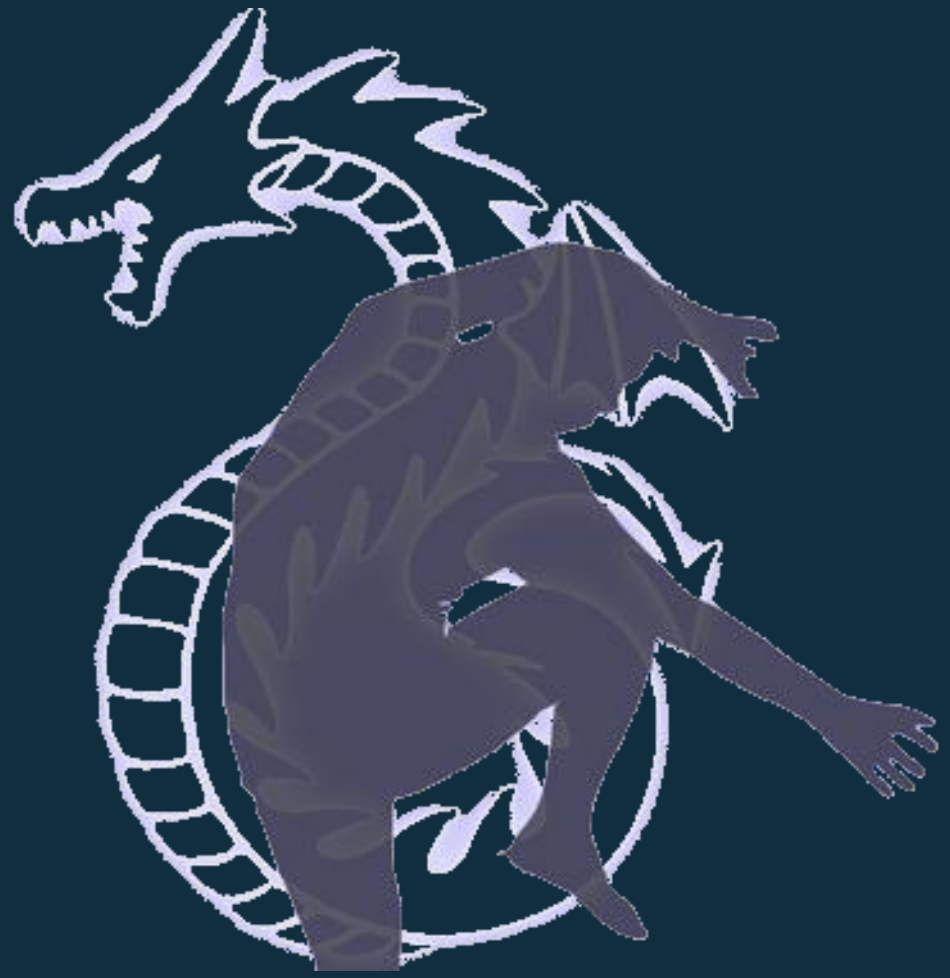
利用成熟的生物信息学工具（专业网站、软件）解决生物信息学问题

——基因组序列分析、序列比对软件（GCG、BLAST、CLUSTAL等）

——系统发育树构造软件（PHYLIP、PALM等……）

——基因芯片检测分析软件（商业软件ScanArray、Array-Pro等 ……）

——分子动力学模拟软件（GROMACS、NAMD等……）



第三式 鸿渐于陆

（3）

概率论基础

(Probability theory)

——随机事件、概率

——随机变量、概率分布

——大数定律、中心极限定理

——几乎用于生物信息学的各个方面

“Most of the problems in computational sequence analysis are essentially statistical.”

——“Biological sequence analysis”

第四式 或跃在渊

（4）

数理统计基础

(Statistical methods)

——样本和统计量（方差、均值……）

——参数估计、假设检验

——基本的统计分析（方差分析、协方差分析、回归分析）

——常用统计软件的运用（SPSS、SAS）

——几乎用于生物信息学的各个方面

第五式 羚羊触藩

（5）

基于频率的组分分析方法和权重矩阵方法

(Composition analysis & weight matrix method)

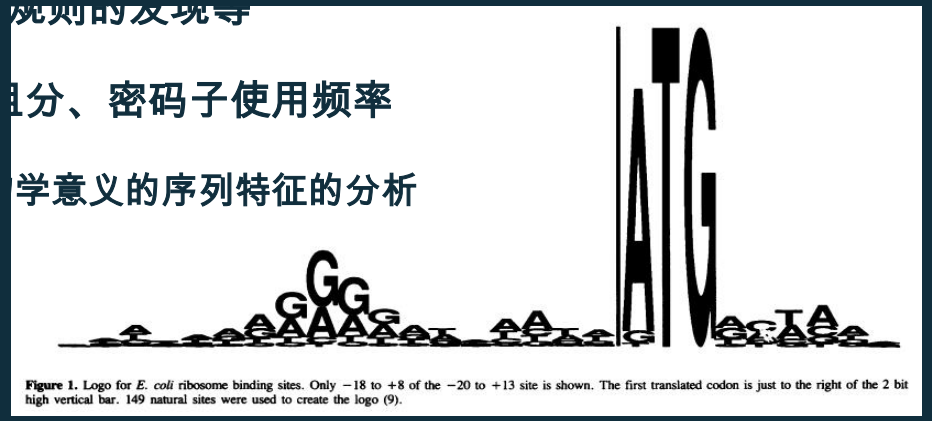


Figure 1. Logo for *E. coli* ribosome binding sites. Only −18 to +8 of the −20 to +13 site is shown. The first translated codon is just to the right of the 2 bit high vertical bar. 149 natural sites were used to create the logo (9).

权重矩阵分析方法举例

——针对序列信号（一段核酸、蛋白），计算每一位点所使用的词汇或叫符号（*碱基、氨基酸*）频率，频率的偏好性反映信号的序列特征（sequence pattern）。

例：人类基因内含子/外显子剪接位点的序列特征分析

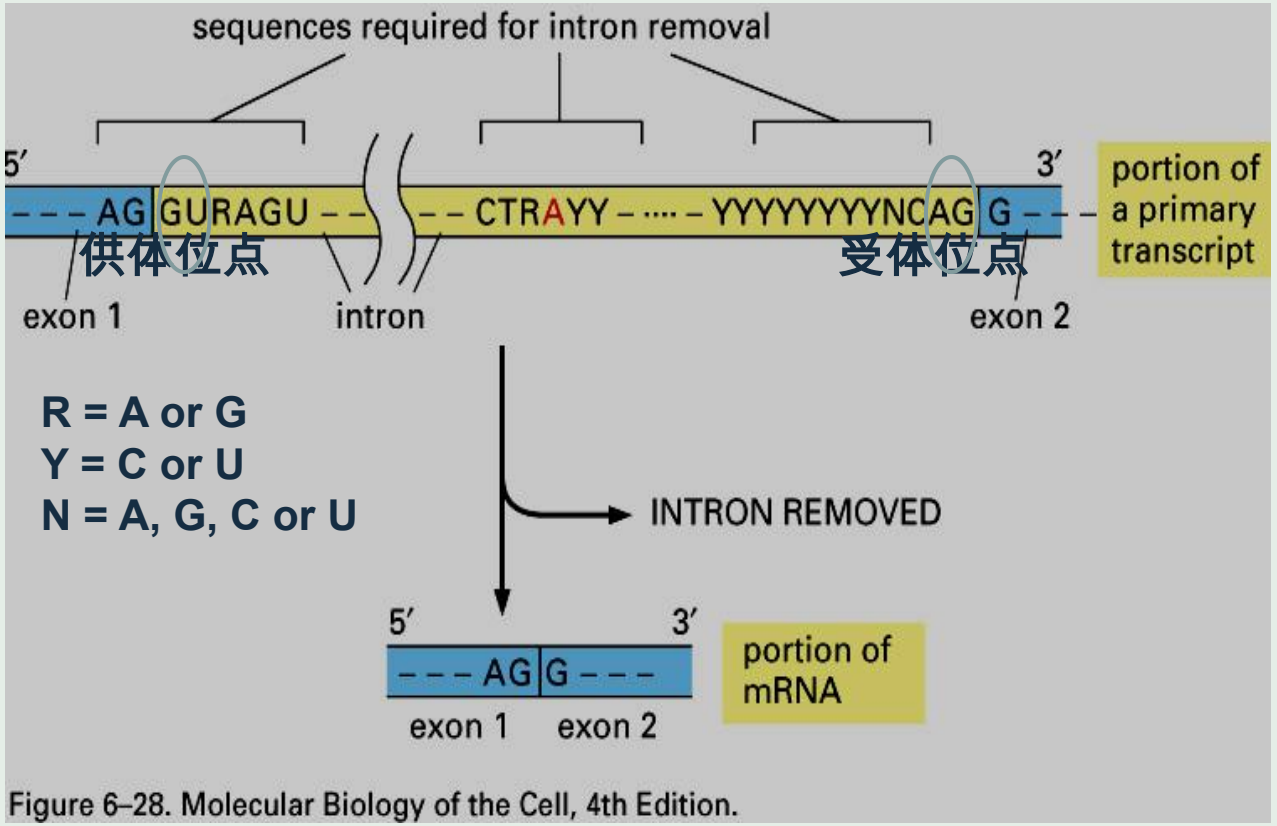


Figure 6–28. Molecular Biology of the Cell, 4th Edition.

Bayesian打分函数用于剪接位点预测的公式

The likelihood that a property value *v* (of a new structure) is drawn from the splicing site is:

$$P(site \mid \upsilon) = \frac{P(\upsilon \mid site)P(site)}{P(\upsilon \mid site)P(site) + P(\upsilon \mid nonsite)P(nonsite)}$$

Score for the overall likelihood of the query sequence being a site is:

$$\sum_{\substack{\text{properties at} \\ \text{associated volumes}}} \log\left(\frac{P(site \mid \upsilon)}{P(site)}\right)$$

Say we have a sequence S= $S_1S_2\ldots S_n$. Then one need to calculate

$$\frac{P(S|\text{splice site})}{P(S|\text{background})}$$

So to look for a donor site in the sequence, we might calculate

第六式 潜龙勿用

（6）

信息论方法

(Information method)

——信息的度量：是信息符号出现何种状态的一种不确定性程度，信息的获得要对不确定性进行否定。
——生物信息的符号如ACGT四种符号，状态空间即其所有可能的排列
——用于结构预测

——信息熵

$$H = -\sum_i p_i \log p_i \qquad H = -\sum_i p_i \log p_i$$

——信息熵*H*刻画了由{p_i}表示的随机试验结果的先验不确定性，或观察到输出时所获得的信息量。

第七式 利涉大川

（7）

期望最大化（EM）方法

(Expectation Maximization)

——EM算法是一种在不完全数据情况下计算极大似然估计或者后验分布的迭代算法。

——适用于具有隐变量的模型和问题，

——用于结构的识别，如Motif识别的MEME方法、HMM中的Baum-Welch算法

第八式 神龙摆尾

（8）

动态规划方法

(Dynamic Programming)

——一种常用的多阶段决策的寻优算法
——动态规划用得最多的方面是DNA序列或者蛋白质序列比对

第九式 密云不雨

（9）

迭代方法

(Iteration)

——迭代的目的通常是在状态空间找到目标函数收敛的稳定解

——在运用模式识别方法时，对系统参数的学习通常要经过迭代来实现

——迭代必须能够不断逼近稳定解

——用于上述某些方法的方法

第十式 突如其来

（10）

回归、拟合、相关性分析、关联分析

(Regression, fitting, correlation & association)

——经典的统计分析方法

——主要目的：描述和预测自变量与因变量间的关系

——用于上述某些方法的方法

第十一式 双龙取水

（11）

判别分析方法

(Discriminant analysis)

——用于判别样品所属类型的统计分析方法

条件：已知研究对象总体的类别数目及其特征（如：分布规律，或各类的训练样本）

目的：判断未知类别的样本的归属类别

——用于基因识别、医学诊断、人类考古学

第十二式 鱼跃于渊

（12）

聚类分析方法

(Clustering method)

——聚类分析（群分析）是实用多元统计分析的一个新分支，正处于发展阶段。理论上尚未完善，但应用十分广泛。实质上是一种分类问题，目的是建立一种分类方法，将一批数据按照特征的亲疏、相似程度进行分类。

——条件：研究对象总体的类别数目未知，也不知总体样本的具体分类情况

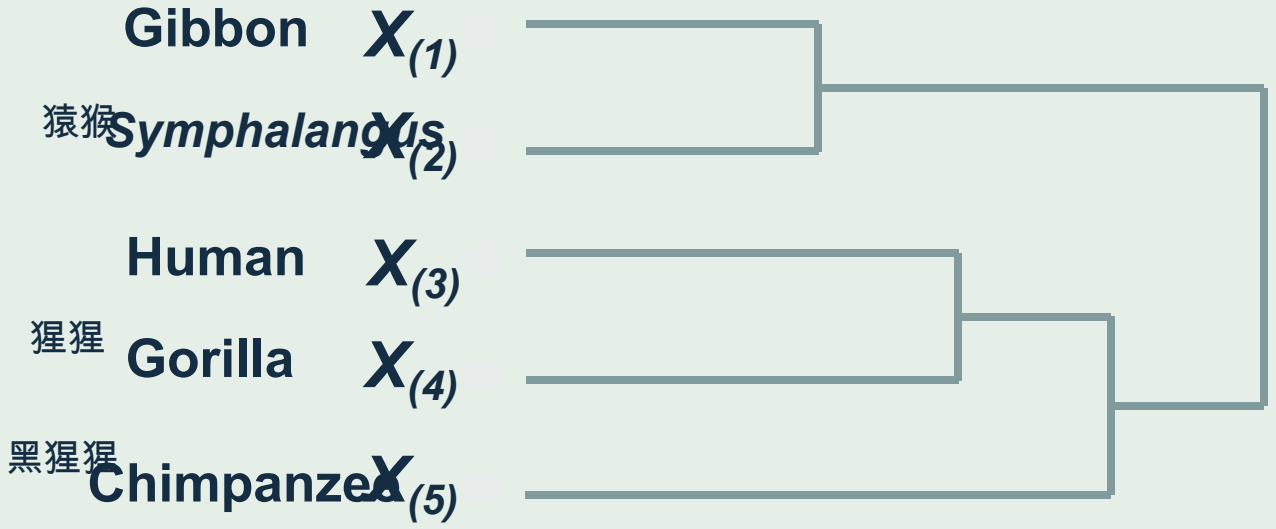
——目的：通过分析，选定描述个体相似程度的统计量、确定总体分类数目、建立分类方法；对研究对象给出合理的分类。(“物以类聚”是聚类分析的基本出发点)

生物信息学研究方法

——定性、经验的分类的局限
分类较粗、数据量小、凭借经验

——谱系聚类法（系统聚类法）、动态聚类法、模糊聚类法

——生物信息学中的聚类分析问题：
根据DNA芯片获得的基因表达数据进行基因聚类（数据量庞大）
蛋白质相互作用网络的分类
根据不同物种的大分子序列进行相似性比较并构建系统发育树



第十三式 震惊百里

（13）

Markov模型的应用

(Markov model)

——Markov过程：从一种状态转移到另一种状态时，过程仅取决于前面n种状态，是一种有序n模型。 n是影响下一个状态选择的状态数。

——最简单的Markov过程是一阶过程，状态的选择完全取决于前一状态，这种选择是依照概率来选择的。

——状态的选择是概率的，而非确定的。故Markov过程本质上是一种随机过程。

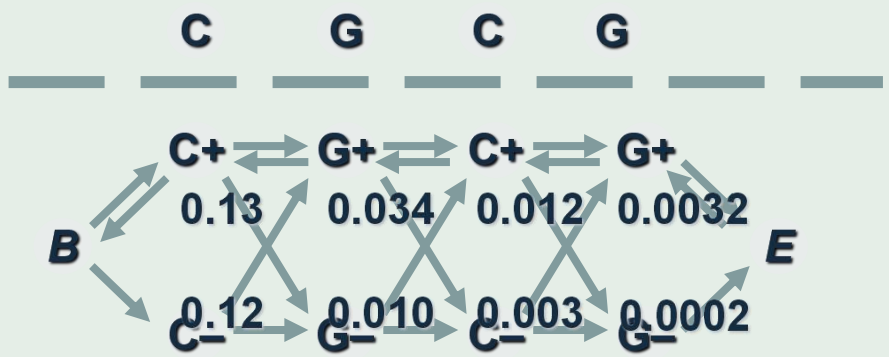
第十四式 损则有孚

（14）

隐Markov模型方法

(HMM method)

——将核苷酸序列看成一个随机序列，DNA序列的编码部分与非编码部分在核苷酸的选用频率上对应着不同的Markov模型。由于这些Markov模型的统计规律是未知的，而HMM能够自动寻找出它们隐藏的统计规律。对于高等生物这样复杂的DNA序列，HMM必须学习不同的基因结构的信号。



语音识别(Speech recognition)

隐Markov模型
(HMM)

光字符识别
(Optical character recognition)

生物序列分析
(Biological sequence analysis)

生物特征识别
(Biometrics)

（1）序列比较与搜寻（尤其是多序列比对）
（2）基因及信号的识别、预测（包括DNA编码与非编码区的识别、真核基因剪接位点信号识别、非编码区的转录调控信号识别、信号肽识别……）
（3）蛋白质二级结构、家族、超家族预测、分类等……

第十五式 时乘六龙

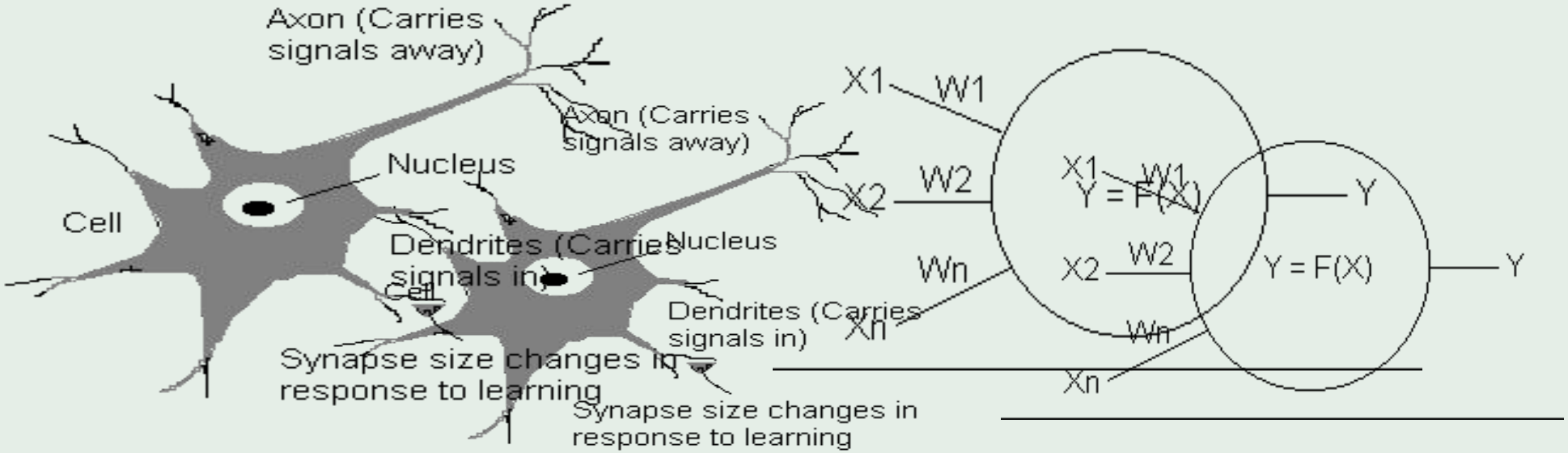
（15）

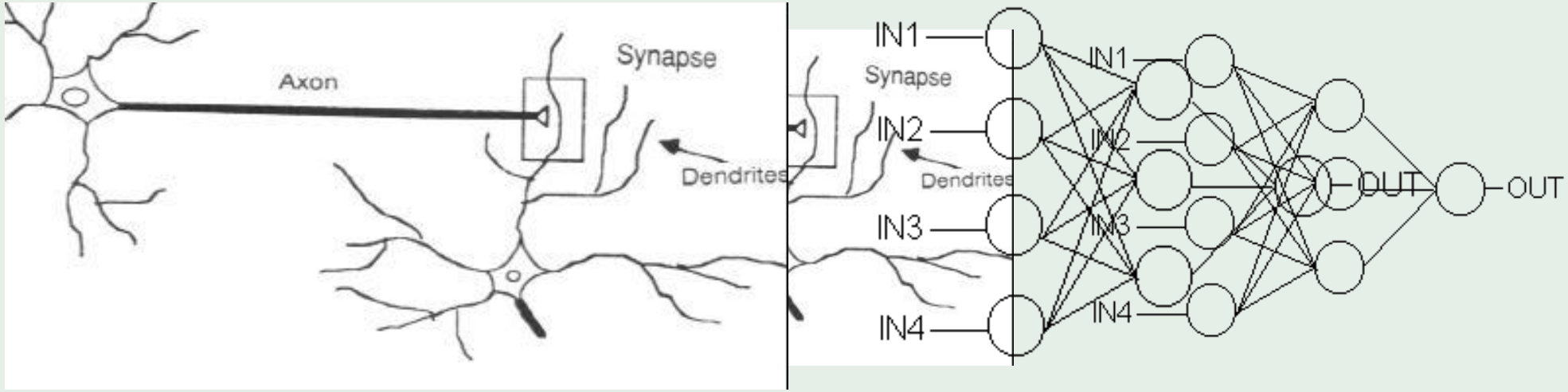
感知器与人工神经网络方法

(Perceptron & ANN method)

——计算机人工神经网络是对大脑神经网络的模拟，在生物信息学研究中，无论是基因识别还是蛋白质结构预测，神经网络都取得了比其它方法更为准确的结果。

生物信息学研究方法


Axon (Carries signals away)
Axon (Carries signals away)
Nucleus
Cell
Dendrites (Carries signals in)
Nucleus
Cell
Dendrites (Carries signals in)
Synapse size changes in response to learning
Synapse size changes in response to learning
X1
W1
X2
W2
Wn
Xn
Y = F(X)
Y
X1
W1
X2
W2
Wn
Xn
Y = F(X)
Y


Axon
Synapse
Dendrites
Synapse
Dendrites
IN1
IN2
IN3
IN4
IN1
IN2
IN3
IN4
OUT
OUT

第十六式 龙战于野

（16）

决策树、支持向量机及其它模式识别方法

(Decision tree & SVM method)

——模式识别是在输入样本中寻找特征并识别对象的一种方法。

——模式识别主要有两种方法，一种是根据统计特征进行识别，另一种是根据对象的结构特征进行识别，而后者常用的方法为句法识别。

——在基因识别中，对于DNA序列上的功能位点和特征信号的识别都需要用到模式识别。

第十七式 履霜冰至

（17）

微分方程的数值方法

(Numerical methods)

——分子动力学模拟：研究生物大分子的构象，主要还是用基于半经验势函数的分子动力学方法，而量子力学则在确定势函数的参数和研究局部性质时起作用。对蛋白质进行动力学研究是利用计算机进行模拟实验的基础。

——分子动力学得到一组动力学微分方程，要求得到初值问题的解。

——微分方程的数值求解：有限差分法、有限元法

十七式合一
亢龙有悔

（18）

最终要诀：各类方法综合运用

All in one!

——综合运用不同的研究方法

——始终面向生物学问题

——知识和技能的学习方法

——文献的查阅和阅读方法

——中、英文论文的写作方法

生物信息学的“东邪西毒南帝北丐中顽童”

——东邪：de Bruijn图算法（基因组拼接）

——西毒：近似算法和似然估计（进化树分析）

——南帝：核函数（基因和物种分类问题）

——北丐：统计建模（序列分析）

——中顽童：深度学习和生成对抗神经网络GAN（判断问题）

Slides credits

——生物信息学研究方法概述：北京大学生物信息中心

——生物统计学：卜东波@中国科学院计算技术研究所，邓明华@北京大学

——神经网络与深度学习: 邱锡鹏@复旦大学

——Introduction to Computational Biology and Bioinformatics: Xiaole Shirley Liu Lab@Havard University

——Combinatorial Methods in Computation Biology: Ken Sung Lab@NUS

——Deep Learning in the Life Sciences: MIT

——Probabilistic Graphical Models: Eric Xing@CMU

——Numerous other leading researchers and leading labs.....

生物信息学与生物统计学：

两种视角，四种技术，两类方法

两种视角

生物学视角
计算视角

四种技术

算法
数据
超算
深度学习

两类方法

经典软件
核心工具

LECTURE 01

1. 算法

Donald Knuth（高德纳）

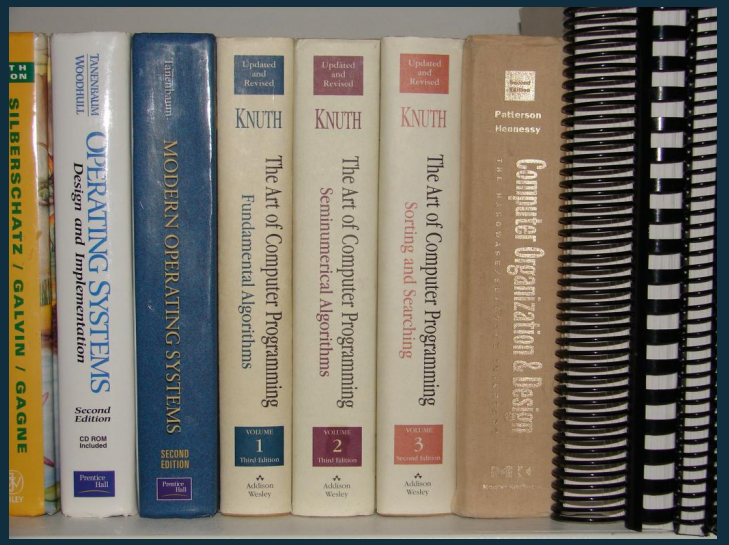
Donald Knuth, the "father of the analysis of algorithms.”

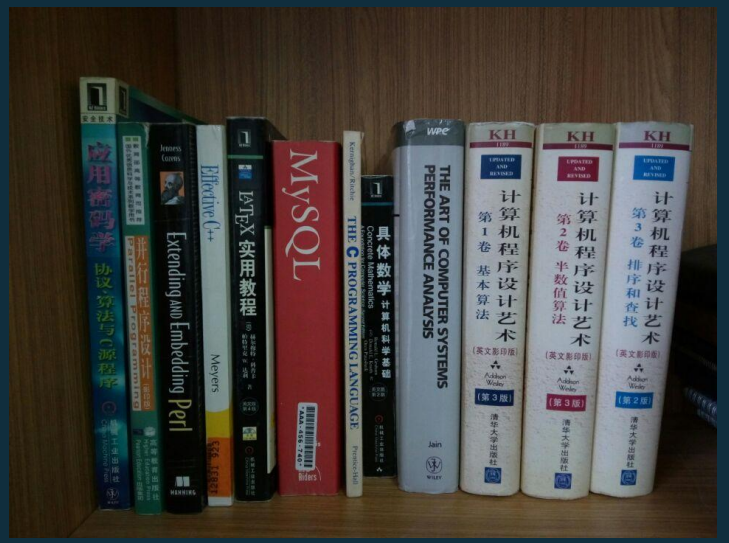
The Art of Computer Programming（计算机程序设计艺术）

Markup	Renders as
`The quadratic formula is $-b \pm \sqrt{b^2 - 4ac} \over 2a$ \bye`	The quadratic formula is $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

“生物信息学为算法研究提供了500年的问题”– Don Knuth

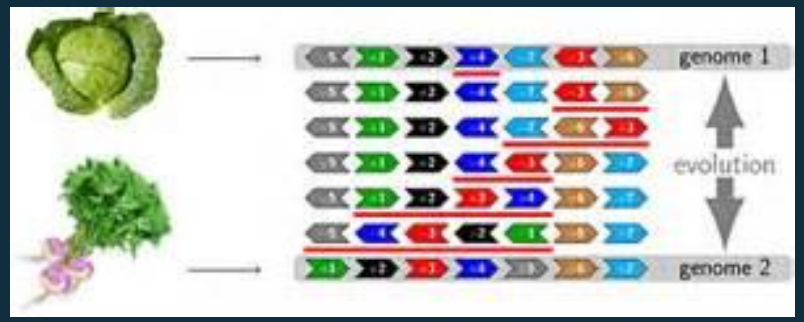
“definitely send me a résumé if you finish this fiendishly difficult book” – Bill Gates





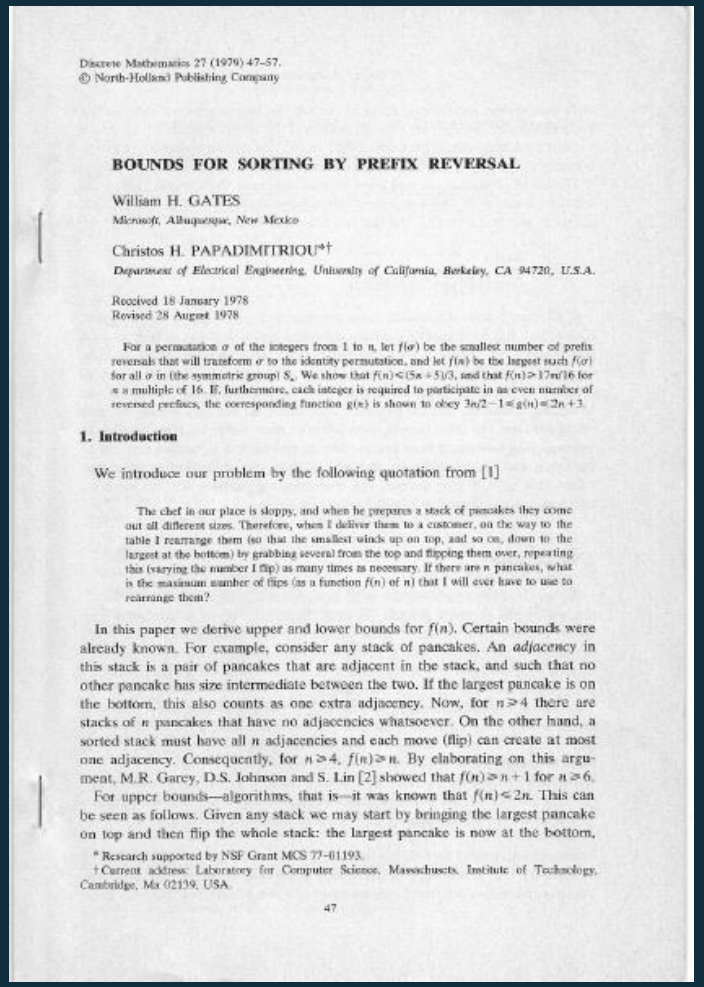
“definitely come to talk about algorithm if you read half of this book” – Kang Ning

Bill Gates（比尔盖茨）



比尔盖茨:下个世界首富出自基因检测领域

Sorting by reversal problem



Discrete Mathematics 27 (1979) 47–57.
© North-Holland Publishing Company

BOUNDS FOR SORTING BY PREFIX REVERSAL

William H. GATES
Microsoft, Albuquerque, New Mexico

Christos H. PAPADIMITRIOU*†
Department of Electrical Engineering, University of California, Berkeley, CA 94720, U.S.A.

Received 18 January 1978
Revised 28 August 1978

For a permutation σ of the integers from 1 to n, let $f(\sigma)$ be the smallest number of prefix reversals that will transform σ to the identity permutation, and let $f(n)$ be the largest such $f(\sigma)$ for all σ in (the symmetric group) S_n. We show that $f(n) \leq (5n+5)/3$, and that $f(n) \geq 17n/16$ for n a multiple of 16. If, furthermore, each integer is required to participate in an even number of reversed prefixes, the corresponding function $g(n)$ is shown to obey $3n/2 - 1 \leq g(n) \leq 2n+3$.

1. Introduction

We introduce our problem by the following quotation from [1]

> The chef in our place is sloppy, and when he prepares a stack of pancakes they come out all different sizes. Therefore, when I deliver them to a customer, on the way to the table I rearrange them (so that the smallest winds up on top, and so on, down to the largest at the bottom) by grabbing several from the top and flipping them over, repeating this (varying the number I flip) as many times as necessary. If there are n pancakes, what is the maximum number of flips (as a function $f(n)$ of n) that I will ever have to use to rearrange them?

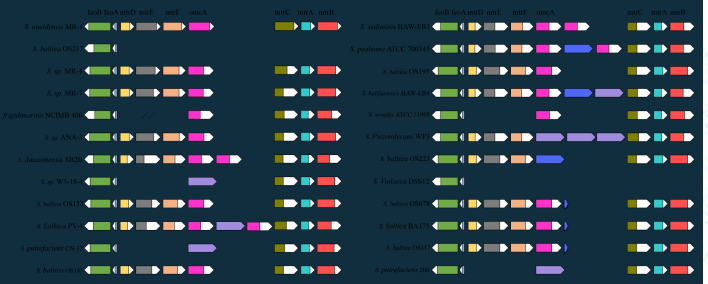
In this paper we derive upper and lower bounds for $f(n)$. Certain bounds were already known. For example, consider any stack of pancakes. An *adjacency* in this stack is a pair of pancakes that are adjacent in the stack, and such that no other pancake has size intermediate between the two. If the largest pancake is on the bottom, this also counts as one extra adjacency. Now, for $n \geq 4$ there are stacks of n pancakes that have no adjacencies whatsoever. On the other hand, a sorted stack must have all n adjacencies and each move (flip) can create at most one adjacency. Consequently, for $n \geq 4$, $f(n) \geq n$. By elaborating on this argument, M.R. Garey, D.S. Johnson and S. Lin [2] showed that $f(n) \geq n+1$ for $n \geq 6$.

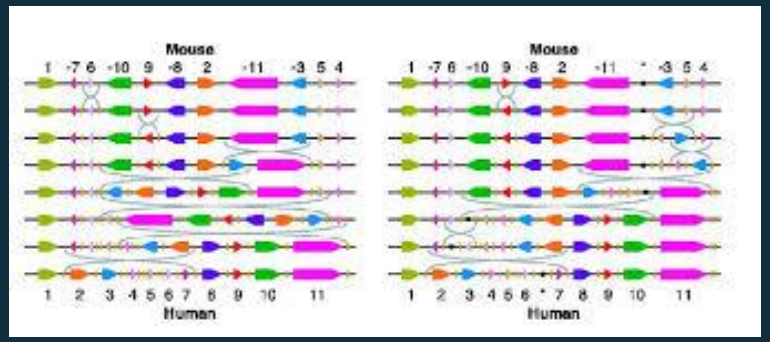
For upper bounds—algorithms, that is—it was known that $f(n) \leq 2n$. This can be seen as follows. Given any stack we may start by bringing the largest pancake on top and then flip the whole stack: the largest pancake is now at the bottom,

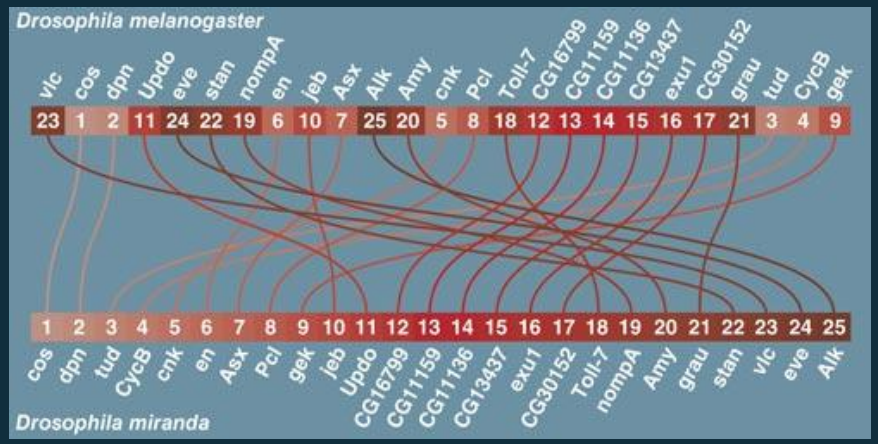
* Research supported by NSF Grant MCS 77-01193.
†Current address: Laboratory for Computer Science, Massachusetts Institute of Technology, Cambridge, Ma 02139, USA.

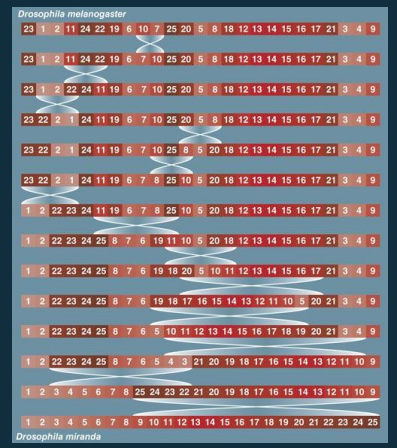
实例

How many reversal steps for this REAL case?

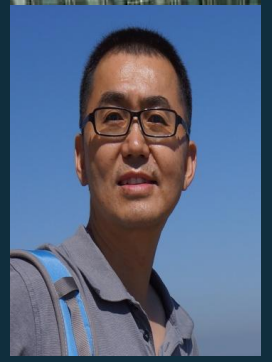








Ming Li（李明）& Tao Jiang（姜涛）



SIAM J. COMPUT.
Vol. 24, No. 5, pp. 1122–1139, October 1995

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012

ON THE APPROXIMATION OF SHORTEST COMMON SUPERSEQUENCES AND LONGEST COMMON SUBSEQUENCES*

TAO JIANG† AND MING LI‡

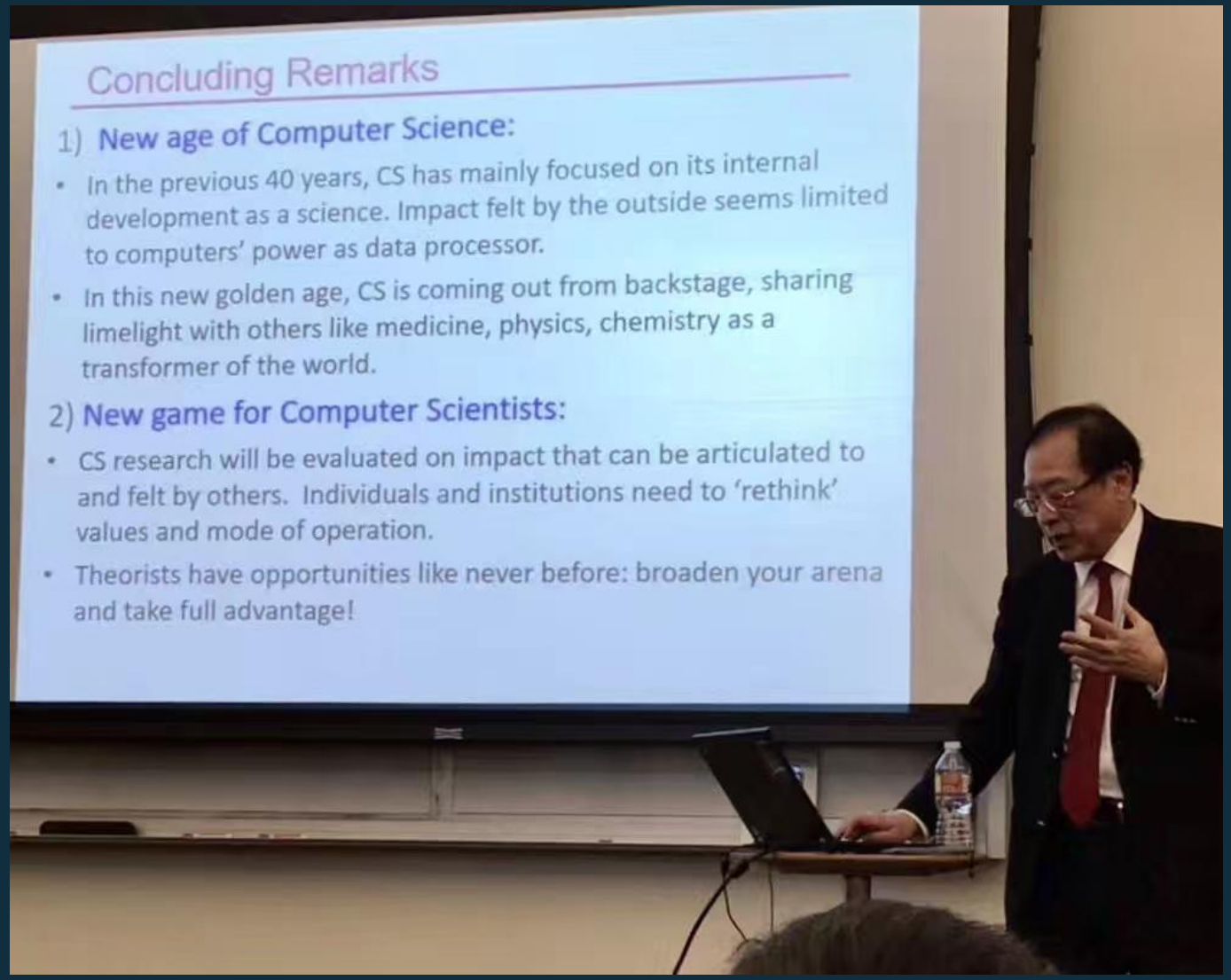
Abstract. The problems of finding shortest common supersequences (SCS) and longest common subsequences (LCS) are two well-known **NP**-hard problems that have applications in many areas, including computational molecular biology, data compression, robot motion planning, and scheduling, text editing, etc. A lot of fruitless effort has been spent in searching for good approximation algorithms for these problems. In this paper, we show that these problems are inherently hard to approximate in the worst case. In particular, we prove that (i) SCS does not have a polynomial-time linear approximation algorithm unless $\mathbf{P} = \mathbf{NP}$; (ii) There exists a constant $\delta > 0$ such that, if SCS has a polynomial-time approximation algorithm with ratio $\log^{\delta} n$, where n is the number of input sequences, then **NP** is contained in $\mathbf{DTIME}(2^{\text{polylog } n})$; (iii) There exists a constant $\delta > 0$ such that, if LCS has a polynomial-time approximation algorithm with performance ratio n^{δ}, then $\mathbf{P} = \mathbf{NP}$. The proofs utilize the recent results of Arora et al. [*Proc. 23rd IEEE Symposium on Foundations of Computer Science*, 1992, pp. 14–23] on the complexity of approximation problems.

In the second part of the paper, we introduce a new method for analyzing the average-case performance of algorithms for sequences, based on Kolmogorov complexity. Despite the above nonapproximability results, we show that near optimal solutions for both SCS and LCS can be found on the average. More precisely, consider a fixed alphabet Σ and suppose that the input sequences are generated randomly according to the uniform probability distribution and are of the same length n. Moreover, assume that the number of input sequences is polynomial in n. Then, there are simple greedy algorithms which approximate SCS and LCS with expected additive errors $O(n^{0.707})$ and $O(n^{1/2+\epsilon})$ for any $\epsilon > 0$, respectively.

Incidentally, our analyses also provide tight upper and lower bounds on the expected LCS and SCS lengths for a set of random sequences solving a generalization of another well-known open question on the expected LCS length for two random sequences [K. Alexander, *The rate of convergence of the mean length of the longest common subsequence*, 1992, manuscript], [V. Chvatal and D. Sankoff, *J. Appl. Probab.*, 12 (1975), pp. 306–315], [D. Sankoff and J. Kruskall, eds., *Time Warps, String Edits, and Macromolecules: The Theory and Practice of Sequence Comparison*, Addison–Wesley, Reading, MA, 1983].

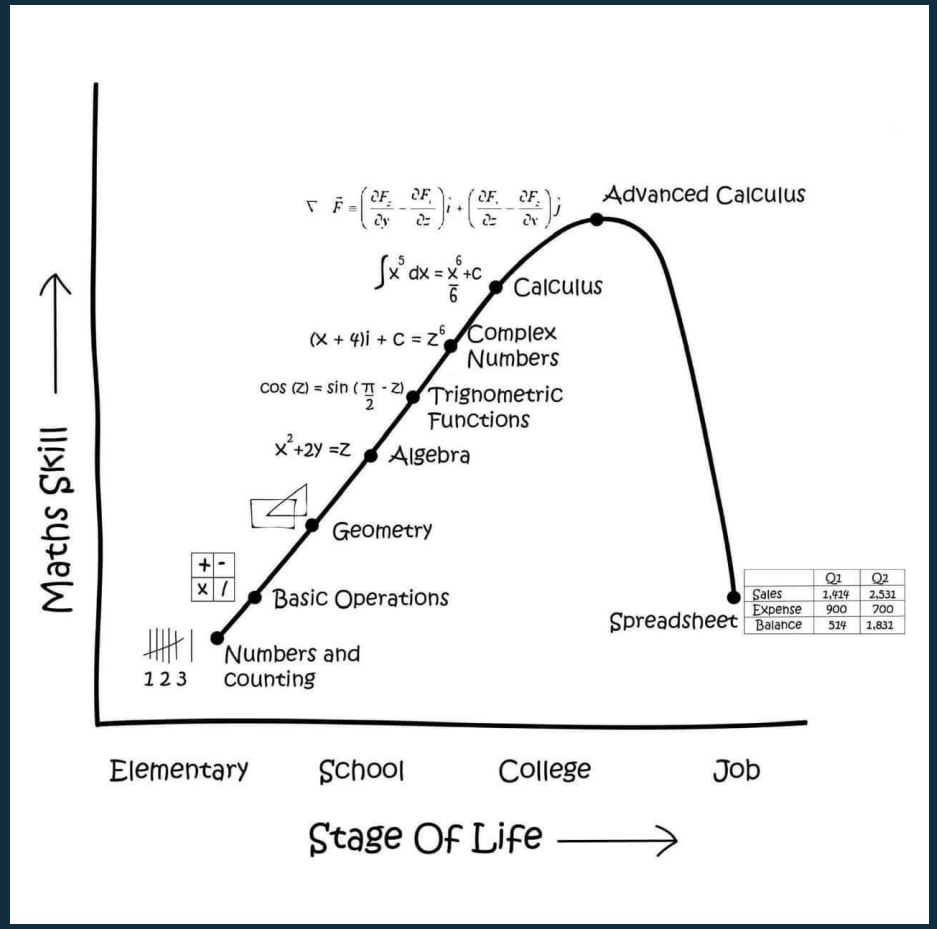
Key words. shortest common supersequence, longest common subsequence, approximation algorithm, **NP**-hardness, average-case analysis, random sequence

姚期智（Andrew Yao）



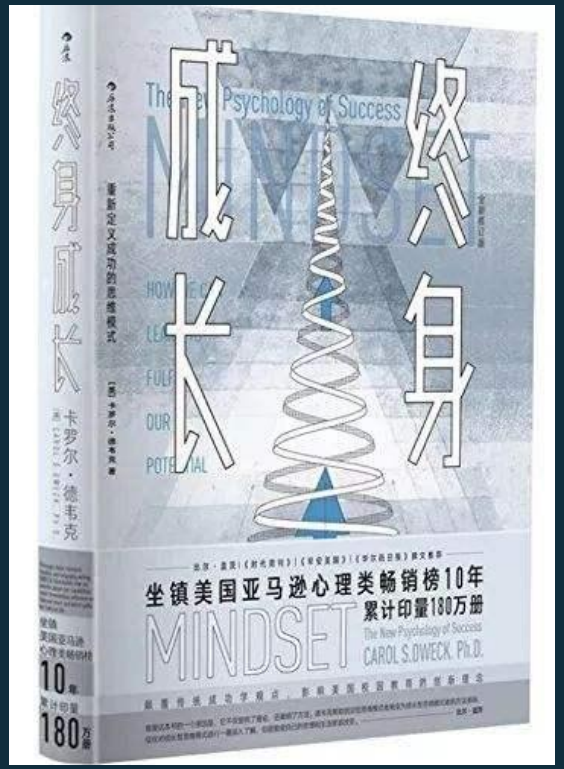
算法

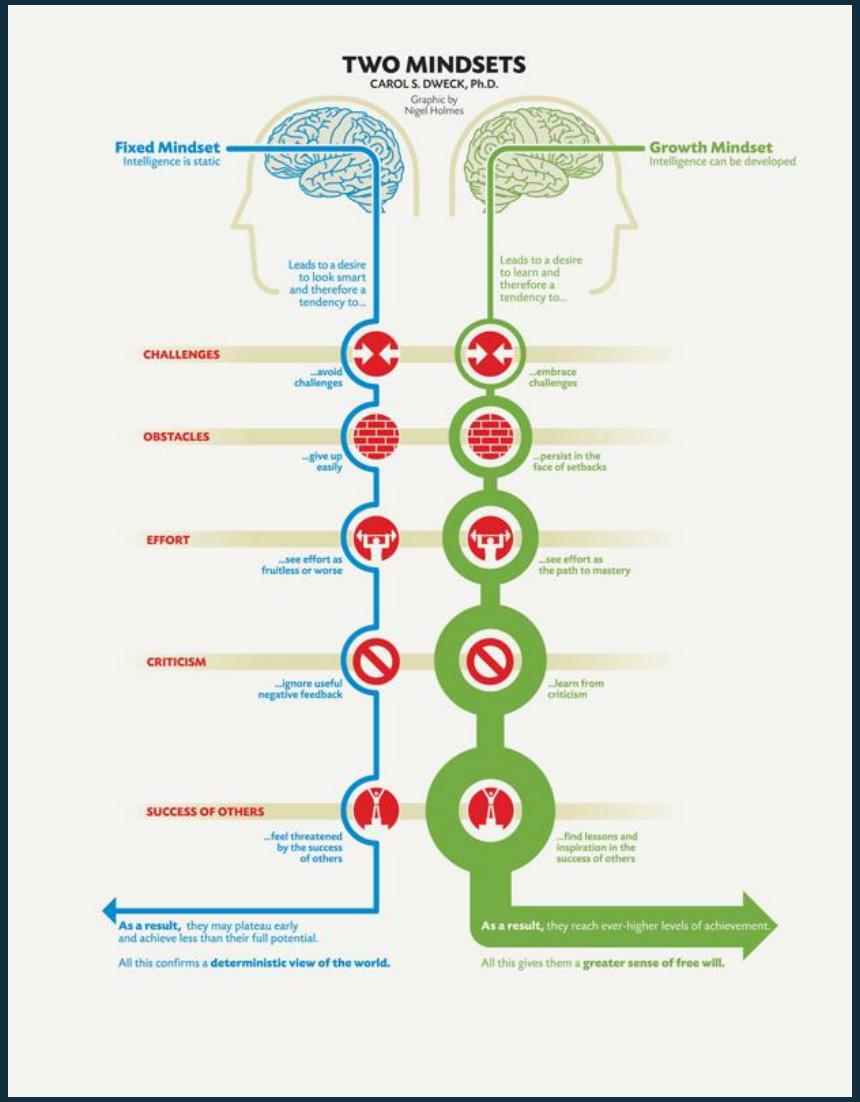
也许觉得算法在今后不会碰到那么多。。。



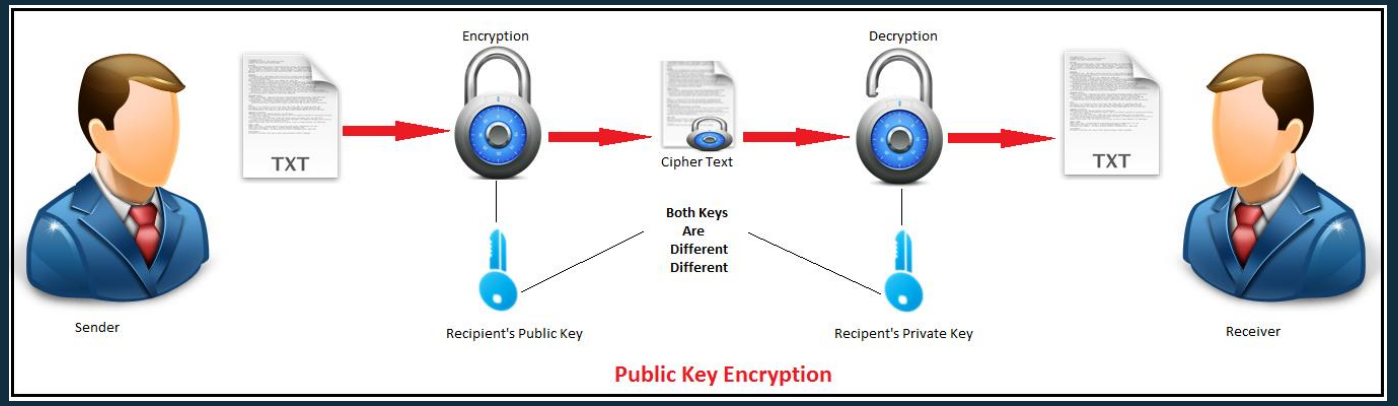
算法

终身学习。。。


终身成长
卡罗尔·德韦克
坐镇美国亚马逊心理类畅销榜10年
累计印量180万册
MINDSET
The New Psychology of Success
CAROL S.DWECK, Ph.D.
10年
180万册


TWO MINDSETS
CAROL S. DWECK, Ph.D.
Graphic by Nigel Holmes
Fixed Mindset
Intelligence is static
Growth Mindset
Intelligence can be developed
Leads to a desire to look smart and therefore a tendency to...
Leads to a desire to learn and therefore a tendency to...
CHALLENGES
...avoid challenges
...embrace challenges
OBSTACLES
...give up easily
...persist in the face of setbacks
EFFORT
...see effort as fruitless or worse
...see effort as the path to mastery
CRITICISM
...ignore useful negative feedback
...learn from criticism
SUCCESS OF OTHERS
...feel threatened by the success of others
...find lessons and inspiration in the success of others
As a result, they may plateau early and achieve less than their full potential.
All this confirms a deterministic view of the world.
As a result, they reach ever-higher levels of achievement.
All this gives them a greater sense of free will.

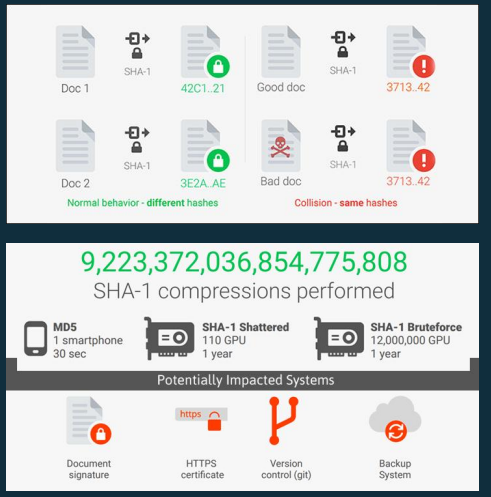
Public key system



RSA algorithm

Adi Shamir, Ron Rivest and Len Adleman

1977



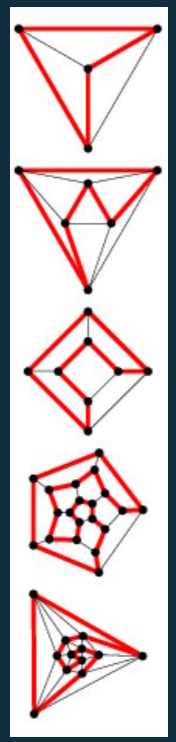
Collision attack

Xiaoyun Wang @ China

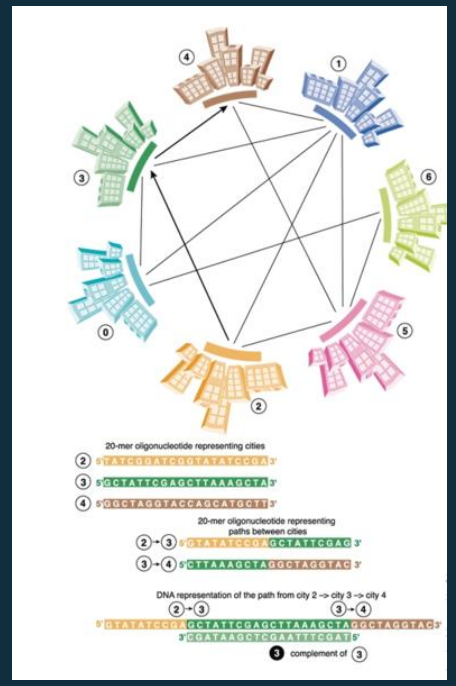
2004

Extension: from statistics back to DNA

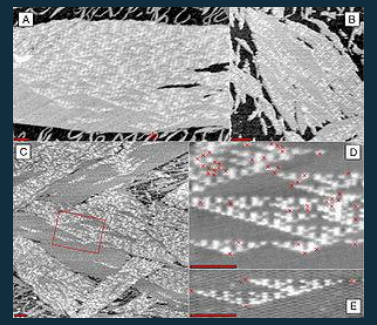
Hamilton path problem



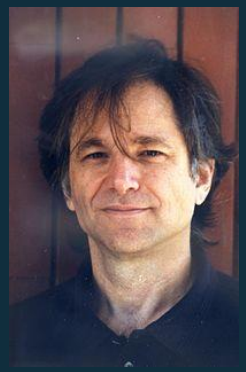
DNA computing



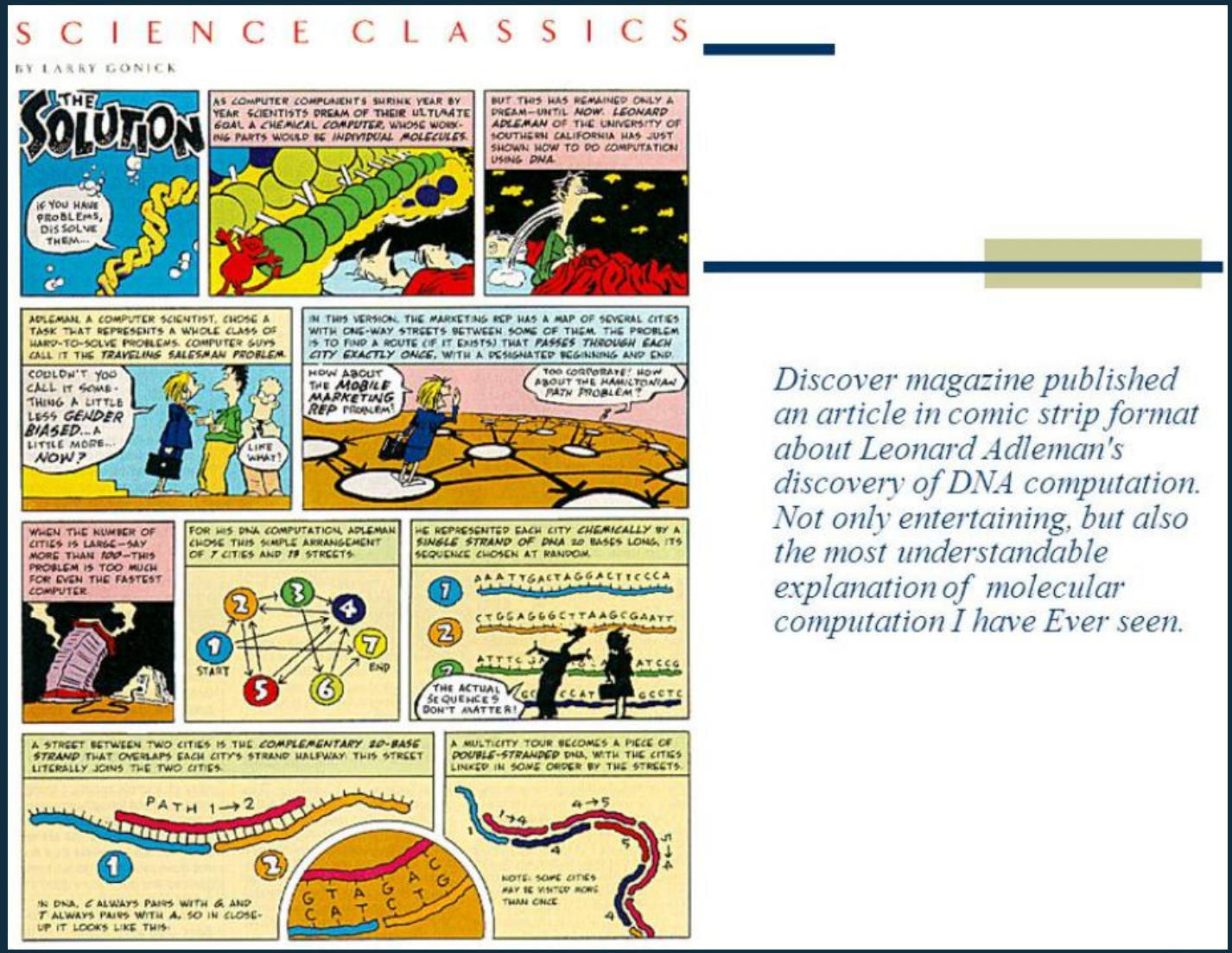
Solution



Leonard Adleman

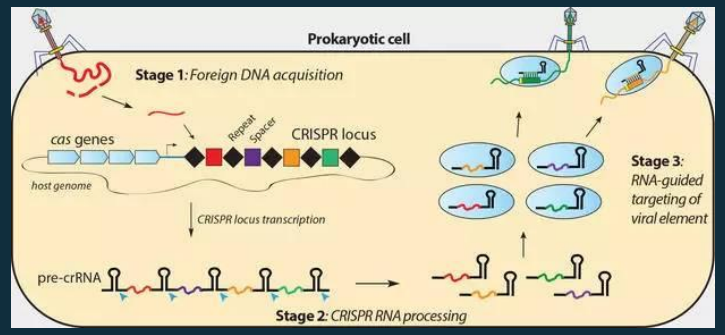


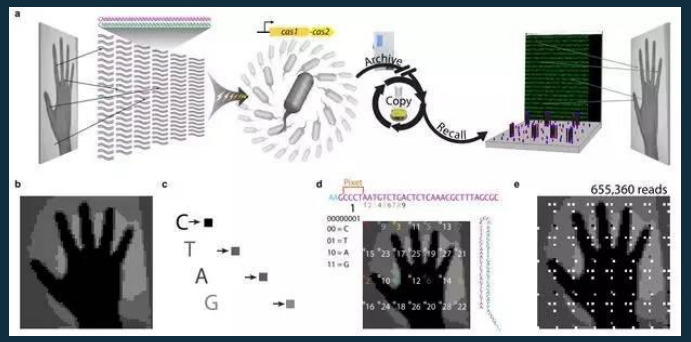
算法

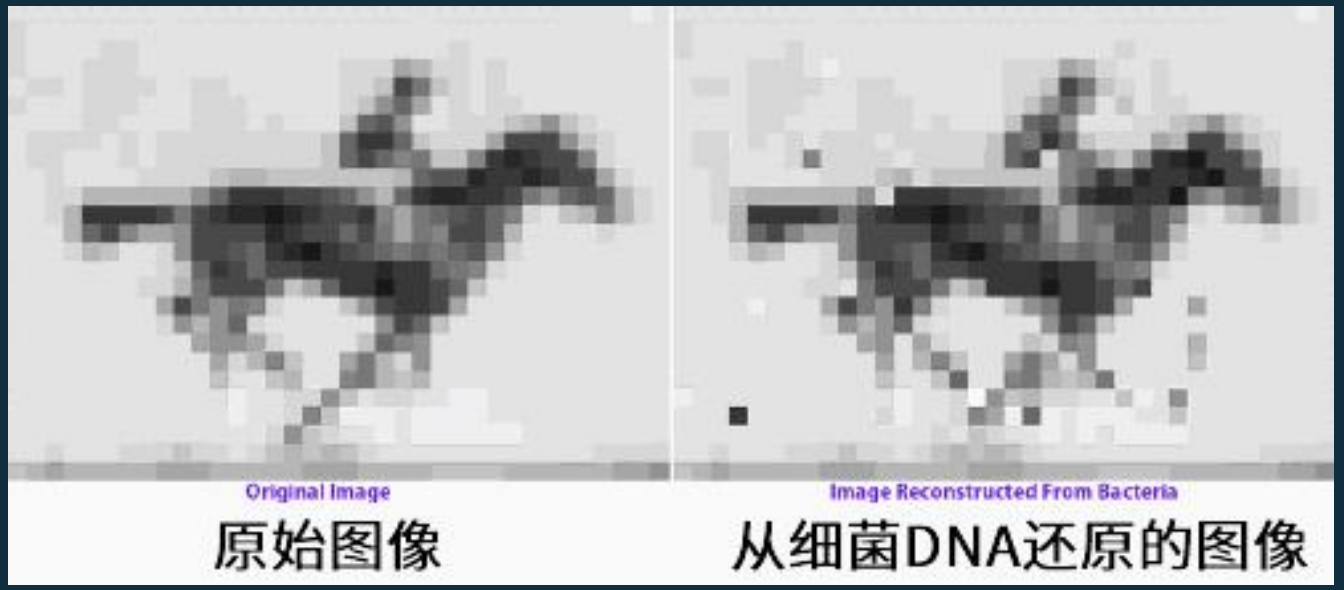


Discover magazine published an article in comic strip format about Leonard Adleman's discovery of DNA computation. Not only entertaining, but also the most understandable explanation of molecular computation I have Ever seen.

Understand it, create it!







CRISPR–Cas encoding of a digital movie into the genomes of a population of living bacteria, Nature, 2017

LECTURE 01

2. 数据

生物数据很大

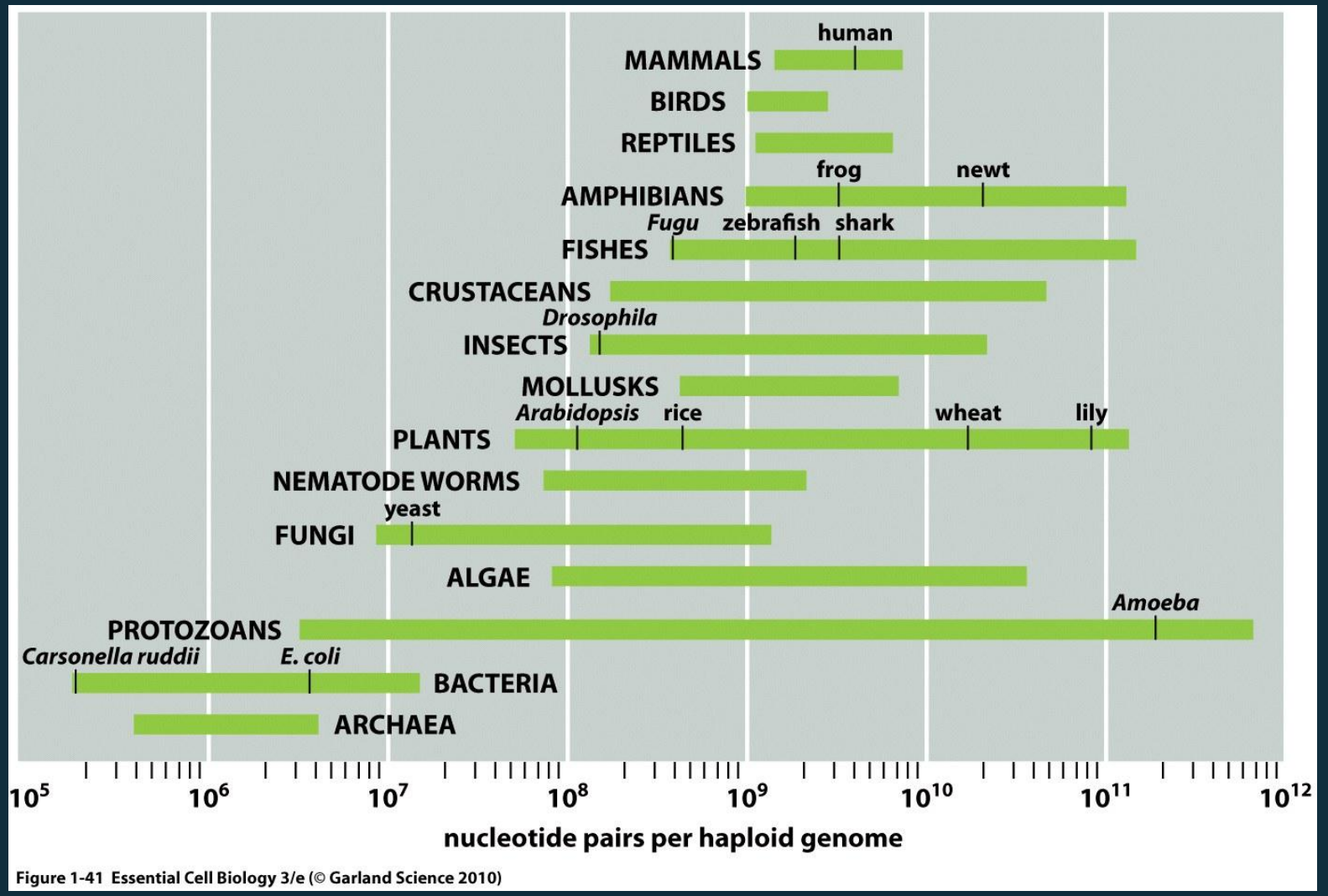
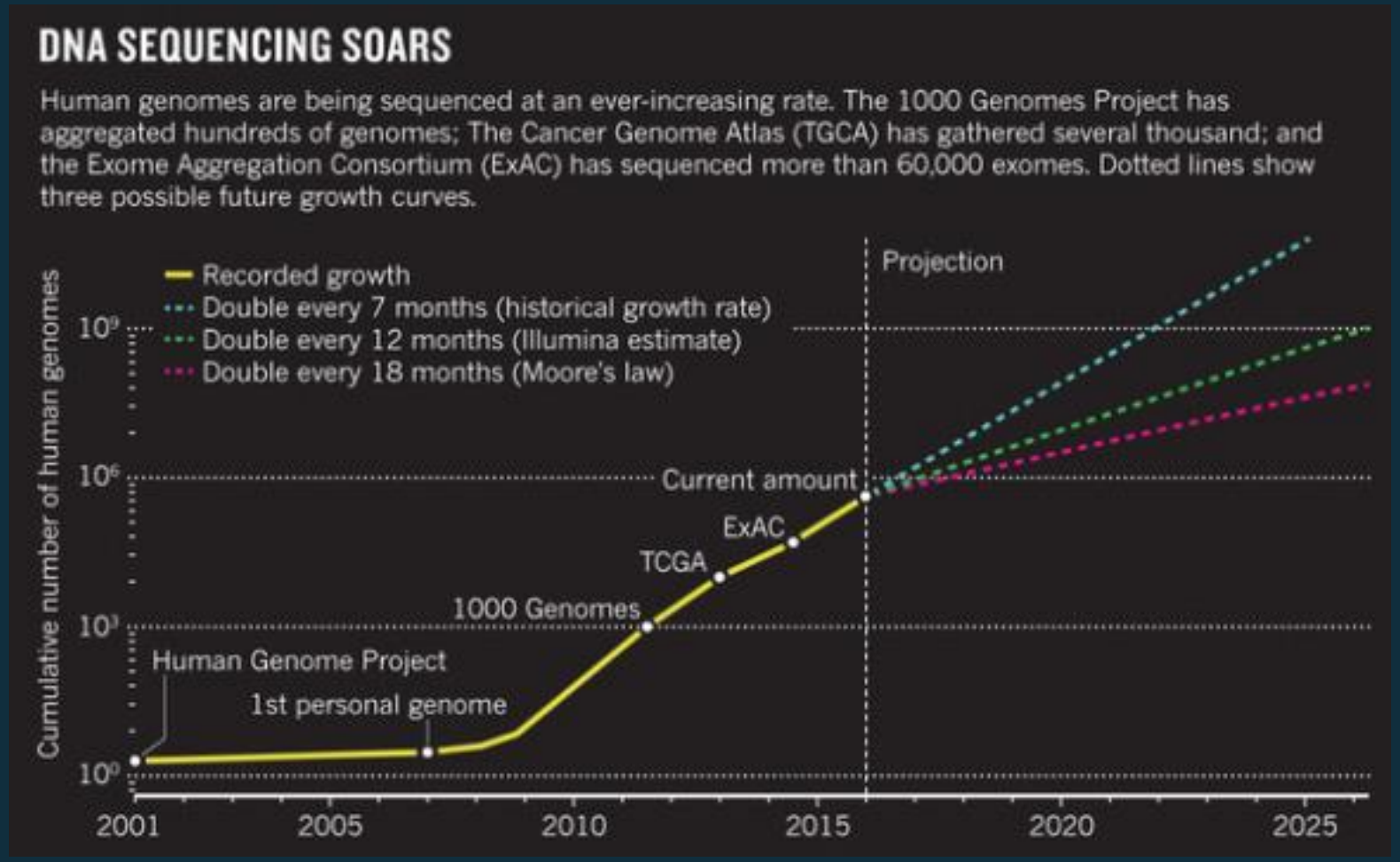
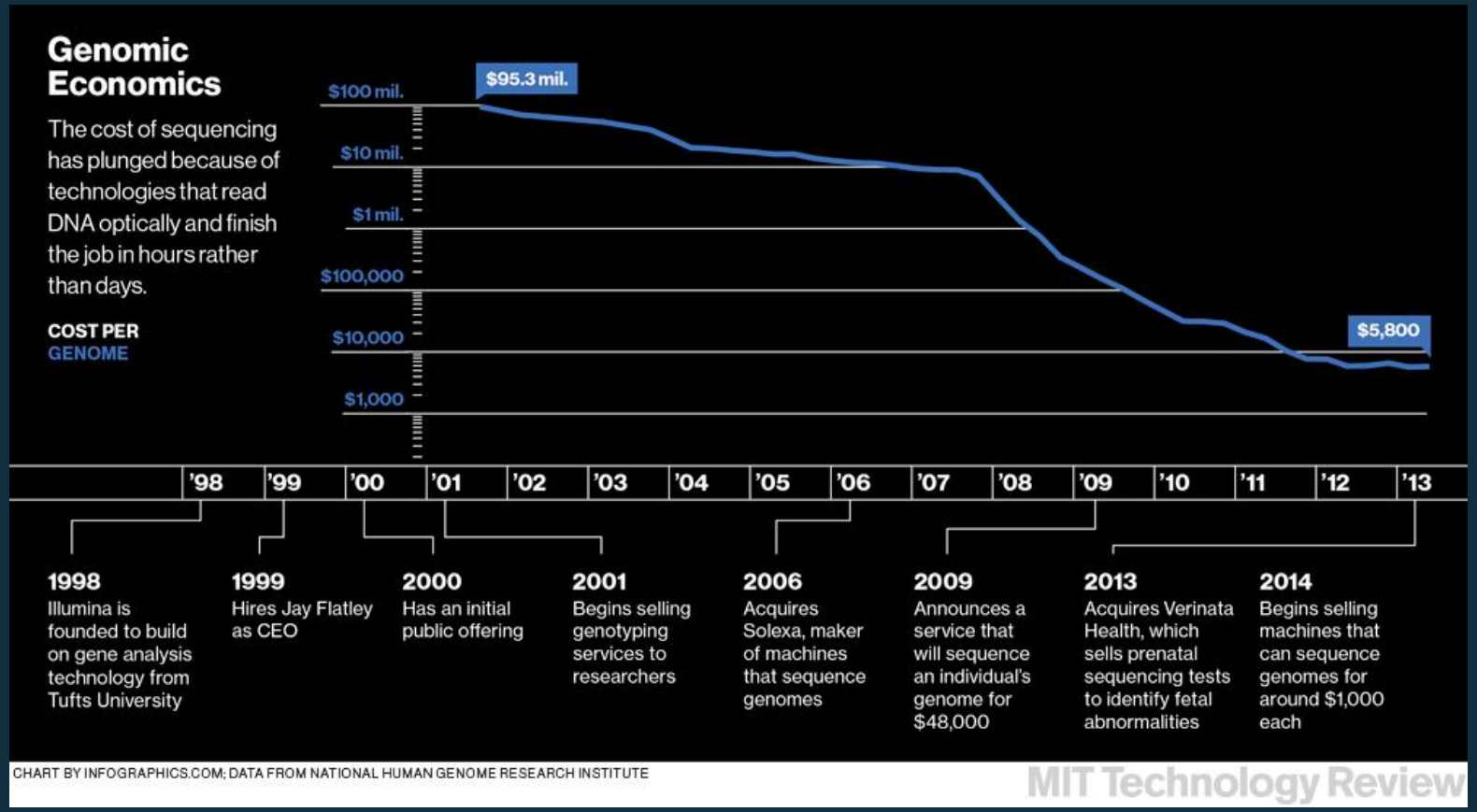


Figure 1-41 Essential Cell Biology 3/e (© Garland Science 2010)

生物数据很大

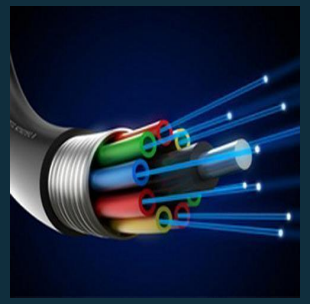

DNA SEQUENCING SOARS
Human genomes are being sequenced at an ever-increasing rate. The 1000 Genomes Project has aggregated hundreds of genomes; The Cancer Genome Atlas (TGCA) has gathered several thousand; and the Exome Aggregation Consortium (ExAC) has sequenced more than 60,000 exomes. Dotted lines show three possible future growth curves.
Recorded growth
Double every 7 months (historical growth rate)
Double every 12 months (Illumina estimate)
Double every 18 months (Moore's law)
Projection
Cumulative number of human genomes
10^9
10^6
10^3
10^0
Current amount
ExAC
TCGA
1000 Genomes
Human Genome Project
1st personal genome
2001
2005
2010
2015
2020
2025

生物数据很大


Genomic Economics
The cost of sequencing has plunged because of technologies that read DNA optically and finish the job in hours rather than days.
COST PER GENOME
$100 mil.
$10 mil.
$1 mil.
$100,000
$10,000
$1,000
$95.3 mil.
$5,800
'98 '99 '00 '01 '02 '03 '04 '05 '06 '07 '08 '09 '10 '11 '12 '13
1998 Illumina is founded to build on gene analysis technology from Tufts University
1999 Hires Jay Flatley as CEO
2000 Has an initial public offering
2001 Begins selling genotyping services to researchers
2006 Acquires Solexa, maker of machines that sequence genomes
2009 Announces a service that will sequence an individual's genome for $48,000
2013 Acquires Verinata Health, which sells prenatal sequencing tests to identify fetal abnormalities
2014 Begins selling machines that can sequence genomes for around $1,000 each
CHART BY INFOGRAPHICS.COM; DATA FROM NATIONAL HUMAN GENOME RESEARCH INSTITUTE
MIT Technology Review

制约数据交换的实际是网络

光纤



北京-->武汉：1Gb/s，5000元/月

Infiniband

服务器之间：~50Gb/s，10万元

快递小哥

北京-->武汉：
(4TB*20)/(60*60*24*2)=463MB/s=3.7Gb/s，200元

他可以多装点，而且次日达可以更快的。。。

LECTURE 01

3. 超算

超级计算机平台

TOP 10 Sites for June 2017

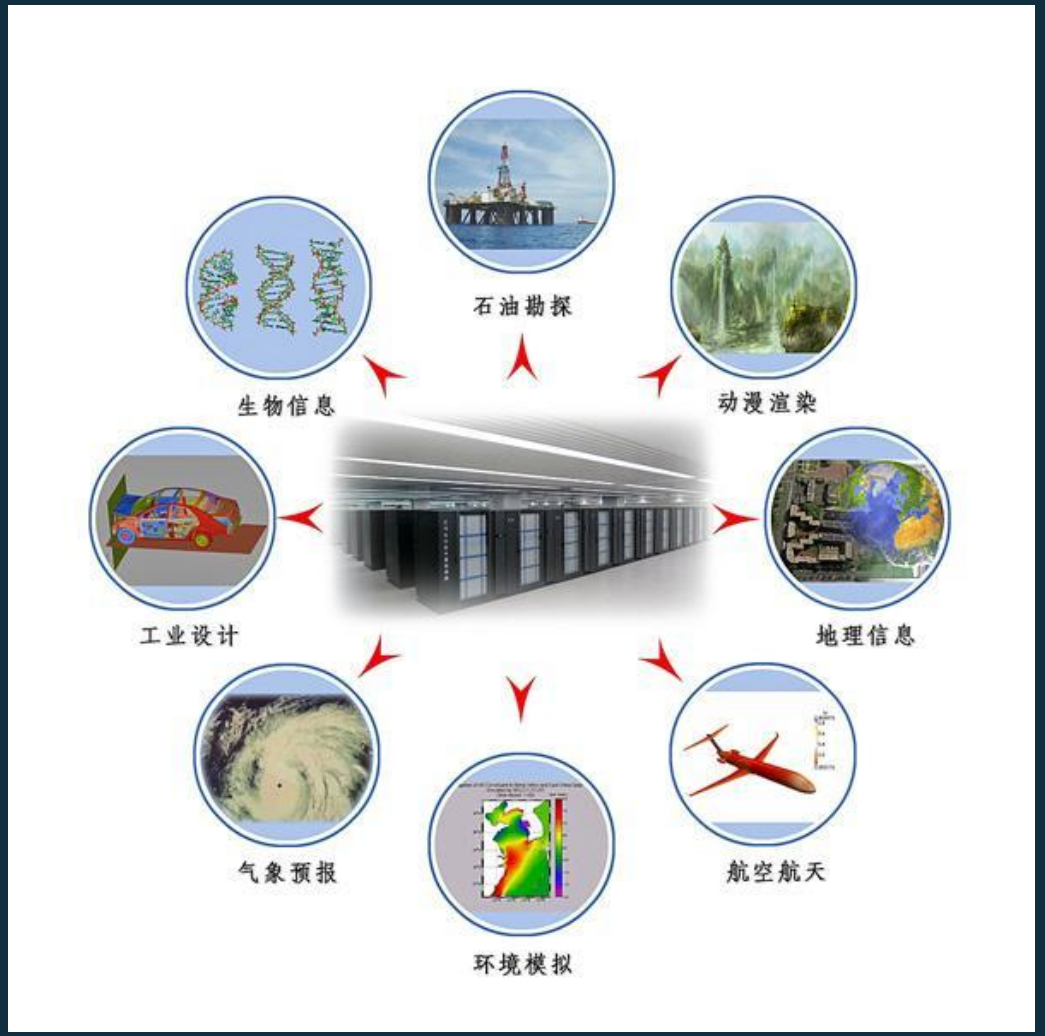
For more information about the sites and systems in the list, click on the links or view the complete list.

1-100 | 101-200 | 201-300 | 301-400 | 401-500

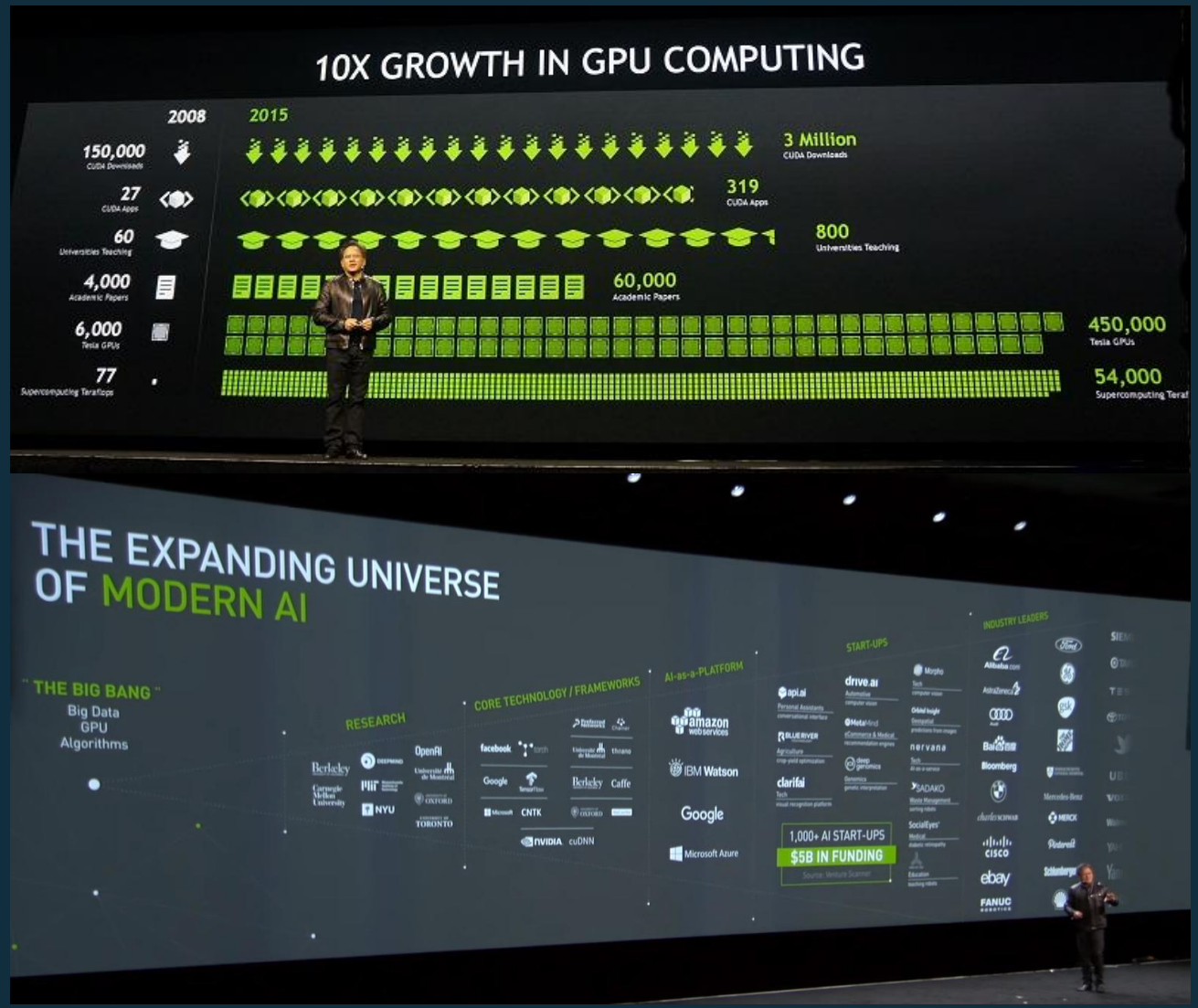
Rank	System	Cores	Rmax (TFlop/s)	Rpeak (TFlop/s)	Power (kW)
1	**Sunway TaihuLight** - Sunway MPP, Sunway SW26010 260C 1.45GHz, Sunway , NRCPC National Supercomputing Center in Wuxi China	10,649,600	93,014.6	125,435.9	15,371
2	**Tianhe-2 (MilkyWay-2)** - TH-IVB-FEP Cluster, Intel Xeon E5-2692 12C 2.200GHz, TH Express-2, Intel Xeon Phi 31S1P , NUDT National Super Computer Center in Guangzhou China	3,120,000	33,862.7	54,902.4	17,808
3	**Piz Daint** - Cray XC50, Xeon E5-2690v3 12C 2.6GHz, Aries interconnect , NVIDIA Tesla P100 , Cray Inc. Swiss National Supercomputing Centre (CSCS) Switzerland	361,760	19,590.0	25,326.3	2,272
4	**Titan** - Cray XK7, Opteron 6274 16C 2.200GHz, Cray Gemini interconnect, NVIDIA K20x , Cray Inc. DOE/SC/Oak Ridge National Laboratory United States	560,640	17,590.0	27,112.5	8,209
5	**Sequoia** - BlueGene/Q, Power BQC 16C 1.60 GHz, Custom , IBM DOE/NNSA/LLNL United States	1,572,864	17,173.2	20,132.7	7,890
6	**Cori** - Cray XC40, Intel Xeon Phi 7250 68C 1.4GHz, Aries interconnect , Cray Inc. DOE/SC/LBNL/NERSC United States	622,336	14,014.7	27,880.7	3,939
7	**Oakforest-PACS** - PRIMERGY CX1640 M1, Intel Xeon Phi 7250 68C 1.4GHz, Intel Omni-Path , Fujitsu Joint Center for Advanced High Performance Computing Japan	556,104	13,554.6	24,913.5	2,719
8	K computer, SPARC64 VIIIfx 2.0GHz, Tofu interconnect , Fujitsu RIKEN Advanced Institute for Computational Science (AICS) Japan	705,024	10,510.0	11,280.4	12,660

超级计算机平台

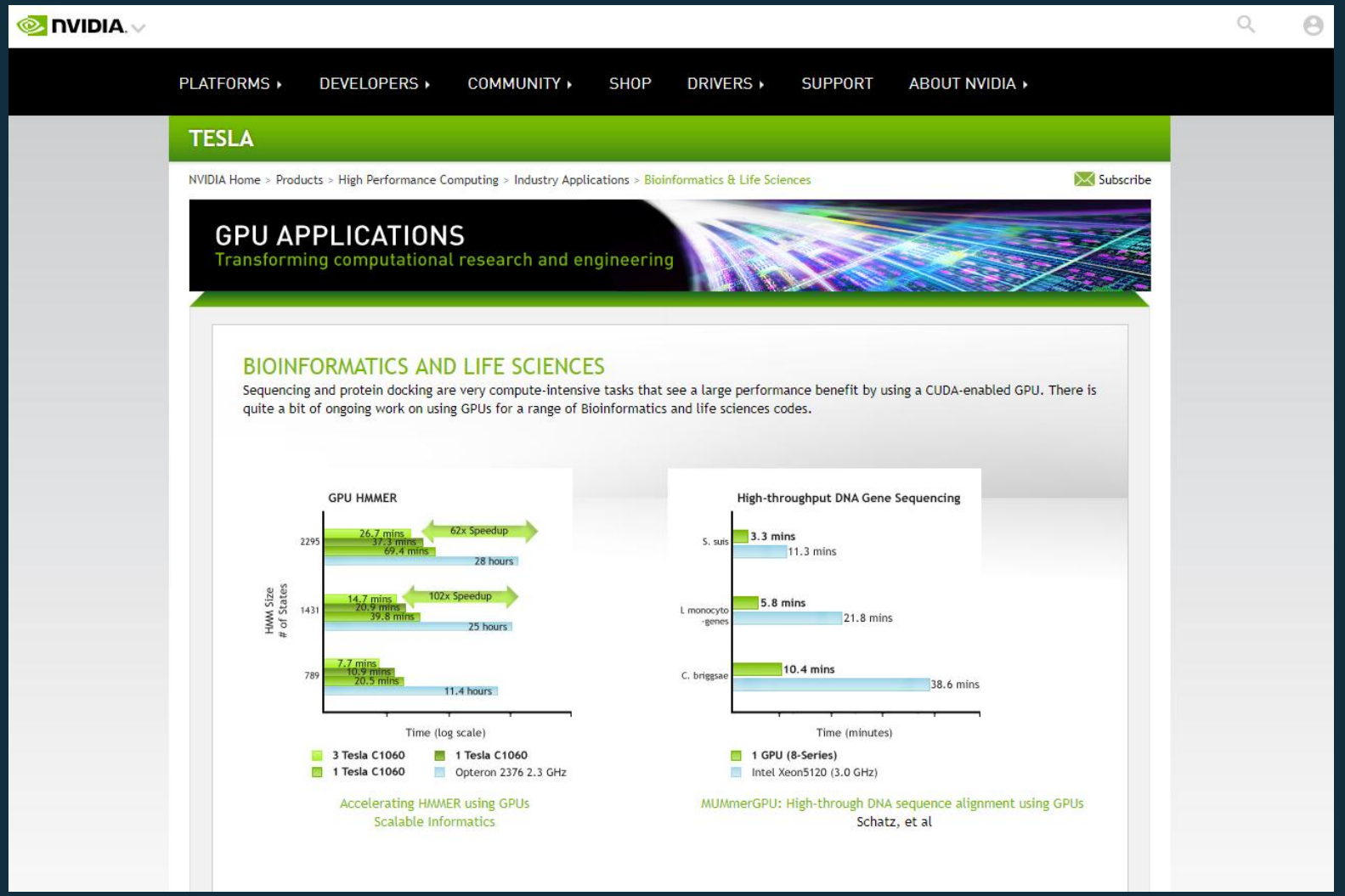
Tianhe-2



GPU计算


10X GROWTH IN GPU COMPUTING
2008
2015
150,000
CUDA Downloads
3 Million
CUDA Downloads
27
CUDA Apps
319
CUDA Apps
60
Universities Teaching
800
Universities Teaching
4,000
Academic Papers
60,000
Academic Papers
6,000
Tesla GPUs
450,000
Tesla GPUs
77
Supercomputing Teraflops
54,000
Supercomputing Tera
THE EXPANDING UNIVERSE
OF MODERN AI
" THE BIG BANG "
Big Data
GPU
Algorithms
RESEARCH
CORE TECHNOLOGY / FRAMEWORKS
AI-as-a-PLATFORM
START-UPS
INDUSTRY LEADERS
IBM Watson
Google
Microsoft Azure
1,000+ AI START-UPS
$5B IN FUNDING

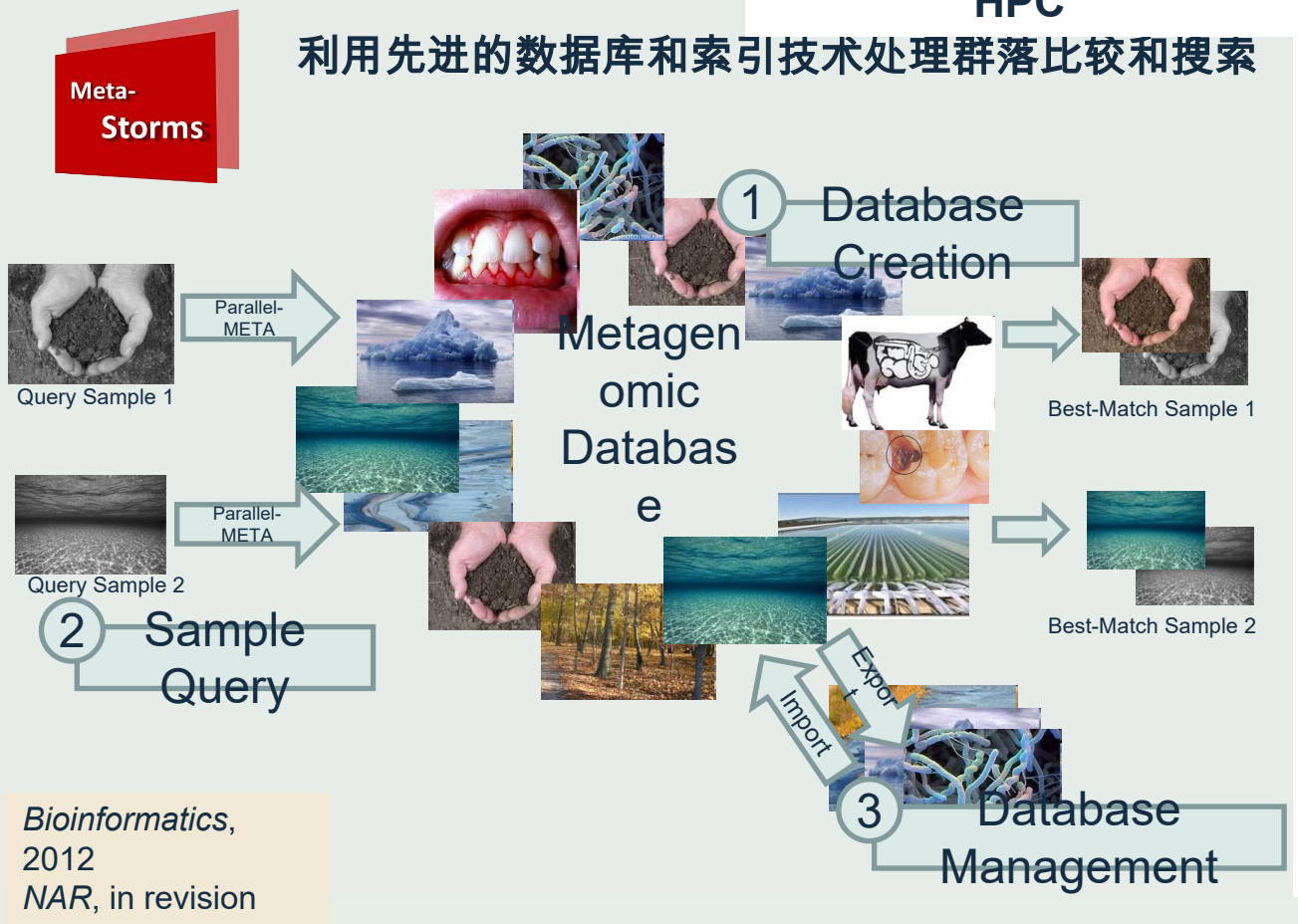
GPU计算


NVIDIA
PLATFORMS
DEVELOPERS
COMMUNITY
SHOP
DRIVERS
SUPPORT
ABOUT NVIDIA
TESLA
NVIDIA Home > Products > High Performance Computing > Industry Applications > Bioinformatics & Life Sciences
Subscribe
GPU APPLICATIONS
Transforming computational research and engineering
BIOINFORMATICS AND LIFE SCIENCES
Sequencing and protein docking are very compute-intensive tasks that see a large performance benefit by using a CUDA-enabled GPU. There is quite a bit of ongoing work on using GPUs for a range of Bioinformatics and life sciences codes.
GPU HMMER
HMM Size # of States
2295
1431
789
26.7 mins
37.3 mins
69.4 mins
28 hours
62x Speedup
14.7 mins
20.9 mins
39.8 mins
25 hours
102x Speedup
7.7 mins
10.9 mins
20.5 mins
11.4 hours
Time (log scale)
3 Tesla C1060
1 Tesla C1060
1 Tesla C1060
Opteron 2376 2.3 GHz
Accelerating HMMER using GPUs
Scalable Informatics
High-throughput DNA Gene Sequencing
S. suis
3.3 mins
11.3 mins
L monocyto -genes
5.8 mins
21.8 mins
C. briggsae
10.4 mins
38.6 mins
Time (minutes)
1 GPU (8-Series)
Intel Xeon5120 (3.0 GHz)
MUMmerGPU: High-through DNA sequence alignment using GPUs
Schatz, et al

Meta-Storms

高性能的生物计算（蓝色）
HPC

利用先进的数据库和索引技术处理群落比较和搜索

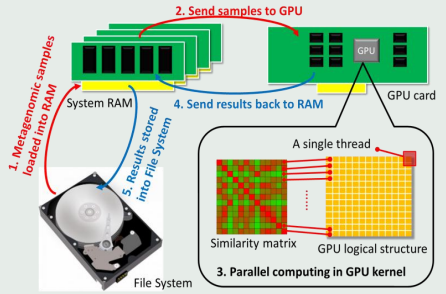


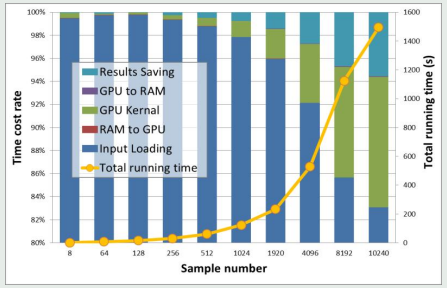
Bioinformatics, 2012
NAR, in revision

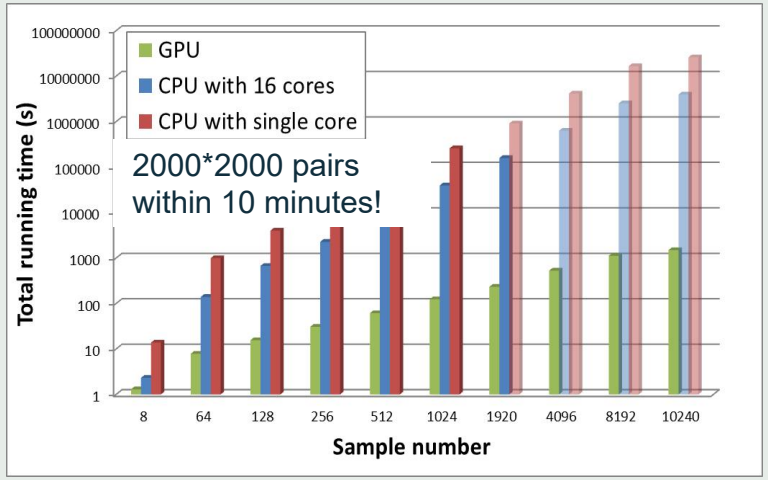
Meta-Storms

高性能的生物计算（蓝色）
HPC

利用并行计算技术
加速大规模群落比
较和搜索







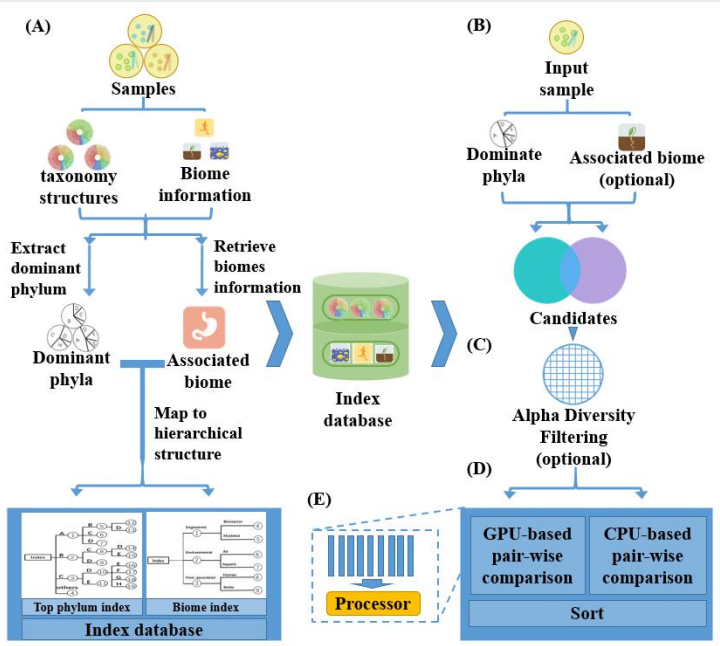
Data source: MG-RAST database

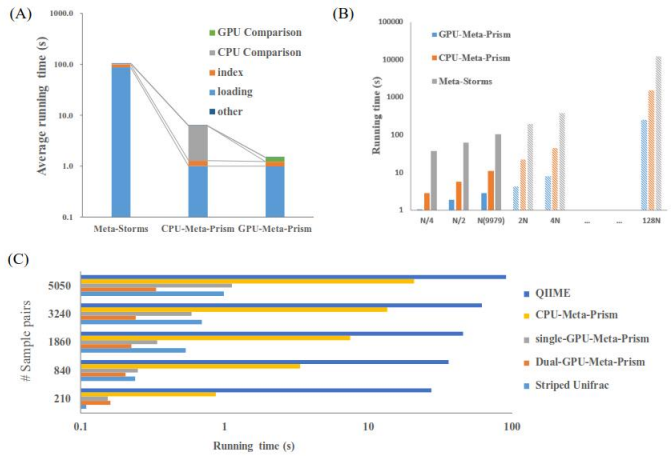
Bioinformatics, 2014
Scientific Reports, 2017

Meta-Prism

高性能的生物计算（蓝色）
HPC

利用双引擎计算，同时
提高准确性和运行速度





Bioinformatics, in press

LECTURE 01

4. 深度学习

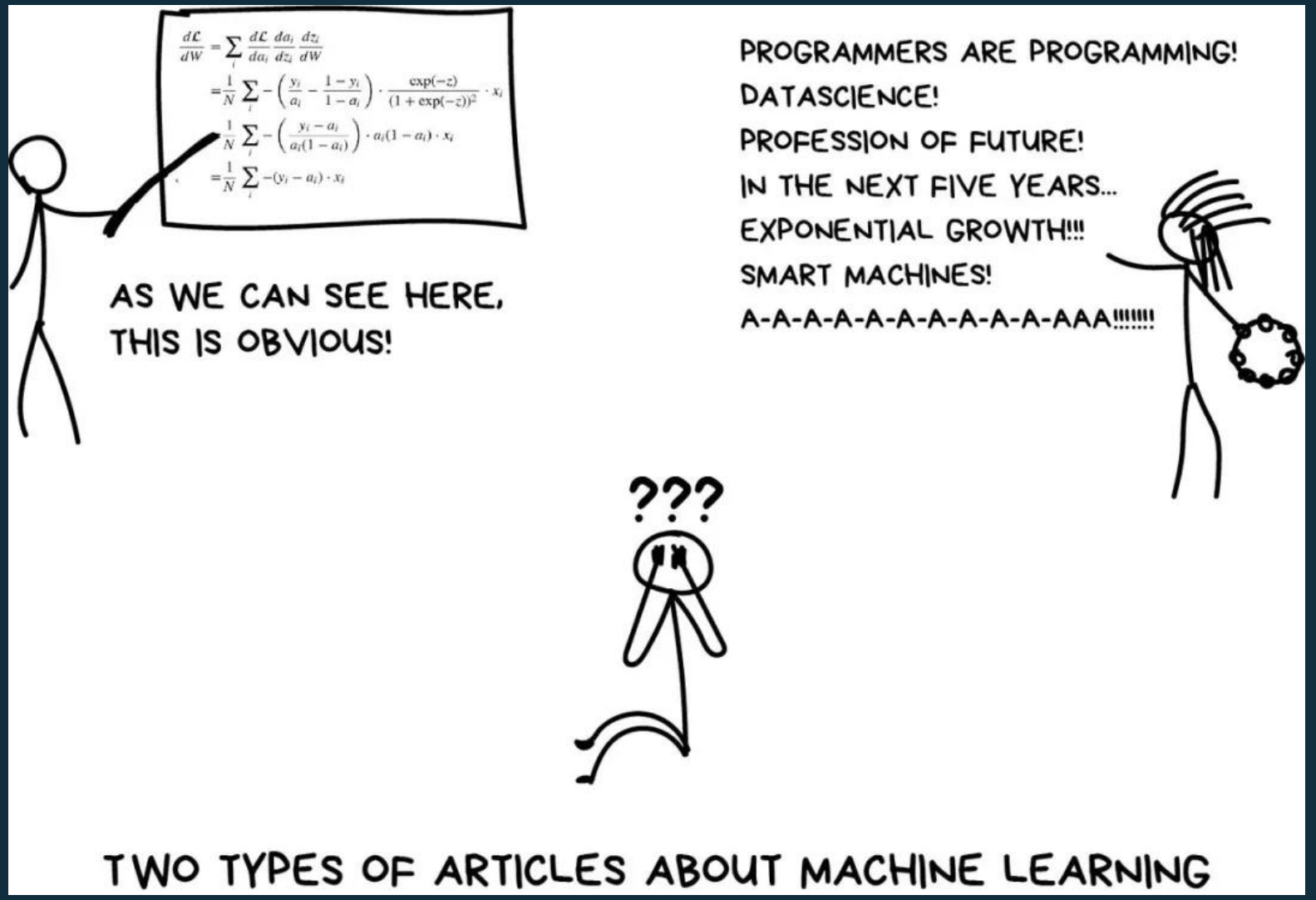
Deep Learning

数据很重要！

算法更重要！

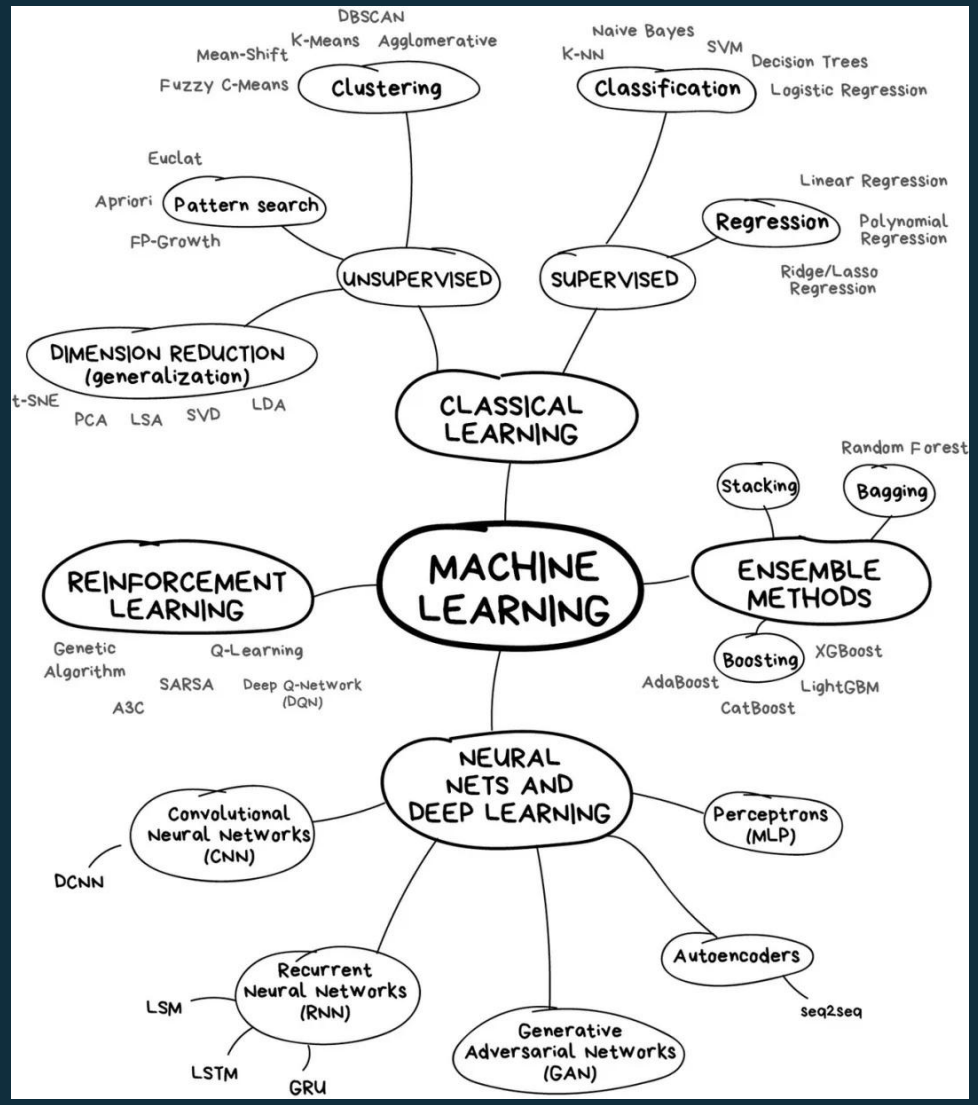
机器学习研究的两种类型

研究问题与研究方法

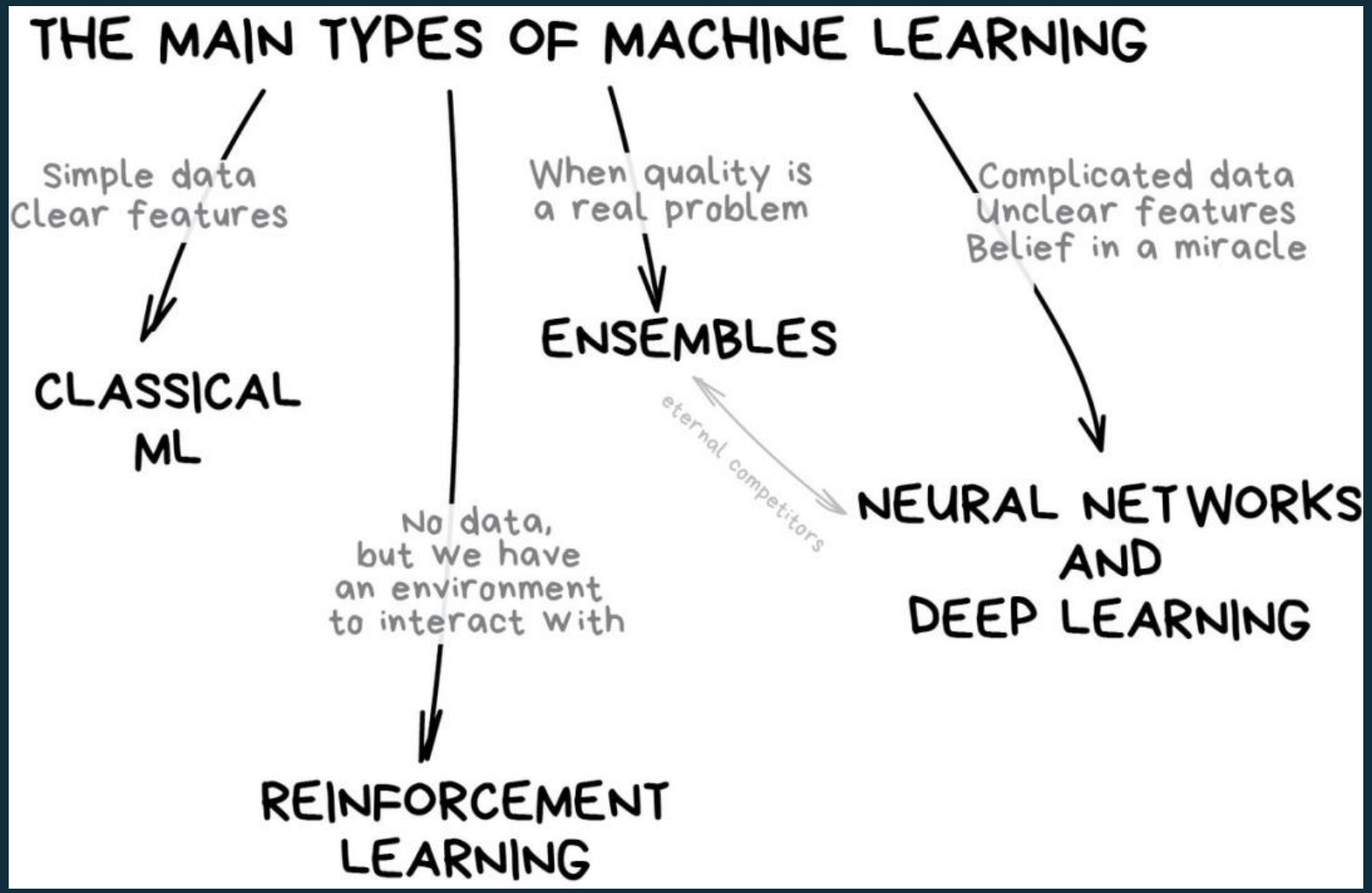


机器学习方法全景

经典方法、集成方法与神经网络



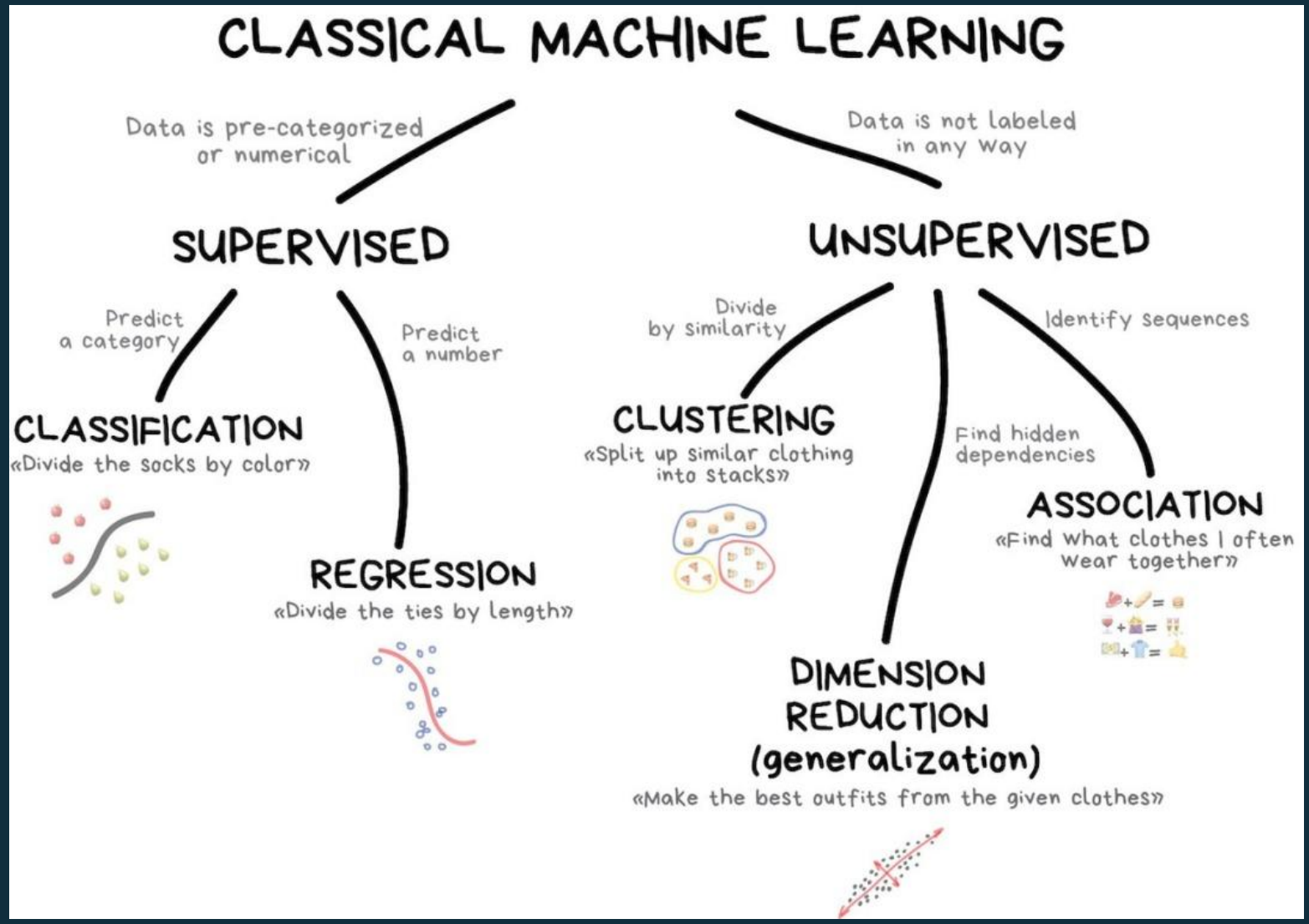
机器学习的主要类型



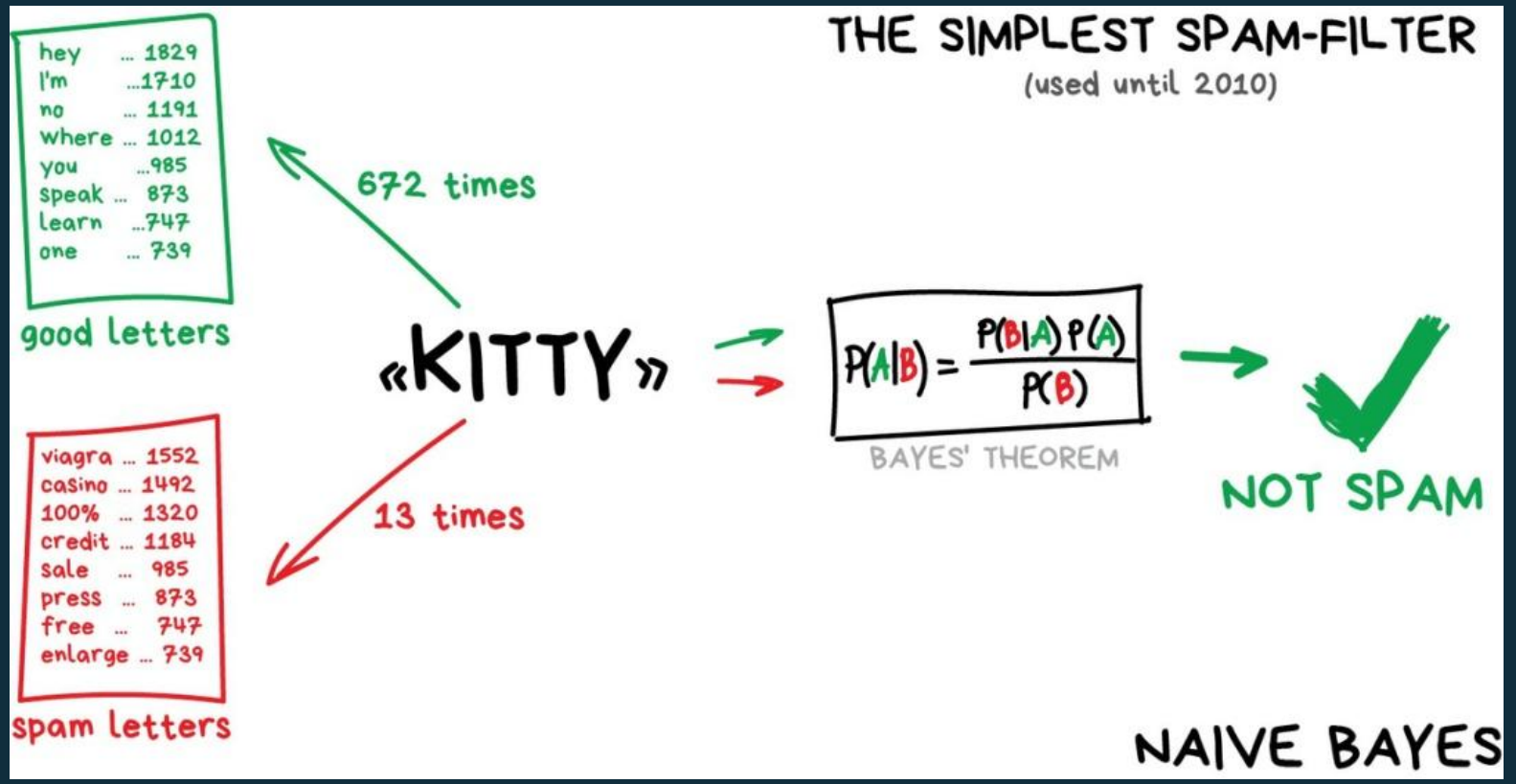
数据、标签与反馈

经典机器学习

监督学习与无监督学习



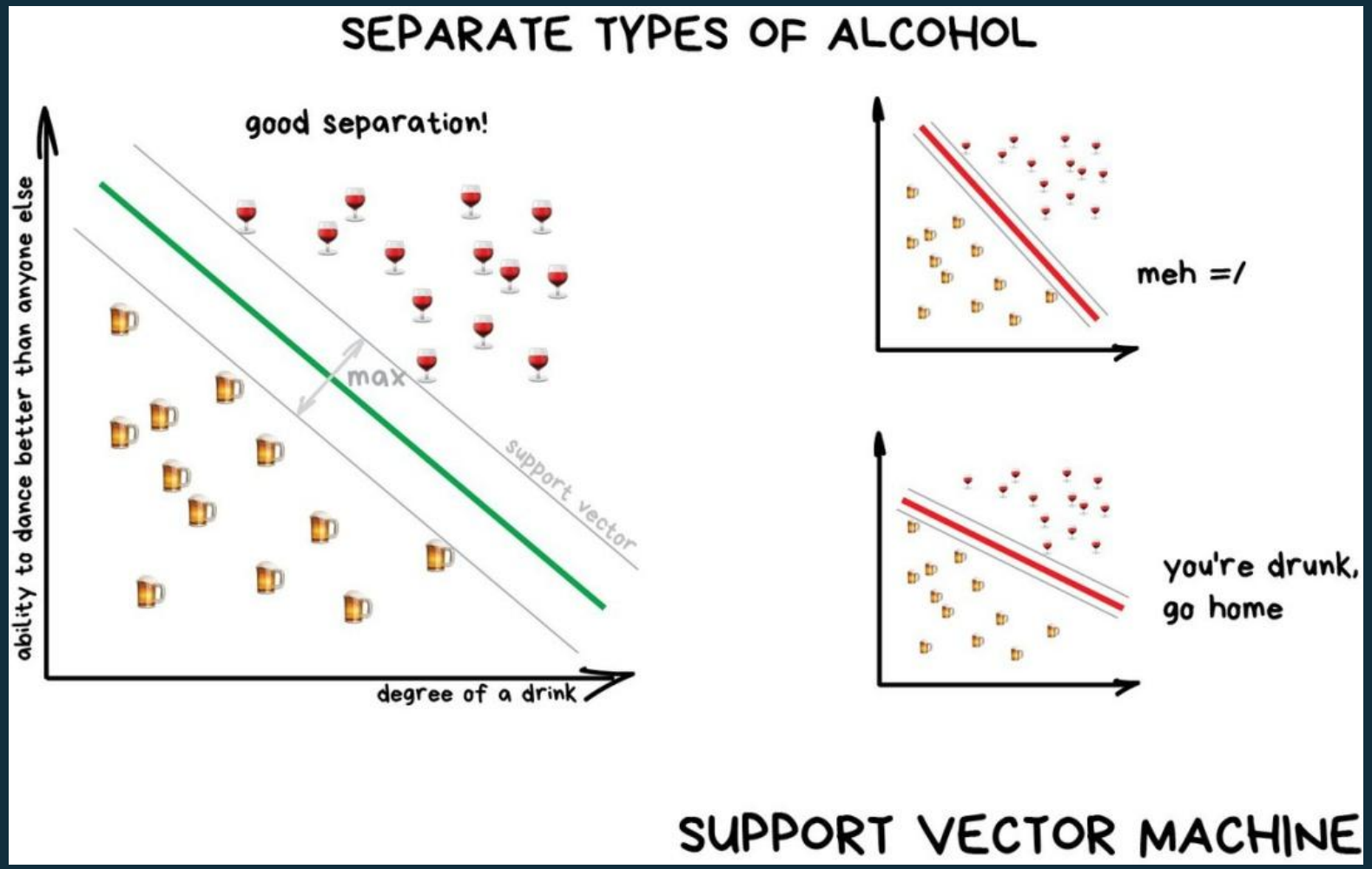
朴素贝叶斯



垃圾邮件过滤示例

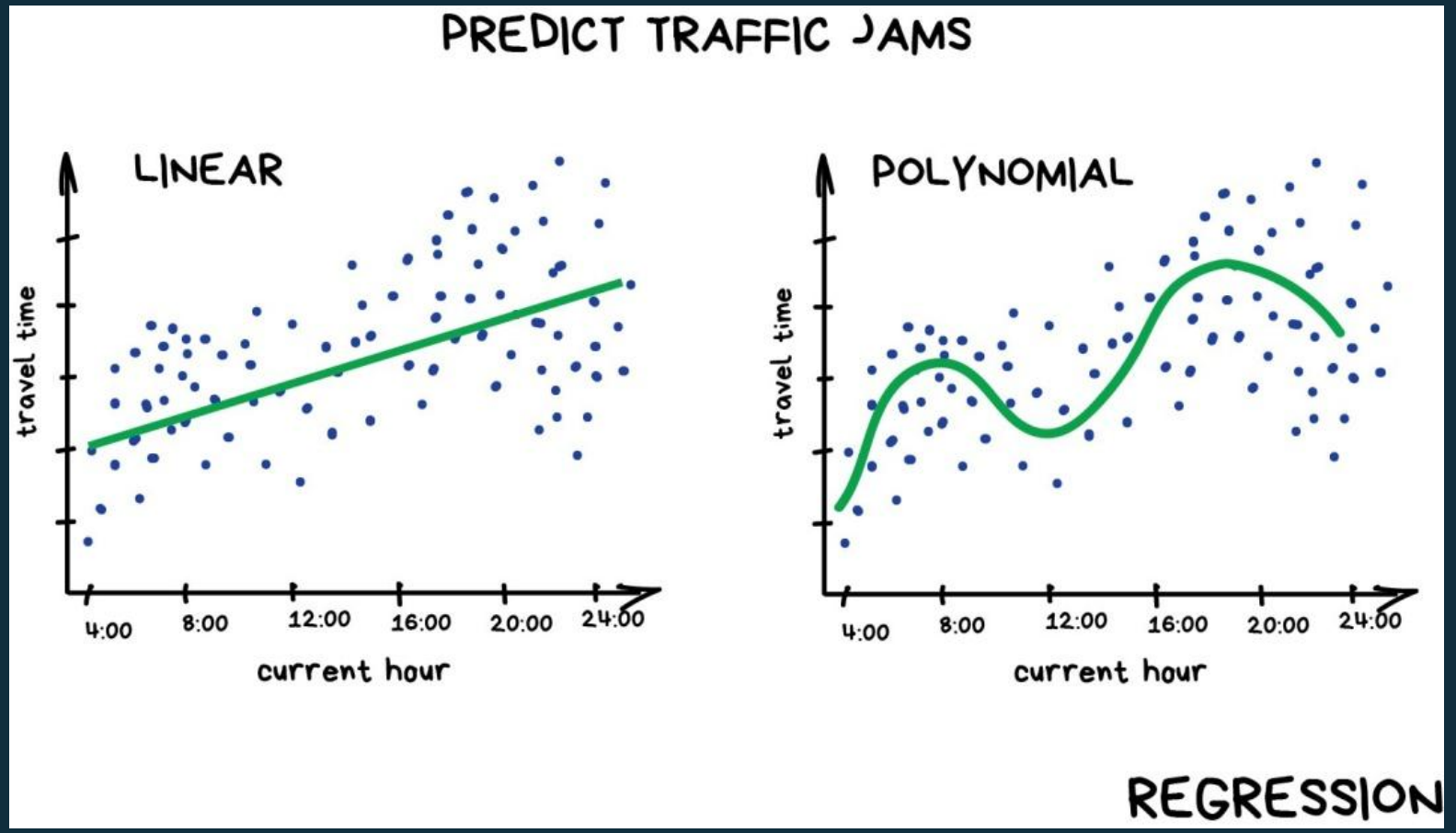
支持向量机

分类与决策边界



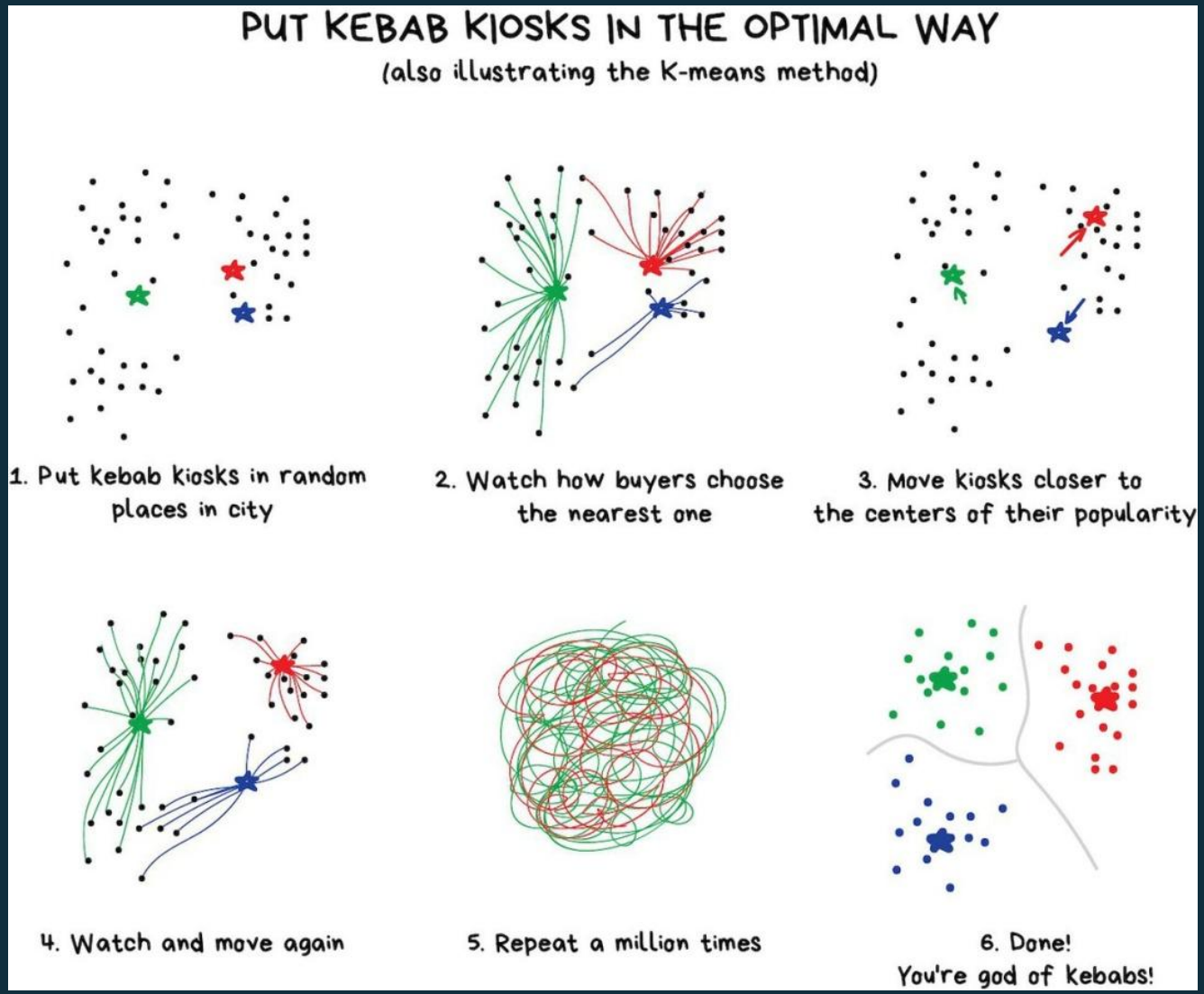
回归

从数据预测数值


PREDICT TRAFFIC JAMS
LINEAR
travel time
4:00
8:00
12:00
16:00
20:00
24:00
current hour
POLYNOMIAL
travel time
4:00
8:00
12:00
16:00
20:00
24:00
current hour
REGRESSION

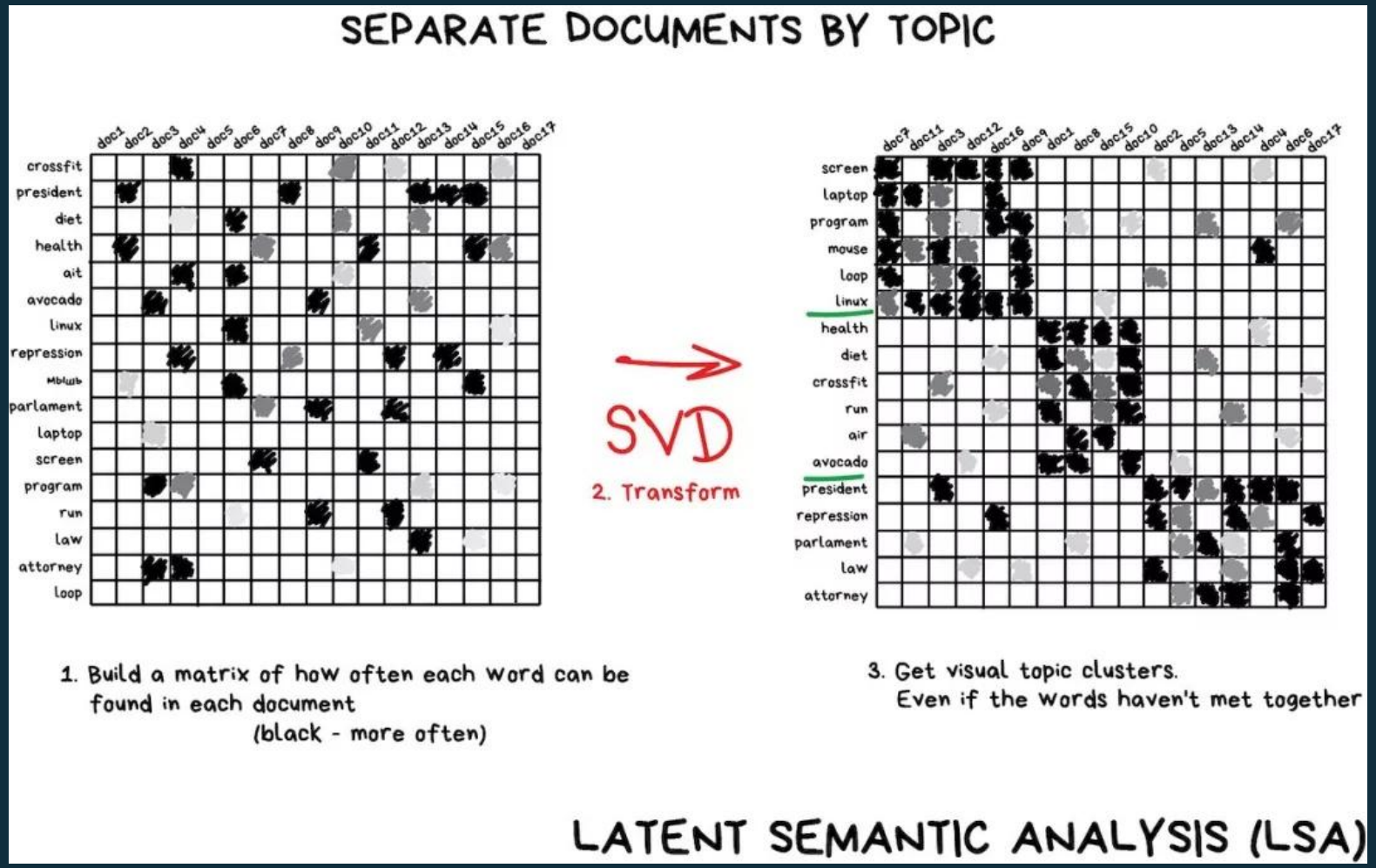
K-means 聚类

无标签数据与质心



潜在语义分析

矩阵与主题

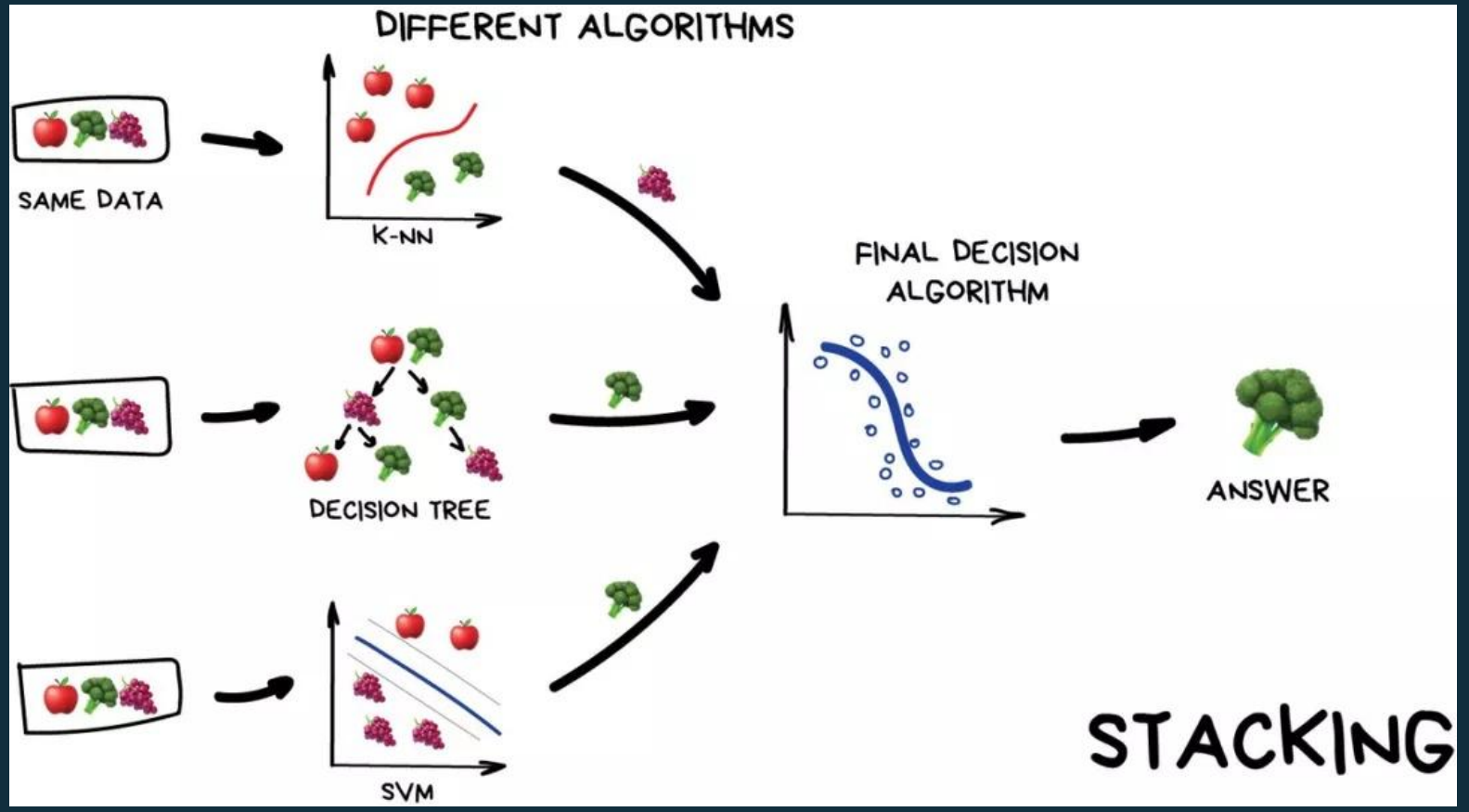


推荐系统

根据已有信息作出推荐

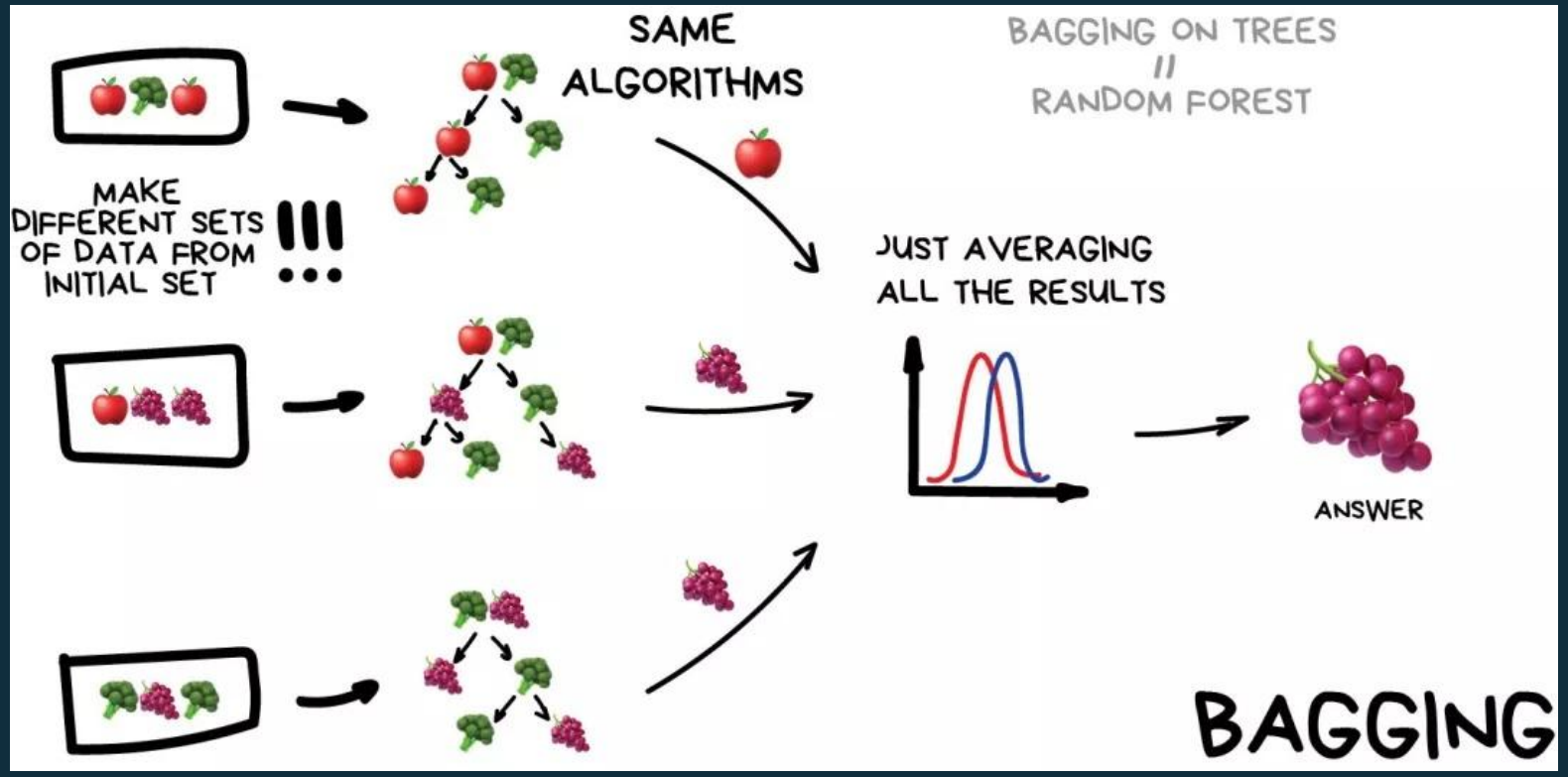
Stacking 集成

组合不同算法的预测



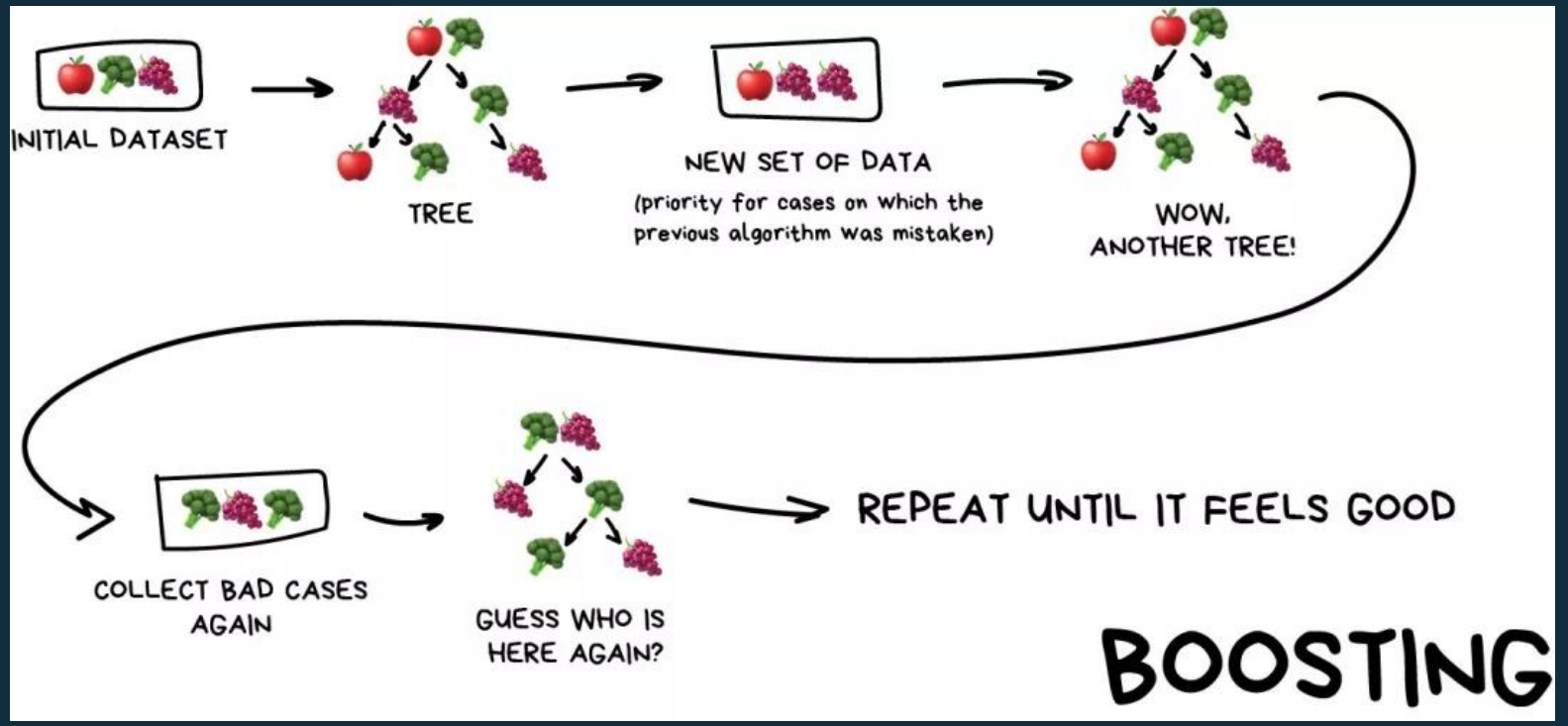
Bagging 集成

同一算法与不同训练样本



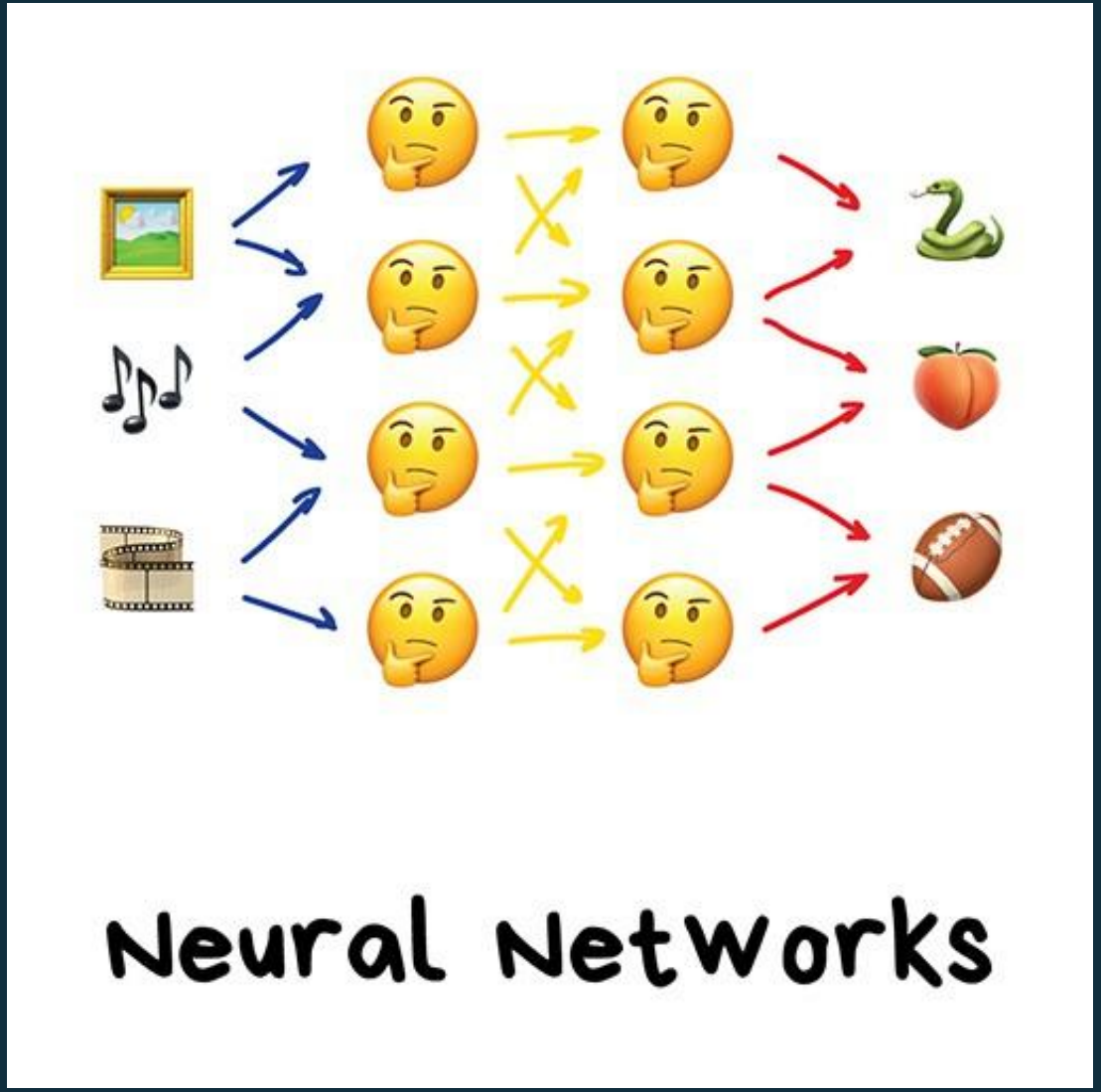
Boosting 集成

逐步改进预测

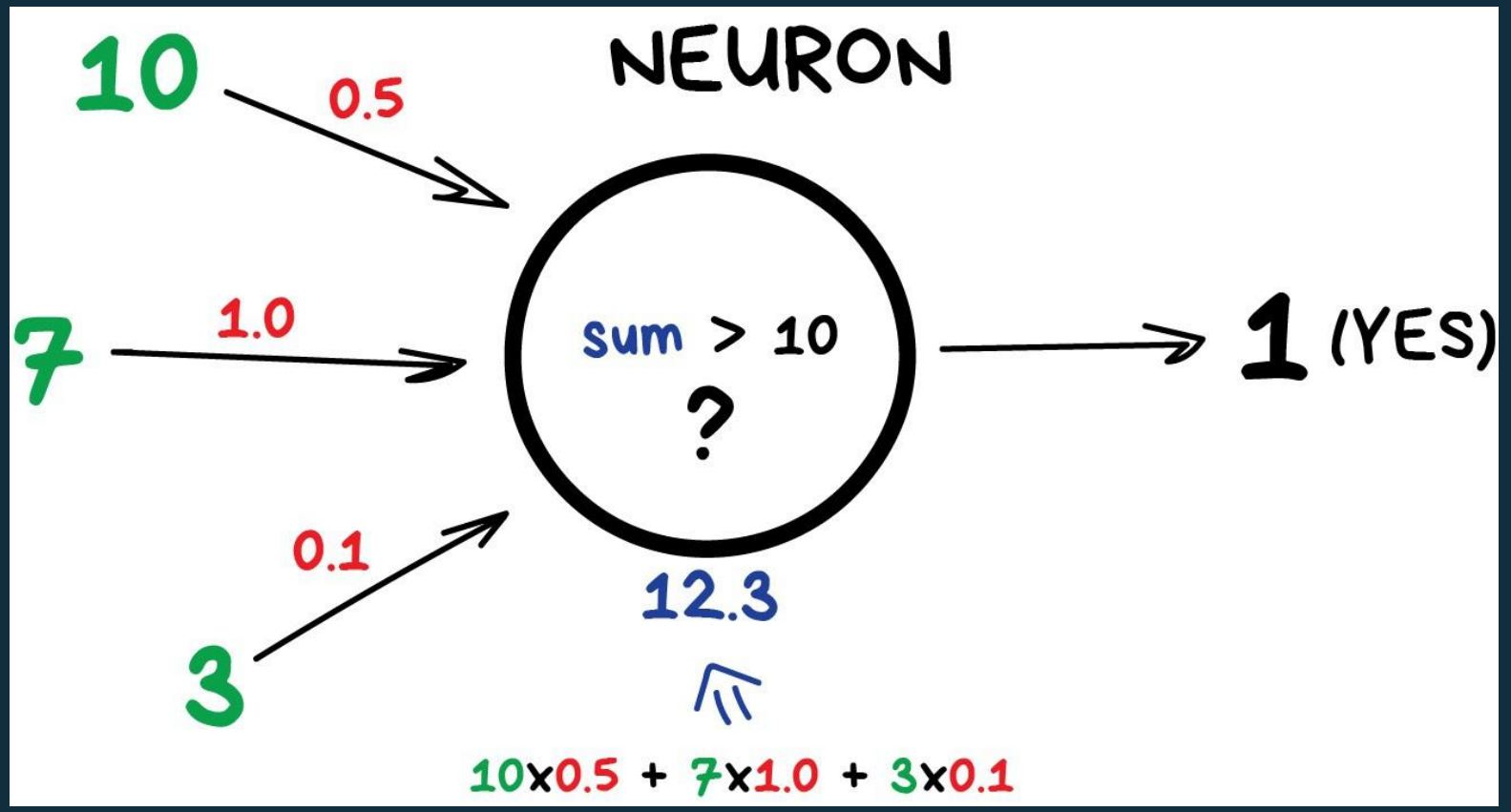


神经网络

从输入到输出

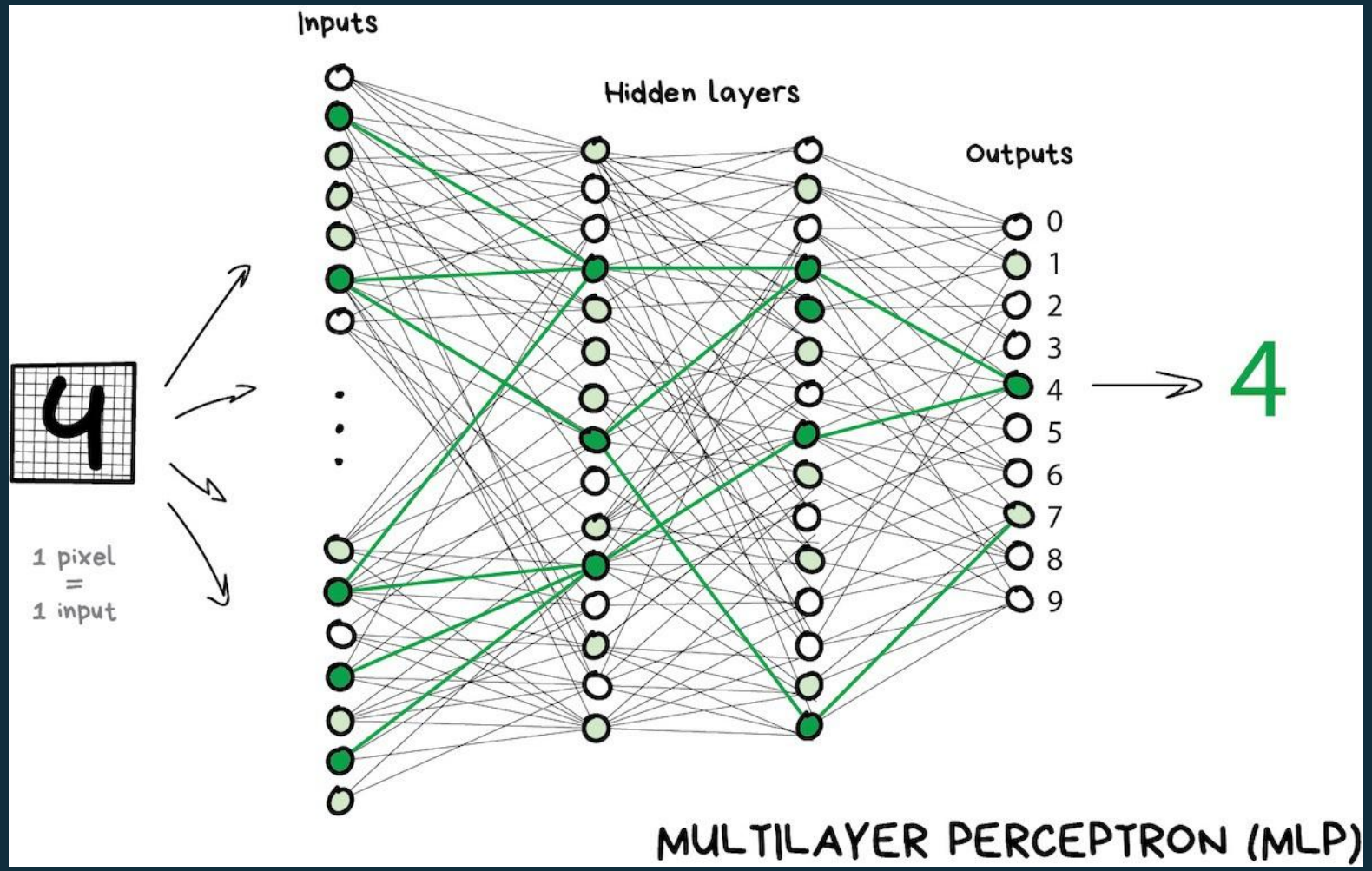


人工神经元



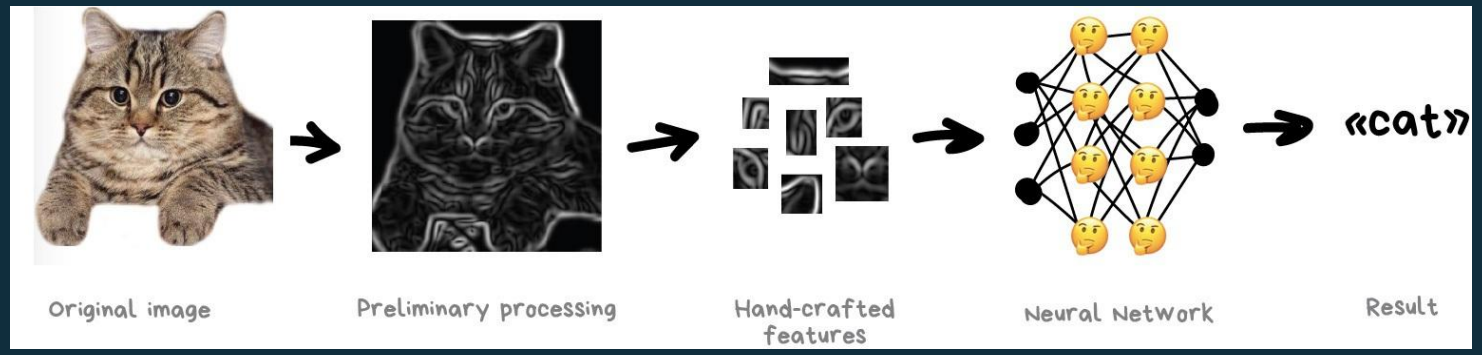
加权求和与输出

多层感知机

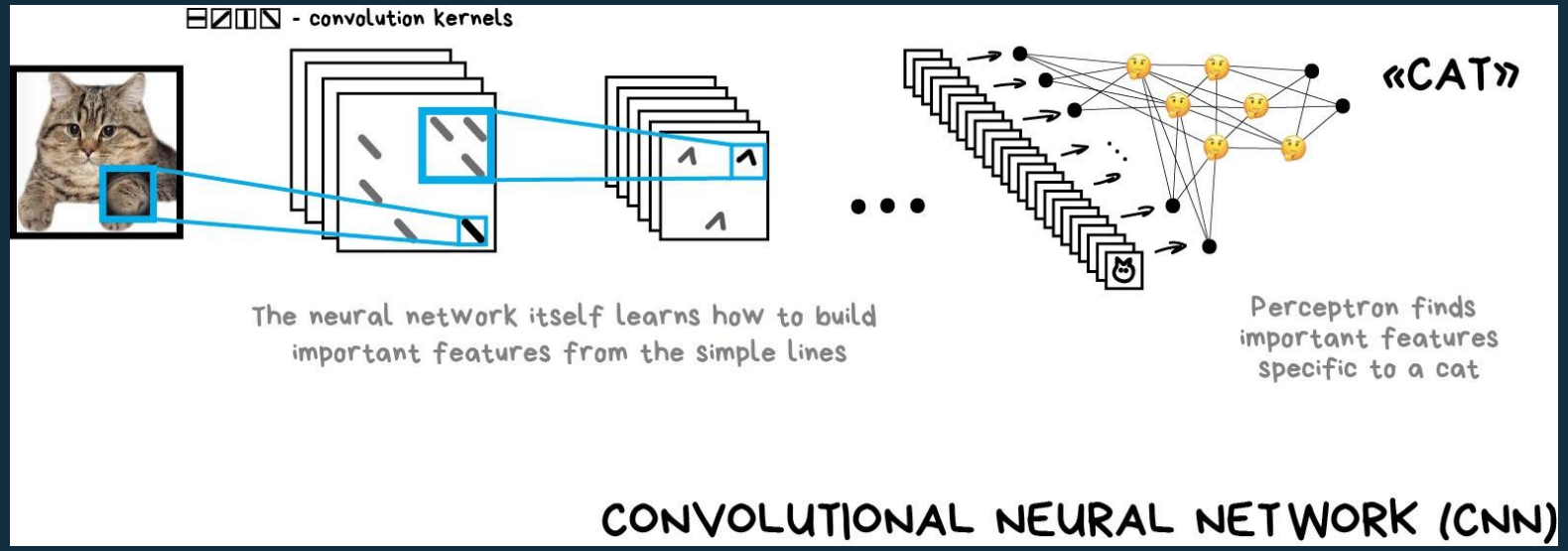


输入层、隐藏层与输出层

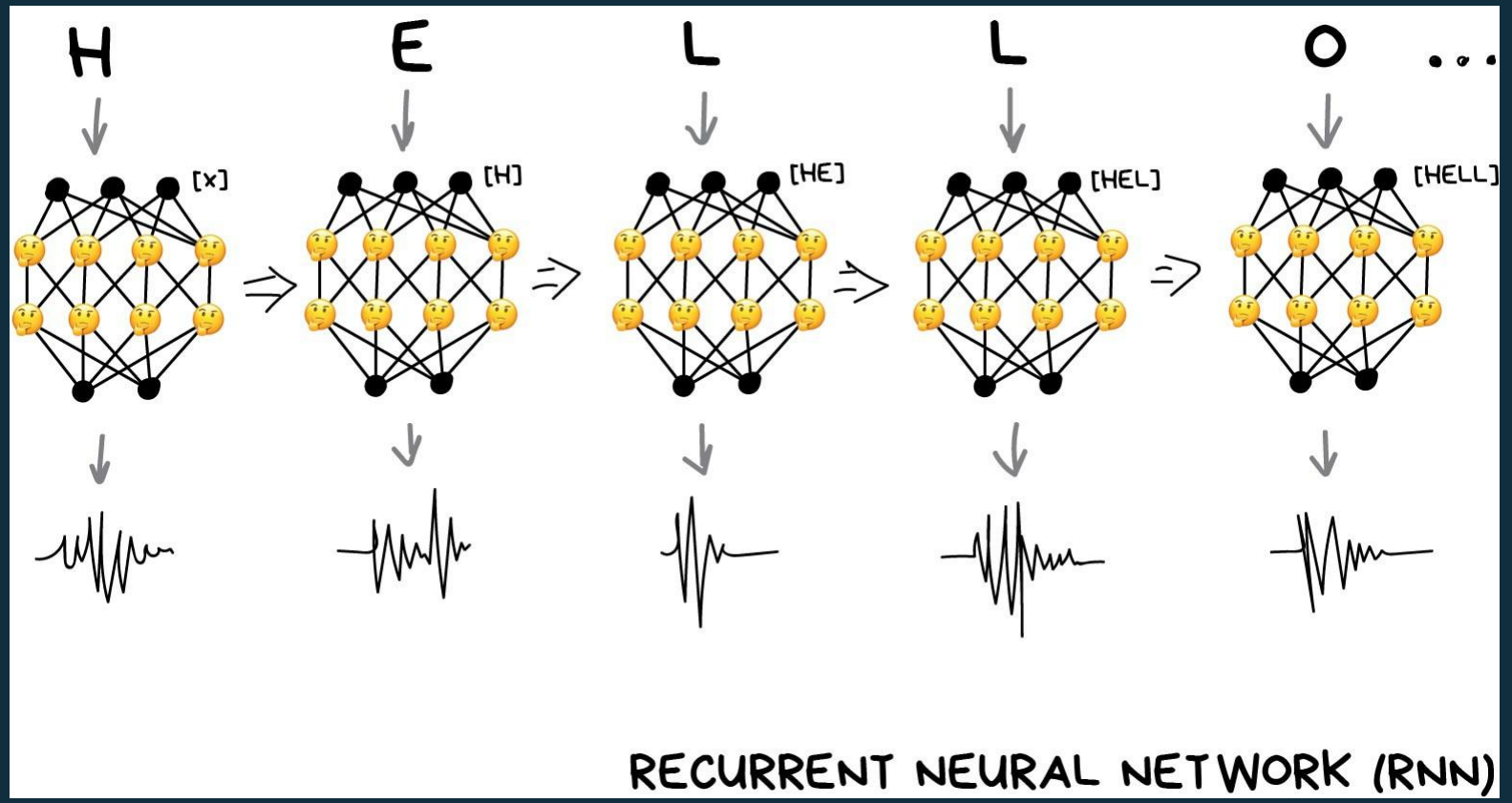
卷积神经网络



图像与特征



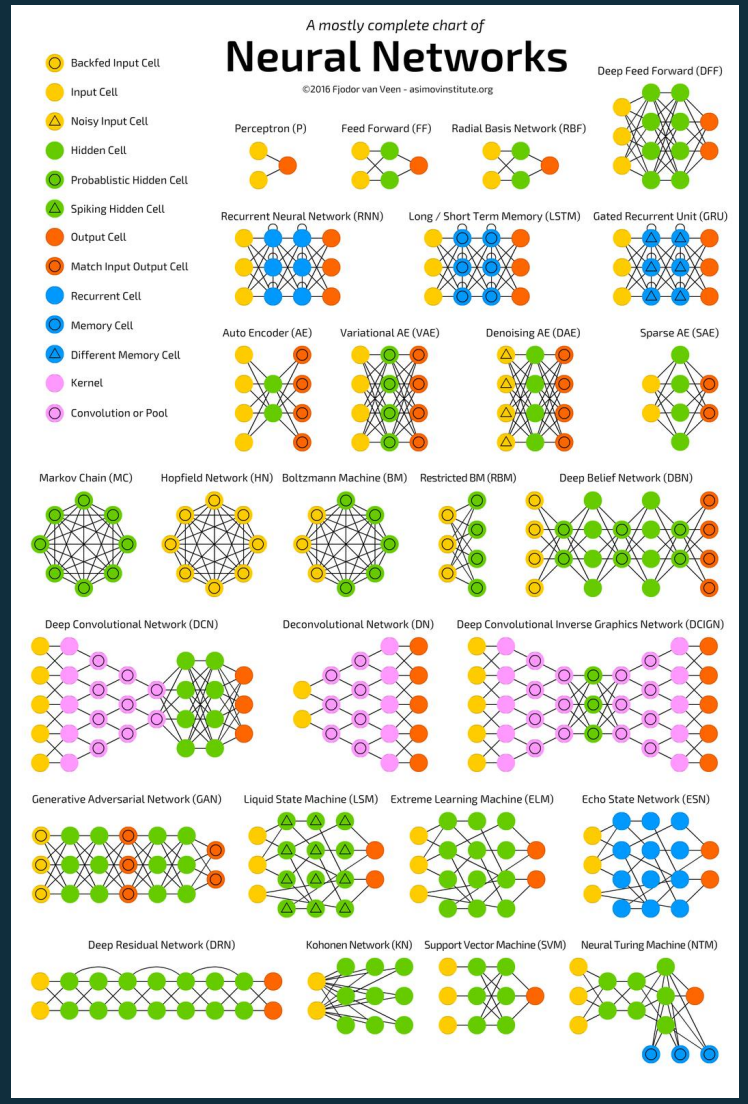
循环神经网络



序列与时间

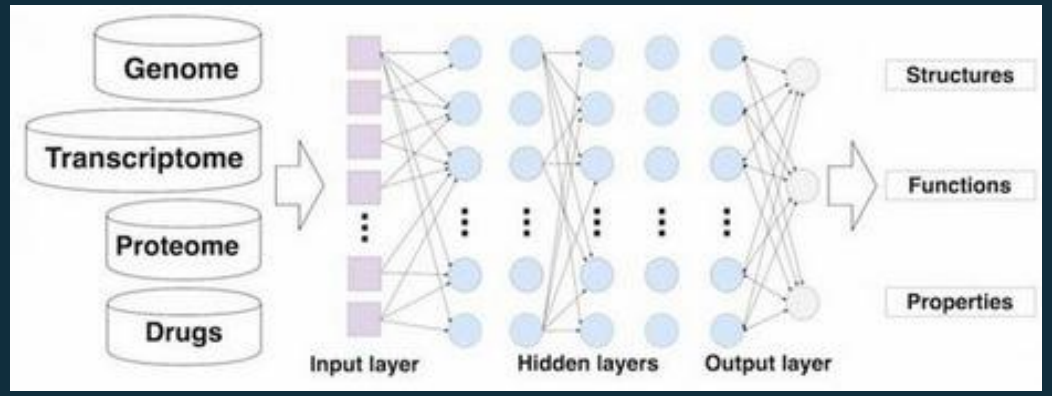
神经网络架构

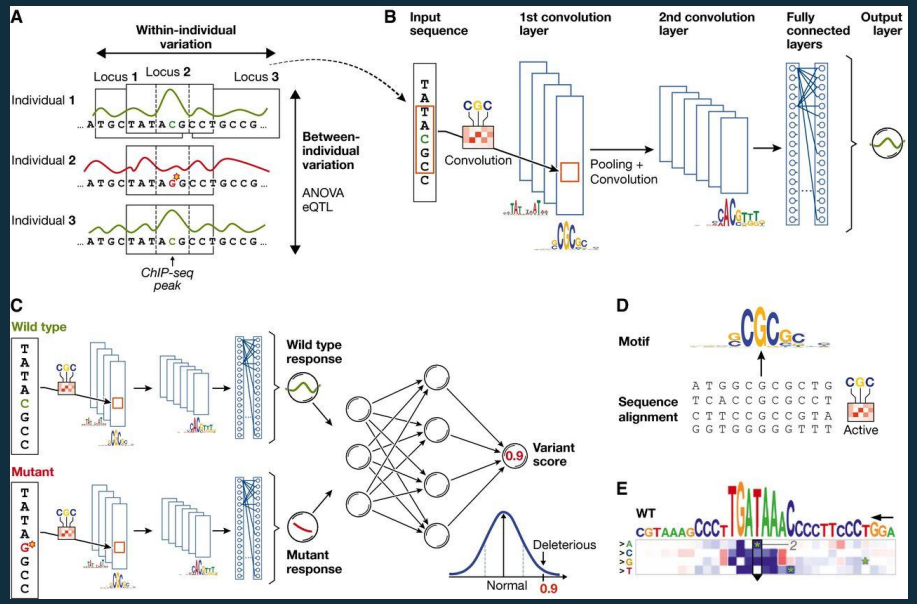
不同连接方式

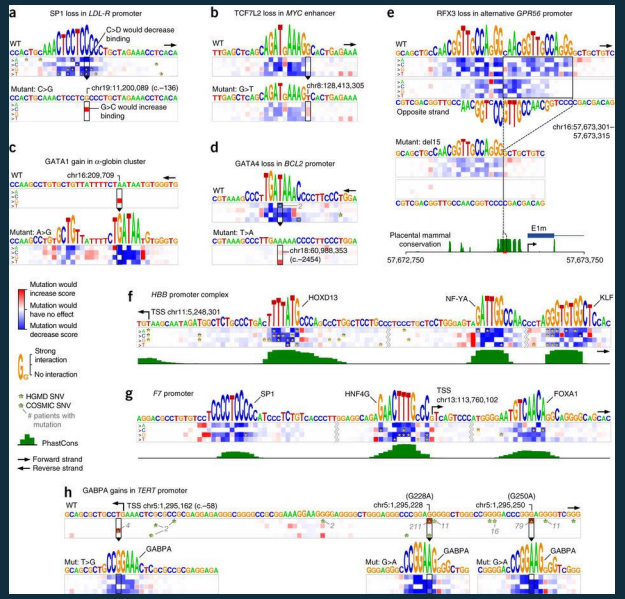


Deep Learning

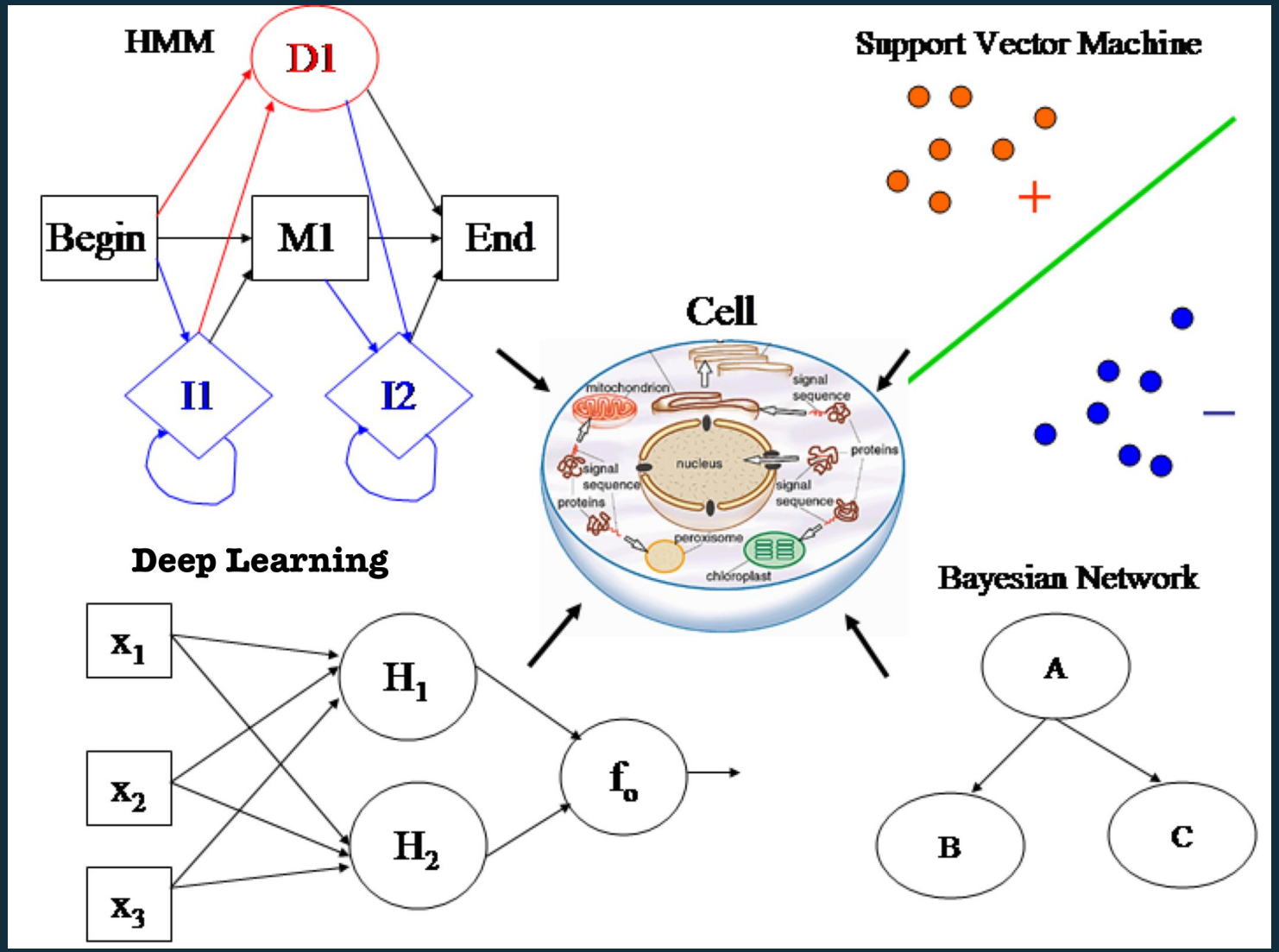
深度学习与生物信息学


Genome
Transcriptome
Proteome
Drugs
Structures
Functions
Properties
Input layer
Hidden layers
Output layer


A
Within-individual variation
Locus 1
Locus 2
Locus 3
Individual 1
Individual 2
Individual 3
Between-individual variation
ANOVA eQTL
ChIP-seq peak
B
Input sequence
1st convolution layer
2nd convolution layer
Fully connected layers
Output layer
Convolution
Pooling + Convolution
C
Wild type
Mutant
Wild type response
Mutant response
Variant score
0.9
Deleterious
Normal
D
Motif
Sequence alignment
Active
E
WT

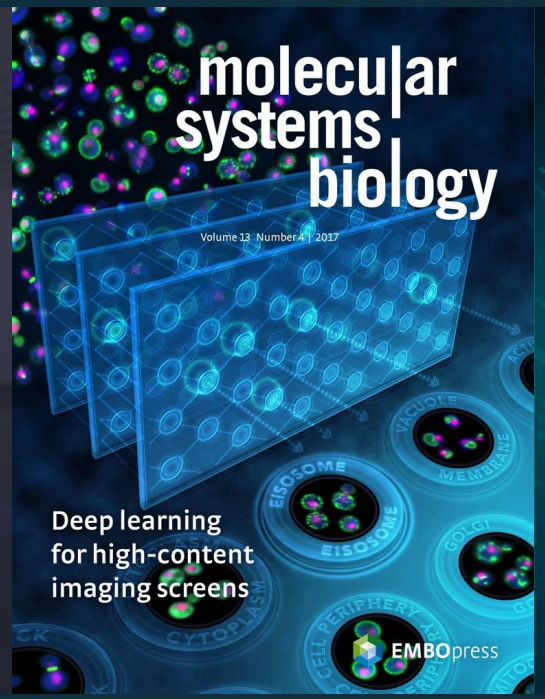

a
SP1 loss in LDL-R promoter
b
TCF7L2 loss in MYC enhancer
e
RFX3 loss in alternative GPR56 promoter
c
GATA1 gain in α-globin cluster
d
GATA4 loss in BCL2 promoter
f
HBB promoter complex
HOXD13
NF-YA
KLF
g
F7 promoter
SP1
HNF4G
FOXA1
h
GABPA gains in TERT promoter
GABPA

深度学习与生物信息学

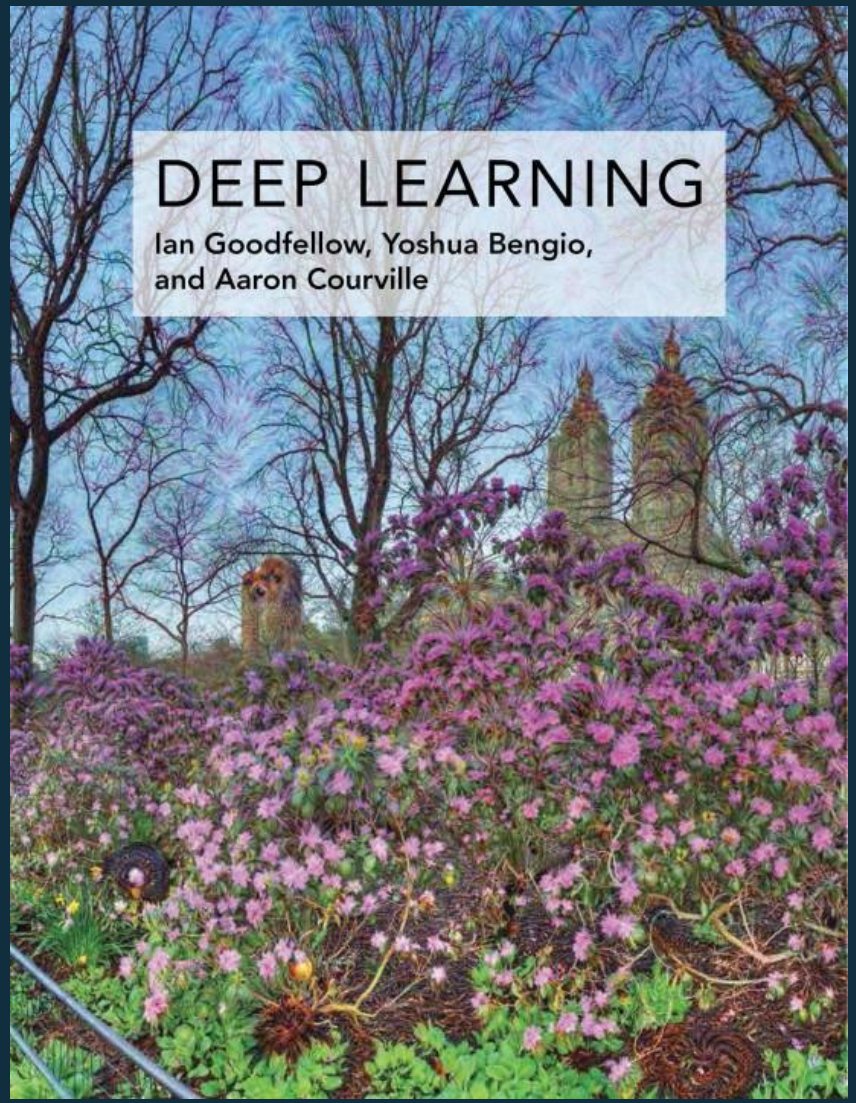

HMM
D1
Begin
M1
End
I1
I2
Support Vector Machine
+
−
Cell
mitochondrion
signal sequence
nucleus
proteins
peroxisome
chloroplast
Deep Learning
x1
x2
x3
H1
H2
fo
Bayesian Network
A
B
C

深度学习与生物信息学

- 高通量测序数据挖掘
- 蛋白和分子对接（药物设计）
- 生物影像分析
- 。。。



深度学习


DEEP LEARNING
Ian Goodfellow, Yoshua Bengio,
and Aaron Courville

生物信息学与生物统计学：

两种视角，四种技术，两类方法

两种视角

生物学视角
计算视角

四种技术

算法
数据
超算
深度学习

两类方法

经典软件
核心工具

LECTURE 01

生物统计学经典软件

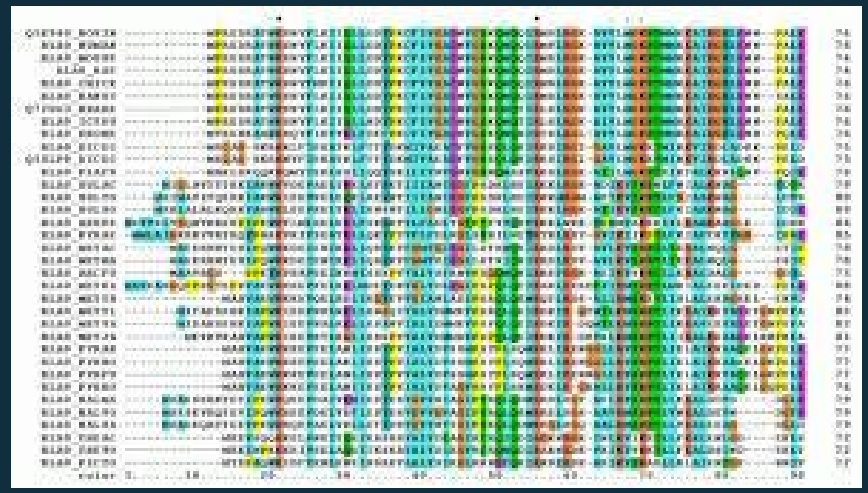
生物统计经典软件

MEME：http://meme-suite.org/

GenScan：http://genes.mit.edu/GENSCAN.html

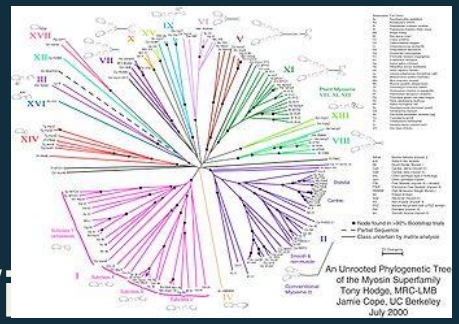
HMMalign：http://www.biology.wustl.edu/gcg/hmmeralign.html

高通量测序数据分析



生物统计经典软件

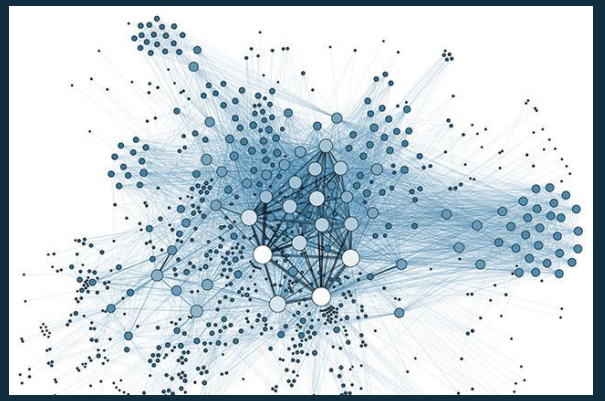
物种和基因进化分析



- iTOL：https://i
- MEGA：http://www.megasoftware.net/

生物统计经典软件

生物分子网络分析



➢Cytoscape：http://www.cytoscape.org/

生物统计经典软件

PCA analysis：http://biit.cs.ut.ee/clustvis/

DREAM Challenge：http://dreamchallenges.org/

降维、分子结构等分析

生物统计经典软件

基因组可视化：Genome Browser，(http://genome.ucsc.edu/), (tracks, annotations, etc.)

序列保守性：WebLogo，(http://weblogo.berkeley.edu/logo.cgi),

基因预测：MEME, (http://meme-suite.org/).

进化树：iTOL, (https://itol.embl.de/),

基因调控网络： GeneNetwork, (http://gn2.genenetwork.org/), Cytoscape, (https://cytoscape.org/),

代谢通路：KEGG, (https://www.kegg.jp/); iPATH, (https://pathways.embl.de/),

蛋白结构与功能：PDB，(http://www.rcsb.org); pFAM, (http://pfam.xfam.org/),

微生物组: EBI Magnify. (https://www.ebi.ac.uk/metagenomics/),

蛋白和小分子互作数据：STITCH，(http://stitch.embl.de/); STRING, (http://string-db.org),

药物数据库：DrugBank, (https://www.drugbank.ca/),

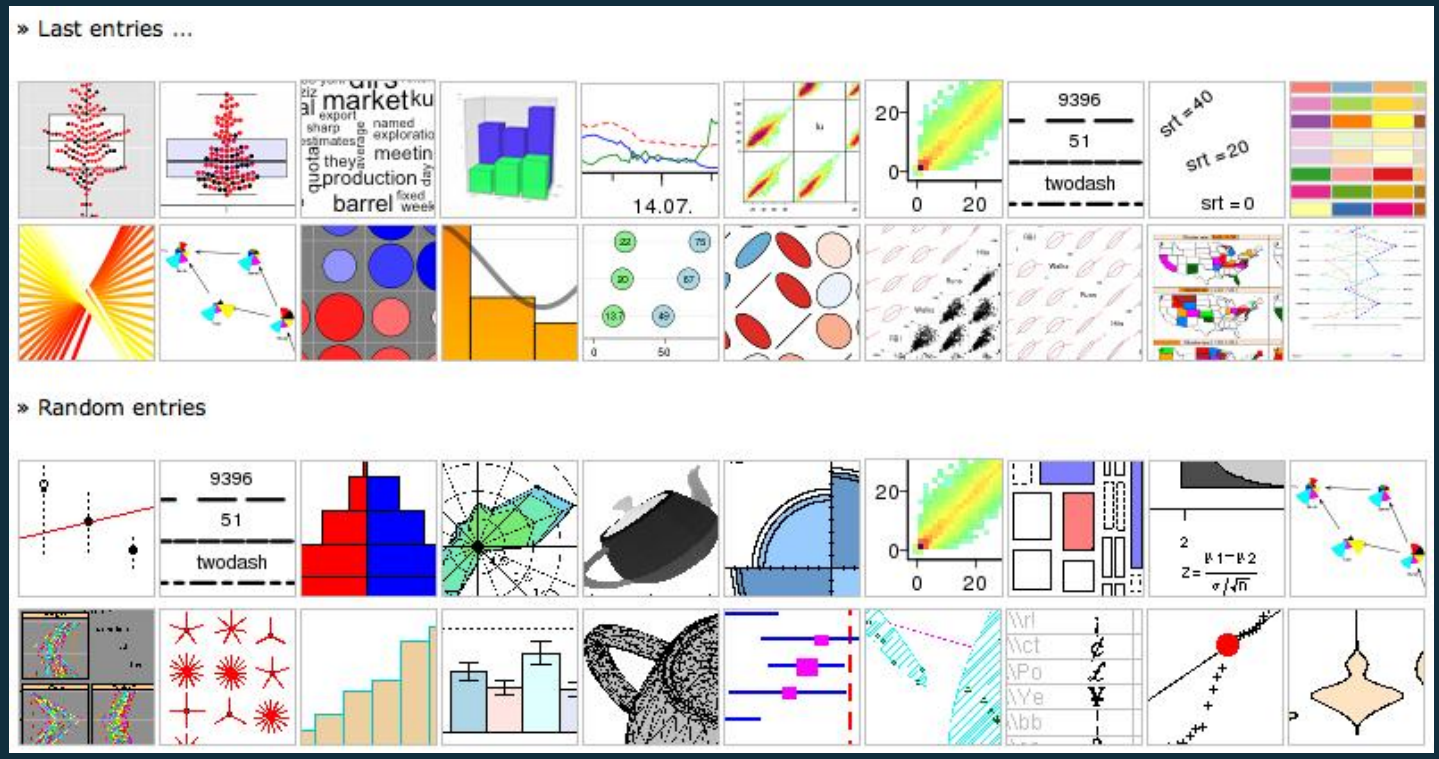
生物数据分析平台：Galaxy, (https://usegalaxy.org/),

生物数据可视化：Echart, (https://www.echartsjs.com/examples/zh/index.html),

LECTURE 01

生物统计学核心工具

R 与生物数据分析



http://blog.revolutionanalytics.com/2011/09/the-r-graph-gallery-goes-social.html

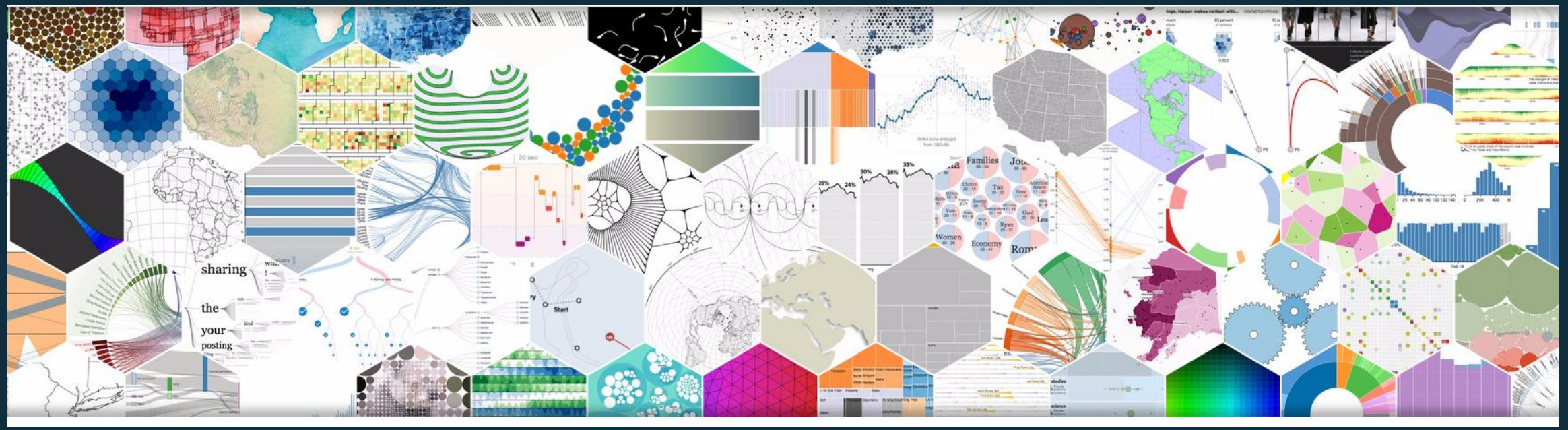
Python 与 Biopython

Python and Biopython：

https://www.python.org/

http://biopython.org/

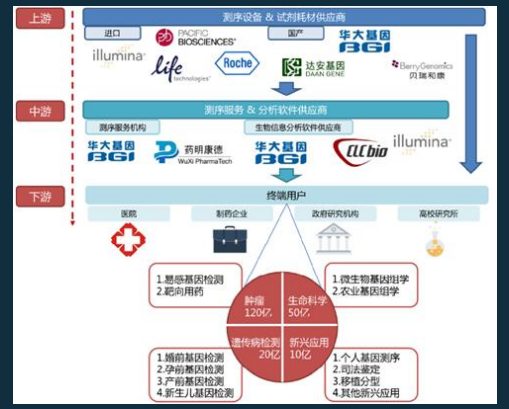
D3.js 数据可视化

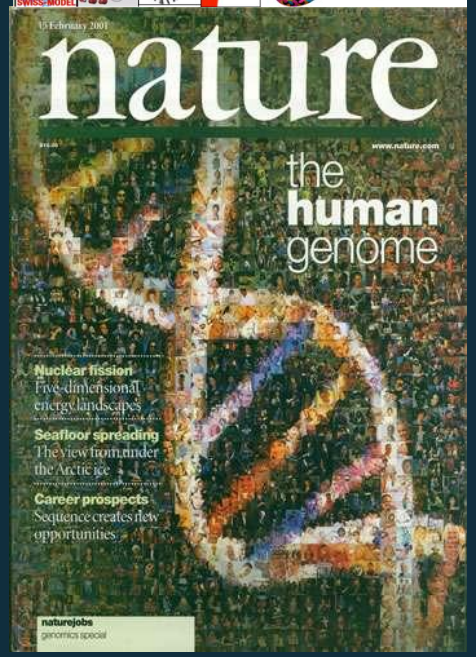


ECharts 数据可视化

Current status（现今态势）

很难找到
与生物信息学没有关系的
生物学和计算科学
研究和应用领域了。。。





Alphabet（谷歌）

Google 的基因组学梦想

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CORRESPONDENCE

23andMe and the FDA

N Engl J Med 2014; 370:2248-2249 | June 5, 2014 | DOI: 10.1056/NEJMc1404692

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To the Editor:

In their Perspective article (March 13 issue),[1] Annas and Elias state that the conflict between the genetic-testing company 23andMe and the Food and Drug Administration (FDA) concerns analytic and clinical validity, clinical utility, and ethical, legal, and social issues. However, their discussion is limited to a domestic U.S. perspective. After a person's raw genetic data have been determined from a DNA sample, the data are stored remotely and can be accessed easily anywhere in the world. For example, in Japan, maternal blood samples from Japanese mothers undergoing

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NATURE BIOTECHNOLOGY | NEWS

FDA approves 23andMe gene carrier test

Nature Biotechnology **33**, 435 (2015) | doi:10.1038/nbt0515-435a

Published online 12 May 2015

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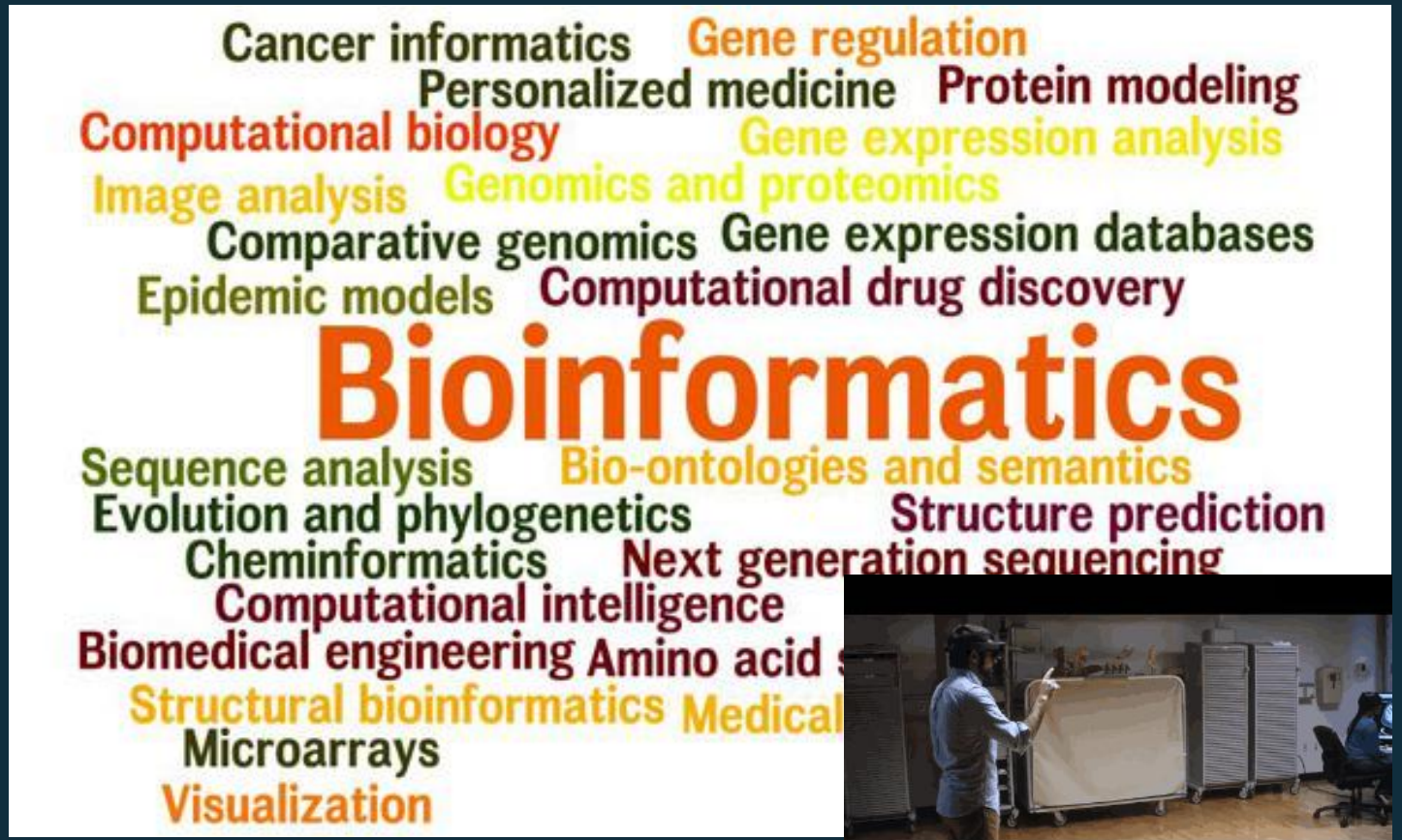
23andMe, based in Mountain View, California, has received word from the US Food and Drug Administration (FDA) that their Bloom syndrome carrier screening test was approved as a class II device. The approval came in February, 15 months after the personal genomics company received a cease and desist letter from the regulator for its genetic tests because the company was dispensing health-related information to consumers without having obtained marketing clearance. The FDA website lists class II devices as moderate risk, requiring some regulatory controls, putting carrier screening tests in the same category as condoms. This turnaround follows the company's

Future（未来）


Cancer informatics
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Gene expression analysis
Image analysis
Genomics and proteomics
Comparative genomics
Gene expression databases
Epidemic models
Computational drug discovery
Bioinformatics
Sequence analysis
Bio-ontologies and semantics
Evolution and phylogenetics
Structure prediction
Cheminformatics
Next generation sequencing
Computational intelligence
Biomedical engineering
Amino acid
Structural bioinformatics
Medical
Microarrays
Visualization

Future（未来）

我们是谁？我们从哪里来？我们到哪里去？